# *The Igusa Square Root*

The Borcherds determinant of $\phi_{0,1}$, its denominator algebra, and the $K3 \times E$ realization problem

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**Abstract**

For $X = K3 \times E$ the theorem-level physical input used here is the protected index. Right-moving localization in the K3 Ramond sector gives

$$Z_{\mathrm{K3}} = 2\phi_{0,1}, \qquad \phi_{0,1} = \sum f(n,l) q^n r^l.$$

Replacing the protected K3 index by its Borcherds half produces only a virtual $K_0$-determinant package. It does not produce a microscopic Hilbert space. Unitarity is used to localize the right-moving K3 supertrace before the index is taken; the index itself is a charge-graded virtual object that no longer requires Hilbert positivity. For virtual super vector spaces $\mathbb{U}_{n,l}$ with $\operatorname{sdim} \mathbb{U}_{n,l} = f(n,l)$, set $\mathcal{V}_{(n,l,m)} = \mathbb{U}_{nm,l}$. In the Borcherds cusp chamber the determinant is

$$\mathcal{D}_X(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

The constant 64 is the theta-constant normalization $[q^{1/2} r^{1/2} s^{1/2}]\Delta_5 = 64$; hence the monic Borcherds product is $D_5 = 64^{-1}\Delta_5$.

The same coefficients are the signed root supermultiplicities in the Gritsenko–Nikulin Borcherds–Kac–Moody denominator algebra. Under

$$\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$$

one has

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z), \qquad \operatorname{smult} \mathfrak{g}_{\Delta_5,\alpha(n,l,m)} = f(nm,l)$$

on the active support. Thus the denominator algebra supplies the algebraic square root target and its target-internal Borcherds bracket. It does not supply a compact BPS Hilbert space, compact Hall correspondences, an orientation, or a BPS operator product/local collision map.

The square root in the title is meant in Dirac's sense. The scalar OP/DT branch $Z^X_{\mathrm{OP/DT}} = -4096\,\Delta_5^{-2}$ is orientation-forgetful: the sign character is squared away. The recognition target is a first-order protected operator/algebra $\mathfrak{D}_X$ with protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = \Delta_5$ and scalar trace $\mathrm{Tr}_{\mathrm{prot}}((-1)^F \mathfrak{D}_X^{-2}) = -4096\,\Delta_5^{-2}$; the orientation/Pfaffian datum is the algebraic input that distinguishes $\Delta_5$ from $-\Delta_5$ and that the squared scalar cannot recover. Constructing $\mathfrak{D}_X$ is the open recognition problem; this paper formulates the target and the orientation-character constraint $\varepsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda^{2,1}_{II})$.

The scalar Behrend-weighted branch is imported from Oberdieck–Pixton, Oberdieck–Pandharipande, and Oberdieck. In OP's normalization $\chi_{10}^{\mathrm{OP}} = D_5^2$, so

$$\chi_{10}^{\mathrm{OP}} = (64^{-1}\Delta_5)^2 = 4096^{-1}\Delta_5^2, \qquad Z_{\mathrm{DT}}^X = Z_{\mathrm{OP}}^X = -\left(\chi_{10}^{\mathrm{OP}}\right)^{-1} = -4096\,\Delta_5^{-2}.$$

The factor $4096 = 64^2$ is a normalization conversion; the minus sign is the OP scalar convention. Neither fixes the automorphic reflection character or the compact orientation monodromy.

A compact realization would have to begin with an additive physical charge lattice $\Gamma_X^{\mathrm{phys}}$, the Gram map $\Pi_X : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$, and a holomorphic $E_3$-factorization algebra on $K3 \times E$. The Gram map is quadratic:

$$\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c').$$

Thus raw Gram descent cannot carry the full BKM bracket. The first structural correction is the normal-ordered extension

$$\widehat{\Gamma}_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c, T) = \Pi_X(c) - T,$$

for which $\overline{\Pi}_X$ is additive. A compact realization would then require a specified formality point, model-dependent homotopy/framing datum, quantum master equation, anomaly control, descent, and finite-first HN completion. It would then specialize along the K3 direction to a chiral object on $\mathrm{Ran}(E)$, solve the $\mathrm{Ran}(E)$-support-locality obstruction in the K3-integrated chiral shadow by mixed local/wrapped correspondences, and identify the Gram-descended protected primitive bracket with $\mathfrak{g}_{\Delta_5}$. This paper proves the virtual determinant, the lattice normal-ordering theorem, and the raw-descent no-go; records the imported scalar and denominator theorems; and formulates that compact Hall/factorization realization as an open recognition problem. The final part records the eight Clery–Gritsenko diagonal-divisor rows and the certificate required before any boundary row becomes a physical host.

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## 1 Introduction and claim strength

The local object comes first. A compact BPS theory on $X = K3 \times E$, if it exists in the form needed here, should produce protected local observables before any trace is taken. After twisting and passing to the protected index, the positive Hilbert space has collapsed to a charge-graded virtual object, but locality has not disappeared: disjoint insertions must still factorize and collisions must remember extensions.

Thus the natural order is

$$\text{protected locality} \longrightarrow \text{physical Mukai charge} \longrightarrow \text{normal-ordered Gram descent} \longrightarrow \text{compact/hybrid factorization} \longrightarrow \text{pri}$$
$$\longrightarrow \text{Borcherds denominator} \longrightarrow \text{OP scalar square} \longrightarrow \text{eight-row host certificates.}$$

*1.0.0.1 Protected locality.* The protected index is theorem-level. Right-moving localization of the unitary K3 sigma model gives

$$Z_{\mathrm{K3}} = 2\phi_{0,1}, \qquad \phi_{0,1} = \sum f(n,l) q^n r^l.$$

The half-index used by the Borcherds lift is represented only in $K_0(\mathrm{sVect}_{\mathrm{fd}})$. Its coefficients are signed superdimensions, not dimensions of a constructed compact D-brane Hilbert space.

*1.0.0.2 Compact and hybrid factorization.* A compact realization would first supply the additive physical charge lattice $\Gamma_X^{\mathrm{phys}}$, the non-additive Gram map $\Pi_X : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$, and a charge-completed holomorphic $E_3$-factorization algebra

$$\mathcal{A}_X^{E_3}, \qquad X = K3 \times E,$$

then a chain-level K3-to-$E$ specialization

$$\mathcal{F}_X \simeq \mathrm{Sp}^{\mathrm{ch}}_{K3,E}(\mathcal{A}_X^{E_3}) \in \mathrm{Fact}(\mathrm{Ran}(E)).$$

This is not reconstructed from the scalar partition function or from the Borcherds product. There is also a geometric obstruction: a one-dimensional subscheme with positive elliptic degree projects onto all of $E$. Hence a naive support-local object on $\mathrm{Ran}(E)$ sees only projection-finite sectors. The full Igusa $s$-direction requires finite-stage local/wrapped correspondence data for positive elliptic degree: the elliptic-degree map $b_R$, rigidified wrapped prequotients with anchors, local-local, ordered local/wrapped, and wrapped-wrapped extension stacks, full flag coherence, quotient-after-correspondence descent, and protected integration with trace degree $s^{b_R}$. A detached global Hecke/Fock factor is only a determinant-level repair. It becomes relevant to the compact realization only after it is represented by those wrapped prequotients and acts through mixed Hall correspondences before the reduced $E$-quotient. This is open.

*1.0.0.3 Primitive recognition.* The microscopic algebra sought from such a compact realization is first the upstairs protected primitive bracket

$$[\ ,\ ] : \mathrm{Prim}^{\mathrm{prot}}_{c}(\mathcal{F}_X) \otimes \mathrm{Prim}^{\mathrm{prot}}_{c'}(\mathcal{F}_X) \longrightarrow \mathrm{Prim}^{\mathrm{prot}}_{c+c'}(\mathcal{F}_X), \qquad c, c' \in \Gamma^{\mathrm{phys}}_X.$$

The recognition certificate proved later is conditional: if the Dirac–Igusa certificate $\mathfrak{K}_X$ exists, including the oriented Hall object, protected integration, cocycle-corrected Gram descent $\overline{\Pi}^{\Theta}_{X,*}$, a conilpotent source coalgebra $C_X$ with finite source filtration and collision coproduct, the bar comparison $b_X$, the cobar map $\Theta_{\mathrm{Kos}}$, presentation-level primitive Lie comparison, no-extra-relations theorem, completed PBW compatibility, and radical quotient, then

$$H^0\,\mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X) \cong \mathfrak{g}_{\Delta_5}$$

and the supplied chiral cobar comparison identifies $\Omega^{\mathrm{ch}}_E C_X$ with the formal current envelope $\mathsf{Den}_{\Delta_5,E}$. This is a recognition criterion, not an existence theorem and not a reconstruction from the determinant.

*1.0.0.4 Borcherds denominator.* The target algebra is imported from the Borcherds–Gritsenko–Nikulin construction. With

$$\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2},$$

the visible active coefficients satisfy

$$\mathrm{smult}\,\mathfrak{g}_{\Delta_5,\alpha(n,l,m)} = f(nm,l), \qquad \mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z).$$

This determines signed root supermultiplicities and the denominator identity. It does not determine compact Hall constants.

*1.0.0.5 OP scalar square.* The reduced scalar branch is imported from Oberdieck–Pixton, Oberdieck–Pandharipande, and Oberdieck:

$$Z^X_{\mathrm{DT}}(T,Q,P) = Z^X_{\mathrm{OP}}(P,Q,T) = -4096\,\Delta_5(Z)^{-2}.$$

The constants are fixed once. The theta-constant form satisfies

$$[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64, \qquad D_5 = 64^{-1}\Delta_5.$$

OP's monic product is $\chi^{\mathrm{OP}}_{10} = D_5^2$, hence

$$\chi^{\mathrm{OP}}_{10} = 4096^{-1}\Delta_5^2, \qquad Z^X_{\mathrm{OP}} = -(\chi^{\mathrm{OP}}_{10})^{-1} = -4096\Delta_5^{-2}.$$

The number 64 is a theta-leading coefficient. The number $-4096$ is the OP scalar normalization. Neither proves the compact orientation character. The comparison

$$\epsilon_o = \nu_{\Delta_5}$$

between geometric orientation monodromy and the automorphic reflection character remains part of the open Dirac–Igusa realization certificate.

*1.0.0.6 Eight-row host certificates.* The diagonal-divisor boundary problem concerns the eight Clery–Gritsenko forms

$$\Delta_5, \Delta_2, \Delta_1, \Delta_{1/2}, \nabla_3, \nabla_2, \nabla_{3/2}, Q_1.$$

Rows 1–4 have Gritsenko–Nikulin denominator theorems. Rows 5–8 have Clery–Gritsenko products and Govindarajan CHL/WKB denominator theorems; $F_7 = \nabla_{3/2}$ is on the required $\mu_4$-multiplier cover. A physical host for any boundary row requires a certificate fixing the modular group or cover, character, divisor, cusp Jacobi inputs, product lattice, Weyl vector, real roots, reflection character, physical charge lattice, protected Gram-index, orientation, reduced scalar theorem if one is claimed, and primitive algebra comparison. Without that certificate the row is only an automorphic product plus any separately proved denominator theorem.

| Structural level | Claim strength | Content |
|---|---|---|
| Virtual determinant | proved here from $K_0$ data plus imported Borcherds product | $\mathcal{D}_X = \Delta_5$. No canonical half-Hilbert space or operator product is constructed. |
| Scalar Behrend square | imported theorem | Oberdieck–Pixton, Oberdieck–Pandharipande, and Oberdieck give $Z^X_{\mathrm{DT}} = Z^X_{\mathrm{OP}} = -4096\Delta_5^{-2}$ in the reduced quotient-by-$E$, Behrend-weighted setting. |
| BKM denominator | imported theorem | Gritsenko–Nikulin give $\mathfrak{g}_{\Delta_5}$ and $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$. |
| Compact Hall/factorization realization | conditional/open | If $\mathfrak{K}_X$ exists, the primitive and chiral comparisons follow by the recognition criterion. Existence, $\mathrm{Ran}(E)$ hybrid locality, protected integration, and orientation monodromy are open. |
| Eight-row host certificates | framework; mostly open | The catalogue and denominator rows are imported where cited. Physical row maps and scalar squares require row-by-row certificates. |

*1.0.0.7 Theorem-status ledger.* The manuscript uses five claim levels.

| Lane | Status | Named content |
|---|---|---|
| Automorphic determinant | proved/imported | Fock determinant plus Borcherds–Gritsenko–Nikulin product gives $\mathcal{D}_X = \Delta_5$. |
| Physical charge descent | proved on lattices | The Mukai–Gram dictionary, primitive lifts, cocycle $B$, and normal-ordered map $\overline{\Pi}_X$ are proved as lattice statements. |
| Raw descent | proved no-go | Raw quadratic $\Pi_X$-pushforward cannot realize the full BKM real-root strings. |
| Compact Dirac–Igusa object | conditional/open | $\mathfrak{D}_X$, strong orientation, Weyl cocycle, hybrid factorization, and primitive recognition are construction data. |
| Scalar OP/DT square | imported | Oberdieck–Pixton and Oberdieck identify the Behrend-weighted reduced scalar branch with $-4096\Delta_5^{-2}$. |

All super vector spaces in the representation-theoretic constructions are over $\mathbb{C}$. Real vector spaces occur only after tensoring lattices with $\mathbb{R}$ to discuss cones, chambers, and period domains. Products may be written over $\Gamma_{\text{eff}}$; the active factors are exactly the triples in $\Gamma_{\text{act}}$, where the exponent $f(nm, l)$ is non-zero.

# Part I

# The automorphic square root

## 2 The physical question

A charged BPS state is not, in the first instance, a local field. It lies in a superselection sector. Its charge is measured at infinity, and its existence depends on a stability chamber. Thus the ordinary local operator algebra of an untwisted four-dimensional theory is not the primary object of this paper.

At the level of the protected index the situation is different. After taking the cohomology of the chosen supercharge, long multiplets cancel. The object which remains is a charge-graded virtual object, not a positive Hilbert space, whose Euler characteristic is stable under deformation. The index remembers the cancellation law of a unitary theory, not its positive inner product.

For $X = K3 \times E$ the expected local object in a compact realization is a supplied holomorphic $E_3$-factorization algebra

$$\mathcal{A}_X^{E_3}, \qquad X = K3 \times E.$$

The auxiliary notation $\Phi_3^{\mathrm{FA}}(X)$, when used, is only a placeholder for this source datum; no such functor is constructed in this paper. No external $\Phi_3^{\mathrm{FA}}$ formalism is used here as an existence theorem unless it supplies the chain-level local observables, factorization maps, descent, anomaly/QME control, and compact charge completion required below. In a compact realization $\mathcal{A}_X^{E_3}$ is the object whose disjoint holomorphic insertions factorize. The elliptic curve $E$ enters only after a supplied chain-level $K3$-specialization whose output is denoted

$$\mathcal{F}_X \in \mathrm{Fact}(\mathrm{Ran}(E)).$$

This is the first compact-realization obstruction. The projection-finite sector, where $p_E(\mathrm{Supp}\, A)$ is a finite subset of $E$, is naturally local on $\mathrm{Ran}(E)$. A one-dimensional sector of positive elliptic degree is not. If $C \subset K3 \times E$ has positive elliptic degree, then $(p_E)_*[C]$ is a positive multiple of $[E]$; hence $p_E(C)$ is a one-dimensional closed subset of the irreducible curve $E$, and so $p_E(C) = E$. Such a class is wrapped or global over $E$, not localized at finitely many points of $E$.

This is the $\mathrm{Ran}(E)$-support-locality obstruction in the K3-integrated chiral shadow. Naive support-local factorization on $\mathrm{Ran}(E)$ sees projection-finite sectors, but by itself it misses the positive $s$-direction of the Igusa product, the direction which records elliptic degree after the OP/Borcherds variable conversion. A compact realization must therefore choose at least one finite categorical repair: projection-finite local factorization, rigidified wrapped prequotients, local-local, ordered local/wrapped, and wrapped-wrapped extension correspondences, full flag coherence, quotient-after-correspondence descent, and protected integration whose $s$-degree is the elliptic degree. A separate global Fock/Hecke mode is insufficient unless it is realized by these wrapped prequotients and coupled to the local sector by mixed Hall actions before the reduced $E$-quotient. This is a realization obstruction. It is not a defect in the Borcherds determinant or in the Gritsenko–Nikulin denominator identity.

Thus, in a realization, the curve-level rule is not imposed externally. It is the one-dimensional chiral shadow of the holomorphic $E_3$-observable: well-separated protected insertions on $E$ multiply, while collisions remember extension and bound-state data. In primitive form this is the expected bracket

$$[\ ,\ ] : \mathrm{Prim}_c^{\mathrm{prot}}(\mathcal{F}_X) \otimes \mathrm{Prim}_{c'}^{\mathrm{prot}}(\mathcal{F}_X) \longrightarrow \mathrm{Prim}_{c+c'}^{\mathrm{prot}}(\mathcal{F}_X), \qquad c, c' \in \Gamma_X^{\mathrm{phys}}.$$

The Igusa square-root problem is to construct or recognize this primitive protected algebra and its Gram descent. The determinant below gives the decategorified signed Gram-index shadow that such a realization would have:

$$\mathrm{sdim}\, \mathrm{Prim}_\gamma^{\mathrm{prot}}(\overline{\Pi}_{X,*}^{\Theta}\mathcal{F}_X) = f(nm, l), \qquad \gamma = (n, l, m).$$

The Borcherds–Kac–Moody denominator algebra gives the algebraic candidate:

$$H^0 \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X) \overset{?}{\cong} \mathfrak{g}_{\Delta_5}.$$

The theorem-level content of the paper is the determinant and denominator comparison. The displayed isomorphism is the compact $E_3$/Hall/chiral realization problem.

The square root is therefore not an analytic ambiguity in $\Delta_5^2$. It is the normalized automorphic determinant of the virtual protected one-particle index. The scalar square forgets the automorphic reflection character; a compact Hall orientation realizing that character is part of the open Dirac–Igusa certificate.

*Schema of the local protected observable algebra*

*Definition* 2.1 *(Local protected observable algebra; proposal-only schema).* For an open $U \subset X = K3 \times E$, let $\mathfrak{M}_c(U)$ denote the derived moduli stack of compactly supported B-branes on $U$ of physical Mukai charge $c \in \Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S$, in the sense of Pantev–Toen–Vaquie–Vezzosi [51], so that $\mathfrak{M}_c(U)$ carries a $(-1)$-shifted symplectic structure; let $\Phi_c$ denote the perverse sheaf of vanishing cycles of any local d-critical chart, in the sense of Brav–Bussi–Dupont–Joyce–Szendroi [9] (BPS sheaf, alias DT sheaf), well defined globally as a pre-D-critical structure on $\mathfrak{M}_c(U)$; let $o_c$ denote the orientation line, a square root of the determinant line of the cotangent complex of $\mathfrak{M}_c(U)$, supplied as part of the Hall datum. The *local protected observable algebra of $X$ on $U$* is the proposal

$$\mathcal{A}_X(U) := \bigoplus_{c \in \Gamma_X^{\mathrm{phys}}} H_*^{\mathrm{BM}}\big(\mathfrak{M}_c(U),\ \Phi_c \otimes o_c\big),$$

graded by physical charge. For disjoint $U_1, \dots, U_k \subset X$, the proposed factorization map is

$$\mathcal{A}_X(U_1) \otimes \cdots \otimes \mathcal{A}_X(U_k) \longrightarrow \mathcal{A}_X(U_1 \sqcup \cdots \sqcup U_k),$$

defined chain-level by Massey–Saito Thom–Sebastiani for vanishing cycles

$$\Phi_{F_1 \oplus \cdots \oplus F_k} \simeq \Phi_{F_1} \boxtimes \cdots \boxtimes \Phi_{F_k}$$

on disjoint d-critical charts, together with the Ext-vanishing $\operatorname{Ext}^*(\mathcal{F}, \mathcal{G}) = 0$ for branes $\mathcal{F}, \mathcal{G}$ with disjoint supports. This is the proposed compact $E_3$ source $\mathcal{A}_X^{E_3}$ of the present paper, in the sense that its disjoint-supports factorization on opens of $X = K3 \times E$ is the holomorphic $E_3$-factorization datum.

*Remark* 2.2. The components are external assumptions: PTVV $(-1)$-shifted symplectic structure on derived moduli of B-branes [51]; BBDJS vanishing-cycle perverse sheaf and its Thom–Sebastiani isomorphism (extending Massey 2001 and Saito 1990 to the d-critical setting) [9]; orientation lines from the Joyce–Upmeier orientation theory [35] or Kontsevich–Soibelman [39]. The factorization map relies on disjoint-supports Ext-vanishing for compactly supported branes (absent for non-compact supports, where one recovers chiral algebra structure, not commutative factorization). Compactness on $K3 \times E$ is open because the relevant moduli of branes are typically non-quasicompact in $\Lambda_S$-charge; the sectorial truncation of Proposition 10.39 is the candidate truncation.

## 3 The one-particle index and its determinant

For the determinant calculation set

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix}, \qquad q = e^{2\pi i z_1}, \qquad r = e^{2\pi i z_2}, \qquad s = e^{2\pi i z_3}.$$

The Gram-index lattice is

$$\Gamma_{\text{ind}} = \mathbb{Z}^3, \qquad \gamma = (n, l, m), \qquad x^\gamma = q^n r^l s^m.$$

The standard cusp polarization is the ordered limit

$$0 < |s| \ll |q| \ll |r|^{-1} \ll 1.$$

It determines the semigroup

$$\Gamma_{\text{eff}} = \{(n, l, m) \mid m > 0,\ n \geq 0,\ l \in \mathbb{Z}\} \cup \{(n, l, 0) \mid n > 0,\ l \in \mathbb{Z}\} \cup \{(0, l, 0) \mid l < 0\}.$$

$m > 0$ is positive because $s$ is the smallest parameter; if $m = 0$, then $n > 0$ is positive because $q$ is next; if $m = n = 0$, positivity is $l < 0$ because $|r|^{-1} \ll 1$. This is a chamber convention for the completed product, not a microscopic charge positivity statement.

*Definition* 3.1 *(Ramond-sector K3 index and its half).* Let $\mathcal{H}_{\text{RR}}(S)$ be the Ramond–Ramond Hilbert space of a unitary $\mathcal{N} = (4, 4)$ sigma model on a K3 surface $S$. Its central charges are

$$c = \overline{c} = 6.$$

This Hilbert space is a supplied worldsheet input for the protected K3 index. It is not a construction of the compact $K3 \times E$ D-brane Hilbert space. Choose a right-moving Ramond supercharge $Q_R$ with

$$Q_R^2 = 0, \qquad \{Q_R, Q_R^\dagger\} = 2\left(\overline{L}_0 - \frac{\overline{c}}{24}\right).$$

The K3 elliptic genus is the protected supertrace

$$Z_{\text{K3}}(\tau, z) = \operatorname{Tr}_{\mathcal{H}_{\text{RR}}(S)} (-1)^{F_L + F_R} q^{L_0 - c/24} \overline{q}^{\overline{L}_0 - \overline{c}/24} r^{J_0}.$$

The BPS space in degree $(n, l)$ is

$$\mathcal{H}^{\text{BPS}}_{n,l}(S) := H^\bullet(\{\psi \mid (L_0 - c/24)\psi = n\psi,\ J_0\psi = l\psi\}, Q_R).$$

Lemma 3.2 *(Right-moving localization).* Then

$$Z_{\text{K3}}(\tau, z) = \sum_{n \geq 0,\ l \in \mathbb{Z}} \operatorname{sdim} \mathcal{H}^{\text{BPS}}_{n,l}(S) q^n r^l.$$

*Proof.* If $\overline{L}_0 - \overline{c}/24 > 0$, the displayed anticommutator pairs states with opposite $(-1)^{F_R}$ parity. Their supertrace cancels. The unpaired states are the $Q_R$-harmonic representatives, hence the graded superdimension of $Q_R$-cohomology gives the elliptic genus. □

*Remark* 3.3 *(Where unitarity enters).* Unitarity is used here before the index is taken: it gives the adjoint relation which pairs positive right-moving energy states and makes the supertrace localize. After passing to the protected index, the object is virtual. A coefficient $f(n,l)$ is a superdimension, not a positive dimension. Negative coefficients record an odd excess in the protected supertrace, or in the chosen $K_0$-representative. They do not determine the two parity dimensions without an additional Hilbert-space or denominator-algebra parity datum.

*Definition* 3.4 *(Grothendieck half-index representative).* Eichler–Zagier normalization [21]:

$$Z_{\mathrm{K3}}(\tau,z) = 2\phi_{0,1}(\tau,z),$$
$$\phi_{0,1}(\tau,z) = \sum_{n\geq 0,\, l\in\mathbb{Z}} f(n,l)q^n r^l.$$

Thus

$$\operatorname{sdim}\mathcal{H}^{\mathrm{BPS}}_{n,l}(S) = 2f(n,l).$$

The arbitrary Grothendieck representative

$$\mathbb{U}_{n,l} \in K_0(\mathrm{sVect}_{\mathrm{fd}}), \qquad \operatorname{sdim}\mathbb{U}_{n,l} = f(n,l),$$

is the normalized half-index class used by the Borcherds lift. It is not asserted to be a canonical half of $\mathcal{H}^{\mathrm{BPS}}_{n,l}(S)$ as a Hilbert space. As a $K_0$-model one may take $\mathbb{U}_{n,l}$ to be $\mathbb{C}_{\bar{0}}^{f(n,l)}$ when $f(n,l)\geq 0$, and $\mathbb{C}_{\bar{1}}^{-f(n,l)}$ when $f(n,l)<0$. No canonical choice of basis, parity lift, or operator product is implied.

$$\phi_{0,1} = r^{-1} + 10 + r + O(q), \qquad \frac{f(0,0)}{2} = 5.$$

Lemma 3.5 *(Fock trace and superdeterminant).* Let

$$V = \bigoplus_{\lambda\in P} V_\lambda$$

be a $P$-graded super vector space with finite-dimensional homogeneous pieces and let $a_\lambda = \operatorname{sdim} V_\lambda$. Put

$$\mathcal{F}(V) = \mathrm{Sym}(V_{\bar{0}}) \otimes \bigwedge(V_{\bar{1}}).$$

As a formal character,

$$\operatorname{sch}\mathcal{F}(V) := \sum_{\mu\in P} \operatorname{sdim}\mathcal{F}(V)_\mu\, x^\mu = \prod_{\lambda\in P}(1-x^\lambda)^{-a_\lambda}.$$

The inverse character is the superdeterminant

$$\operatorname{sdet}_V(1-x) = \prod_{\lambda\in P}(1-x^\lambda)^{a_\lambda}.$$

*Proof.*

$$V_\lambda = \mathbb{C}_{\bar{0}}: \qquad \operatorname{sch}\mathrm{Sym}(V_\lambda) = (1-x^\lambda)^{-1}.$$
$$V_\lambda = \mathbb{C}_{\bar{1}}: \qquad \operatorname{sch}\bigwedge(V_\lambda) = 1-x^\lambda.$$

Tensor product over a homogeneous basis proves the product. □

When $P = \Gamma_{\mathrm{eff}}$, all products below are taken in the cusp-completed semigroup algebra determined by $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$. Every coefficient receives only finitely many contributions in this completion.

$$x^{(n,l,m)} = q^n r^l s^m.$$

Form now the virtual Borcherds determinant package. The third index $m$ is the second-quantized/Borcherds degree. The positive $m$ sector carries the Hecke–DMVV exponents $f(nm, l)$. The $m = 0$ sector is a cusp/Weyl boundary package carrying the repeated boundary indices $f(0, l)$. Neither sector is a microscopic one-particle Hilbert space.

$$\mathcal{V}^{\mathrm{bulk}} := \bigoplus_{\substack{(n,l,m)\in\Gamma_{\mathrm{eff}}\\ m>0}} \mathbb{U}_{nm,l}, \qquad \mathcal{V}^{\mathrm{cusp}} := \bigoplus_{(n,l,0)\in\Gamma_{\mathrm{eff}}} \mathbb{U}_{0,l}.$$

$$\mathcal{V} = \mathcal{V}^{\mathrm{bulk}} \oplus \mathcal{V}^{\mathrm{cusp}}, \qquad \mathcal{V}_{(n,l,m)} = \mathcal{V}_{n,l,m} := \mathbb{U}_{nm,l}, \qquad \operatorname{sdim} \mathcal{V}_{n,l,m} = f(nm, l).$$

This is a definition in $K_0$, not a microscopic state space. The $m > 0$ sector is the $K_0$-level virtual half of the DMVV product [19], obtained by replacing $Z_{\mathrm{K3}} = 2\phi_{0,1}$ by $\phi_{0,1}$. The normalized Fock character in the Borcherds chamber is

$$\operatorname{sch} \mathcal{F}(\mathcal{V}) = \prod_{\gamma\in\Gamma_{\mathrm{eff}}} (1 - x^\gamma)^{-\operatorname{sdim} \mathcal{V}_\gamma} = \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{-f(nm,l)}.$$

$$\operatorname{sdet}_{\mathcal{V}}(1 - x) = \prod_{\gamma\in\Gamma_{\mathrm{eff}}} (1 - x^\gamma)^{\operatorname{sdim} \mathcal{V}_\gamma} = \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

The sector $m > 0$ is the normalized half of the second-quantized DMVV index. The terms with $m = 0$ are cusp and Weyl-chamber boundary factors of the Borcherds product; the determinant reads only their signed indices $f(0, l)$. The normalized cusp-form section below proves the leading coefficient which fixes the Weyl monomial

$$v_0(Z) = 64 q^{1/2} r^{1/2} s^{1/2}.$$

$$v_0 \notin \bigoplus_{\gamma\in\Gamma_{\mathrm{eff}}} \mathcal{V}_\gamma.$$

*Definition 3.6 (Normalized $K_0$-determinant).*

$$\mathcal{D}_X(Z) = v_0(Z) \operatorname{sdet}_{\mathcal{V}}(1 - x) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

$$\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}}(Z) = \mathcal{D}_X(Z)^{-1} = \frac{1}{64} q^{-1/2} r^{-1/2} s^{-1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{-f(nm,l)}.$$

The object constructed here is the virtual $K_0$-determinant; compact BPS meaning is part of the separate realization problem.

Proposition 3.7 *(Determinant normalization).*

$$[q^{1/2} r^{1/2} s^{1/2}]\, \mathcal{D}_X = 64,$$

$$\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}} = \frac{1}{64} q^{-1/2} r^{-1/2} s^{-1/2} \operatorname{sch} \mathcal{F}(\mathcal{V}),$$

$$\mathcal{D}_X = 64 q^{1/2} r^{1/2} s^{1/2} \operatorname{sdet}_{\mathcal{V}}(1 - x).$$

*Proof.*

$$\operatorname{sch}\mathcal{F}(\mathcal{V}) = \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1-q^n r^l s^m)^{-f(nm,l)}, \qquad \operatorname{sdet}_{\mathcal{V}}(1-x) = \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1-q^n r^l s^m)^{f(nm,l)}.$$

$$v_0(Z)\operatorname{sdet}_{\mathcal{V}}(1-x) = \mathcal{D}_X, \qquad v_0(Z)^{-1}\operatorname{sch}\mathcal{F}(\mathcal{V}) = \mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}}.$$

□

Proposition 3.8 *(Virtuality of the square root).* The displayed construction produces a class

$$[\mathcal{V}] \in K_0(\Gamma_{\mathrm{eff}}\text{-graded finite-dimensional super vector spaces})$$

together with its Fock determinant. By itself this determinant does not construct a Hilbert space, a vertex algebra, a holomorphic $E_3$-factorization algebra, or a chiral factorization algebra whose character is $\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}}$.

All denominator statements use only the equalities

$$\operatorname{sdim}\mathcal{V}_{(n,l,m)} = f(nm,l), \qquad \mathcal{D}_X = v_0\operatorname{sdet}_{\mathcal{V}}(1-x).$$

The denominator algebra and formal current envelope are constructed later from the Borcherds root datum, not from a canonical half of the K3 Hilbert space.

*Proof.* By definition, $\mathcal{V}_{(n,l,m)} = \mathbb{U}_{nm,l}$ is specified only through its class in $K_0(\mathrm{sVect}_{\mathrm{fd}})$ and through the integer $\operatorname{sdim}\mathbb{U}_{nm,l} = f(nm,l)$. The Fock trace and superdeterminant lemma depends only on these signed dimensions. No operation in the construction supplies composition maps, operator products, locality, or a Hilbert completion. □

Proposition 3.9 *(Dirac/Pfaffian recognition target).* The class $[\mathcal{V}] \in K_0$ and its Fock determinant $\mathcal{D}_X = v_0\operatorname{sdet}_{\mathcal{V}}(1-x) = \Delta_5$ record signed Gram-degree multiplicities and a formal infinite-product; they do not produce an operator with a spectrum. The structural upgrade demanded by the protected-locality side of the constellation is an *open recognition target*:

> construct from the compact $K3\times E$ Dirac–Igusa certificate of Definition 10.44 a first-order protected operator/algebra $\mathfrak{D}_X$ acting on a chiral protected module, whose charge-graded Pfaffian satisfies
>
> $$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = \mathcal{D}_X = \Delta_5.$$
>
> Its scalar trace, in the OP normalization of Definition 11.3 below, satisfies
>
> $$\mathrm{Tr}_{\mathrm{prot}}\big((-1)^F(\mathfrak{D}_X)^{-2}\big) = -4096\,\Delta_5^{-2}.$$

Existence of $\mathfrak{D}_X$ is open; it would convert the $K_0$ identity of Proposition 3.8 into a first-order protected statement. The target carries the sign-square ambiguity of Conjecture 5.3 and the orientation monodromy of Open Problem 5.7; choosing a square root of $(\mathfrak{D}_X)^{-2}$ requires the orientation datum of that open problem and is not fixed by the OP scalar.

*Definition* 3.10 *(Finite type-II Pfaffian wall certificate).* A finite certificate for the type-II Pfaffian local model consists, for each of the three simple type-II walls $\mathbb{H}_{\partial_i}$, $i = 1, 2, 3$, of the following data.

(Pf1) A generic wall point $x_i$ in the reduced compact Hall sector and an $s_{\delta_i}$-equivariant real analytic slice $(U_i, u_i)$ through $x_i$, with

$$\mathbb{H}_{\delta_i} \cap U_i = \{u_i = 0\}, \qquad s_{\delta_i}(u_i) = -u_i.$$

(Pf2) A splitting of the reduced determinant complex on the slice into its tangent determinant factor and a normal factor

$$K^{\mathrm{red}}|_{U_i} \simeq K_i^{\mathrm{tan}} \oplus K_i^{\mathrm{nor}},$$

with $s_{\delta_i}$ preserving the tangent determinant factor.

(Pf3) An oriented quasi-isomorphism of the normal factor to a finite sum of odd rank-one Pfaffian crossings

$$K_i^{\mathrm{nor}} \simeq \bigoplus_{j=1}^{N_i^{\mathrm{Pf}}} \left[\mathbb{R} \xrightarrow{u_i} \mathbb{R}\right].$$

In the induced trivialization the normal Pfaffian section is $u_i^{N_i^{\mathrm{Pf}}}$ times a unit.

(Pf4) The three normal-mode ranks $N_i^{\mathrm{Pf}} = N_{\delta_i}^{\mathrm{Pf}}$ are

$$N_1^{\mathrm{Pf}} = N_2^{\mathrm{Pf}} = N_3^{\mathrm{Pf}} = 1.$$

This is a finite geometric certificate: three wall charts, three anti-invariance checks $s_{\delta_i}(u_i) = -u_i$, three tangent-factor compatibility checks, and three odd Pfaffian normal-mode ranks. These ranks are local Dirac/Pfaffian data; they are not read from the automorphic divisor. The certificate constructs neither the global reduced orientation line nor the Weyl lift. It uses no scalar normalization of $\Delta_5$, no factor 64, and no OP/DT scalar sign.

Proposition 3.11 *(Local obstruction certificate for* (O2)*).* Fix one of the three simple type-II roots $\delta \in \{\delta_1, \delta_2, \delta_3\}$. A geometric proof of the rank-one Pfaffian model at $\mathbb{H}_\delta$ is not a consequence of $(\delta, \delta) = 2$, of the automorphic divisor order, or of the Maass character. In the reduced compact Hall framework it must supply the following local deformation-obstruction calculation.

(L1) Let $x_\delta = [C_\delta]$ be a generic wall object. An $s_\delta$-equivariant real analytic d-critical/Kuranishi chart

$$(U_\delta, x_\delta, u_\delta, K_\delta^{\mathrm{red}}) \longrightarrow \mathfrak{M}_{C_\delta}^{\mathrm{red}}$$

is supplied, where $K_\delta^{\mathrm{red}}$ is the real form of the cosection-reduced self-Ext complex

$$\mathrm{RHom}_{\mathrm{red}}(C_\delta, C_\delta)$$

in that chart, the wall is exactly $U_\delta \cap \{u_\delta = 0\}$, and

$$s_\delta(u_\delta) = -u_\delta, \qquad T_{x_\delta}(U_\delta \cap \mathbb{H}_\delta) = T_{x_\delta} U_\delta^{s_\delta}.$$

(L2) A splitting in the $s_\delta$-equivariant derived category

$$K_\delta^{\mathrm{red}} \simeq K_\delta^{\mathrm{tan}} \oplus K_\delta^{\mathrm{nor}}$$

whose tangent factor carries the wall tangent determinant and whose normal factor is the anti-invariant normal determinant factor. Equivalently, the extension class

$$\eta_\delta \in \mathrm{Ext}^1_{D^{s_\delta}(U_\delta)}(K_\delta^{\mathrm{nor}}, K_\delta^{\mathrm{tan}})$$

vanishes after the reduced $E$-quotient and the chosen quotient orientation trivializations are imposed.

(L3) In an oriented trivialization of the normal Pfaffian line,

$$\mathrm{Pf}(K_\partial^{\mathrm{nor}}) = v_\partial\, u_\partial^{N_\partial^{\mathrm{Pf}}}, \qquad v_\partial(x_\partial) \neq 0, \qquad s_\partial(v_\partial) = v_\partial,$$

and $K_\partial^{\mathrm{nor}}$ is oriented quasi-isomorphic to

$$\bigoplus_{j=1}^{N_\partial^{\mathrm{Pf}}} \left[\mathbb{R} \xrightarrow{u_\partial} \mathbb{R}\right].$$

The rank-one assertion used in (O2) is the further equality

$$N_\partial^{\mathrm{Pf}} = 1.$$

If (L1)–(L3) hold, then the local Hall/Pfaffian sign is

$$s_\partial^* \mathrm{Pf}(K_\partial^{\mathrm{nor}}) = (-1)^{N_\partial^{\mathrm{Pf}}} \mathrm{Pf}(K_\partial^{\mathrm{nor}}),$$

while the squared determinant section is fixed. If the unit $v_\partial$ has a nontrivial reflection character, that character multiplies the displayed sign and belongs to the orientation-transport datum $(\mathrm{O1})^+$. If the splitting obstruction $\eta_\partial$ does not vanish, the tangent and normal determinants cannot be separated and no rank-one normal sign has been computed. If $N_\partial^{\mathrm{Pf}} \neq 1$, the order-one automorphic divisor is at best a target check and the local Hall model required by (O2) is false or incomplete. Thus the exact missing inputs are the three charts, the three reduced self-Ext complexes, the three equivariant splittings, the three invariant Pfaffian units, and the three rank computations $N_{\partial_i}^{\mathrm{Pf}} = 1$. In any compact source claim these local data must also come from the same finite hybrid correspondence system that supplies the positive elliptic-degree sector; this is the compatibility obstruction of Proposition 10.21.

Theorem 3.12 *(Conditional Dirac–Pfaffian normalization and orientation character).* Suppose the following data items are granted on the reduced compact critical Hall/CoHA sector of $X = K3 \times E$ of Definition 5.4.

(D0) *Compact first-order certificate.* A finite-stage first-order $D_0$-certificate in the sense of Definition 10.40 is supplied. Thus, for every $R$ in the chosen cofinal HN subsystem, the datum contains a finite Hall/factorization stage $D_{0,R}$ on the finite support $\widehat{\Gamma}_R$, a finite target test window $\Gamma_R^{\mathrm{test}}$, successor transition maps whose composites give the smaller-stage maps, source/observable comparison maps, and a vanishing stage residual tuple $\mathfrak{D}_{D_0,R}$. Only after these finite data, transition cocycles, and support/parity residuals vanish may one form the HN-completed state object. The completed object is part of the supplied certificate: it carries its charge completion by $\widehat{\Gamma}_X$, a source observable shadow on $\mathrm{Ran}^{\mathrm{hyb}}(E)$, a protected first-order operator/algebra $\mathfrak{D}_X$, a Pfaffian line with normalized section, and a $\Theta_\Pi$-corrected descent whose primitive algebra is $\Gamma_{\mathrm{gram}}$-graded with signed superdimensions

$$\mathrm{sdim}\, P^{\Pi}_{X,(n,l,m)} = f(nm, l) \qquad ((n,l,m) \in \Gamma_{\mathrm{eff}}).$$

This is an assumption of the theorem. The proved lattice part is the normal-ordered homomorphism $\overline{\Pi}_X$; the imported target part is the Borcherds–Gritsenko–Nikulin denominator algebra; the compact state-side construction remains open unless the finite stages, transition maps, source/observable comparison, and residual trivializations of Definition 10.40 are supplied.

(O1) *Strong multiplicative reduced orientation datum.* There exists a Joyce–Upmeier-type orientation line [35]

$$o_C \in \mathrm{Pic}(\mathfrak{M}_X^{or}), \qquad o_C^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(C, C),$$

in the sense of Definition 4.3, satisfying

$$o_{C\oplus C'} \simeq o_C \otimes o_{C'} \quad \text{compatibly with the reduced extension correspondences,}$$

and equipped with the quotient-descent package for the reduced $E$-quotient. The Weyl-equivariant lift is a separate additional datum, isolated in $(\mathrm{O1})^+$, and is transported by Gram-projection $\Pi_X : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$. Here descent under the reduced $E$-quotient is part of the datum and means the simultaneous vanishing, on every stratum used by the Hall correspondences, of

$$\alpha_C^{\mathrm{red}} \in H^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2), \qquad \alpha_C^{E,\mathrm{free}} = a_1(C)u_1 + a_2(C)u_2 \in H^2(BE; \mathbb{F}_2),$$

and of the finite-stabilizer classes

$$\beta_C^{E,N} \in H^2(BE[N]; \mathbb{F}_2).$$

These are degree-two gerbe obstruction classes. Their vanishing is not the same datum as a trivial finite-stabilizer action on an already linearized line. After the degree-two class has been null-trivialized, the residual choices of finite-stabilizer linearization form an $H^1(BE[N]; \mathbb{F}_2)$-torsor; the trivial quotient orientation requires the zero point of this torsor. This $H^1$-choice is part of the quotient-orientation package transported in $(\mathrm{O1})^+$. Thus (O1) is not merely an ordinary Picard-line assertion: it consists of an ordinary square root in Pic, null-trivializations of the reduced and equivariant degree-two gerbes, and an equivariant line whose finite-stabilizer degree-one characters

$$\lambda_C^{E,N} \in H^1(BE[N]; \mathbb{F}_2)$$

are all chosen to be zero after the corresponding gerbes have vanished. For $N = 2$, with $E[2] = \langle e_1, e_2 \rangle$ and $e_3 = e_1 + e_2$, this is the Klein-four condition

$$\beta_C^{E,E[2]} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2, \qquad (b_{20}, b_{11}, b_{02}) = (r_1, r_1 + r_2 + r_3, r_2),$$

where

$$\mathrm{res}_{\langle e_i \rangle} \beta_C^{E,E[2]} = r_i \tau_i^2.$$

If this class vanishes, an $E[2]$-linearization still has character

$$\lambda_C^{E[2]} = \lambda_1 x_1 + \lambda_2 x_2 \in H^1(BE[2]; \mathbb{F}_2),$$

with cyclic restrictions

$$\rho_1 = \lambda_1, \qquad \rho_2 = \lambda_2, \qquad \rho_3 = \lambda_1 + \lambda_2.$$

Trivial $E[2]$-linearization means $\rho_1 = \rho_2 = \rho_3 = 0$. The bits $r_i$ live in degree two; the bits $\rho_i$ live in degree one. For even $N$ with $2^a \parallel N$, the mod-2 residual is a class on the two-primary stabilizer

$$E[N]_{(2)} \simeq (\mathbb{Z}/2^a)^2.$$

The case $a = 1$ is the displayed Klein-four calculation. For $a \geq 2$,

$$H^*(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2, \quad x_i^2 = 0,$$

and hence

$$H^2(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2) = \mathbb{F}_2\langle y_1, x_1x_2, y_2\rangle.$$

Writing

$$\beta^{E,N}_{C,(2)} = A_1^{(N)} y_1 + A_{12}^{(N)} x_1x_2 + A_2^{(N)} y_2,$$

the order-two cyclic restrictions see

$$\mathrm{res}_{\langle h_1\rangle}\beta^{E,N}_{C,(2)} = A_1^{(N)}\tau_1^2, \quad \mathrm{res}_{\langle h_2\rangle}\beta^{E,N}_{C,(2)} = A_2^{(N)}\tau_2^2, \quad \mathrm{res}_{\langle h_3\rangle}\beta^{E,N}_{C,(2)} = (A_1^{(N)} + A_2^{(N)})\tau_3^2,$$

where $h_i = 2^{a-1}g_i$ for a basis $g_1, g_2$ of $(\mathbb{Z}/2^a)^2$ and $h_3 = h_1 + h_2$. The coefficient $A_{12}^{(N)}$ is invisible to these cyclic order-two restrictions. Thus the $E[2]$ vector $(r_1, r_2, r_3)$ detects only the two-torsion residual; it is not a computation of $\beta^{E,N}_C$ for even $N \geq 4$. For $a \geq 2$, even after $\beta^{E,N}_{C,(2)} = 0$, the remaining two-primary linearization choices form

$$H^1(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2) = \mathbb{F}_2\langle x_1, x_2\rangle,$$

and the trivial linearization requires that character to vanish on the chosen two-primary generators. Invariance of the underlying line under translations of $E$ is not a substitute for these quotient-linearization data.

(O1)$^+$ *Weyl-equivariant determinant-line transport.* The orientation line of (O1) is equipped with coherent lifts of the three type-II wall transports

$$\tau_i : o_C \longrightarrow s^*_{\partial_i} o_C, \qquad i = 1, 2, 3,$$

but the word "coherent" is load-bearing. A choice of lifts over words in the simple reflections is first only a projective determinant-line action: for $w, w' \in W^{(2)}(\Lambda^{2,1}_{II})$ one has a composition defect

$$w'^*\tau_w \circ \tau_{w'} = (-1)^{c_o(w,w')}\tau_{ww'}, \qquad c_o \in Z^2\Big(W^{(2)}(\Lambda^{2,1}_{II});\mathbb{F}_2\Big).$$

Its class

$$[c_o] \in H^2\Big(W^{(2)}(\Lambda^{2,1}_{II});\mathbb{F}_2\Big)$$

is the Weyl determinant-line cocycle obstruction. The datum (O1)$^+$ requires $[c_o] = 0$, a choice of 1-cochain killing $c_o$, and normalization $\tau_i^2 = 1$ for $i = 1, 2, 3$. Only then do the lifts satisfy the Coxeter presentation of $W^{(2)}(\Lambda^{2,1}_{II})$, or equivalently give an honest action on the oriented Pfaffian line over the wall-complement of the reduced compact Hall sector. This is a geometric determinant-line monodromy datum. It does not include a scalar formula; $\tau_i$ must also transport the quotient-orientation package

$$\mathfrak{q}_C := \Big(\alpha^{\mathrm{red}}_C, \alpha^{E,\mathrm{free}}_C, \{(\beta^{E,N}_C, \lambda^{E,N}_C)\}_{N\geq 2}\Big)$$

and the chosen trivializations of these classes between the charge strata lying over Weyl-related Gram degrees. Equivalently, after the degree-two classes in $\mathfrak{q}_C$ vanish and the zero finite characters $\lambda^{E,N}_C = 0$ are selected, choose null-trivializations

$$\theta_C = \Big(\theta^{\mathrm{red}}_C, \theta^{E,\mathrm{free}}_C, \{\theta^{E,N}_C\}_{N\geq 2}\Big).$$

For finite stabilizers $\theta_C^{E,N}$ includes both the null-trivialization of $\beta_C^{E,N} \in H^2(BE[N])$ and the choice of the zero $H^1(BE[N])$-linearization character. For a simple type-II reflection $s_{\delta_i}$, the failure of $\tau_i$ to preserve the quotient trivialization is the torsor difference

$$\omega_{i,C} \in H^1(\mathfrak{M}^{\mathrm{red}}_{s_{\delta_i}C}; \mathbb{F}_2) \oplus \bigoplus_{N\geq 2} H^1(BE[N]; \mathbb{F}_2).$$

Here $s_{\delta_i}C$ denotes the chosen target stratum over the Weyl-reflected Gram degree. The connected $BE$-linearization contributes no $H^1$-term because $H^1(BE; \mathbb{F}_2) = 0$. The datum $(\mathrm{O1})^+$ includes the vanishing of all $\omega_{i,C}$ and the Coxeter coherence of the resulting transports. Without this compatibility the reflection sign is a sign of an ordinary square-root line, not a sign on the reduced $E$-quotient Hall sector. The passage from three generator signs to a character is then the pure group-theoretic Lemma 3.17, after the projective and quotient-torsor obstruction package of Lemma 3.18 has vanished.

(O2) *Pfaffian local model on type-II walls.* The finite wall certificate of Definition 3.10 is supplied. Proposition 3.11 records the geometric computation required to remove this hypothesis. The geometric reflection sign

$$\varepsilon_0 : W^{(2)}(\Lambda^{2,1}_{II}) \longrightarrow \{\pm 1\}$$

of Open Problem 5.7 is computed from a local Pfaffian normal form at a generic point of each type-II wall $\mathbb{H}_\delta$. Namely, the reduced determinant complex is assumed to split, after removing the tangent determinant factor, as

$$K^{\mathrm{nor}}_\delta \simeq \bigoplus_{j=1}^{N^{\mathrm{Pf}}_\delta} \big[\mathbb{R} \xrightarrow{u_\delta} \mathbb{R}\big],$$

where $u_\delta$ is a real local equation of the wall, $s_\delta(u_\delta) = -u_\delta$, and the Pfaffian section is $u_\delta^{N^{\mathrm{Pf}}_\delta}$ times a unit. For the three simple type-II walls the supplied normal-mode model has

$$N^{\mathrm{Pf}}_{\delta_1} = N^{\mathrm{Pf}}_{\delta_2} = N^{\mathrm{Pf}}_{\delta_3} = 1.$$

This local normal form is part of the hypothesis. The Borcherds divisor order and the Maass character give an independent automorphic target check after the local model is supplied; they do not construct the Hall orientation line and do not use scalar normalization.

Then:

(i) (*Pfaffian normalization.*) The supplied first-order object has protected Pfaffian equal to the theta-normalized Borcherds determinant:

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) \;=\; \mathcal{D}_X \;=\; \Delta_5.$$

Its scalar trace, in the OP normalization of Definition 11.3, satisfies

$$\mathrm{Tr}_{\mathrm{prot}}\big((-1)^F\, \mathfrak{D}_X^{-2}\big) \;=\; -4096\, \Delta_5^{-2}.$$

(ii) (*Three-sign identification and Weyl character.*) For the three type-II simple reflections $\delta_i$ of the Weyl chamber $\mathcal{P}_{II}$ of Section The Weyl chamber and the Weyl vector (notation $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$),

$$\varepsilon_0(s_{\delta_i}) \;=\; -1 \;=\; \nu_{\Delta_5}(s_{\delta_i}), \qquad i = 1, 2, 3,$$

and therefore

$$\varepsilon_0\big|_{W^{(2)}(\Lambda^{2,1}_{II})} \;=\; \nu_{\Delta_5}\big|_{W^{(2)}(\Lambda^{2,1}_{II})}.$$

The three $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ chamber-permuting reflections $s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_2+f_3}$ have automorphic Maass value +1, but no Hall orientation sign on them is asserted unless the oriented Hall action is separately extended from $W^{(2)}(\Lambda^{2,1}_{II})$ to the semidirect product $W^{(2)}(\Lambda^{2,1}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II})$.

(iii) (*Categorical refinement.*) Under (D0), (O1), (O1)$^+$, and (O2), the supplied chiral protected first-order theory refines the virtual $K_0$-determinant package $[\mathcal{V}]$, $\mathcal{D}_X$ of Proposition 3.8. Its decategorification is the Borcherds product

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)\big|_{\mathfrak{c}_\infty} = 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5,$$

and its orientation monodromy character is the Maass character $\nu_{\Delta_5}$ restricted to $W^{(2)}(\Lambda^{2,1}_{II})$.

$D_0$ has the status split of Theorem 10.43: the normal-ordered lattice part is proved and the denominator target is imported, but the compact finite stages, transition maps, and obstruction trivializations of $D_0$ are open on the reduced $K3 \times E$ sector. The orientation clauses (O1), (O1)$^+$, and (O2) are likewise open on that sector; on Calabi–Yau threefolds (O1) is Joyce–Song [33], the multiplicative orientation cocycle on $\mathrm{Perf}(X)$ is Joyce–Upmeier [35] and Joyce–Tanaka–Upmeier [34], on Calabi–Yau 4-fold analogues the unreduced orientation exists by Borisov–Joyce / Cao–Gross–Joyce / Bojko–Borisov–Joyce [8], [50], [49], and rank-one $\mathrm{Pin}^c$-orientation sign conventions in DT/PT Hilbert-scheme models are compatible with the BBDJS/Joyce–Upmeier framework and with the Cao–Kool–Monavari Calabi–Yau 4-fold orientation package [12]; non-compact and Joyce–Song extensions are Bojko [4]. These sources supply the orientation technology and partial local inputs. They do not by themselves prove the cosection-reduced $K3 \times E$ quotient orientation, its $E[2]$-linearization, or its type-II Weyl-equivariant lift. The cosection-induced parity flip $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd}^{\mathrm{std}} + 1$ of the Maulik–Toda / Oberdieck reduced obstruction theory is absorbed into (O1) as a $\mathbb{Z}/2$-grading shift $o_C^{\mathrm{red}} = o_C^{\mathrm{std}} \otimes \Pi$ of the BBDJS [9] line under cosection contraction of the PTVV [51] $(-1)$-shifted symplectic form by $\sigma_{K_3}$ (clause (P4) of Open Problem 5.7). This parity shift is part of the reduced orientation datum and may be compatible with the fermionic grading used in the protected supertrace. It is not the source of the OP scalar minus sign and it does not determine the Hall reflection character. The latter must be computed from monodromy of the orientation line around the type-II walls. The theorem therefore has certificate form: granted (D0), (O1), (O1)$^+$, and (O2), the normalization and character follow.

*Proof of Theorem 3.12. Step 1 (Pfaffian product from the supplied first-order datum).* By (D0), the compact first-order object $\mathfrak{D}_X$ and its normal-ordered primitive multiplicities are supplied. By Definition 4.3, the orientation line of a stable charge-$c$ object satisfies $o_C^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(C, C)$. The hypothesis (O1) promotes this to an extension-multiplicative system: on every direct sum $C \oplus C'$, $o_{C\oplus C'} \simeq o_C \otimes o_{C'}$, compatibly with the Hall extension stack and descended under the reduced $E$-quotient. Thus the supplied $\mathfrak{D}_X$ has a protected Pfaffian product whose exponent in normal-ordered Gram degree $(n, l, m)$ is

$$\mathrm{sdim}\, P^{\Pi}_{X,(n,l,m)} = f(nm, l).$$

With the leading cusp normalization fixed by the theta coefficient, the graded product is

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = v_0\ \mathrm{sdet}_{\mathcal{V}}(1 - x) \quad \text{with} \quad v_0 = 64 q^{1/2} r^{1/2} s^{1/2},$$

where sdet is the superdeterminant of Proposition 3.8. The theorem does not construct $\mathfrak{D}_X$; it identifies the Pfaffian of a supplied object satisfying (D0) and (O1).

*Step 2 (Identification* $\mathrm{Pf}_{\mathrm{prot}} = \Delta_5$*.)* Proposition 4.5 (Borcherds–Gritsenko–Nikulin) identifies $v_0 \operatorname{sdet}_{\mathcal{V}}(1-x)$ with the Igusa cusp form $\Delta_5$ at the level of Fourier coefficients. Step 1 promotes this from a formal product equality to an equality of Pfaffian sections, since orientation data resolves the $\pm$-ambiguity inherent in any supplied square root: the Borcherds product comes with the sign convention $c_\Delta(1,1,1) = 64$ (positive leading coefficient, Borcherds normalization), and the Pfaffian section inherits the same sign on the unique top-cohomology generator of the isotropic-cusp expansion of Section The normalized cusp form. Hence $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = \Delta_5$ as a section of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$.

*Step 3 (Scalar square in OP normalization).* Squaring forgets the orientation: $\big(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)\big)^2 = \det \mathrm{RHom}_{\mathrm{red}}(C,C)|_{\mathrm{tot}} = \Delta_5^2$, as sections of $\mathcal{L}^{10}$ (the squared automorphic line is the trivially-charactered determinant, since $\Delta_5^2 = \Delta_{10}$ is scalar by Section The normalized cusp form). The protected scalar trace is then $\mathrm{Tr}_{\mathrm{prot}}((-1)^F \mathfrak{D}_X^{-2}) = C_{\square,X}\,\Delta_5^{-2}$ for some normalization scalar $C_{\square,X}$; the explicit $C_{\square,X} = -4096$ follows from Proposition 11.1 of Oberdieck–Pixton, which gives $Z^X_{\mathrm{OP}} = -(\chi_{10}^{\mathrm{OP}})^{-1} = -4096\,\Delta_5^{-2}$, together with the arithmetic $-4096 = -(64)^2$ of the squared theta-leading coefficient and the OP scalar sign convention recorded in Definition 11.3.

*Step 4 (Orientation-character identity (ii)).* By (O2), each type-II simple reflection $s_{\delta_i}$ acts on $o_C$ by the parity of its oriented real normal modes. Lemma 3.13 is the Hall-side local calculation: under the supplied Pfaffian normal form $N^{\mathrm{Pf}}_{\delta_i} = 1$, hence $\varepsilon_o(s_{\delta_i}) = (-1)^{N^{\mathrm{Pf}}_{\delta_i}} = -1$. Lemma 8.2 is the independent automorphic calculation: $\nu_{\Delta_5}(s_{\delta_i}) = -1$ for the same three simple reflections, and $+1$ on the three chamber-permuting $\mathrm{Aut}(\mathcal{P}_{II})$-reflections. The latter $+1$-values are automorphic target data only; they are not Hall orientation signs unless a semidirect-product Hall lift is supplied.

The passage from the three simple Hall signs to a character on the full type-II Weyl group is purely group-theoretic only after $(\mathrm{O1})^+$ has killed the projective determinant-line cocycle and the quotient trivialization torsors of Lemma 3.18. Then $(\mathrm{O1})^+$ gives an honest lifted Coxeter action on the reduced quotient orientation. By Lemma 3.17,

$$W^{(2)}(\Lambda^{2,1}_{II})_{\mathrm{ab}} \simeq (\mathbb{Z}/2)^3,$$

with one generator for each $s_{\delta_i}$. Therefore two $\{\pm 1\}$-valued characters of $W^{(2)}(\Lambda^{2,1}_{II})$ that agree on $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$ agree everywhere. Hence $\varepsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda^{2,1}_{II})$.

*Step 5 (Categorical refinement (iii)).* The chiral protected theory is supplied by (D0). Steps 1–4 identify its decategorification and character: the class $[\mathcal{V}] \in K_0(\mathrm{sVect}_{\mathrm{fd}})$ is the decategorified primitive content, the formal product $v_0 \operatorname{sdet}_{\mathcal{V}}(1-x)$ is the Pfaffian section $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)$, and the formal character equality is the orientation-character identity of (ii). At the chosen isotropic cusp $\mathfrak{c}_\infty$ this decategorification is the Borcherds product, and the orientation monodromy character is $\nu_{\Delta_5}$ on the type-II Weyl group as computed. □

Lemma 3.13 *(Conditional Pfaffian signs on the type-II simple walls).* Under the compact first-order certificate (D0), the strong orientation datum (O1), the Weyl-equivariant extension $(\mathrm{O1})^+$, and the Pfaffian local model (O2) of Theorem 3.12, for each $\delta_i \in \{\delta_1, \delta_2, \delta_3\}$ the orientation-monodromy reflection sign is

$$\varepsilon_o(s_{\delta_i}) = (-1)^{N^{\mathrm{Pf}}_{\delta_i}} = -1, \qquad N^{\mathrm{Pf}}_{\delta_i} = 1.$$

Here $N^{\mathrm{Pf}}_{\delta_i}$ is the number of supplied odd real normal modes in the local Pfaffian normal form. It is not the automorphic divisor order, although the independent target order agrees in the three simple

type-II cases. The lemma makes no Hall-orientation claim for the three chamber-permuting reflections in $\mathrm{Aut}(\mathcal{P}_{II})$.

*Proof. Local Pfaffian calculation.* Hypothesis (O2) supplies, at a generic point of each $\mathbb{H}_{\delta_i}$, a real local equation $u_i = u_{\delta_i}$ with $s_{\delta_i}(u_i) = -u_i$ and a normal determinant-complex splitting

$$K^{\mathrm{nor}}_{\delta_i} \simeq \left[\mathbb{R} \xrightarrow{u_i} \mathbb{R}\right].$$

The normal Pfaffian section is $u_i$ times a unit. Transport by the reflection sends $u_i$ to $-u_i$, so it multiplies the Pfaffian line by $-1$; after squaring, the determinant section $u_i^2$ is fixed. Hence one supplied odd real normal mode gives

$$\varepsilon_0(s_{\delta_i}) = (-1)^{N^{\mathrm{Pf}}_{\delta_i}} = (-1)^1 = -1, \qquad i = 1, 2, 3.$$

*Independent automorphic target labels.* The three roots $\delta_1 = 2f_2 - f_3, \delta_2 = 2f_{-2} - f_3, \delta_3 = f_3$ have Gram pairing $(\delta_i, \delta_i) = 2$, $(\delta_i, \delta_j) = -2$ for $i \neq j$, by the matrix displayed in Section The Weyl chamber and the Weyl vector. All three are real simple roots of the Lorentzian Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$. In the Gram-index chart of Definition 8.3 (using $\alpha(\gamma) = 2nf_2 - lf_3 + 2mf_{-2}$), the three simple walls correspond to the lattice triples

$$\delta_1 \;\leftrightarrow\; (n, l, m) = (1, 1, 0), \qquad \delta_2 \;\leftrightarrow\; (0, 1, 1), \qquad \delta_3 \;\leftrightarrow\; (0, -1, 0),$$

matching the K3 elliptic-genus support $f(0, \pm 1) = 1, f(0, 0) = 10, f(nm, l) = 0$ for $4nm - l^2 < -1$. Thus the automorphic divisor order

$$d^{\mathrm{aut}}_{\delta_i} := \mathrm{ord}_{\mathbb{H}_{\delta_i}}(\Delta_5)$$

is 1 for $i = 1, 2, 3$. This target order is a check against the supplied local model; it is not a construction of the compact Hall orientation line.

The computation is:

| $\delta_i$ | Gram label | $N^{\mathrm{Pf}}_{\delta_i}$ | $d^{\mathrm{aut}}_{\delta_i}$ | $\varepsilon_0$ |
|---|---|---|---|---|
| $\delta_1 = 2f_2 - f_3$ | $(n, l, m) = (1, 1, 0), f(0, 1) = 1$ | 1 | 1 | $-1$ |
| $\delta_2 = 2f_{-2} - f_3$ | $(n, l, m) = (0, 1, 1), f(0, 1) = 1$ | 1 | 1 | $-1$ |
| $\delta_3 = f_3$ | $(n, l, m) = (0, -1, 0), f(0, -1) = 1$ | 1 | 1 | $-1$ |

□

*Remark* 3.14 *(Automorphic target check via the Maass character).* The argument below verifies the target character $\nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})}$ using only the Maass multiplier and the Borcherds product. It does not construct the Hall orientation character $\varepsilon_0$. The comparison $\varepsilon_0 = \nu_{\Delta_5}$ remains a geometric statement supplied by (O1), (O1)$^+$, and (O2). No scalar normalization enters this sign comparison: replacing $\Delta_5$ by $c\Delta_5$, $c \in \mathbb{C}^\times$, changes neither the Maass character nor the divisor orders. The factor 64 only names the monic Borcherds product $D_5 = 64^{-1}\Delta_5$, and the OP/DT scalar $-4096$ is unused.

*Step 1 (Maass-side coboundary).* Define $\varepsilon\colon W^{(2)}(\Lambda^{2,1}_{II}) \to \mathbb{Z}/2$ by $\varepsilon(g) := \frac{1}{2}(1 - \nu_{\Delta_5}(g))$, so $(-1)^{\varepsilon(g)} = \nu_{\Delta_5}(g)$. Maass's character is a homomorphism [42], [30, Section 3], [29], hence

$$c'(g, h) := \varepsilon(g) + \varepsilon(h) - \varepsilon(gh) \equiv \tfrac{1}{2}\left(1 - \nu_{\Delta_5}(g)\nu_{\Delta_5}(h)\nu_{\Delta_5}(gh)^{-1}\right) \equiv 0 \pmod 2.$$

The displayed value of $\nu_{\Delta_5}$ on the six generators is Lemma 8.2: $-1$ on $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$, $+1$ on $s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_2+f_3}$; equivalently $\varepsilon(s_{\delta_i}) = 1$ and $\varepsilon(s_{f_{-2}-f_2}) = \varepsilon(s_{f_2-f_3}) = \varepsilon(s_{f_2+f_3}) = 0$.

*Step 2 (Verification on the nine pairs of three-wall generators).* For the three-wall computation the only nontrivial check is on the $3 \times 3$ table of pairs $(g, h) \in \{s_{\delta_1}, s_{\delta_2}, s_{\delta_3}\}^{\times 2}$. The relations used here are only $s_{\delta_i}^2 = 1$ and the fact that $\nu_{\Delta_5}$ is a character. For $i \neq j$ the Coxeter exponent is $m_{ij} = \infty$, so there is no braid relation to impose:

- Diagonal $(g, g) = (s_{\delta_i}, s_{\delta_i})$ for $i = 1, 2, 3$ (three pairs): $gh = 1$, $\varepsilon(gh) = 0$, so $c'(g, h) = 1 + 1 - 0 = 2 \equiv 0 \pmod 2$.
- Off-diagonal $(g, h) = (s_{\delta_i}, s_{\delta_j})$, $i \neq j$ (six pairs): since $\nu_{\Delta_5}$ is a character, $\nu_{\Delta_5}(gh) = (-1)(-1) = +1$, giving $\varepsilon(gh) = 0$ and $c'(g, h) = 1 + 1 - 0 = 2 \equiv 0 \pmod 2$.

All nine pairs vanish modulo 2; $c'$ is a coboundary on the three-wall block, with primitive $\varepsilon$. This is an automorphic calculation only.

*Step 3 (Divisor-exponent target check).* Borcherds' product theorem [31, Theorem 2.1, (2.1)–(2.3)], [30, Theorem 4.1] writes

$$D_5 = 64^{-1}\Delta_5 = \mathrm{Borch}(\phi_{0,1}) = q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_{\mathrm{eff}}^+} (1 - q^n r^l s^m)^{f(nm,l)}$$

in the type-II cusp polarization $(q, r, s) = (e^{2\pi i z_1}, e^{2\pi i z_2}, e^{2\pi i z_3})$. This fixes the automorphic divisor orders

$$d_\delta^{\mathrm{aut}} := \mathrm{ord}_{\mathbb{H}_\delta}(\Delta_5).$$

For the three type-II roots these orders are

$$d_{\delta_1}^{\mathrm{aut}} = f(0, 1) = 1, \qquad d_{\delta_2}^{\mathrm{aut}} = f(0, 1) = 1, \qquad d_{\delta_3}^{\mathrm{aut}} = f(0, -1) = 1.$$

Together with Lemma 8.2, this gives the automorphic target signs $\nu_{\Delta_5}(s_{\delta_i}) = -1 = (-1)^{d_{\delta_i}^{\mathrm{aut}}}$. The symbol $d_\delta^{\mathrm{aut}}$ is deliberately distinct from the Pfaffian normal-mode rank $N_\delta^{\mathrm{Pf}}$ in (O2).

The chamber-permuting roots

$$\alpha_1 = f_{-2} - f_2, \qquad \alpha_2 = f_2 - f_3, \qquad \alpha_3 = f_2 + f_3$$

are type-I roots in $\Lambda^{2,1} \setminus \Lambda_{II}^{2,1}$. In the type-II parametrization $\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$, the coefficients of $f_2$ and $f_{-2}$ must be even. Thus $k = 1$ does not contribute to any $\alpha_i$. For $2\alpha_1$ one obtains either $(-1, 0, 1)$ or $(1, 0, -1)$, outside $\Gamma_{\mathrm{eff}}$. For $2j\alpha_2$ one obtains $(j, 2j, 0)$, and for $2j\alpha_3$ one obtains $(j, -2j, 0)$; the $q^0$-row of $\phi_{0,1}$ has support only at $l = 0, \pm 1$, so $f(0, \pm 2j) = 0$ for $j \geq 1$ [21, §2.2]. Hence

$$\mathrm{ord}_{\mathbb{H}_{\alpha_i}}(\Delta_5) = 0, \qquad i = 1, 2, 3,$$

and Lemma 8.2 gives automorphic monodromy $+1$ around these chamber-permuting reflections.

*What this check proves.* It proves the automorphic target values

$$\nu_{\Delta_5}(s_{\delta_i}) = -1, \qquad \nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_2+f_3}) = +1.$$

It does not prove that a compact Hall orientation line transports with these signs. That bridge is exactly the geometric content of the Pfaffian local model and Weyl-equivariant orientation cocycle.

*Remark* 3.15 *(Comparison of the two proofs).* The proof of Lemma 3.13 and the Maass-character check have different logical roles. The first computes the Hall orientation signs from the supplied Pfaffian local model (O2). The second computes the automorphic target signs of $\nu_{\Delta_5}$. The equality $\varepsilon_0 = \nu_{\Delta_5}$ follows

only after the local Dirac model supplies the Pfaffian normal-mode parity and the Weyl-equivariant orientation cocycle transports it as a character. The Borcherds product and Maass character supply the target character but not the Hall orientation character.

*Remark* 3.16 *(Imaginary roots are not Weyl reflections).* The Weyl group of the Borcherds–Kac–Moody datum is generated by real simple roots only. Imaginary simple roots contribute to the Weyl–Kac–Borcherds denominator through the additive correction coefficients $m(a)$ and $\tau(a)$ of Gritsenko–Nikulin [31, Theorem 2.1, (2.1)–(2.3)], [30, Sections 3–4]; they do not define reflections $s_a$ in $W^{(2)}(\Lambda^{2,1}_{II})$. Thus the orientation character

$$\varepsilon_o : W^{(2)}(\Lambda^{2,1}_{II}) \to \{\pm 1\}$$

is a real-root reflection datum. A compact Hall theory may also have local divisor monodromy around imaginary components of the Borcherds divisor, but that monodromy is not a Weyl character and is not asserted here. In particular, no formula $\varepsilon_o(s_a) = (-1)^{m(a)}$ is available for imaginary $a$, and the coefficients $m(a)$ must not be identified with the visible product exponents $f(nm, l)$ without the Gritsenko–Nikulin additive-side computation.

Lemma 3.17 *(Pure group-theoretic Weyl sign extension).* For any signs $\epsilon_i \in \{\pm 1\}$, $i = 1, 2, 3$, there is a unique character

$$\chi_\epsilon : W^{(2)}(\Lambda^{2,1}_{II}) \longrightarrow \{\pm 1\}$$

with $\chi_\epsilon(s_{\delta_i}) = \epsilon_i$. Hence a supplied Weyl-equivariant determinant-line transport datum $(O1)^+$, together with local Pfaffian signs $\epsilon_i = (-1)^{N^{\mathrm{Pf}}_{\delta_i}}$ from (O2), induces a unique orientation character. For $N^{\mathrm{Pf}}_{\delta_1} = N^{\mathrm{Pf}}_{\delta_2} = N^{\mathrm{Pf}}_{\delta_3} = 1$, this character is the determinant character on $W^{(2)}(\Lambda^{2,1}_{II})$.

*Proof.* The Cartan matrix of the type-II simple reflections, by $(\delta_i, \delta_i) = 2$ and $(\delta_i, \delta_j) = -2$ for $i \neq j$ (Section The Weyl chamber and the Weyl vector), is

$$\begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

Thus $A_{ij}A_{ji} = 4$ for $i \neq j$, so the Coxeter exponents are $m_{ij} = \infty$. The Coxeter presentation of the type-II reflection group is consequently the universal rank-three Coxeter group

$$W^{(2)}(\Lambda^{2,1}_{II}) \simeq \langle s_{\delta_1}, s_{\delta_2}, s_{\delta_3} \mid s^2_{\delta_1} = s^2_{\delta_2} = s^2_{\delta_3} = 1 \rangle.$$

Its abelianization is

$$W^{(2)}(\Lambda^{2,1}_{II}) / [W^{(2)}, W^{(2)}] \simeq (\mathbb{Z}/2)^3,$$

one $\mathbb{Z}/2$ per Coxeter generator $s_{\delta_i}$.

The only defining relations are $s^2_{\delta_i} = 1$. Any assignment $s_{\delta_i} \mapsto \epsilon_i \in \{\pm 1\}$ respects these relations, since $\epsilon_i^2 = 1$, and therefore extends uniquely to a $\{\pm 1\}$-valued character, equivalently to a homomorphism factoring through $(\mathbb{Z}/2)^3$. This is pure group theory. Applied to a supplied determinant-line action from $(O1)^+$, the scalars computed by (O2) on the three generators give the orientation character. If all three $\epsilon_i = -1$, the character sends every Coxeter generator to $-1$, hence is the determinant character. □

Lemma 3.18 *(Weyl quotient-orientation obstruction package).* Let $W = W^{(2)}(\Lambda_{II}^{2,1})$. Suppose (O1) supplies an ordinary orientation line together with the full reduced quotient package

$$\theta_C = \left(\theta_C^{\mathrm{red}}, \theta_C^{E,\mathrm{free}}, \{(\theta_C^{E,N}, \lambda_C^{E,N} = 0)\}_{N\geq 2}\right)$$

on every charge stratum used by the compact/wrapped Hall correspondences. A choice of simple wall transports $\tau_i$ gives an honest Weyl-equivariant orientation on the reduced quotient if and only if the following two independent obstructions vanish.

$$[c_o] = 0 \in H^2(W; \mathbb{F}_2)$$

with a killing cochain $b \in C^1(W; \mathbb{F}_2)$ for which the renormalized lifts $\widetilde{\tau}_w = (-1)^{b(w)}\tau_w$ satisfy $\widetilde{\tau}_i^2 = 1$ for the three simple reflections; and

$$\omega_{i,C} = 0 \in H^1(\mathfrak{M}^{\mathrm{red}}_{s_{\delta_i}C}; \mathbb{F}_2) \oplus \bigoplus_{N\geq 2} H^1(BE[N]; \mathbb{F}_2)$$

for every simple reflection and every relevant charge stratum, with the word defects obeying their natural torsor cocycle law. Since the type-II Weyl group has no braid relations among the three simple reflections, the generator-square constraints are the only Coxeter relation constraints after the projective cocycle is killed. Killing $[c_o]$ gives an action on the ordinary determinant line; killing the $\omega_{i,C}$ gives an action on the quotient orientation with its finite-stabilizer zero characters. Neither condition is implied by the Maass character $\nu_{\Delta_5}$.

*Proof.* The chosen transports over words in the generators define a central $\{\pm 1\}$-extension of $W$ precisely when their composition defect is a 2-cocycle

$$w'^{*}\tau_w \circ \tau_{w'} = (-1)^{c_o(w,w')}\tau_{ww'}.$$

Changing the word lifts by $(-1)^{b(w)}$ replaces $c_o$ by $c_o + \delta b$. Hence the projective determinant-line action becomes an honest action on the ordinary orientation line exactly when $[c_o]$ is zero and the chosen splitting is normalized on the relations. By Lemma 3.17, the type-II Coxeter presentation is the free product of three copies of $\mathbb{Z}/2$. Thus the only relation-level normalizations are $\widetilde{\tau}_i^2 = 1$.

After the ordinary line action is honest, the quotient null-trivializations form torsors under

$$H^1(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2) \oplus \bigoplus_{N\geq 2} H^1(BE[N]; \mathbb{F}_2).$$

The difference between the transported trivialization and the chosen trivialization on the reflected stratum is exactly $\omega_{i,C}$. Composition of transports gives the usual torsor cocycle law $\omega_{ww',C} = w'^{*}\omega_{w,C} + \omega_{w',C}$. Therefore vanishing on the three generators, together with the square coherences already imposed above, propagates to every word in $W$. Conversely, any honest Weyl action on the reduced quotient orientation preserves the chosen null-trivializations and zero finite characters, so both $c_o$ and all $\omega_{i,C}$ vanish after the corresponding choices. The construction uses only determinant-line and quotient-torsor data, not the scalar automorphic character. □

Corollary 3.19 *(Semidirect wall-transport obstruction).* Let

$$G = W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \mathrm{Aut}(\mathcal{P}_{II})$$

and suppose (O1) has been supplied on the reduced quotient strata. An extension of the type-II Weyl orientation transport from $W$ to $G$ is an additional datum. It exists on the reduced quotient orientation if and only if the projective determinant-line cocycle for the chosen lifts to $G$,

$$c_o^G \in Z^2(G;\mathbb{F}_2),$$

has trivial class in $H^2(G;\mathbb{F}_2)$, the chosen killing cochain normalizes the Coxeter relations in $W$ and the $S_3$-relations in $\mathrm{Aut}(\mathcal{P}_{II})$, and the quotient-torsor defects

$$\omega_{g,C} \in H^1(\mathfrak{M}^{\mathrm{red}}_{gC};\mathbb{F}_2) \oplus \bigoplus_{N\geq 2} H^1(BE[N];\mathbb{F}_2)$$

vanish for the chosen generators $g$ of $G$. The $S_3$-part must transport the full quotient package

$$\left(\alpha^{\mathrm{red}}, \alpha^{E,\mathrm{free}}, \{(\beta^{E,N}, \lambda^{E,N} = 0)\}_{N\geq 2}\right)$$

between chamber-permuted strata.

Consequently the Maass values

$$\nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_2+f_3}) = +1$$

are only automorphic target values. They do not construct a Hall orientation sign on the semidirect-product generators. Without the vanishing above, the manuscript asserts the Hall orientation character only on $W^{(2)}(\Lambda^{2,1}_{II})$, under (O1), (O1)$^+$, and (O2).

*Proof.* The proof is the same obstruction calculation as Lemma 3.18, with $W$ replaced by $G$. A projective action on the determinant line is a central $\mathbb{F}_2$-extension, hence is classified by $[c_o^G]$. A splitting must respect all relations of the semidirect presentation, not only the three generator squares in $W$. After the ordinary determinant-line action is honest, the chosen quotient trivializations remain torsors under the displayed $H^1$-groups; their failure to be preserved is exactly $\omega_{g,C}$. Vanishing on a generating set, with the relation coherences, propagates to all of $G$. The Maass character is a scalar automorphic character and has no map to these determinant-line or quotient-torsor classes. □

*Remark* 3.20. The table values of $\nu_{\Delta_5}$ on the six generators (Lemma 8.2), the determinant identity $\mathcal{D}_X = \Delta_5$ (Proposition 4.5), and the OP scalar $-4096\,\Delta_5^{-2}$ (Proposition 11.1) are unconditional. The Pfaffian recognition (i), the orientation-character identity (ii), and the categorical upgrade (iii) are conditional on (O1), (O1)$^+$, and (O2). Lemma 3.17 identifies the strongest algebraic consequence of a supplied Weyl-equivariant lift: the sign factors through the $(\mathbb{Z}/2)^3$ abelianization of $W^{(2)}(\Lambda^{2,1}_{II})$. Lemma 3.18 separates this group-theoretic sign extension from the projective determinant-line and quotient-torsor obstructions that must vanish before there is an honest Weyl action on the reduced quotient orientation. Open: (O1) for the reduced $E$-quotiented Weyl-equivariant compact critical $K3 \times E$ Hall/CoHA sector. Joyce–Upmeier [35], Joyce–Tanaka–Upmeier [34], and Borisov–Joyce–Bojko [8] cover the unreduced $\mathrm{Perf}(X)$, and [35] covers shifted symplectic stacks, but the reduced $E$-quotient with type-II Weyl-equivariant lift is not in the literature; (O2) is the supplied Pfaffian local normal form on type-II walls of that sector, not a consequence of the automorphic product. The local obstruction certificate of Proposition 3.11 names the missing chart, splitting, invariant-unit, and rank-one normal-mode computations. The squared scalar branch $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ is orientation-forgetful; rescaling $\Delta_5$ by a nonzero scalar leaves $\nu_{\Delta_5}$ invariant. The three type-II rows match Borcherds automorphic divisor orders $d^{\mathrm{aut}}_{\delta_i} = 1$ via $f(nm, l)$ at $(1,1,0)$, $(0,1,1)$, $(0,-1,0)$; the independent automorphic check at the three chamber-permuting type-I roots yields divisor order 0 from the explicit support of $\phi_{0,1}$ [21, §2.2].

*Open Problem* 3.21 *(Protected one-particle lift).* Construct, if it exists, a protected one-particle object with a $\Gamma_{\mathrm{eff}}$-graded super-vector-space decomposition

$$\mathcal{H}^{\mathrm{1p,ch}} = \bigoplus_{\gamma \in \Gamma_{\mathrm{eff}}} \mathcal{H}^{\mathrm{1p,ch}}_{\gamma}$$

with protected superdimensions

$$\mathrm{Tr}_{\mathcal{H}^{\mathrm{1p,ch}}_{(n,l,m)}} (-1)^F = f(nm, l).$$

The determinant construction uses only the virtual classes $\mathbb{U}_{n,l}$ and their signed dimensions. The algebraic object constructed from the same charge grading and the same signed multiplicities is the Borcherds denominator algebra. Such a lift would still be weaker than the holomorphic $E_3$-factorization algebra, because it would not by itself supply the collision law.

## 4 The normalized cusp form

Freitag [22] and Van der Geer [56]: the even-weight scalar subring is

$$M_*^{\mathrm{even}}(\mathbf{Sp}_4(\mathbb{Z})) = \mathbb{C}[E_4, E_6, \Delta_{10}, \Delta_{12}],$$

$$\mathrm{wt}(E_4, E_6, \Delta_{10}, \Delta_{12}) = (4, 6, 10, 12).$$

$$\Delta_{10} = \Delta_5^2, \qquad \mathrm{wt}(\Delta_5) = 5, \qquad \Delta_5|_5 g = \nu_{\Delta_5}(g)\Delta_5.$$

$$\Delta_5 = \prod_{\substack{a,b \in (\mathbb{Z}/2\mathbb{Z})^2 \\ {}^t ab \equiv 0 \mod 2}} \nu_{a,b},$$

$$\nu_{a,b}(z) = \sum_{l \in \mathbb{Z}^2} \exp\big(\pi i (z[l + \tfrac{1}{2}a] + {}^t bl)\big), \qquad z[x] = {}^t xzx.$$

For $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Z})$, Maass's character [42]

$$\nu_{\Delta_5} : \mathbf{Sp}_4(\mathbb{Z}) \to \mathbb{C}^\times, \qquad |\nu_{\Delta_5}| = 1,$$

is determined on the standard generators by

$$\nu_{\Delta_5}\begin{pmatrix} 0 & \mathbf{I}_2 \\ -\mathbf{I}_2 & 0 \end{pmatrix} = 1,$$

$$\nu_{\Delta_5}\begin{pmatrix} \mathbf{I}_2 & B \\ 0 & \mathbf{I}_2 \end{pmatrix} = (-1)^{b_1+b_2+b_3},$$

$$\nu_{\Delta_5}\begin{pmatrix} {}^t A^{-1} & 0 \\ 0 & A \end{pmatrix} = (-1)^{(1+a_1+a_4)(1+a_2+a_3)+a_1a_4}$$

with $A = \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \in \mathbf{GL}_2(\mathbb{Z})$ and $B = \begin{pmatrix} b_1 & b_2 \\ b_2 & b_3 \end{pmatrix} \in M_{2\times 2}(\mathbb{Z})$.

$\Delta_5$ is a weight-5 form with character $\nu_{\Delta_5}$, and $\Delta_5^2 = \Delta_{10}$ is scalar.

For

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix},$$

$$\Delta_5(Z) = \sum_{\substack{n,l,m \equiv 1 \bmod 2 \\ 4nm - l^2 > 0 \\ n,m > 0}} c_\Delta(n,l,m) \exp\big(\pi i (n z_1 + l z_2 + m z_3)\big),$$

with

$$c_\Delta(1,1,1) = 64, \qquad c_\Delta(n,l,m) \in 64\mathbb{Z}.$$

Proposition 4.1 *(Isotropic cusp coefficient).*

$$1 + \frac{1}{64} \sum_{t \geq 1} c_\Delta(1+2t,1,1) q^t = \prod_{k \geq 1} (1-q^k)^9.$$

*Proof.*

$$\Delta_5(Z) = \sum_{\substack{m > 0 \\ m \equiv 1 \bmod 2}} \phi_{5,m}(z_1,z_2) \exp(\pi i m z_3).$$

$$\phi_{5,1} \in J_{5,\frac{1}{2}}(\nu_{\Delta_5}).$$

$$\nu_{\mathrm{II}}(z_1,z_2) = \sum_{n \in \mathbb{Z}} (-1)^n \exp\left(\frac{\pi i}{4}(2n+1)^2 z_1 + \pi i (2n+1) z_2\right).$$

$$\nu_{\mathrm{II}} = -q^{1/8} r^{-1/2} \prod_{n \geq 1} (1 - q^{n-1} r)(1 - q^n r^{-1})(1 - q^n),$$

$$q = \exp(2\pi i z_1), \qquad r = \exp(2\pi i z_2).$$

$$\psi_{5,\frac{1}{2}} = \eta(z_1)^9 \nu_{\mathrm{II}}(z_1,z_2), \qquad \eta(z_1) = q^{1/24} \prod_{n \geq 1} (1-q^n).$$

$$\frac{1}{64} \phi_{5,1}(z_1,z_2) = \psi_{5,\frac{1}{2}}(z_1,z_2) = -q^{1/2} r^{-1/2} \prod_{n \geq 1} (1 - q^{n-1} r)(1 - q^n r^{-1})(1-q^n)^{10}.$$

$$1 + \frac{1}{64} \sum_{t \geq 1} c_\Delta(1+2t,1,1) q^t = \prod_{k \geq 1} (1-q^k)^9.$$

□

*Definition 4.2 (Massless divisor and determinant sign).*

$$\mathcal{H}_{\mathrm{diag}} = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mid a, b \in \mathbb{H}_1 \right\}.$$

Gritsenko–Cléry [28], $N = 1$:

$$\operatorname{supp} \operatorname{div}(\Delta_5) = \mathbf{Sp}_4(\mathbb{Z}) \cdot \mathcal{H}_{\mathrm{diag}}, \qquad \operatorname{mult}_{\mathcal{H}_{\mathrm{diag}}}(\Delta_5) = 1.$$

$$\mathcal{M}_{\mathrm{massless}} := \operatorname{supp} \operatorname{div}(\Delta_5), \qquad \epsilon_{\det} := \nu_{\Delta_5}.$$

$$\Delta_5|_5 g = \nu_{\Delta_5}(g) \Delta_5, \qquad \Delta_5(s_{\delta_i} Z) = -\Delta_5(Z).$$

The displayed sign on the type-II diagonal reflections $s_{\delta_i}$, $i = 1,2,3$, is the Maass-character verification carried out in Lemma 8.2 below; the $\nu_{\Delta_5}(s_{\delta_i}) = -1$ line of that lemma is the input used here.

*Definition* 4.3 *(Orientation line of the protected sector).* Let $C$ be a stable object in the reduced compact critical Hall/CoHA sector envisaged in Definition 5.4, with derived self-Ext line after cosection reduction

$$\det \mathrm{RHom}_{\mathrm{red}}(C, C) \in \mathrm{Pic}(\mathfrak{M}_X^{or}).$$

By an *orientation line* of the protected sector is meant a line bundle

$$o_C \in \mathrm{Pic}(\mathfrak{M}_X^{or})$$

together with the squaring isomorphism

$$o_C^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(C, C),$$

fitting into a strong orientation datum compatible with the extension correspondences of Definition 5.4. This is the existence frame for orientations on critical and $(-1)$-shifted symplectic moduli; on Calabi–Yau threefolds the orientation-data formalism is Joyce–Song [33], and on Calabi–Yau 4-folds it is the Borisov–Joyce / Oh–Thomas construction [8], [50], [49]. The $K3 \times E$ case is neither of these: its reduced compact critical sector is not constructed in this paper, and the existence of a square-root line $o_C$ for that sector is one of the unresolved data items of Open Problem 5.7. No such line is constructed here.

*Remark* 4.4 *(Polarization is independent of orientation).* The identity

$$B(c, c) = 2\Pi_X(c)$$

is the polarization identity for the quadratic Gram map $\Pi_X$. It is independent of the parity split of a protected cohomology representative and independent of the choice of an orientation line. Orientation data enters later, when one tries to promote a determinant section to a Pfaffian section and to compute its monodromy under wall reflections. It does not change the Gram polarization. In particular no Bott-periodicity convention, Grothendieck–Witt sign convention, or choice of Bott element is used to prove $B(c, c) = 2\Pi_X(c)$. Such conventions may rigidify an already constructed orientation theory; they do not supply the lattice polarization, the reduced $E$-quotient linearization, or the compact Pfaffian sign.

Proposition 4.5 *(Borcherds–Gritsenko–Nikulin identification).* By the Borcherds–Gritsenko–Nikulin product formula for $\phi_{0,1}$,

$$\mathcal{D}_X(Z) = \Delta_5(Z).$$

*Proof.* The singular-theta product theorem of Borcherds [6] gives

$$64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}$$

as the theta-normalized lift of $\phi_{0,1}$, with

$$\mathrm{wt} = f(0, 0)/2 = 5, \qquad c_\Delta(1, 1, 1) = 64.$$

The Lorentzian Kac–Moody identification of this product with the Igusa cusp form is the corrected denominator identity of Gritsenko–Nikulin [30]:

$$64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

$$\mathcal{D}_X(Z) = \Delta_5(Z).$$

□

*Remark 4.6 (Normalization of the square root).* The constants in the square-root product are fixed by the following chain:

| Object | Fixed value |
|---|---|
| Fourier variables | $q = e^{2\pi i z_1},\ r = e^{2\pi i z_2},\ s = e^{2\pi i z_3}$ |
| Borcherds input | $Z_{\mathrm{K3}} = 2\phi_{0,1}$ |
| Weight | $\mathrm{wt}\ \mathrm{Borch}_\theta(\phi_{0,1}) = f(0,0)/2 = 5$ |
| Igusa leading coefficient | $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$ |
| Monic product | $D_5 = 64^{-1}\Delta_5 = q^{1/2}r^{1/2}s^{1/2}\prod(1-q^n r^l s^m)^{f(nm,l)}$ |
| Determinant normalization | $\mathcal{D}_X = \Delta_5, \quad v_0 = 64q^{1/2}r^{1/2}s^{1/2}$ |
| Virtual inverse | $\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}} = 64^{-1}q^{-1/2}r^{-1/2}s^{-1/2}\,\mathrm{sch}\,\mathcal{F}(\mathcal{V})$ |
| BKM denominator | $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 64^{-1}\Delta_5(2Z)$ |
| Scalar square | $Z^X_{\mathrm{OP}} = -4096\Delta_5^{-2}$; in general $Z^X_\square = C_\square\Delta_5^{-2}$, hence $(\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}})^2 = Z^X_\square/C_\square$ |

Proposition 4.7 *(Borcherds–Gritsenko weight identity at $N = 1$).* Let $\phi_{0,1} \in J^{\mathrm{weak}}_{0,1}$ be the weight-zero index-one weak Jacobi form with Fourier expansion $\phi_{0,1}(z_1, z_2) = \sum_{n,l} f(n,l)\, q^n r^l$. The theta-normalized Borcherds lift at $N = 1$ is $\Delta_5$, and the Borcherds weight reads

$$\mathrm{wt}(\Delta_5) = \frac{f(0,0)}{2} = 5 = \frac{c_1(0)}{2},$$

with $c_1(0) := f(0,0) = 10$. Since the weight is integral, no metaplectic cover is needed for the weight factor.

*Proof.* The Borcherds singular-theta regularisation of $\phi_{0,1}$ produces a meromorphic Siegel modular form of weight $f(0,0)/2$; at $N = 1$ the paramodular group is $\mathbf{Sp}_4(\mathbb{Z})$ by Gritsenko–Nikulin [30]. The input Jacobi form has $f(0,0) = 10$ (Eichler–Zagier [21], elliptic genus of K3), hence the output weight is 5. Identifying $\mathrm{Borch}_\theta(\phi_{0,1})$ with $\Delta_5$ via the Igusa identification ($\mathcal{D}_X = \Delta_5$) fixes $\mathrm{wt}(\Delta_5) = 5$. Setting $c_1(0) := f(0,0)$ exhibits the row ($N = 1,\ \mathrm{wt} = 5,\ c_1(0) = 10$) of the Gritsenko–Cléry catalogue [28]; the remaining integral and half-integral Clery–Gritsenko rows are listed in [28, Theorem 1.4]. Integrality $\mathrm{wt}(\Delta_5) \in \mathbb{Z}$ means that no metaplectic cover is required for the weight factor. The Hodge automorphic line $\mathcal{L}^5$ remains nontrivial, and the Maass character $\nu_{\Delta_5}$ is a finite-order character of $\mathbf{Sp}_4(\mathbb{Z})$. □

# Part II

# Physical charge and normal-ordered descent

## 5 D6–D2–D0 PROTECTED-INDEX DICTIONARY

Only the OP/DT scalar branch requires an elliptically fibered K3 surface with a section. The Borcherds determinant above used only the universal K3 elliptic genus.

Let $S$ be an elliptically fibered K3 surface

$$\pi : S \to \mathbb{P}^1$$

with a section $s_1$, and let $E$ be an elliptic curve. Set

$$X = S \times E.$$

Let $F \in \mathrm{Pic}(S)$ be the fibre class and put

$$\beta_h = s_1 + hF, \qquad h \geq 0.$$

Type IIA D6–D2–D0 charge:

$$\gamma(h, d, n) = (1;\ (\beta_h, d);\ n).$$

Write $\Omega_X(h, d, n)$ for the putative protected D6–D2–D0 index attached to this charge in a compact physical realization. The microscopic trace is not constructed here; the desired datum would be:

$$\overset{?}{\exists}\ \mathcal{H}_{\gamma(h,d,n)}, \qquad \mathrm{Tr}_{\mathcal{H}_{\gamma(h,d,n)}}(-1)^F = \Omega_X(h, d, n).$$

$$\mathrm{Hilb}^n(X, (\beta_h, d)) = \{Y \subset X \mid [Y] = (\beta_h, d),\ \chi(\mathcal{O}_Y) = n\}.$$

$$E \curvearrowright X, \qquad (x, e') \mapsto (x, e + e').$$

Lemma 5.1 *(D6–D2–D0 Mukai–Gram dictionary).* Let $X = S \times E$ with $S$ K3 and $E$ elliptic. Let $Y \subset X$ be a one-dimensional subscheme with class $[Y] = (\beta, d) \in H_2(S, \mathbb{Z}) \oplus H_2(E, \mathbb{Z})$ and $\chi(\mathcal{O}_Y) = n \in \mathbb{Z}$. Let $\mathcal{I}_Y \in D^b(X)$ be its ideal sheaf. Decompose the Mukai vector

$$v_X(\mathcal{I}_Y) = \mathrm{ch}(\mathcal{I}_Y) \cdot \sqrt{\mathrm{td}(X)}$$

against the Künneth basis $\Lambda_S \otimes \{1_E, \omega_E\}$ where $\Lambda_S = H^*(S, \mathbb{Z})$ and $\omega_E \in H^2(E, \mathbb{Z})$ is the fundamental class. Then

$$v_X(\mathcal{I}_Y) = (1,\ 0,\ 1 - d) \otimes 1_E + (0,\ -\beta,\ -n) \otimes \omega_E.$$

With

$$P_Y = (1, 0, 1 - d), \qquad Q_Y = (0, -\beta, -n),$$

the Gram-degree triple is

$$\left(\frac{Q_Y^2}{2},\ Q_Y \cdot P_Y,\ \frac{P_Y^2}{2}\right) = \left(\frac{\beta^2}{2},\ n,\ d - 1\right).$$

In particular, for $\beta = \beta_h = s_1 + hF$ with $\beta_h^2 = 2h - 2$,

$$\Pi_X(Q_Y, P_Y) = (h - 1, n, d - 1).$$

*Proof.* Since $X = S \times E$ is a Calabi–Yau threefold,

$$\mathrm{ch}(\mathcal{I}_Y) = 1 - \mathrm{ch}(\mathcal{O}_Y).$$

For a one-dimensional sheaf on a Calabi–Yau threefold,

$$\mathrm{ch}_2(\mathcal{O}_Y) = \mathrm{PD}[Y], \qquad \int_X \mathrm{ch}_3(\mathcal{O}_Y) = \chi(\mathcal{O}_Y) = n.$$

The Todd square root factorizes as

$$\sqrt{\mathrm{td}(X)} = \pi_S^* \sqrt{\mathrm{td}(S)}\, \pi_E^* \sqrt{\mathrm{td}(E)} = \pi_S^*(1, 0, 1) \cup 1,$$

since $\sqrt{\mathrm{td}(E)} = 1$ and $\sqrt{\mathrm{td}(S)} = (1, 0, 1)$ in Mukai notation on a K3 surface. Decomposing the curve class, the K3-vertical component $\beta \times \{\mathrm{pt}\}$ contributes $\beta^\vee \otimes \omega_E$, while the elliptic component $d\{\mathrm{pt}\} \times E$ contributes $d[\mathrm{pt}_S] \otimes 1_E$. Thus

$$\mathrm{ch}(\mathcal{O}_Y) = (0, 0, d) \otimes 1_E + (0, \beta, n) \otimes \omega_E.$$

Multiplying

$$1 - \mathrm{ch}(\mathcal{O}_Y) = (1, 0, -d) \otimes 1_E + (0, -\beta, -n) \otimes \omega_E$$

by $(1, 0, 1) \otimes 1_E$ gives

$$v_X(\mathcal{I}_Y) = (1, 0, 1 - d) \otimes 1_E + (0, -\beta, -n) \otimes \omega_E,$$

because $\beta^\vee[\mathrm{pt}_S] = 0$ on the K3 surface.

The Mukai pairing on $\Lambda_S$ is

$$(r, D, s) \cdot (r', D', s') = D \cdot D' - rs' - r's.$$

Therefore

$$\frac{Q_Y^2}{2} = \frac{\beta^2}{2}, \qquad Q_Y \cdot P_Y = (-\beta) \cdot 0 - 0(1 - d) - 1(-n) = n,$$

and

$$\frac{P_Y^2}{2} = \frac{-2(1)(1 - d)}{2} = d - 1.$$

For $\beta = \beta_h$, $\beta_h^2 = 2h - 2$, this gives $\Pi_X(Q_Y, P_Y) = (h - 1, n, d - 1)$. □

The Gram-degree triple $(Q^2/2,\ Q \cdot P,\ P^2/2) = (h - 1,\ n,\ d - 1)$ identifies the Hilbert/PT charge with a triple in the Mukai lattice of $X$; this is the Gram-index data used by the Borcherds product and by the OP normalization of Theorem 11.2. The hypothesis $\beta_h^2 = 2h - 2$ of Theorem 11.2 is exactly the $Q^2/2 = h - 1$ line of this lemma.

Proposition 5.2 *(Reduced scalar quotient integration).* The reduced DT/PT side used in this paper is the coefficientwise scalar Behrend-weighted quotient integration

$$\mathbf{DT}^X_{n,(\beta,d)} = \int_{\mathrm{Hilb}^n(X,(\beta,d))/E} \nu_{\mathrm{Hilb}}\, de,$$

and

$$\mathsf{P}^X_{n,(\beta,d)} = \int_{P_n(X,(\beta,d))/E} \nu_P\, de$$

for the stable-pair moduli space $P_n(X,(\beta,d))$. It lands in numbers, coefficient by coefficient. Oberdieck's reduced stable-pair quotient invariant agrees with the reduced virtual-class incidence definition, and his reduced DT/PT theorem identifies the two reduced quotient series for nonzero K3 class [46, Theorem 3(i)]. This is scalar integration. It is not a Hall product, not a CoHA integration map, and not an orientation character.

*Proof.* Behrend's constructible function gives the weighted Euler characteristic used in Donaldson–Thomas theory [2], [53]. The $E$-action on $X = S \times E$ forces the ordinary unreduced invariant to vanish, so the reduced theory integrates over the quotient by $E$. Bryan's Hilbert-scheme normalization and Oberdieck's stable-pair normalization are exactly the quotient Behrend-weighted Euler characteristics [10], [46]. Oberdieck proves the quotient stable-pair definition equals the reduced incidence invariant and proves the reduced DT/PT comparison for nonzero K3 class. These statements produce integers indexed by $(\beta,d,n)$; no algebra object or primitive projection is constructed. □

*Conjecture* 5.3 *(Protected D-brane index).* For the D6–D2–D0 charge

$$\gamma(h,d,n) = (1;\ (\beta_h,d);\ n),$$

if such a compact physical Hilbert space is supplied, the conjectural enumerative dictionary identifies its protected trace with the Behrend-weighted Hilbert-scheme quotient invariant:

$$\Omega_X(h,d,n) := \mathrm{Tr}_{\mathcal{H}_{\gamma(h,d,n)}}(-1)^F \overset{\text{phys}}{=} \mathbf{DT}^X_{n,(\beta_h,d)}.$$

This physical identification is not used in the scalar theorem below.

*Definition* 5.4 *(Protected Hall integration criterion).* A protected Hall or critical CoHA integration map refining Proposition 5.2 would have to supply the following data before comparison with $\mathfrak{g}_{\Delta_5}$: an extension-closed compact Hall/CoHA sector for the reduced $E$-quotient, oriented critical extension correspondences, vanishing-cycle coefficient systems, HN-finite sector completions, protected integration compatible with the completed Hall product and wall crossing, a primitive projection, and a lifted Weyl action on the oriented completed theory.

If these data exist, the test is rigid: protected primitives must have signed dimensions

$$\mathrm{sdim}\,\mathrm{Prim}_{(n,l,m)} = f(nm,l),$$

their bracket must be the Borcherds–Gritsenko–Nikulin bracket, and the induced orientation character must restrict on the type-II Weyl group as

$$\epsilon_0|_{W^{(2)}(\Lambda^{2,1}_{II})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})}.$$

General CoHA technology supplies a language for such conditions [39], [17]; it does not construct this compact $K3 \times E$ datum.

LEMMA 5.5 *(Brauer-class parity criterion for Pin$^c$-lift gerbes).* Let $C$ be a protected Mukai charge on the reduced $K3 \times E$ sector, and let $\alpha^{\mathrm{red}}_C$ and $\alpha^E_C$ be the reduced Pin$^c$-lift gerbe class and the $E$-equivariant descent obstruction of Open Problem 5.7. On any projection-finite stratum on which:

(i) the reduced self-Ext determinant is identified with a K3 chamber polarization line $L$, and
(ii) $c_1(L)^2/2 \equiv 0 \pmod 2$, and

(iii) the $E$-descent parity $c_1(\text{degree}) \cup c_1(\text{level})$ is even,

one has

$$\alpha_C^{\mathrm{red}} = 0, \qquad \alpha_C^E = 0$$

on that stratum. In particular this proves the claimed vanishing on rank-two chambers satisfying the displayed parity and on HN strata whose graded factors reduce to rank-one/rank-two pieces on which the ordinary and quotient orientation classes have already vanished. It does not prove the length-one stable higher-rank case.

*Proof.* The BBDJS orientation gerbe of the reduced self-Ext complex is the second Stiefel–Whitney class of the determinant line, equivalently the mod-two Brauer class of the square-root gerbe [9]. On a stratum satisfying the first hypothesis this class is represented by

$$\frac{c_1(L)^2}{2} \bmod 2.$$

The second hypothesis therefore gives $\alpha_C^{\mathrm{red}} = 0$. The equivariant Serre spectral sequence for the free $E$-translation quotient identifies the connected descent obstruction with $c_1(\text{degree}) \cup c_1(\text{level}) \bmod 2$, so the third hypothesis gives $\alpha_C^E = 0$. Additivity of orientation gerbes on HN direct sums, from the Joyce–Upmeier / Joyce–Tanaka–Upmeier multiplicative orientation formalism [34, 35], then transports the vanishing to HN strata whose graded factors all lie in the proved rank-one/rank-two sectors; off-diagonal self-Ext blocks occur in dual pairs and contribute even Stiefel–Whitney class. No induction over a stable length-one higher-rank factor is used. □

*Remark* 5.6. Lemma 5.5 reads the parity off the Chern integrand of the chamber polarization; Lemma 5.17 reads it off the HN-filtration data. Both are reduction criteria: they agree with the rank-one base case (P1) and with each other on the rank-two/HN-reducible strata after the rank-one finite-data checks of (P1) have been performed, but neither closes stable higher-rank length-one strata.

*Open Problem* 5.7 *(Orientation monodromy character).* The geometric orientation monodromy character

$$\epsilon_o : W^{(2)}(\Lambda_{II}^{2,1}) \longrightarrow \{\pm 1\},$$

expected to satisfy

$$\epsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})},$$

is not produced by normalizing $\Delta_5$. Multiplying $\Delta_5$ by a nonzero scalar leaves its Maass character unchanged. The character must instead be produced by transporting the square root of the determinant line around the simple diagonal wall reflections; the expected local model is Pfaffian, in the sense made precise as hypothesis (O2) of Theorem 3.12: a simple real-root reflection reverses one oriented normal mode, so its square is scalar while the square root changes sign.

(P1) *Rank-one D6–D2–D0 stratum: ordinary Pin$^c$ input and quotient checks.* On the rank-one D6–D2–D0 stratum $\mathfrak{M}_C^{\mathrm{red},(1)}$ of physical Mukai charge $C = (1, \beta, n)$, the desired reduced orientation line $o_C^{\mathrm{red},(1)} \in \mathrm{Pic}(\mathfrak{M}_C^{\mathrm{red},(1)})$ is reduced to four pieces of data: an ordinary Pin$^c$ lift of the cosection-reduced self-Ext frame bundle, a connected $BE$-equivariant linearization class, finite-stabilizer gerbe classes on the small-orbit strata, and the residual finite-stabilizer $H^1$-linearization characters after those gerbes have been null-trivialized. The standard orientation technologies [9, 12, 34, 35] provide the local Pin$^c$

frame and sign-rigidification model used in Lemma 5.15. They do not, without the following quotient computations, construct the reduced $E$-quotient orientation. The ordinary obstruction class is

$$\alpha_C^{\mathrm{red}} := w_2\big(\det \mathrm{RHom}_{\mathrm{red}}(C,C)\big) + \mathrm{cs}^* w_2\big(\det(\mathrm{coker}\ \mathrm{cs})\big) \ \in\ H^2(\mathfrak{M}_C^{\mathrm{red}};\mathbb{Z}/2)$$

of the reduced orientation gerbe. The connected quotient class has the form

$$\alpha_C^{E,\mathrm{free}} = a_1(C)u_1 + a_2(C)u_2 \in H^2(BE;\mathbb{F}_2),$$

with $u_1, u_2$ as in Lemma 5.8. On the $N = 2$ small-orbit stratum the finite residual is

$$\beta_C^{E,E[2]} = b_{20}(C)x_1^2 + b_{11}(C)x_1x_2 + b_{02}(C)x_2^2 \in H^2(BE[2];\mathbb{F}_2).$$

After $\beta_C^{E,E[2]} = 0$, the chosen two-torsion linearization has a degree-one character

$$\lambda_C^{E[2]} = \lambda_1(C)x_1 + \lambda_2(C)x_2 \in H^1(BE[2];\mathbb{F}_2).$$

Thus rank-one (O1) through the two-torsion small-orbit boundary is proved exactly when

$$\alpha_C^{\mathrm{red}} = 0, \qquad a_1(C) = a_2(C) = 0, \qquad b_{20}(C) = b_{11}(C) = b_{02}(C) = 0, \qquad \lambda_1(C) = \lambda_2(C) = 0.$$

The first equality is the ordinary $\mathrm{Pin}^c$ orientation calculation. The remaining bits are quotient gerbe and linearization data. The three finite-sector bits $r_1, r_2, r_3$ of Proposition 5.12 are equivalent to the three coefficients $b_{20}, b_{11}, b_{02}$, but they do not see the connected $BE$-class $a_1u_1 + a_2u_2$ or the degree-one finite character $\lambda_C^{E[2]}$. Thus the current three-bit criterion cannot by itself be promoted to a proof of rank-one (O1). Rank-one (O1) through the two-torsion boundary is proved only after the ordinary class, the two connected coefficients, the three finite cyclic degree-two residuals, and the finite degree-one character have all been computed to vanish. None of these vanishing statements is implied by ordinary translation invariance or by the OP scalar square. Full rank-one descent through all small-orbit strata also requires $\beta_C^{E,N} = 0$ and the trivial $H^1(BE[N];\mathbb{F}_2)$ linearization character for every even $N \geq 4$, with the two-primary residual described in Lemma 5.13; the $E[2]$ bits do not imply those vanishings. Higher-rank semistable strata remain open except where the HN criteria below reduce them to already-oriented factors.

(P2) *Type-II diagonal reflections: wall-character formula conditional on the orientation existence.* Conditional on (P1) and on the multiplicative orientation cocycle of Joyce–Upmeier [35] and Joyce–Tanaka–Upmeier [34] extending equivariantly across the reduced $E$-quotient, and on the Pfaffian local model (O2) of Theorem 3.12, equivalently the three local calculations isolated in Proposition 3.11, the induced orientation character satisfies $\epsilon_0(s_{\delta_i}) = -1 = \nu_{\Delta_5}(s_{\delta_i})$, $i = 1, 2, 3$, so $\epsilon_0 = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$. The three $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ chamber-permuting reflections have Maass value +1, but Hall signs on them require a separately supplied semidirect-product lift. This is exactly conclusion (ii) of Theorem 3.12, with the explicit row-by-row calculation in Lemma 3.13; conditional on (O1), $(\mathrm{O1})^+$, and (O2).

(P3) *Higher-rank obstruction.* The required task is to compute the obstruction classes

$$\alpha_C^{\mathrm{red}} \ \in\ H^2(\mathfrak{M}_C^{\mathrm{red}};\mathbb{Z}/2), \qquad \alpha_C^{E} \ \in\ H_E^2(\mathfrak{M}_C^{\mathrm{red}};\mathbb{F}_2),$$

where $\alpha_C^{\mathrm{red}}$ is the $\mathrm{Pin}^c$-lift gerbe class of the reduced derived self-Ext frame bundle on $\mathfrak{M}_C^{\mathrm{red}}$, and $\alpha_C^{E}$ is the equivariant-linearization obstruction to descent of the orientation along the free $E$-translation quotient.

On the rank-one stratum it should be computed, not inferred from the OP scalar, by a direct Chern-class calculation in the $E_2^{2,0}$-term of the equivariant Serre spectral sequence $\mathfrak{M}_C^{\mathrm{red}} \to \mathfrak{M}_C^{\mathrm{red}}/E \to BE$ of Lemma 5.8. One must then prove their joint vanishing on the Harder–Narasimhan-completed semistable stack $\mathfrak{M}_C^{\mathrm{red,ss,HN}}$. The small-orbit gerbe contribution $\overline{\beta}_C^E$ of Lemma 5.8(iv) from the $N \geq 2$ torsion locus $K3 \times (E/E[N]) \subset X$ is governed by the finite-stabilizer criterion of Lemma 5.9: the band at $N = 2$ is the Klein-four $BE[2]$ (not cyclic $B\mu_2$), and vanishing requires a trivial $E[2]$-equivariant linearization of the orientation line, detected on the three nontrivial cyclic subgroups. Granted the ordinary $\mathrm{Pin}^c$ gerbe vanishing, the connected $BE$-linearization vanishing, and this finite-stabilizer vanishing, (O1) would follow from the Joyce–Upmeier [35] multiplicative cocycle plus the supplied $\mathrm{Pin}^c$ lift; granted (O2), the wall-character formula of (P2) would follow by Theorem 3.12. On higher-rank semistable strata the joint vanishing of $\alpha_C^{\mathrm{red}}$ and $\alpha_C^E$ is reduced by Lemma 5.17 to HN strata whose graded factors are already oriented; stable higher-rank length-one strata and even finite-stabilizer strata not covered by a trivial linearization remain open. The $\mathrm{Pin}^c$-lift program of [35], [34], [12], [4], [9] supplies the analytic technology on the cosection-twisted reduced $E$-quotient; non-compact and Joyce–Song extensions are Bojko [4].

(P4) *Cosection-induced parity flip from $\sigma_{K3}$:* $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd}^{\mathrm{std}} + 1$*; absorbed into (O1) as a $\mathbb{Z}/2$-grading shift on $o_C$.* The Maulik–Toda / Oberdieck reduced obstruction theory contracts the PTVV [51] $(-1)$-shifted symplectic form $\omega_{\mathfrak{M}}$ on $\mathfrak{M}_C(X)$ against the cosection $\mathrm{cs}\colon \mathrm{Obs}_{\mathfrak{M}_C} \to \mathcal{O}_{\mathfrak{M}_C}$ generated by Mukai dilation along $\sigma_{K3}$, and the restriction $\omega_{\mathfrak{M}}^{\mathrm{red}}$ to the cosection vanishing locus $\mathfrak{M}_C^{\mathrm{red}} \hookrightarrow \mathfrak{M}_C$ is non-degenerate on the OP/DT D6–D2–D0 stratum, with virtual dimension shifted by one, $\mathrm{vd}_C^{\mathrm{red}} = \mathrm{vd}_C^{\mathrm{std}} + 1$. Under the BBDJS [9] Pin-to-Spin lift of the BBDJS line of Definition 4.3, this dimension shift propagates as a $\mathbb{Z}/2$-grading shift on the orientation line:

$$o_C^{\mathrm{red}} \;=\; o_C^{\mathrm{std}} \otimes \Pi,$$

where $\Pi$ is the parity-flip line generated by the cosection cokernel. This shift is *coherent* in the sense that it commutes with the extension correspondences of Definition 5.4 and so does not perturb the Joyce–Upmeier multiplicative cocycle of (P2) once the parity is fixed once-and-for-all by the Oberdieck monic normalization [46]. The parity-flip is absorbed into (O1): the strong orientation line $o_C$ demanded by Theorem 3.12 is, on the reduced sector, $o_C^{\mathrm{red}}$, not the unreduced BBDJS line, and the squaring isomorphism is to the reduced derived self-Ext $\det \mathrm{RHom}_{\mathrm{red}}(C,C) = \mathrm{cofib}(\sigma_{K3}\colon \mathrm{RHom}(C,C)[-2] \to \mathrm{RHom}(C,C))$. Operationally this parity shift belongs to the reduced orientation datum and to any eventual Berezinian convention for a compact first-order object. The overall minus in $Z_{\mathrm{OP}}^X = -4096\,\Delta_5^{-2}$ is the scalar OP/DT branch convention. It may be compatible with a chosen fermionic grading, but it is not evidence for the Hall orientation character. The reflection character $\epsilon_o$ must be computed from monodromy of the orientation line around the type-II walls. The (P3) residual is the joint vanishing of $\alpha_C^{\mathrm{red}} + \alpha_C^E$ on the HN-completed semistable stack with $\mathrm{rk}\, C \geq 2$ and on the small-orbit $BE[N]$ finite-stabilizer strata. Lemma 5.17 and Lemma 5.5 give conditional reductions for the HN-reducible and rank-two parity pieces; Lemma 5.9 gives the $N = 2$ Klein-four detection criterion. Stable higher-rank length-one strata and nontrivial finite-stabilizer linearizations remain genuine open obligations.

Lemma 5.8 *(Connected and finite-stabilizer $E$-descent of the orientation line).* Let $C \in \Gamma_X^{\mathrm{phys}}$ and let $\mathfrak{M}_C^{\mathrm{red}}$ be the cosection-reduced critical Hall stack of Definition 5.4. The translation action of $E$ on $X = K3 \times E$ lifts to an action on $\mathfrak{M}_C^{\mathrm{red}}$, free outside the small-orbit locus

$$\mathfrak{M}_C^{\mathrm{red,small}} = \bigcup_{N \geq 2} \mathfrak{M}_C^{\mathrm{red}}\big(K3 \times (E/E[N])\big) \;\subset\; \mathfrak{M}_C^{\mathrm{red}},$$

and on the free locus produces a quotient stack $\overline{\mathfrak{M}}_C^{\mathrm{red}} = \mathfrak{M}_C^{\mathrm{red}}/E$. Suppose the unreduced orientation $o_C^{\mathrm{red}}$ of (P1) of Open Problem 5.7 exists on the free locus. Its descent to a line $\overline{o}_C^{\mathrm{red}} \in \mathrm{Pic}(\overline{\mathfrak{M}}_C^{\mathrm{red}})$ is governed by the equivariant Serre spectral sequence

$$E_2^{p,q} = H^p(BE;\, H^q(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{Z}/2)) \implies H_E^{p+q}(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{Z}/2)$$

attached to the fibration $\mathfrak{M}_C^{\mathrm{red}} \to \overline{\mathfrak{M}}_C^{\mathrm{red}} \to BE$. The connected $E$-quotient and the finite stabilizer strata have different cohomology and must be treated separately.

(i) (*Connected free $E$-descent.*) Topologically $E^{\mathrm{top}} \simeq T^2$, hence

$$BE \simeq BT^2 \simeq K(\mathbb{Z}^2, 2), \qquad H^*(BE; \mathbb{F}_2) = \mathbb{F}_2[u_1, u_2], \quad |u_i| = 2.$$

Therefore

$$H^1(BE; \mathbb{F}_2) = 0, \qquad H^2(BE; \mathbb{F}_2) = \mathbb{F}_2 u_1 \oplus \mathbb{F}_2 u_2.$$

In particular there is no connected-torus obstruction in bidegree

$$E_2^{1,1} = H^1(BE; \mathbb{F}_2) \otimes H^1(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2) = 0.$$

The degree-two obstruction on the free locus has only the two standard components

$$E_2^{2,0} = H^2(BE; \mathbb{F}_2) \otimes H^0(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2)$$

and

$$E_2^{0,2} = H^0(BE;\, H^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2)).$$

Thus the free-locus descent obstruction is a class

$$\alpha_C^{E,\mathrm{free}} \in H_E^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2)$$

whose connected-torus part lies in $\mathbb{F}_2 u_1 \oplus \mathbb{F}_2 u_2$, not in a rank-three $\langle \alpha^2, \alpha\beta, \beta^2 \rangle$ group.

(ii) (*Rank-one free-locus criterion.*) On the rank-one stratum $\mathfrak{M}_C^{\mathrm{red},(1)}$, $C = (1, \beta, n)$, descent of $o_C^{\mathrm{red},(1)}$ through the free $E$-quotient is equivalent to the vanishing of the two components in (i): the ordinary reduced orientation gerbe in $E_2^{0,2}$, and the connected-torus equivariant linearization class in $E_2^{2,0}$. If a direct Chern-class calculation identifies the latter with a class of the form

$$c_1(\mathrm{degree})u_1 + c_1(\mathrm{level})u_2 \in H^2(BE; \mathbb{F}_2),$$

then the class vanishes exactly when those two mod-2 coefficients vanish. The OP scalar normalization by itself does not prove this equivariant linearization statement; it is a separate orientation calculation.

(iii) (*Finite stabilizer descent.*) On the small-orbit locus $\mathfrak{M}_C^{\mathrm{red}}(K3 \times (E/E[N]))$ the $E$-quotient is the stacky orbifold $[\mathfrak{M}_C^{\mathrm{red}}(K3 \times (E/E[N]))/(E/E[N])]$ banded by $BE[N]$, not by $B\mu_N$ unless a cyclic subgroup has been chosen. The orientation line may pick up a finite-stabilizer gerbe class

$$\beta_C^{E,N} \in H^2(BE[N]; \mathbb{F}_2).$$

For $N = 2$,

$$E[2] \simeq (\mathbb{Z}/2)^2, \qquad H^*(BE[2]; \mathbb{F}_2) = \mathbb{F}_2[x_1, x_2], \quad |x_i| = 1,$$

so

$$H^2(BE[2];\mathbb{F}_2) = \mathbb{F}_2\langle x_1^2, x_1x_2, x_2^2\rangle.$$

This rank-three group belongs to the finite two-torsion stabilizer, not to connected $BE$. If $N$ is odd, the positive-degree mod-2 cohomology of $BE[N]$ vanishes because $E[N]$ has odd order. For even $N$, the residual class is detected on the 2-primary part and must be computed stratum by stratum from the equivariant linearization of the orientation line. More precisely, if $2^a \parallel N$ and $a \geq 2$, then

$$H^*(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2, \quad x_i^2 = 0,$$

so

$$\beta^{E,N}_{C,(2)} = A_1^{(N)} y_1 + A_{12}^{(N)} x_1x_2 + A_2^{(N)} y_2.$$

This is not determined by the $N = 2$ Klein-four vector.

(iv) (*Combined criterion.*) The orientation line descends over the full reduced quotient only if the ordinary reduced orientation gerbe, the connected-torus equivariant linearization class, and every finite-stabilizer gerbe class $\beta^{E,N}_C$ vanish on the relevant strata. After each finite gerbe has been null-trivialized, the chosen stabilizer linearization has a residual character

$$\lambda^{E,N}_C \in H^1(BE[N];\mathbb{F}_2),$$

and the quotient orientation requires this character to be zero. Translation invariance of the underlying ordinary line is not enough: a fixed line can carry a non-trivial equivariant character.

*Proof.* The cohomology calculation in (i) is the standard classifying space computation for a compact torus:

$$BT^2 = K(\mathbb{Z}^2, 2), \qquad H^*(BT^2;\mathbb{F}_2) = \mathbb{F}_2[u_1, u_2], \quad |u_i| = 2.$$

Hence $H^1(BE;\mathbb{F}_2) = 0$, and the bidegree $(1, 1)$ term in the equivariant Serre spectral sequence vanishes. The only total-degree two terms on the free connected quotient are $(2, 0)$ and $(0, 2)$, as displayed. This proves the connected separation.

For (ii), descent of a square-root line through a free group quotient is exactly an equivariant linearization problem together with the ordinary gerbe-trivialization problem. The spectral sequence in (i) records these two classes. A proposed rank-one Chern-class formula is therefore a criterion: once the two coefficients of $u_1, u_2$ and the ordinary gerbe class vanish, the line descends on that stratum. The scalar identity $Z^X_{\mathrm{OP/DT}} = -4096\Delta_5^{-2}$ has no information about a choice of $E$-equivariant linearization, so it cannot by itself prove the criterion.

For (iii), the stabilizer of the $N$-fold small-orbit stratum is the finite group $E[N] \simeq (\mathbb{Z}/N)^2$. At $N = 2$ this is the Klein-four group. Its mod-2 cohomology is the polynomial ring on the two degree-one character classes, hence its degree-two part has basis $x_1^2, x_1x_2, x_2^2$. If $N$ is odd, transfer kills positive-degree $\mathbb{F}_2$-cohomology because $|E[N]| = N^2$ is invertible in $\mathbb{F}_2$. Even $N$ reduces to its 2-primary stabilizer. This proves the finite-stabilizer typing and separates it from connected $BE$.

Clause (iv) is the standard descent criterion for a line on a stack quotient: ordinary square-root gerbe, connected equivariant linearization, finite-stabilizer degree-two gerbes, and the residual degree-one stabilizer characters must all vanish. A line fixed as an ordinary line can still carry a non-trivial character of the stabilizer group, so ordinary translation invariance does not imply equivariant descent. □

LEMMA 5.9 *(Klein-four criterion for the small-orbit gerbe residual).* Let $C \in \Gamma_X^{\mathrm{phys}}$ and adopt the notation of Lemma 5.8. The two-torsion subgroup $E[2] \subset E$ is the Klein-four group $\mathbb{Z}/2 \times \mathbb{Z}/2$, generated by the three nontrivial two-torsion points $e_1, e_2, e_3 \in E[2]$ (any two are independent). The small-orbit stratum at $N = 2$ is $\mathfrak{M}_C^{\mathrm{red}}(K3 \times (E/E[2]))$, banded by $BE[2]$, not $B\mu_2$; the cohomological residual of clause (iv) of Lemma 5.8 at $N = 2$ is the class

$$\beta_C^{E,E[2]} \in H^2(BE[2];\mathbb{Z}/2) \cong (\mathbb{Z}/2)^3,$$

with cohomology basis

$$H^*(BE[2];\mathbb{Z}/2) = \mathbb{Z}/2[\,c_1(L_1), c_1(L_2)\,], \quad H^2(BE[2];\mathbb{Z}/2) = \mathbb{Z}/2\langle c_1(L_1)^2\rangle \oplus \mathbb{Z}/2\langle c_1(L_2)^2\rangle \oplus \mathbb{Z}/2\langle c_1(L_1) \smile c_1(L_2)$$

where $L_1, L_2$ are the two character lines of the dual $E[2]^\vee \simeq E[2]$, and the three nontrivial cyclic subgroups $\mu_2^{(i)} = \langle e_i \rangle \subset E[2]$, $i = 1, 2, 3$, restrict the basis to the three generators of $H^2(B\mu_2^{(i)};\mathbb{Z}/2) \cong \mathbb{Z}/2$ componentwise. Write

$$\beta_C^{E,E[2]} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2, \qquad x_j = c_1(L_j),$$

and let $\tau_i$ be the degree-one generator of $H^*(B\mu_2^{(i)};\mathbb{Z}/2)$. Define the three cyclic residual bits $r_i \in \mathbb{Z}/2$ by

$$\mathrm{res}_{\mu_2^{(i)}}\big(\beta_C^{E,E[2]}\big) = r_i\tau_i^2.$$

With $e_3 = e_1 + e_2$,

$$r_1 = b_{20}, \qquad r_2 = b_{02}, \qquad r_3 = b_{20} + b_{11} + b_{02},$$

hence

$$b_{20} = r_1, \qquad b_{02} = r_2, \qquad b_{11} = r_1 + r_2 + r_3.$$

Thus the $N = 2$ finite-stabilizer residual vanishes if and only if the three cyclic residual bits $r_1, r_2, r_3$ vanish. On the rank-one D6–D2–D0 input $C = (1, \beta, n)$, the following hypotheses are sufficient, and in the finite $E[2]$-sector necessary, for the $N = 2$ residual to vanish and for the chosen $E[2]$-linearization to be trivial:

(K1) (*Künneth split.*) The reduced orientation line factors as a tensor product $o_C^{\mathrm{red}} = o_{K3} \otimes o_E$ on the protected sector, where $o_{K3}$ is the $\mathrm{Pin}^c$ orientation of the K3 PT factor and $o_E$ is the $\mathrm{Pin}^c$ orientation of the elliptic fibration factor, by the Künneth factorization of the cosection-twisted Joyce–Tanaka–Upmeier multiplicative cocycle [35],[34] on $\mathfrak{M}_C^{\mathrm{red}}$ viewed as a relative Hilbert scheme over the elliptic base, and the Cao–Kool–Monavari $\mathrm{Pin}^c$ lift on the K3 factor [12].

(K2) (*K3 factor.*) The action of $E[2]$ on $X = K3 \times E$ is fibrewise translation on the second factor and trivial on the first. The supplied factorwise equivariant lift on the K3 orientation factor is required to be the trivial $E[2]$-lift; this removes no elliptic finite-stabilizer obstruction.

(K3) (*Elliptic factor: finite gerbe and character data.*) Translation by $e_i \in E[2]$ fixes the underlying elliptic orientation line $o_E$ as an ordinary line. This fixedness does not decide the equivariant lift. First one must compute the three degree-two gerbe bits

$$r_i(C) = \mathrm{res}_{\mu_2^{(i)}}(\beta_C^{E,E[2]})/\tau_i^2 \in \mathbb{Z}/2, \qquad i = 1, 2, 3.$$

The degree-two vanishing hypothesis is

$$r_1(C) = r_2(C) = r_3(C) = 0.$$

After this vanishing has supplied a null-trivialized gerbe, the set of $E[2]$-linearizations is an

$$H^1(BE[2];\mathbb{F}_2) = \mathbb{F}_2\langle x_1, x_2\rangle$$

torsor. For a supplied linearization write its character as

$$\lambda_C^{E[2]} = \lambda_1 x_1 + \lambda_2 x_2,$$

or, on the three cyclic inertia sectors,

$$\operatorname{res}_{\mu_2^{(i)}} \lambda_C^{E[2]} = \rho_i \tau_i, \qquad \rho_1 = \lambda_1, \quad \rho_2 = \lambda_2, \quad \rho_3 = \lambda_1 + \lambda_2.$$

The required trivial-linearization hypothesis is

$$\rho_1(C) = \rho_2(C) = \rho_3(C) = 0.$$

Equivalently, the $\mathrm{Pin}^c$ sign character

$$w_1(o_E^{E[2]})\colon E[2] \longrightarrow \mathbb{Z}/2$$

vanishes on $e_1, e_2, e_3$. This is a degree-one equivariant-linearization hypothesis, distinct from the degree-two gerbe bits $r_i$, and not a consequence of ordinary translation invariance alone.

(K4) If (K1), (K2), and (K3) hold, then the degree-two bits $r_i(C)$ and the degree-one character bits $\rho_i(C)$ vanish for every $\mu_2^{(i)} \subset E[2]$. Since

$$b_{20} = r_1, \qquad b_{02} = r_2, \qquad b_{11} = r_1 + r_2 + r_3,$$

the three coefficients of $\beta_C^{E,E[2]}$ vanish, and therefore

$$\beta_C^{E,E[2]} = 0 \ \in \ H^2(BE[2];\mathbb{Z}/2).$$

The additional equations $\rho_i = 0$ select the trivial point of the $H^1(BE[2];\mathbb{F}_2)$-torsor of linearizations.

Thus $\beta_C^{E,E[2]} = 0$ and the finite $E[2]$-linearization is trivial under the stated hypotheses. For odd $N$, the finite-stabilizer obstruction vanishes cohomologically by Lemma 5.8. For even $N \geq 4$, the obstruction is detected on the 2-primary finite stabilizer and remains subject to the same equivariant linearization calculation; it is not closed by ordinary translation invariance.

*Proof.* The small-orbit stratum at $N = 2$ is the locus on which the $E$-action factors through the quotient $E \to E/E[2]$, so its isotropy is exactly $E[2]$ and the orbifold quotient $[\mathfrak{M}_C^{\mathrm{red}}(K_3 \times (E/E[2]))/(E/E[2])]$ is banded by $BE[2]$ (the inertia of the residual stabiliser), *not* by $B\mu_2$; this corrects the cyclic typing of clause (iv) of Lemma 5.8 at $N = 2$, which identified the band by its order without recording its abstract group structure. Since $E[2] \simeq (\mathbb{Z}/2)^2$ is the Klein-four group, its mod-2 cohomology is the polynomial ring on two degree-one generators $c_1(L_1), c_1(L_2)$ with $L_1, L_2 \in \widehat{E[2]}$ the two characters (Cartan–Eilenberg folklore for the cohomology of an elementary abelian 2-group), recovering $H^2(BE[2];\mathbb{Z}/2) \cong (\mathbb{Z}/2)^3$ as stated.

The three nontrivial cyclic subgroups $\mu_2^{(i)} = \langle e_i\rangle$ are detected as follows: writing $E[2]$ additively with basis $\{e_1, e_2\}$ and $e_3 = e_1 + e_2$, the inclusion $\mu_2^{(1)} \hookrightarrow E[2]$ restricts $L_1$ to the sign character of $\mu_2^{(1)}$

and $L_2$ to the trivial character; the inclusion $\mu_2^{(2)} \hookrightarrow E[2]$ restricts $L_2$ to the sign character of $\mu_2^{(2)}$ and $L_1$ to the trivial character; the inclusion $\mu_2^{(3)} \hookrightarrow E[2]$ restricts both $L_1, L_2$ to the sign character of $\mu_2^{(3)}$, so $c_1(L_1)$ and $c_1(L_2)$ restrict to the same generator and $c_1(L_1) \smile c_1(L_2)$ restricts to its square. The three restriction maps are therefore jointly injective on the rank-three space $H^2(BE[2]; \mathbb{Z}/2)$, so a class in $H^2(BE[2]; \mathbb{Z}/2)$ vanishes if and only if its image in each $H^2(B\mu_2^{(i)}; \mathbb{Z}/2)$ vanishes. This is the standard Klein-four detection by cyclic restriction.

Equivalently, for $\beta = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2$, the three restrictions are

$$\mathrm{res}_1(\beta) = b_{20}\tau_1^2, \qquad \mathrm{res}_2(\beta) = b_{02}\tau_2^2, \qquad \mathrm{res}_3(\beta) = (b_{20} + b_{11} + b_{02})\tau_3^2.$$

This gives the displayed conversion between the coefficient vector $(b_{20}, b_{11}, b_{02})$ and the cyclic residual vector $(r_1, r_2, r_3)$.

(K1). The Künneth factorization $o_C^{\mathrm{red}} = o_{K3} \otimes o_E$ is the underlying splitting of the multiplicative orientation cocycle of [35],[34] along the product factorization $X = K3 \times E$ on the protected sector, where the moduli of stable pairs decompose as a relative Hilbert scheme over the elliptic base. The factorwise $\mathrm{Pin}^c$ lift on the K3 factor is a supplied input, modeled on the BBDJS/Joyce–Upmeier and Cao–Kool–Monavari orientation packages [9, 12, 35]; the elliptic ordinary factor is the orientation of the cotangent bundle $\Omega_E^1$, which is trivial of rank one and trivialized by the flat translation-invariant one-form $\mathrm{d}z$.

(K2). The translation $t_{e_i} : X \to X$ acts as the identity on the K3 factor and as translation by $e_i$ on the elliptic factor. Pulling back along $t_{e_i}$ does not move the K3 factor. The trivial $E[2]$-lift on $o_{K3}$ is therefore a supplied factorwise linearization choice, not a consequence that settles the elliptic quotient descent.

(K3). The cotangent bundle $\Omega_E^1$ is trivial, trivialized by $\mathrm{d}z$; translation acts trivially on $\mathrm{d}z$. This proves only fixedness of the underlying ordinary line. It does not determine the equivariant lift. There are two separate finite calculations: the degree-two gerbe residual $\mathrm{res}\, \beta_C^{E,E[2]} = r_i\tau_i^2$, and, only after this residual has been null-trivialized, the degree-one character $\mathrm{res}\, \lambda_C^{E[2]} = \rho_i\tau_i$ of the chosen line linearization. Hypothesis (K3) is the explicit finite computation $r_i = 0$ and $\rho_i = 0$ for all three cyclic inertia sectors.

(K4). Combining (K1), (K2), (K3), the three cyclic degree-two residuals and degree-one character bits vanish. The coefficient conversion above then gives $b_{20} = b_{11} = b_{02} = 0$, hence $\beta_C^{E,E[2]} = 0 \in H^2(BE[2]; \mathbb{Z}/2)$; the equations $\rho_i = 0$ give the trivial $E[2]$-linearization. The odd-$N$ cases vanish for the cohomological reason of Lemma 5.8. Even-$N$ cases for $N \geq 4$ require the corresponding 2-primary equivariant linearization calculation. □

*Remark* 5.10 *(Orbifold-stratum subtlety).* The Künneth factorization (K1) is constructed on the smooth locus of $\mathfrak{M}_C^{\mathrm{red}}$; on the orbifold stratum $K3 \times \{e_i\} \subset X$ it requires the orbifold extension of the Joyce–Tanaka–Upmeier multiplicative cocycle to the inertia stack of the Klein-four banded quotient. Joyce–Tanaka–Upmeier [34] provide the orientation technology for finite-band quotient stacks. In the present Klein-four case the required check is concrete: the inertia decomposition into the three nontrivial cyclic sectors must carry the same trivial equivariant linearization. Equivalently, the degree-two equalities $r_i(C) = 0$ and the degree-one equalities $\rho_i(C) = 0$ of clause (K4) must both hold. Once that is verified, the orbifold stratum introduces no additional obstruction beyond the cyclic-sector criterion.

*Remark* 5.11 *(Two-torsion routes which do not replace the finite criterion).* There are three natural attempts to turn the $N = 2$ small-orbit calculation into a direct vanishing theorem. They are useful checks, but none replaces Lemma 5.9 and Corollary 5.14.

*Hyperelliptic route.* One may try to identify the stabilizer with the hyperelliptic involution $[-1]$ on $E$ and prove vanishing from a trivial action on Picard data. This is not a theorem in the present generality. The action of $[-1]$ on $\mathrm{Pic}^0(E)$ is inversion, not pointwise triviality, and ordinary fixedness of the line does not select the equivariant lift. The route becomes valid only after one supplies the actual $E[2]$-linearized orientation line and proves that its degree-one character $\lambda_C^{E[2]} \in H^1(BE[2];\mathbb{F}_2)$ is zero. That is precisely the $\rho_i = 0$ part of the finite criterion.

*Wu-class route.* A direct Stiefel–Whitney computation can identify the degree-two gerbe class with the $w_2$-class of the finite stabilizer representation, and Wu's formula expresses it as

$$w_2 = v_2 + Sq^1 v_1.$$

This gives an independent route to the bits $(r_1, r_2, r_3)$ once the reduced tangent/self-Ext complex and its $E[2]$-linearization have been constructed. It still does not compute the residual $H^1(BE[2];\mathbb{F}_2)$-character of the chosen linearization. Hence it can prove a degree-two gerbe vanishing, not strong reduced orientation.

*Orbifold Pin$^c$ route.* One can attempt to build the lift on the inertia stack of the four $E[2]$-fixed sectors by combining the K3 Pin$^c$ package of Cao–Kool–Monavari [12], the Joyce–Upmeier/Joyce–Tanaka–Upmeier orientation cocycle [34, 35], and Toën's orbifold Riemann–Roch [54]. This is a legitimate conditional construction target, not a completed proof here. It requires an inertia-stack Pin$^c$ lift on each cyclic sector, an age-correction parity calculation, multiplicative gluing across the smooth and orbifold strata, and trivial residual character under the whole Klein-four stabilizer. The $N = 2$ age calculation does not imply the higher even $E[N]$ calculation; for $N \geq 4$ the mixed two-primary class $A_{12}^{(N)} x_1 x_2$ and the degree-one characters $\lambda_i^{(N)}$ remain separate obstructions.

Thus the stable theorem in the manuscript is the finite Klein-four criterion: degree-two gerbe bits and degree-one linearization characters are both named and must both vanish. The three routes above are cross-checks or conditional construction programmes for those entries. They are not used as unconditional vanishing theorems.

Proposition 5.12 *(Rank-one two-torsion gerbe and linearization package).* On the rank-one D6–D2–D0 stratum $\mathfrak{M}^{\mathrm{red},(1)}_{(1,\beta,n)}$, assume the ordinary reduced Pin$^c$ orientation line has been constructed. The remaining $N = 2$ finite-stabilizer gerbe obstruction to descent of this line through the small-orbit quotient is the vector of three degree-two bits

$$r_i = \mathrm{res}_{\mu_2^{(i)}}(\beta_C^{E,E[2]})/\tau_i^2 \in \mathbb{Z}/2, \qquad i = 1, 2, 3.$$

Equivalently,

$$\beta_C^{E,E[2]} = 0 \quad \Longleftrightarrow \quad (r_1, r_2, r_3) = (0, 0, 0).$$

After this class has been null-trivialized, the triviality of the $E[2]$-linearization is a separate degree-one question. The set of linearizations is an $H^1(BE[2];\mathbb{F}_2)$-torsor; for a supplied linearization $\lambda_C^{E[2]}$, write

$$\mathrm{res}_{\mu_2^{(i)}} \lambda_C^{E[2]} = \rho_i \tau_i.$$

Trivial $E[2]$-linearization means $(\rho_1, \rho_2, \rho_3) = (0, 0, 0)$. The $r_i$ are not the $\rho_i$: they live in different cohomological degrees.

(C) *Cyclic-sector character, after null-trivialization.* For a genuine $E[2]$-linearized orientation line on the inertia sector,

$$\rho_i = \frac{1-\chi_i}{2}, \qquad \chi_i = \text{the scalar by which } e_i \text{ acts on } o_C^{\mathrm{red}}|_{I_{e_i}\mathfrak{M}}.$$

This computes the degree-one linearization character, not the degree-two gerbe class.

(W) *Stiefel–Whitney/Wu.* The degree-two bit is the coefficient of $\tau_i^2$ in the equivariant reduced obstruction class:

$$\mathrm{res}_{\mu_2^{(i)}}\, w_2^{E[2]}\Big(\mathbb{T}_C^{\mathrm{red}}\Big) = r_i\tau_i^2, \qquad w_2 = v_2 + Sq^1 v_1,$$

where $\mathbb{T}_C^{\mathrm{red}}$ is the reduced virtual tangent/self-Ext complex on the cyclic fixed sector and $v_j$ are its Wu classes.

(P) *Inertia-stack Pin$^c$.* The degree-two bit is the Pin$^c$ central holonomy corrected by the mod-2 orbifold Riemann–Roch age:

$$r_i = \mathrm{hol}_{\ker(\mathrm{Pin}^c\to SO)}\Big(\widetilde{P}^{\mathrm{red}}_{\mathrm{Pin}^c}|_{I_{e_i}\mathfrak{M}}\Big)(e_i) + \mathrm{age}^{\mathrm{red}}_{e_i} \pmod 2.$$

Thus the rank-one two-torsion small-orbit sector is closed only after both vectors vanish:

$$(r_1, r_2, r_3) = (0,0,0), \qquad (\rho_1, \rho_2, \rho_3) = (0,0,0).$$

The present manuscript computes neither vector; the proposition is the finite obstruction and linearization target, not a vanishing theorem.

*Proof.* Lemma 5.9 identifies

$$H^2(BE[2];\mathbb{F}_2) = \mathbb{F}_2\langle x_1^2, x_1x_2, x_2^2\rangle$$

and proves that restriction to the three cyclic subgroups $\mu_2^{(i)} = \langle e_i\rangle$ is jointly injective in degree two. Hence $\beta_C^{E,E[2]}$ vanishes if and only if its three restrictions $r_i\tau_i^2$ vanish.

For a line over the quotient stack, after the degree-two gerbe class has been null-trivialized, the restriction to $\mu_2^{(i)}$ is a one-dimensional equivariant line on $[I_{e_i}\mathfrak{M}/\mu_2^{(i)}]$. The character $\chi_i = \pm 1$ is trivial exactly when the degree-one coefficient $\rho_i$ is zero; this is the formula in (C). The degree-two residual is a different class. It is the second Stiefel–Whitney obstruction of the equivariant reduced tangent/self-Ext complex, and Wu's identity gives $w_2 = v_2 + Sq^1v_1$, giving (W). Finally, a Pin$^c$ lift of the cyclic-sector frame records the central $\mathbb{Z}/2$-holonomy of $\ker(\mathrm{Pin}^c \to SO)$; orbifold Riemann–Roch contributes the age correction on the fixed inertia sector. This is (P). The formulae (W) and (P) compute the residual degree-two class $\mathrm{res}_{\mu_2^{(i)}}\beta_C^{E,E[2]}$, while (C) computes the residual degree-one choice after that class has been null-trivialized. □

LEMMA 5.13 *(Even finite-stabilizer residuals beyond two torsion).* Let $N$ be even and write $2^a \parallel N$. On the $N$-fold small-orbit stratum of the reduced $E$-quotient, the positive-degree mod-2 finite-stabilizer obstruction is the two-primary class

$$\beta_{C,(2)}^{E,N} \in H^2(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2).$$

For $a = 1$ this is the Klein-four class $\beta_C^{E,E[2]}$ of Lemma 5.9. For $a \geq 2$,

$$H^*(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2) = \Lambda(x_1,x_2)\otimes\mathbb{F}_2[y_1,y_2], \qquad |x_i| = 1, \quad |y_i| = 2, \quad x_i^2 = 0,$$

and therefore

$$H^2(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2) = \mathbb{F}_2\langle y_1, x_1x_2, y_2\rangle.$$

Write

$$\beta^{E,N}_{C,(2)} = A_1^{(N)} y_1 + A_{12}^{(N)} x_1x_2 + A_2^{(N)} y_2.$$

Let $g_1, g_2$ be the standard generators of $(\mathbb{Z}/2^a)^2$, put $h_i = 2^{a-1}g_i$, $i = 1, 2$, and $h_3 = h_1 + h_2$. For the three order-two cyclic subgroups $\langle h_i\rangle$,

$$\operatorname{res}_{\langle h_1\rangle} \beta^{E,N}_{C,(2)} = A_1^{(N)}\tau_1^2, \qquad \operatorname{res}_{\langle h_2\rangle} \beta^{E,N}_{C,(2)} = A_2^{(N)}\tau_2^2,$$

$$\operatorname{res}_{\langle h_3\rangle} \beta^{E,N}_{C,(2)} = (A_1^{(N)} + A_2^{(N)})\tau_3^2.$$

The mixed coefficient $A_{12}^{(N)}$ is invisible to all cyclic restrictions, not only to the three order-two subgroups above. It is detected by the rank-two quotient $(\mathbb{Z}/2^a)^2 \twoheadrightarrow (\mathbb{Z}/2)^2$, which pulls the mixed class $\bar{x}_1\bar{x}_2$ back to $x_1x_2$. Therefore the three $E[2]$ bits $r_1, r_2, r_3$ do not determine the even $N \geq 4$ residual. Vanishing for $N \geq 4$ is a separate $2^a$-primary equivariant-linearization calculation on the small-orbit stratum. After $\beta^{E,N}_{C,(2)} = 0$, triviality of the chosen two-primary linearization is a further degree-one condition:

$$\lambda^{E,N}_C = \lambda_1^{(N)} x_1 + \lambda_2^{(N)} x_2 \in H^1(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2)$$

must vanish. This character is detected on the cyclic subgroups generated by $g_1$ and $g_2$, not by the order-two subgroups $\langle h_i\rangle$.

*Proof.* The odd part of $E[N] \simeq (\mathbb{Z}/N)^2$ has order invertible in $\mathbb{F}_2$, so transfer identifies the positive-degree $\mathbb{F}_2$-cohomology with that of the two-primary factor $(\mathbb{Z}/2^a)^2$. If $a = 1$, this is $E[2]$ and the cohomology ring is $\mathbb{F}_2[x_1, x_2]$, $|x_i| = 1$, giving the Klein-four basis $\langle x_1^2, x_1x_2, x_2^2\rangle$. If $a \geq 2$, the cyclic calculation gives

$$H^*(B\mathbb{Z}/2^a;\mathbb{F}_2) \cong \Lambda(x) \otimes \mathbb{F}_2[y], \qquad |x| = 1, \quad |y| = 2, \quad x^2 = 0.$$

Kunneth for the product gives the degree-two basis $\langle y_1, x_1x_2, y_2\rangle$.

For the restriction formulas, restrict first to the order-two subgroup $\langle h_1\rangle = \langle 2^{a-1}g_1\rangle$. The degree-one class $x_1 \in H^1(B\mathbb{Z}/2^a;\mathbb{F}_2)$ restricts to zero because the generator $h_1$ maps to $2^{a-1} \equiv 0 \pmod 2$ under the mod-two character, while the degree-two cyclic extension class $y_1$ restricts to the generator $\tau_1^2 \in H^2(B\langle h_1\rangle;\mathbb{F}_2)$. The same calculation holds for $h_2$. For $h_3 = 2^{a-1}(g_1 + g_2)$, both $y_1$ and $y_2$ restrict to $\tau_3^2$, while $x_1$ and $x_2$ restrict to zero. Hence $x_1x_2$ restricts to zero on all three order-two cyclic subgroups, and the displayed formulas follow.

The quotient $(\mathbb{Z}/2^a)^2 \twoheadrightarrow (\mathbb{Z}/2)^2$ identifies $H^1(B(\mathbb{Z}/2)^2;\mathbb{F}_2)$ with the span of $x_1, x_2$; hence $\bar{x}_1\bar{x}_2$ inflates to $x_1x_2$. The mixed coefficient is therefore a genuine rank-two equivariant-linearization obstruction. It is invisible on every cyclic subgroup: for a cyclic subgroup of order 2, the degree-one classes $x_1, x_2$ restrict to zero as above; for a cyclic subgroup of order $2^b$ with $b \geq 2$, the restrictions of $x_1, x_2$ are scalar multiples of the single degree-one generator $x$, and $x_1x_2$ restricts to a scalar multiple of $x^2 = 0$ in $H^2(B\mathbb{Z}/2^b;\mathbb{F}_2)$. This is a different stabilizer calculation from the $N = 2$ Klein-four residual, so the vanishing of $(r_1, r_2, r_3)$ supplies no vanishing theorem for $N \geq 4$. Finally, when the degree-two class has vanished, the residual two-primary linearisations are classified by

$$H^1(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2) = \mathbb{F}_2\langle x_1, x_2\rangle.$$

The order-two subgroups generated by $h_i = 2^{a-1}g_i$ do not detect this character, since $x_j(h_i) = 0$. Triviality must be checked on the full two-primary generators, for instance $g_1, g_2$. □

COROLLARY 5.14 *(Finite-stabilizer obstruction to* (O1)*).* Fix a small-orbit stratum of the reduced $E$-quotient with stabilizer $E[N]$. Assume that the ordinary reduced square-root gerbe $\alpha_C^{\mathrm{red}}$ and the connected free quotient class $\alpha_C^{E,\mathrm{free}}$ have already been null-trivialized on this stratum. Then the finite-stabilizer part of the strong reduced orientation (O1) exists with trivial quotient linearization if and only if the following finite obstruction tuple vanishes.

(i) If $N$ is odd, there is no positive-degree mod-2 finite-stabilizer obstruction:

$$H^{>0}(BE[N];\mathbb{F}_2) = 0.$$

(ii) If $N = 2$, the tuple is

$$\mathfrak{o}_C^{E[2]} = (r_1, r_2, r_3; \rho_1, \rho_2, \rho_3),$$

where

$$\mathrm{res}_{\langle e_i\rangle}\, \beta_C^{E,E[2]} = r_i\tau_i^2, \qquad \mathrm{res}_{\langle e_i\rangle}\, \lambda_C^{E[2]} = \rho_i\tau_i.$$

Equivalently,

$$\beta_C^{E,E[2]} = 0 \quad \text{and} \quad \lambda_C^{E[2]} = 0.$$

(iii) If $N$ is even and $2^a \parallel N$ with $a \geq 2$, the tuple is

$$\mathfrak{o}_C^{E[N]} = (A_1^{(N)}, A_{12}^{(N)}, A_2^{(N)}; \lambda_1^{(N)}, \lambda_2^{(N)}),$$

where

$$\beta_{C,(2)}^{E,N} = A_1^{(N)} y_1 + A_{12}^{(N)} x_1x_2 + A_2^{(N)} y_2$$

and

$$\lambda_C^{E,N} = \lambda_1^{(N)} x_1 + \lambda_2^{(N)} x_2 \in H^1(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2).$$

Thus the $E[2]$ three-bit calculation proves only the $N = 2$ degree-two gerbe vanishing. It does not prove the higher even finite-stabilizer class $A_{12}^{(N)} x_1x_2$, and it does not prove the residual degree-one linearization character. Any non-zero entry of $\mathfrak{o}_C^{E[N]}$ obstructs (O1) on that small-orbit stratum. No scalar automorphic character, Borcherds divisor order, or OP/DT scalar square changes this finite quotient obstruction.

*Proof.* For a finite stabilizer $G = E[N]$, the Borel descent spectral sequence

$$E_2^{p,q} = H^p(BG;\, H^q(\mathfrak{M}_C^{\mathrm{red}};\mathbb{F}_2)) \Longrightarrow H_G^{p+q}(\mathfrak{M}_C^{\mathrm{red}};\mathbb{F}_2)$$

puts the finite square-root obstruction in total degree two. After that class is null-trivialized, linearizations of the resulting line form an $H^1(BG;\mathbb{F}_2)$-torsor. Hence descent with the trivial quotient orientation is equivalent to the simultaneous vanishing of the degree-two finite gerbe and the selected degree-one character. The odd case is transfer. The $N = 2$ case is Lemma 5.9 together with Proposition 5.12. The higher even case is Lemma 5.13. □

LEMMA 5.15 *(Rank-one Pin-lift sign rigidification conditional on a supplied lift).* Assume the BBDJS $\mathrm{Pin}^c$ lift [9] of the **SO**-frame bundle of $\det \mathrm{RHom}_{\mathrm{red}}(C,C)$ on $\mathfrak{M}^{\mathrm{red},(1)}_{(1,\beta,n)}$ has been constructed after the ordinary reduced gerbe class has vanished. Its underlying **SO**-bundle determines it only up to a global $\mathbb{Z}/2$ ambiguity $\eta_C \in H^0(\mathfrak{M}^{\mathrm{red},(1)}_{(1,\beta,n)};\mathbb{Z}/2)$, the kernel of the surjection $\mathrm{Pin}^c \twoheadrightarrow \mathbf{SO}$ acting on global sections of the lift. On the rank-one D6–D2–D0 stratum, this residual sign is rigidified by the following data, jointly:

(S1) the Joyce–Upmeier orientability cocycle [35], [34] on the Hall extension correspondences, which forces a multiplicative compatibility $\eta_{C\oplus C'} = \eta_C\,\eta_{C'}\,(-1)^{\langle C,C'\rangle_{\mathrm{Mukai}}}$ under direct sum on the rank-one locus;

(S2) the requirement that the Pfaffian section $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)$ of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ supplied in Theorem 3.12 restricts at the unique top-cohomology generator of the isotropic-cusp expansion to the monic Borcherds normalization $D_5 = 64^{-1}\Delta_5$ of Proposition 4.5; equivalently, the leading Fourier coefficient of $\mathrm{Pf}_{\mathrm{prot}}$ at the isotropic cusp $\mathfrak{c}_\infty$ equals +1 (positive orientation, monic);

(S3) compatibility, after squaring, with the OP/DT scalar sign $C_{\square,X} = -4096$ of Proposition 11.1, i.e. $(\mathrm{Pf}_{\mathrm{prot}})^2|_{\text{rank-one}} = \Delta_5^2$ with positive leading coefficient $c_\Delta(1,1,1)^2 = 64^2 = 4096$, the absolute value of $C_{\square,X}$, and the OP scalar sign $\mathrm{sign}(C_{\square,X}) = -1$ absorbed into the fermionic shift $(-p_{\mathrm{DT}})^n$ of the OP/DT chamber, not into the orientation line.

Then $\eta_{(1,\beta,n)} = +1$ for the supplied ordinary $\mathrm{Pin}^c$ lift, and the orientation line $o^{\mathrm{red},(1)}_{(1,\beta,n)}$ is the unique square root of $\det \mathrm{RHom}_{\mathrm{red}}(C,C)$ whose Pfaffian section is monic at $\mathfrak{c}_\infty$, subject to the separate connected $BE$-linearization and finite $E[2]$-residual vanishing of Lemmas 5.8 and 5.9. This lemma fixes the residual global sign; it does not prove the quotient linearization.

*Proof.* The $\mathrm{Pin}^c \twoheadrightarrow \mathbf{SO}$ sequence has kernel $\mathbb{Z}/2$, so on a connected component the global lift is unique up to multiplying by a constant element of this kernel; on the disconnected components of $\mathfrak{M}^{\mathrm{red},(1)}_{(1,\beta,n)}$ (indexed by the discrete Mukai-charge label $(1,\beta,n)$, connected within each fixed charge by the Beauville–Mukai construction of [12]) the ambiguity is the function $\eta\colon \pi_0\mathfrak{M}^{\mathrm{red},(1)} \to \{\pm 1\}$.

*(S1)* fixes the multiplicative behaviour of $\eta$ under direct sum: by Joyce–Upmeier [35, Theorem 5.1], the orientation cocycle on the extension correspondence $\mathfrak{M}^{\mathrm{red},(1)}_C \times \mathfrak{M}^{\mathrm{red},(1)}_{C'} \to \mathfrak{M}^{\mathrm{red},(1)}_{C+C'}$ is the Mukai parity $(-1)^{\langle C,C'\rangle_{\mathrm{Mukai}}}$, and the BBDJS $\mathrm{Pin}^c$ lift respects this cocycle by [9, §5.4] together with the cosection-twist compatibility of [12]. Hence $\eta$ is a *group* homomorphism modulo the Mukai parity, and is determined by its value on a single generating component.

*(S2)* fixes $\eta$ on that generator: the Pfaffian section is a section of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$, and Proposition 4.5 together with the square-root normalization table of Remark Normalization of the square root require the leading Fourier coefficient of $\mathrm{Pf}_{\mathrm{prot}}$ at $\mathfrak{c}_\infty$ to be +1 (the monic $D_5 = 64^{-1}\Delta_5$ being the canonical Borcherds output). The coefficient 64 is the theta-normalization converting the monic Borcherds product $D_5$ to $\Delta_5$; it is not the square of an orientation line. The orientation line chooses the sign of a supplied first-order Pfaffian section, not the scalar prefactor. This pins $\eta_{(1,\beta,n)} \in \{+1,-1\}$ to a specific value.

*(S3)* verifies consistency: squaring an orientation line forgets $\eta$, so the only sign data surviving after squaring are $(c_\Delta(1,1,1))^2 = 4096$ (positive, automatic) and the fermionic-shift sign $\mathrm{sign}(C_{\square,X}) = -1$, which lives in the OP/DT chamber of Proposition 11.1 (the $(-p_{\mathrm{DT}})^n$ factor) and does not constrain $\eta$. So (S3) is consistent with (S2) but does not refine it. Combining (S1) and (S2), $\eta_{(1,\beta,n)} = +1$ on the generating component; multiplicatively extended via Mukai parity, this determines $\eta$ on all components of the rank-one stratum.

The choice is determinate once the ordinary lift and quotient linearizations have been supplied. This is the analogue, on the cosection-twisted reduced $K3 \times E$ rank-one sector, of the BBDJS Pin lift sign convention of [9, §5.4] and the Joyce–Upmeier rigidification by Pfaffian normalization data [35]. It is not a proof that $\alpha^{E,\mathrm{free}}_C$ or $\beta^{E,E[2]}_C$ vanishes. □

*Remark* 5.16. The same data (S1), (S2), (S3) rigidify $\eta_C$ on a higher-rank semistable stratum only after the $\mathrm{Pin}^c$-lift gerbe class $\alpha^{\mathrm{red}}_C$ of (P3) and the required quotient linearization classes have already vanished.

Lemma 5.17 *(HN-reducible Pin$^c$-lift gerbe criterion).* Let $C$ be a higher-rank protected charge and let $\tau = (C_1, \dots, C_\ell)$ be an HN-type of length $\ell \geq 2$ in the reduced semistable stack. On the corresponding HN stratum, the reduced orientation gerbe satisfies

$$\alpha_C^{\mathrm{red}}|_\tau = \sum_i \alpha_{C_i}^{\mathrm{red}} \bmod 2,$$

with off-diagonal self-Ext blocks contributing even class. Therefore $\alpha_C^{\mathrm{red}}|_\tau = 0$ on every HN stratum whose graded factors lie in the already-oriented rank-one sector or in a rank-two sector covered by Lemma 5.5. If, in addition, the $E$-equivariant parity

$$\sum_i d_E(C_i)\,\mathrm{rk}(\det C_i)$$

is even, then the joint class $\alpha_C^{\mathrm{red}} + \alpha_C^E$ vanishes on that HN stratum, subject to the finite-stabilizer criterion of Lemma 5.9 on any small-orbit boundary. Stable higher-rank length-one HN strata are not covered by this lemma.

*Proof.* The HN stratification expresses a length-$\ell$ stratum as an extension stack over the product of the moduli of its graded factors. The Joyce–Upmeier and Joyce–Tanaka–Upmeier multiplicative orientation formalism [34, 35] identifies the orientation gerbe of the direct-sum target with the tensor product of the gerbes of the graded pieces. The off-diagonal reduced self-Ext blocks occur in dual pairs under the shifted-symplectic pairing, so their Stiefel–Whitney contribution is even. This gives the displayed additivity formula. The stated vanishing follows immediately when each graded factor lies in a sector whose orientation gerbe has already been trivialized.

The connected $E$-descent obstruction is computed by the equivariant Serre spectral sequence of the free translation quotient; on an HN stratum its parity is the sum of the elliptic rotation numbers of the graded factors. If that sum is even, the connected obstruction vanishes. The remaining torsion part is the small-orbit finite stabilizer gerbe; the rank-one $N = 2$ detection criterion is isolated in Lemma 5.9. No induction on stable higher-rank length-one factors is used. □

## 6 Physical charge and normal-ordered Gram descent

The Hall charge is additive. The Igusa degree is not. This section isolates the exact defect and removes it by normal ordering. The result is a lattice theorem, not yet a chain-level Hall strictification.

*Definition* 6.1 *(Dyonic Mukai lattice and quadratic Gram projection).* Let $S$ be a complex K3 surface and put

$$\Lambda_S := H^*(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z}),$$

the Mukai lattice in the sense of Mukai (1987): even, unimodular, signature $(4, 20)$, rank 24, equipped with the Mukai pairing $\langle (r, c, s), (r', c', s') \rangle_{\mathrm{Muk}} = c \cdot c' - rs' - r's$. The physical microscopic charge lattice for the type-II compactification on $X = S \times E$ is the additive abelian group

$$\Gamma_X^{\mathrm{phys}} := \Lambda_S \oplus \Lambda_S, \qquad c = (Q, P) \in \Gamma_X^{\mathrm{phys}},$$

graded electric $\oplus$ magnetic. The Gram-index lattice is

$$\Gamma_{\mathrm{gram}} := \mathbb{Z}^3.$$

The Gram map is the quadratic projection

$$\Pi_X : \Gamma_X^{\mathrm{phys}} \longrightarrow \Gamma_{\mathrm{gram}}, \qquad (Q,P) \longmapsto \big(\tfrac{(Q,Q)}{2}, (Q,P), \tfrac{(P,P)}{2}\big).$$

The bilinear obstruction to additivity of $\Pi_X$ is the symmetric cocycle

$$B : \Gamma_X^{\mathrm{phys}} \times \Gamma_X^{\mathrm{phys}} \longrightarrow \Gamma_{\mathrm{gram}}, \qquad B(c,c') = \big((Q,Q'), (Q,P') + (Q',P), (P,P')\big),$$

satisfying $\Pi_X(c+c') = \Pi_X(c) + \Pi_X(c') + B(c,c')$ and $B(c,c) = 2\Pi_X(c)$; $[B] = 0$ in $H^2_{\mathrm{sym}}(\Gamma_X^{\mathrm{phys}}; \Gamma_{\mathrm{gram}})$ via Lemma 6.9. The central extension

$$\widehat{\Gamma}_X := \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}, \qquad (c,T) \star (c',T') := \big(c+c',\ T+T'+B(c,c')\big),$$

has the normal-ordered Gram homomorphism

$$\overline{\Pi}_X(c,T) = \Pi_X(c) - T.$$

It is the additive abelian group on which a Hall product graded by physical charge and normal-ordered Gram invariants can be formulated.

*Remark* 6.2. The quadratic-projection identity $\Pi_X(c+c') = \Pi_X(c) + \Pi_X(c') + B(c,c')$ expands $(Q+Q', Q+Q')/2, (Q+Q', P+P'), (P+P', P+P')/2$. Cohomologically the central extension is split ($[B] = 0$, Remark 6.10), but the splitting requires the quadratic primitive $\Pi_X$, not a linear cochain. Dyonic charges with distinct $(Q,P)$ but identical $(n,l,m)$ sit in the same Gram sector; the Hall bracket distinguishes them through the physical grading on $\Gamma_X^{\mathrm{phys}}$ before passing to Gram invariants.

The preceding determinant was constructed in one cusp chamber. Duality does not preserve the displayed chamber inequalities in the same coordinates; it transports the chamber, the Fourier expansion, and the automorphic line together. The invariant object is

$$\mathcal{D}_X = \Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5}),$$

with cusp expansion

$$\mathcal{D}_X|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

For

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix},$$

put

$$q = e^{2\pi i z_1}, \qquad r = e^{2\pi i z_2}, \qquad s = e^{2\pi i z_3}.$$

For

$$T(n,l,m) = \begin{pmatrix} n & l/2 \\ l/2 & m \end{pmatrix}, \qquad n,l,m \in \mathbb{Z},$$

put

$$(T(n,l,m), Z)_{\mathrm{tr}} = \mathrm{tr}(T(n,l,m)Z) = n z_1 + l z_2 + m z_3.$$

Then

$$x^{(n,l,m)} = \exp(2\pi i (T(n,l,m), Z)_{\mathrm{tr}}) = q^n r^l s^m.$$

*Definition 6.3 (Gram-index notation).*

$$\Gamma_{\mathrm{gram}} := \Gamma_{\mathrm{ind}} := \mathbb{Z}^3, \qquad \Gamma_{\mathrm{act}} \subset \Gamma_{\mathrm{eff}} \subset \Gamma_{\mathrm{gram}}.$$

Here $\Gamma_{\mathrm{eff}}$ is the cusp-positive semigroup and $\Gamma_{\mathrm{act}}$ is the non-zero Jacobi-support locus. This is notation for Gram invariants and Fourier indices. It is not the microscopic charge lattice; the latter is $\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S$ of Definition 6.1, and the Gram-index lattice is the image of the quadratic projection $\Pi_X : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$. A Hall algebra cannot be graded by $\Gamma_{\mathrm{ind}}$ directly because $\Pi_X$ is quadratic; the correct grading lattice is the central extension $\widehat{\Gamma}_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}$ (Definition 6.1). The grading on $\mathcal{A}_{\mathrm{den}}$ by $\Gamma_{\mathrm{ind}}$ is then the normal-ordered image of a physical $\widehat{\Gamma}_X$-grading on a hypothetical Hall source. The chain-level lift of this image is part of the open Hall datum.

Proposition 6.4 *(Gram matrix convention for protected charges).* Assume a protected charge grading with primitive electric-magnetic pair $Q, P$ in an even lattice. Set

$$n = \frac{(Q,Q)}{2}, \qquad l = (Q,P), \qquad m = \frac{(P,P)}{2}.$$

Then

$$\begin{pmatrix} (Q,Q) & (Q,P) \\ (P,Q) & (P,P) \end{pmatrix} = 2T(n,l,m),$$

and the Gram discriminant is

$$\det(2T(n,l,m)) = 4nm - l^2.$$

The genus-two chemical potential $Z$ pairs with the same matrix:

$$\exp(2\pi i \ \mathrm{tr}(TZ)) = q^n r^l s^m.$$

Thus the Borcherds monomial records the Gram invariants of the protected charge grading, once such a physical grading has been supplied.

*Proof.* By hypothesis the compactification charge lattice is even, so $(Q,Q)/2$ and $(P,P)/2$ are integral and $(Q,P) \in \mathbb{Z}$. The matrix identity and its determinant are immediate. The final equality is the Fourier character attached to the chemical potential $Z$. The Harvey–Moore and Dijkgraaf–Verlinde–Verlinde interpretation of this Fourier character as a physical charge grading is additional physical input [32], [20]. □

*Definition 6.5 (Additive physical charge lattice and Gram shadow).* Let $X = S \times E$, with $S$ a K3 surface, and let

$$\Lambda_S := \widetilde{H}(S,\mathbb{Z}) = H^0(S,\mathbb{Z}) \oplus H^2(S,\mathbb{Z}) \oplus H^4(S,\mathbb{Z})$$

with Mukai pairing

$$(r,D,s)\cdot(r',D',s') = D\cdot D' - rs' - r's.$$

In the even $K3 \times E$ Mukai sector, every charge decomposes as

$$v_X = P \otimes 1_E + Q \otimes \omega_E, \qquad Q, P \in \Lambda_S,$$

where $\int_E \omega_E = 1$. The additive physical charge lattice is

$$\Gamma_X^{\mathrm{phys}} := \Lambda_S \oplus \Lambda_S, \qquad c = (Q,P).$$

For exact triangles or Hall extensions the class $c$ is additive. The Igusa variables are the Gram shadow

$$\Pi_X : \Gamma_X^{\mathrm{phys}} \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \Pi_X(Q,P) = \left(\frac{Q^2}{2},\, Q\cdot P,\, \frac{P^2}{2}\right).$$

The map $\Pi_X$ is an invariant map to Fourier/root degrees; it is not the Hall charge lattice.

Lemma 6.6 *(Primitive lift of every Gram triple).* Suppose $\Lambda_S$ contains two orthogonal hyperbolic planes

$$U_1 \oplus U_2 = \mathbb{Z}e_1 \oplus \mathbb{Z}f_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}f_2,$$

with $e_i^2 = f_i^2 = 0$, $e_i \cdot f_i = 1$, and $U_1 \perp U_2$. Then every Gram triple $(n,l,m) \in \mathbb{Z}^3$ has a primitive saturated physical lift $c = (Q,P) \in \Gamma_X^{\mathrm{phys}}$ with

$$\Pi_X(Q,P) = (n,l,m).$$

One may take

$$Q = e_1 + nf_1, \qquad P = lf_1 + e_2 + mf_2.$$

*Proof.* The hyperbolic-plane pairings give

$$Q^2 = 2n, \qquad P^2 = 2m, \qquad Q\cdot P = l.$$

Therefore $\Pi_X(Q,P) = (n,l,m)$. The sublattice $\mathbb{Z}Q + \mathbb{Z}P \subset \Lambda_S$ is primitive because the coefficient matrix of $Q, P$ in the ordered basis $(e_1, f_1, e_2, f_2)$ contains the minor

$$\det\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$$

using the $e_1, e_2$ rows. Hence the lift is primitive saturated. □

*Remark* 6.7. The obstruction in the compact Hall problem is therefore not pointwise existence of physical charge planes with a prescribed Igusa degree. Every formal Borcherds degree has such a primitive lift. The obstruction is functorial compatibility with Hall addition and the primitive bracket.

Lemma 6.8 *(Gram additivity defect).* For $c_i = (Q_i, P_i) \in \Gamma_X^{\mathrm{phys}}$,

$$\Pi_X(c_1 + c_2) = \Pi_X(c_1) + \Pi_X(c_2) + B(c_1, c_2),$$

where

$$B(c_1,c_2) = \big(Q_1\cdot Q_2,\ Q_1\cdot P_2 + Q_2\cdot P_1,\ P_1\cdot P_2\big) \in \Gamma_{\mathrm{gram}}.$$

*Proof.* Expand

$$\frac{(Q_1+Q_2)^2}{2}, \qquad (Q_1+Q_2)\cdot(P_1+P_2), \qquad \frac{(P_1+P_2)^2}{2}.$$

The three cross terms are exactly the three components of $B(c_1,c_2)$. □

Lemma 6.9 *(Symmetric bilinear 2-cocycle and on-the-nose associativity of the Gram-corrected charge product).* Let $S$ be a complex K3 surface and $\Lambda_S = H^*(S,\mathbb{Z})$ the Mukai lattice (even unimodular, signature $(4,20)$). Let $\Gamma_X^{\mathrm{phys}} := \Lambda_S \oplus \Lambda_S$, $\Gamma_{\mathrm{gram}} := \mathbb{Z}^3$, and let $\Pi_X : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$, $B : \Gamma_X^{\mathrm{phys}} \times \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$ be as in Definition 6.1, namely, with $c = (Q,P)$,

$$\Pi_X(Q,P) = \big(\tfrac{(Q,Q)}{2},\ (Q,P),\ \tfrac{(P,P)}{2}\big), \qquad B((Q,P),(Q',P')) = \big((Q,Q'),(Q,P')+(Q',P),(P,P')\big).$$

Then:

(i) $B$ is symmetric and $\mathbb{Z}$-bilinear, and takes integral values: $B \in \mathrm{Hom}_{\mathbb{Z}}\big(S^2\Gamma_X^{\mathrm{phys}},\Gamma_{\mathrm{gram}}\big) \subset \mathrm{Hom}_{\mathbb{Z}}\big(\Gamma_X^{\mathrm{phys}} \otimes_{\mathbb{Z}} \Gamma_X^{\mathrm{phys}},\Gamma_{\mathrm{gram}}\big)$.

(ii) (Polarisation.) $B(c,c) = 2\Pi_X(c)$ for every $c \in \Gamma_X^{\mathrm{phys}}$.

(iii) (Symmetric Hochschild 2-cocycle.) $B$ is normalised, $B(c,0) = B(0,c) = 0$, and the additive coboundary

$$(\partial B)(a,b,c) := B(b,c) - B(a+b,c) + B(a,b+c) - B(a,b)$$

vanishes identically. Hence $[B] \in H^2_{\mathrm{sym}}\big(\Gamma_X^{\mathrm{phys}};\Gamma_{\mathrm{gram}}\big)$ is well defined.

(iv) (Star-associativity on the nose.) The product on $\widehat{\Gamma}_X := \Gamma_X^{\mathrm{phys}} \oplus \Gamma_{\mathrm{gram}}$,

$$(c,T) \star (c',T') := \big(c+c',\ T+T'+B(c,c')\big),$$

is associative and commutative; the unit is $(0,0)$ and the inverse of $(c,T)$ is $(-c,\ -T+B(c,c)) = (-c,\ -T+2\Pi_X(c))$. Thus $(\widehat{\Gamma}_X,\star)$ is an abelian group, and the sequence

$$0 \longrightarrow \Gamma_{\mathrm{gram}} \xrightarrow{T\mapsto(0,T)} \widehat{\Gamma}_X \xrightarrow{\mathrm{pr}_1} \Gamma_X^{\mathrm{phys}} \longrightarrow 0$$

is a central extension of abelian groups with cocycle $B$.

(v) (Normal-ordered Gram map.) The map

$$\overline{\Pi}_X : \widehat{\Gamma}_X \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c,T) = \Pi_X(c) - T,$$

is a homomorphism from $(\widehat{\Gamma}_X,\star)$ to the additive group $\Gamma_{\mathrm{gram}}$.

(vi) (Splitting; $[B] = 0$.) The Gram-shadow section $s : \Gamma_X^{\mathrm{phys}} \to \widehat{\Gamma}_X$, $s(c) := (c,\Pi_X(c))$, is a homomorphism for $\star$; equivalently $B = -\partial\Pi_X$ in $B^2_{\mathrm{sym}}(\Gamma_X^{\mathrm{phys}};\Gamma_{\mathrm{gram}})$, where $(\partial\Pi_X)(c,c') = \Pi_X(c) + \Pi_X(c') - \Pi_X(c+c')$. In particular $[B] = 0$, the extension is split as an extension of abelian groups, and $(c,T) \mapsto (c, T - \Pi_X(c))$ is an isomorphism $\widehat{\Gamma}_X \xrightarrow{\sim} \Gamma_X^{\mathrm{phys}} \oplus \Gamma_{\mathrm{gram}}$ of abelian groups carrying $\star$ to componentwise addition.

(vii) (Quadratic obstruction.) $B$ is not the coboundary of any *linear* 1-cochain $L : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$: if $B(c,c') = L(c) + L(c') - L(c+c')$ for an additive $L$, then setting $c' = c$ and using (ii) gives $2\Pi_X(c) = 2L(c) - L(2c) = 0$, i.e. $\Pi_X \equiv 0$, contradiction. Trivialising $[B]$ requires the quadratic refinement $\Pi_X$; the defect is a polarization-correction class, not a separate Heisenberg extension datum.

*Proof.* Write $c = (Q, P)$, $c' = (Q', P')$, $c'' = (Q'', P'')$ with $Q, Q', Q'', P, P', P'' \in \Lambda_S$, and abbreviate $(\,\cdot\,,\,\cdot\,) : \Lambda_S \times \Lambda_S \to \mathbb{Z}$ by juxtaposition, $QQ' = (Q, Q')$, so that $B(c, c') = (QQ',\ QP' + Q'P,\ PP')$.

*(i) Symmetry, bilinearity, integrality.* The components $QQ'$ and $PP'$ are the symmetric Mukai pairing on $\Lambda_S$, so each is symmetric and $\mathbb{Z}$-bilinear. The middle component is the polarisation $(c, c') \mapsto QP' + Q'P$ of the symmetric biadditive form $(c, c') \mapsto QP' + Q'P$; swapping $(Q, P) \leftrightarrow (Q', P')$ returns $Q'P + QP' = QP' + Q'P$, so it is symmetric, and $\mathbb{Z}$-bilinear in each slot by bilinearity of $(\,\cdot\,,\,\cdot\,)$. Integrality of every component follows from the unimodular even structure on $\Lambda_S$.

*(ii) Polarisation.* Set $c' = c$: the cross terms reduce to $QQ$, $QP + QP = 2QP$, $PP$, so $B(c, c) = (Q^2, 2QP, P^2) = 2\Pi_X(c)$.

*(iii) Cocycle.* Bilinearity gives normalisation $B(c, 0) = B(0, c) = 0$ immediately. For the cocycle identity, observe that any symmetric $\mathbb{Z}$-bilinear form $\beta : G \times G \to A$ on an abelian group $G$ with values in a trivial $G$-module $A$ satisfies, by bilinearity in the first slot,

$$\beta(a + b, c) = \beta(a, c) + \beta(b, c), \qquad \beta(a, b + c) = \beta(a, b) + \beta(a, c).$$

Substituting into $(\delta\beta)(a, b, c) = \beta(b, c) - \beta(a + b, c) + \beta(a, b + c) - \beta(a, b)$,

$$(\delta\beta)(a, b, c) = \beta(b, c) - \beta(a, c) - \beta(b, c) + \beta(a, b) + \beta(a, c) - \beta(a, b) = 0.$$

Applied componentwise to $B$, this yields $\delta B = 0$. Hence $B \in Z^2_{\mathrm{sym}}(\Gamma_X^{\mathrm{phys}}; \Gamma_{\mathrm{gram}})$.

*(iv) Associativity, identity, inverse.* Compute the two bracketings of $(c, T) \star (c', T') \star (c'', T'')$:

$$\big((c, T) \star (c', T')\big) \star (c'', T'') = \big(c + c' + c'',\ T + T' + T'' + B(c, c') + B(c + c', c'')\big),$$
$$(c, T) \star \big((c', T') \star (c'', T'')\big) = \big(c + c' + c'',\ T + T' + T'' + B(c', c'') + B(c, c' + c'')\big).$$

The associator is the difference of the central components, $\mathrm{Ass}(c, c', c'') = B(c, c') + B(c + c', c'') - B(c', c'') - B(c, c' + c'')$, which equals $-(\delta B)(c, c', c'') = 0$ by (iii); equivalently, it vanishes by bilinearity of $B$. Commutativity follows from symmetry of $B$. The identity computation $(c, T) \star (0, 0) = (c, T + B(c, 0)) = (c, T)$ and the inverse computation $(c, T) \star (-c, -T + B(c, c)) = (0, B(c, c) - B(c, c)) = (0, 0)$ (using $B(c, -c) = -B(c, c)$ by bilinearity) finish the abelian-group structure. That $B$ is the cocycle of the displayed central extension is direct from the formula for $\star$ and normalisation $B(c, 0) = B(0, c) = 0$.

*(v) Normal-ordered Gram map.* Using Lemma 6.8,

$$\begin{aligned}
\overline{\Pi}_X((c, T) \star (c', T')) &= \overline{\Pi}_X(c + c', T + T' + B(c, c')) \\
&= \Pi_X(c + c') - T - T' - B(c, c') \\
&= \Pi_X(c) + \Pi_X(c') - T - T' \\
&= \overline{\Pi}_X(c, T) + \overline{\Pi}_X(c', T').
\end{aligned}$$

Thus $\overline{\Pi}_X$ is additive.

*(vi) Splitting.* By Lemma 6.8, $\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c')$, so

$$s(c) \star s(c') = (c + c',\ \Pi_X(c) + \Pi_X(c') + B(c, c')) = (c + c',\ \Pi_X(c + c')) = s(c + c').$$

Thus $s$ is a homomorphism, and $B(c,c') = \Pi_X(c+c') - \Pi_X(c) - \Pi_X(c') = -(\delta\Pi_X)(c,c')$ shows $B = -\delta\Pi_X$, so $[B] = 0$. The change of variables $(c,T) \mapsto (c, T - \Pi_X(c))$ is bijective on the underlying set with inverse $(c,T) \mapsto (c, T + \Pi_X(c))$, and carries $\star$ to componentwise addition:

$$(c, T - \Pi_X(c)) + (c', T' - \Pi_X(c')) = (c+c',\ T + T' - \Pi_X(c) - \Pi_X(c')) = (c+c',\ (T + T' + B(c,c')) - \Pi_X(c+c')),$$

matching the change of variables applied to $(c+c', T+T'+B(c,c')) = (c,T) \star (c',T')$.

*(vii) Quadratic obstruction.* Suppose $B = \delta L$ for some additive $L : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$, i.e. $B(c,c') = L(c) + L(c') - L(c+c')$. Setting $c' = c$, $B(c,c) = 2L(c) - L(2c)$; but $L$ additive gives $L(2c) = 2L(c)$, so $B(c,c) = 0$, contradicting $B(c,c) = 2\Pi_X(c)$ for any $c = (Q,P)$ with $(Q,Q), (Q,P), (P,P)$ not all zero. Hence no linear trivialisation exists; the quadratic primitive $\Pi_X$ is necessary.

*Scope of the chain-level claim.* Clauses (i)–(vii) prove an ordinary set-level and $H^0$-level polarization correction. They do not prove that the protected Hall chain complex has no higher associators. In particular no acyclicity of a bar complex $B_*(\widehat{\Gamma}_X)$ is asserted: the bar construction of a discrete abelian group records group homology, and cannot be used to kill Stasheff coherences. Chain-level strictness of the Hall product therefore remains part of the orientation/coherence hypotheses in Theorem 10.27; what this lemma supplies is only the explicit quadratic polarization correction $B = -\delta\Pi_X$ and the split degree monoid on which a later cocycle-corrected descent can act. □

*Remark* 6.10. The preceding lemma records the additive degree correction. It proves that the Gram defect is the coboundary of the quadratic refinement $\Pi_X$, and that the corrected degree monoid is split. It does not strictify the Hall chain complex. A chain-level Hall product still requires the multiplicative orientation, cyclic equivariance, and associator data listed in Theorem 10.27; absent those data the lift is only $A_\infty$-coherent, and the obstruction is tracked by the supplied cochain $\Theta$, not by acyclicity of a group bar complex.

*Definition* 6.11 *(Normal-ordered Gram-corrected charge extension).* Set

$$\widehat{\Gamma}_X := \Gamma_X^{\mathrm{phys}} \oplus \Gamma_{\mathrm{gram}}.$$

For $c, c' \in \Gamma_X^{\mathrm{phys}}$ and $T, T' \in \Gamma_{\mathrm{gram}}$, define

$$(c,T) \star (c',T') := (c+c',\ T + T' + B(c,c')).$$

The section

$$s : \Gamma_X^{\mathrm{phys}} \longrightarrow \widehat{\Gamma}_X, \qquad s(c) = (c, \Pi_X(c))$$

is additive for $\star$. The normal-ordered Gram map is

$$\overline{\Pi}_X : \widehat{\Gamma}_X \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c,T) = \Pi_X(c) - T.$$

Thus the compact Hall object, if constructed, is graded upstairs by $\Gamma_X^{\mathrm{phys}}$, or by the refinement $\widehat{\Gamma}_X$ after the normal-ordering counterterm is recorded. The $\Gamma_{\mathrm{gram}}$-graded Igusa object is obtained only after normal-ordered Gram pushforward along $\overline{\Pi}_X$, or by a chain-level categorification of the same subtraction.

*Proof.* By Lemma 6.9(i),(iii),(iv), $\star$ is associative and commutative, the identity is $(0,0)$, and the inverse of $(c,T)$ is $(-c, -T + B(c,c))$; the section $s$ is the splitting of clause (vi). Clause (v) of the same lemma proves that $\overline{\Pi}_X$ is a homomorphism. □

Theorem 6.12 *(Raw Gram descent cannot realize the full BKM real-root strings).* Let

$$P = \bigoplus_{c \in \Gamma_X^{\mathrm{phys}}} P_c$$

be a $\Gamma_X^{\mathrm{phys}}$-graded Lie superalgebra with

$$[P_c, P_{c'}] \subset P_{c+c'}.$$

Suppose a raw pushforward along the quadratic map $\Pi_X : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$ were a $\Gamma_{\mathrm{gram}}$-graded Lie superalgebra structure on all bracket channels needed for the real-root strings of $\mathfrak{g}_{\Delta_s}$. Then every non-zero bracket channel $[P_c, P_{c'}]$ contributing to that raw pushforward must satisfy

$$B(c, c') = 0.$$

For the type-II real simple roots of the Gritsenko–Nikulin algebra, this condition is incompatible with the non-zero second real-root strings $2\delta_i + \delta_j, i \neq j$. Hence the full BKM bracket cannot be obtained by raw $\Pi_X$-descent. It requires the normal-ordered extension $\widehat{\Gamma}_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}$ and the homomorphism $\overline{\Pi}_X(c, T) = \Pi_X(c) - T$.

*Proof.* If the raw pushforward is $\Gamma_{\mathrm{gram}}$-graded, then a non-zero bracket of classes of raw Gram degrees $\Pi_X(c)$ and $\Pi_X(c')$ must land in raw Gram degree $\Pi_X(c) + \Pi_X(c')$. But the upstairs bracket lands in physical degree $c + c'$, whose raw Gram shadow is

$$\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c').$$

Thus every non-zero bracket channel visible in a raw pushforward must have $B(c, c') = 0$.

Let $c_i, c_j$ be physical lifts of two distinct type-II real simple root degrees $\gamma_i, \gamma_j$, so

$$\Pi_X(c_i) = \gamma_i, \qquad \Pi_X(c_j) = \gamma_j, \qquad \gamma_i \neq 0.$$

The Gritsenko–Nikulin real simple roots have Cartan matrix

$$\begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The real-root string through distinct simple roots contains non-zero brackets in degrees $\gamma_i + \gamma_j$ and $2\gamma_i + \gamma_j$; this is the low-degree real-root information recorded in Proposition 10.46. The first non-zero bracket would force

$$B(c_i, c_j) = 0.$$

The second non-zero bracket is obtained by bracketing once more with the $c_i$-degree generator, so raw Gram additivity would force

$$B(c_i, c_i + c_j) = 0.$$

By bilinearity and polarization,

$$B(c_i, c_i + c_j) = B(c_i, c_i) + B(c_i, c_j) = 2\Pi_X(c_i) + 0 = 2\gamma_i.$$

Since $\gamma_i \neq 0$ in the torsion-free lattice $\Gamma_{\mathrm{gram}}$, this is non-zero. Contradiction.

The contradiction uses only the raw quadratic map $\Pi_X$. It does not exclude special isolated $B$-isotropic bracket channels. It excludes a functorial raw pushforward realizing the full $\mathfrak{g}_{\Delta_5}$ real-root bracket. The normal-ordered extension removes exactly this obstruction because $\overline{\Pi}_X$ is additive on $\widehat{\Gamma}_X$. □

*Remark* 6.13 *(Normal-ordered primitive recognition target).* The interface with the denominator algebra is therefore not the raw map $\Pi_X$. It is the corrected route

$$\Gamma_X^{\mathrm{phys}} \longrightarrow \widehat{\Gamma}_X \xrightarrow{\overline{\Pi}_X} \Gamma_{\mathrm{gram}} \xrightarrow{\alpha} \Lambda_{II}^{2,1}.$$

At chain level this becomes the descent $\overline{\Pi}^{\Theta}_{X,*}$ of Open Problem 10.23. The primitive recognition problem is to construct a protected primitive Hall algebra whose normal-ordered descended bracket is the Borcherds–Kac bracket of $\mathfrak{g}_{\Delta_5}$. Raw Gram pushforward is not an admissible recognition target.

*Definition* 6.14 *(Gram-index pairing).* For

$$\gamma = (n, l, m), \qquad \gamma' = (n', l', m'),$$

define

$$\langle \gamma, \gamma' \rangle_{\mathrm{ind}} = 2(nm' + n'm) - ll'.$$

The Gram discriminant is

$$\Delta_{\mathrm{gram}}(\gamma) = \langle \gamma, \gamma \rangle_{\mathrm{ind}} = 4nm - l^2.$$

*Remark* 6.15 *(Chamber transport).* The pairing $\langle\ ,\ \rangle_{\mathrm{ind}}$ is the pairing on Gram invariants. The order on $\Gamma_{\mathrm{eff}}$ is a Fock polarization at one cusp. A duality element transports this polarization to the corresponding product chamber at the transported cusp.

# Part III

# Denominator algebra and compact Dirac–Igusa realization

## 7 The exterior-square model of the automorphic group

The form $\Delta_5$ is an automorphic form on the Siegel space. The physical U-duality interpretation of the same arithmetic group is external input from the $K3 \times E$ compactification; it is not needed for the lattice calculation below. The calculation records the first-principles bridge from the Siegel variables to the type-IV Lorentzian lattice on which the denominator algebra is written.

$$\Lambda^4 = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}e_3 \oplus \mathbb{Z}e_4.$$

$$g \in \mathbf{SL}_4(\mathbb{Z}) : \qquad \wedge^2 g : \wedge^2\Lambda^4 \longrightarrow \wedge^2\Lambda^4.$$

$$u \wedge v = (u, v)e_1 \wedge e_2 \wedge e_3 \wedge e_4.$$

$$\left(\wedge^2\Lambda^4, (\ ,\ )\right) \simeq \mathrm{II}_{3,3}.$$

Fix

$$q_0 = e_1 \wedge e_3 + e_2 \wedge e_4.$$

For $x, y \in \Lambda^4$, define

$$-x \wedge y \wedge q_0 = B_{q_0}(x, y)e_1 \wedge e_2 \wedge e_3 \wedge e_4.$$

$$\{g \in \mathbf{SL}(\Lambda^4) \mid (\wedge^2 g)q_0 = q_0\} \simeq \mathbf{Sp}_4(\mathbb{Z}).$$

Set

$$\Lambda^{3,2} = q_0^{\perp} \subset \wedge^2\Lambda^4.$$

$$f_1 = e_1 \wedge e_2, \quad f_2 = e_2 \wedge e_3, \quad f_3 = e_1 \wedge e_3 - e_2 \wedge e_4, \quad f_{-2} = e_4 \wedge e_1, \quad f_{-1} = e_4 \wedge e_3.$$

$$(f_1, f_{-1}) = (f_2, f_{-2}) = -1, \qquad (f_3, f_3) = 2.$$

$$\Lambda^{3,2} \simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2], \qquad \Lambda^{(1,1)} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

$$\mathbb{H}^{\mathbf{IV}} = \{Z \in \mathbb{P}(\Lambda^{3,2} \otimes \mathbb{C}) \mid (Z, Z) = 0,\ (Z, \overline{Z}) < 0\}.$$

$$\mathbb{H}^{\mathbf{IV}}_{+} = \{{}^t(z_2^2 - z_1 z_3,\ z_3,\ z_2,\ z_1,\ 1)z_0 \in \mathbb{H}^{\mathbf{IV}} \mid \mathrm{Im} z_1 > 0\}.$$

Writing $y_i = \mathrm{Im} z_i$,

$$(Z, \overline{Z}) < 0 \quad \Longleftrightarrow \quad y_1 y_3 - y_2^2 > 0.$$

$$\mathbb{H}^{\mathbf{IV}}_{+} \simeq \left\{ \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix} \in \mathbb{H}_2 \right\}.$$

Let $\mathbf{O}_{\mathbb{R}}(\Lambda^{3,2})_+$ be the index-two subgroup preserving $\mathbb{H}^{\mathbf{IV}}_{+}$. Its kernel on $\mathbb{H}^{\mathbf{IV}}_{+}$ is $\{\pm \mathbf{I}_5\}$, and

$$\mathbf{O}_{\mathbb{R}}(\Lambda^{3,2})_+ = \{\pm \mathbf{I}_5\}\mathbf{SO}_{\mathbb{R}}(\Lambda^{3,2})_+,$$

where $\mathbf{SO}_{\mathbb{R}}(\Lambda^{3,2})_+$ has real spin-norm 1. Put

$$\mathbf{O}(\Lambda^{3,2})_+ = \mathbf{O}_{\mathbb{R}}(\Lambda^{3,2})_+ \cap \mathbf{O}(\Lambda^{3,2})$$

and

$$\mathbf{SO}(\Lambda^{3,2})_+ = \mathbf{SO}_{\mathbb{R}}(\Lambda^{3,2})_+ \cap \mathbf{O}(\Lambda^{3,2}).$$

$$M = \begin{pmatrix} m_1 & m_2 \\ m_2 & m_3 \end{pmatrix},$$

$$\wedge^2(g_0) = \wedge^2\left(\begin{pmatrix} 0 & \mathbf{I}_2 \\ -\mathbf{I}_2 & 0 \end{pmatrix}\right) = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\wedge^2(g_M) = \wedge^2\left(\begin{pmatrix} \mathbf{I}_2 & M \\ 0 & \mathbf{I}_2 \end{pmatrix}\right) = \begin{pmatrix} 1 & -m_1 & 2m_2 & -m_3 & m_2^2 - m_1 m_3 \\ 0 & 1 & 0 & 0 & m_3 \\ 0 & 0 & 1 & 0 & m_2 \\ 0 & 0 & 0 & 1 & m_1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\wedge^2(g_A) = \wedge^2\left(\begin{pmatrix} {}^tA^{-1} & 0 \\ 0 & A \end{pmatrix}\right) = \det(A)\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & a_1^2 & -2a_1a_2 & a_2^2 & 0 \\ 0 & -a_1a_3 & a_1a_4 + a_2a_3 & -a_2a_4 & 0 \\ 0 & a_3^2 & -2a_3a_4 & a_4^2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Lemma 7.1. The exterior-square representation is

$$\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \to \mathbf{SO}_+(\Lambda^{3,2}) \simeq \mathbf{O}(\Lambda^{3,2})_+/\{\pm\mathbf{I}_5\}.$$

For every $g \in \mathbf{Sp}_4(\mathbb{Z})$,

$$\begin{tikzcd} \mathbb{H}_2 \arrow[r] \arrow[d] & \mathbb{H}_2 \arrow[d] \\ \mathbb{H}_+^{\mathbf{IV}} \arrow[r, "g\wedge g"] & \mathbb{H}_+^{\mathbf{IV}} \end{tikzcd}$$

*Proof.* The Pfaffian pairing is preserved by $\wedge^2\mathbf{SL}_4$. The subgroup fixing

$$q_0 = e_1 \wedge e_3 + e_2 \wedge e_4$$

is the group preserving the alternating form $B_{q_0}$, namely $\mathbf{Sp}_4(\mathbb{Z})$. Its action preserves $q_0^\perp = \Lambda^{3,2}$ and the displayed component of the type-**IV** domain. The kernel on $q_0^\perp$ is $\{\pm\mathbf{I}_4\}$. The exceptional isomorphism $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}_{\mathbb{R}}(3,2)_+$ identifies the image with $\mathbf{SO}_+(\Lambda^{3,2})$. The displayed matrices are the integral generators on both sides. Substitution of

$${}^t(z_2^2 - z_1z_3, z_3, z_2, z_1, 1)z_0$$

into the two actions produces the commutative square. □

*Definition 7.2 (Automorphic duality datum).*

$$G_{\mathrm{aut}} = \mathbf{PSp}_4(\mathbb{Z}) \simeq \mathbf{SO}_+(\Lambda^{3,2}).$$

The virtual determinant $\mathcal{D}_X = \Delta_5$ is not an invariant scalar function on $\mathbb{H}_2$. It is a section of the automorphic line

$$\mathcal{L}^5 \otimes \nu_{\Delta_5}$$

with transformation law

$$\Delta_5(g \cdot Z) = \nu_{\Delta_5}(g) \det(CZ + D)^5 \Delta_5(Z), \qquad g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Z}).$$

The scalar square is a section of $\mathcal{L}^{10}$:

$$\Delta_5^2(g \cdot Z) = \det(CZ + D)^{10} \Delta_5^2(Z).$$

For every scalar-square branch,

$$Z_{\square}^X(g \cdot Z) = \det(CZ + D)^{-10} Z_{\square}^X(Z).$$

## 8 The Weyl chamber and the Weyl vector

Fix the primitive hyperbolic sublattice

$$\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2] \simeq \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2} \subset \Lambda^{3,2}$$

and its type-II sublattice

$$\Lambda_{II}^{2,1} = \{mf_2 + lf_3 + nf_{-2} \in \Lambda^{2,1} \mid m \equiv n \equiv 0 \pmod 2\}.$$

Change polarization on the same Gram-index lattice:

$$\Gamma_{\mathrm{ind}} \xrightarrow{\alpha} \Lambda_{II}^{2,1}, \qquad \alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}.$$

The cusp polarization gives the Fock determinant. The denominator polarization gives the Borcherds chamber.

Extend $\phi \in \mathbf{O}(\Lambda^{2,1})$ by the identity on $(\Lambda^{2,1})^\perp$:

$$\mathbf{O}(\Lambda^{2,1}) \hookrightarrow \mathbf{O}(\Lambda^{3,2}).$$

For the component-preserving subgroup attached to the positive cone below,

$$\Delta_5 \in M_5(\mathbf{O}(\Lambda^{2,1})_+, \nu_{\Delta_5}).$$

For a primitive element $\alpha \in \Lambda^{2,1}$ satisfying $(\alpha, \alpha) > 0$ and

$$(\alpha, \alpha) \mid 2(\Lambda^{2,1}, \alpha),$$

Kac reflection [37]:

$$s_\alpha : x \mapsto x - 2\frac{(x, \alpha)}{(\alpha, \alpha)}\alpha.$$

$$s_\alpha(\alpha) = -\alpha, \qquad s_\alpha|_{\alpha^\perp} = \mathbf{I}.$$

For $(\alpha, \alpha) = 2$:

$$s_\alpha : x \mapsto x - (x, \alpha)\alpha.$$

Lemma 8.1. Let $(\alpha, \alpha) = 2$. For

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

[30, Theorem 2.1 and Section 4]; equivalently [31, Section 5.1.1, case (t=1,II,0)].

$$\Delta_5(s_\alpha Z) = -\Delta_5(Z)$$

when $\alpha \in \{\delta_1, \delta_2, \delta_3\}$. For

$$\alpha \in \{f_{-2} - f_2,\ f_2 - f_3,\ f_2 + f_3\},$$
$$\Delta_5(s_\alpha Z) = \Delta_5(Z)$$

*Proof.* Let

$$\overline{U} = \wedge^2 \begin{pmatrix} {}^tU^{-1} & 0 \\ 0 & U \end{pmatrix}, \qquad U \in \mathbf{GL}_2(\mathbb{Z}).$$

Required reflections:

$$s_{f_{-2}-f_2} = -\overline{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}, \qquad s_{f_3} = -\overline{\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}, \qquad s_{f_2-f_3} = -\overline{\begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}}.$$

Maass's character has value $-1$ on $s_{\delta_i}$ and $+1$ on the three other reflection classes. □

The time-like cone is

$$C(\Lambda^{2,1}) = \{x \in \Lambda^{2,1} \otimes \mathbb{R} | (x, x) < 0\}.$$

Let $C(\Lambda^{2,1})_+$ be the component for which

$$\Omega(C(\Lambda^{2,1})_+) := \Lambda^{2,1} \otimes \mathbb{R} + iC(\Lambda^{2,1})_+ \subset \mathbb{H}_+^{\mathbf{IV}}.$$

For

$$z = z_3 f_2 + z_2 f_3 + z_1 f_{-2} \in \Omega(C(\Lambda^{2,1})_+),$$

the type **IV** point is

$$Z = {}^t(z_2^2 - z_1 z_3,\ z_3,\ z_2,\ z_1,\ 1) z_0.$$

Let $\mathbf{O}(\Lambda^{2,1})_+$ be the index-two subgroup fixing $C(\Lambda^{2,1})_+$. It acts discretely on

$$C(\Lambda^{2,1})_+ / \mathbb{R}_{>0}.$$

For $(\alpha, \alpha) > 0$,

$$\mathbb{H}_\alpha = \{\mathbb{R}_{>0} x \in C(\Lambda^{2,1})_+ / \mathbb{R}_{>0} \mid (x, \alpha) = 0\},$$

and

$$\mathbb{H}_{\alpha,+} = \{\mathbb{R}_{>0} x \in C(\Lambda^{2,1})_+ / \mathbb{R}_{>0} \mid (x, \alpha) \leq 0\}.$$

Let $W(\Lambda^{2,1})$ be the subgroup generated by these positive-norm reflections:

$$W(\Lambda^{2,1}) \subset \mathbf{O}(\Lambda^{2,1})_+.$$

Nikulin's theorem for $\Lambda^{(1,1)} \oplus [2]$ [45]:

$$\mathbf{O}(\Lambda^{2,1})_+ = W^{(2)}(\Lambda^{2,1}),$$

where $W^{(2)}$ is generated by reflections in square-2 vectors. For $k > 0$ put

$$\Delta^{(k)}(\Lambda^{2,1}) = \{\delta \in \Lambda^{2,1} \mid \delta \text{ primitive}, \ (\delta, \delta) = k, \ s_\delta \in \mathbf{O}(\Lambda^{2,1})\}.$$

Each $\delta \in \Delta^{(2)}(\Lambda^{2,1})$ has one of two divisibilities:

- Type I: $(\delta, \Lambda^{2,1}) = \mathbb{Z}$
- Type II: $(\delta, \Lambda^{2,1}) = 2\mathbb{Z}$

The type sublattices are

$$\Lambda_I^{2,1} = \{mf_2 + lf_3 + nf_{-2} \in \Lambda^{2,1} \mid m + l + n \equiv 0 \pmod 2\},$$

and the already defined $\Lambda_{II}^{2,1}$. Type I roots lie in $\Lambda_I^{2,1}$; type II roots lie in $\Lambda_{II}^{2,1}$. The reflection subgroups are normal, with indices

$$[W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda_I^{2,1})] = 2, \qquad [W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda_{II}^{2,1})] = 6.$$

The chambers are

$$\mathcal{P}' = \bigcap_{\delta \in \mathcal{P}'_{prim}} \mathbb{H}_{\delta,+}, \qquad \mathcal{P}' \in \{\mathcal{P}, \mathcal{P}_I, \mathcal{P}_{II}\},$$

with walls

$$\mathcal{P}_{prim} = \{f_2 - f_3, \ f_{-2} - f_2, \ f_3\},$$
$$\mathcal{P}_{I,prim} = \{f_2 - f_3, \ f_{-2} - f_2, \ f_2 + f_3\},$$
$$\mathcal{P}_{II,prim} = \{\delta_1, \delta_2, \delta_3\}.$$

For

$$\mathrm{Aut}(\mathcal{P}') = \{g \in \mathbf{O}(\Lambda^{2,1})_+ \mid g\mathcal{P}' = \mathcal{P}'\},$$
$$\mathrm{Aut}(\mathcal{P}) = 1, \qquad \mathrm{Aut}(\mathcal{P}_I) = \langle s_{f_3} \rangle, \qquad \mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3,$$

with $\mathrm{Aut}(\mathcal{P}_{II})$ generated by $s_{f_2 - f_3}$ and $s_{f_{-2} - f_2}$. The chamber decomposition is

$$\mathbf{O}(\Lambda^{2,1})_+ \simeq W^{(2)}(\Lambda^{2,1}) \simeq W^{(2)}(\Lambda_I^{2,1}) \rtimes \mathrm{Aut}(\mathcal{P}_I) \simeq W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \mathrm{Aut}(\mathcal{P}_{II}).$$

Lemma 8.2 *(Multiplier restriction).* • for $w \in W^{(2)}(\Lambda_I^{2,1})$ and $a \in \mathrm{Aut}(\mathcal{P}_I)$,

$$\Delta_5(w \cdot aZ) = \det(a)\Delta_5(Z)$$

- for $w \in W^{(2)}(\Lambda_{II}^{2,1})$ and $a \in \mathrm{Aut}(\mathcal{P}_{II})$,

$$\Delta_5(w \cdot aZ) = \det(w)\Delta_5(Z)$$

Equivalently,

$$\nu_{\Delta_5}(s_{\delta_i}) = -1, \qquad i = 1, 2, 3,$$

and

$$\nu_{\Delta_5}(s_{f_{-2} - f_2}) = \nu_{\Delta_5}(s_{f_2 - f_3}) = \nu_{\Delta_5}(s_{f_2 + f_3}) = +1.$$

*Proof.* The Maass multiplier of the product of the ten even theta constants is the character displayed in Section The normalized cusp form [42]. Under the exterior-square identification $\mathbf{PSp}_4(\mathbb{Z}) \simeq \mathbf{SO}_+(\Lambda^{3,2})$, Gritsenko–Nikulin compute its restriction to the reflective chambers [30, Section 3], [29]: it is $-1$ on the three type-II simple reflections and trivial on the three $\mathrm{Aut}(\mathcal{P}_{II})$-reflections. This is a character computation, not a normalization computation: multiplying $\Delta_5$ by any nonzero scalar leaves the displayed values unchanged. Thus the six generator values are exactly

$$\nu_{\Delta_5}(s_{\delta_i}) = -1, \qquad \nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_2+f_3}) = +1.$$

For the type-II chamber decomposition

$$W^{(2)}(\Lambda^{2,\mathrm{I}}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II}),$$

the first line says that the restriction to $W^{(2)}(\Lambda^{2,\mathrm{I}}_{II})$ is the determinant character, while the second says that the $\mathrm{Aut}(\mathcal{P}_{II})$-factor acts trivially. The type-I statement is the parallel restriction in the decomposition

$$W^{(2)}(\Lambda^{2,\mathrm{I}}_{I}) \rtimes \mathrm{Aut}(\mathcal{P}_{I}),$$

where the nontrivial chamber automorphism contributes $\det(a)$. □

Set

$$\Delta(\Lambda^{2,\mathrm{I}}_{II})_+ = \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3$$

and

$$\Delta(\Lambda^{2,\mathrm{I}}_{II})^*_+ = \{x \in \Lambda^{2,\mathrm{I}} \otimes \mathbb{R} | (x, \delta_i) \leq 0\}.$$

For the finite-volume chamber $\mathcal{P}_{II}$, the dual cone obeys

$$\Delta(\Lambda^{2,\mathrm{I}}_{II})^*_+ \subset \overline{C(\Lambda^{2,\mathrm{I}})_+} \subset \Delta(\Lambda^{2,\mathrm{I}}_{II})_+.$$

The Weyl vector is determined by

$$(\rho, \delta_i) = -\frac{(\delta_i, \delta_i)}{2} = -1, \qquad i = 1, 2, 3.$$

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

$$\rho = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_2 + \frac{1}{2}\delta_3 = f_2 - \frac{1}{2}f_3 + f_{-2}.$$

Also

$$\rho \in C(\Lambda^{2,\mathrm{I}})_+ = C(\Lambda^{2,\mathrm{I}}_{II})_+.$$

Fix

$$z = z_1 f_{-2} + z_2 f_3 + z_3 f_2 \in \Lambda^{2,\mathrm{I}} \otimes \mathbb{R} + iC(\Lambda^{2,\mathrm{I}})_+ = \Lambda^{2,\mathrm{I}}_{II} \otimes \mathbb{R} + iC(\Lambda^{2,\mathrm{I}}_{II})_+.$$

The dual lattice is

$$(\Lambda^{2,\mathrm{I}}_{II})^* = \mathbb{Z}\frac{1}{2}f_2 + \mathbb{Z}\frac{1}{2}f_3 + \mathbb{Z}\frac{1}{2}f_{-2} = \frac{1}{2}\Lambda^{2,\mathrm{I}}.$$

*Definition* 8.3 *(Gram-index degrees in the Weyl chamber).* For $\gamma = (n, l, m) \in \Gamma_{\mathrm{ind}}$ set

$$\alpha(\gamma) = \alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2} \in \Lambda_{II}^{2,1}.$$

With

$$z = z_1 f_{-2} + z_2 f_3 + z_3 f_2,$$

the character of the Gram-index degree is

$$x^\gamma = q^n r^l s^m = \exp(-\pi i(\alpha(\gamma), z)).$$

For $\gamma' = (n', l', m')$,

$$(\alpha(\gamma), \alpha(\gamma')) = -2\langle \gamma, \gamma' \rangle_{\mathrm{ind}}, \qquad (\alpha(\gamma), \alpha(\gamma)) = -2(4nm - l^2).$$

Thus the Borcherds degree is the Lorentzian image of the Gram-index grading. The Weyl chamber and real walls are part of the Gritsenko–Nikulin/Nikulin root datum. The cusp polarization fixed in Section 6 selects the corresponding product expansion, and the active Jacobi support gives the nonzero visible product exponents in that chamber.

$$E_0 := \rho.$$

$$q^{1/2} r^{1/2} s^{1/2} = \exp(-\pi i(\rho, z)).$$

$$\exp(-2\pi i(\alpha(\gamma), z)), \qquad \exp(-2\pi i(\rho, z)).$$

*Definition* 8.4 *(Active support and Borcherds cone).* In the standard cusp chamber, the active support is

$$\Gamma_{\mathrm{act}} = \{\gamma = (n, l, m) \in \Gamma_{\mathrm{eff}} \mid f(nm, l) \neq 0\}.$$

The inactive triples in $\Gamma_{\mathrm{eff}} \setminus \Gamma_{\mathrm{act}}$ are ordering data only; their product exponents are zero. The product does not prove absence of root spaces with zero signed superdimension. The proof-grade Borcherds cone is the real cone generated by the active positive Gram-index degrees after the Lorentzian degree map:

$$C_{\mathrm{Borch}} := \mathbb{R}_{\geq 0}\, \alpha(\Gamma_{\mathrm{act}}) \subset \Lambda_{II}^{2,1} \otimes \mathbb{R}.$$

The real-root hyperplanes are the walls used by the denominator identity in this polarization.

Lemma 8.5 *(Active support gives the type-II chamber cone).* For $\gamma = (n, l, m) \in \Gamma_{\mathrm{act}}$,

$$\alpha(\gamma) = n\delta_1 + m\delta_2 + (n + m - l)\delta_3,$$

with

$$n \geq 0, \qquad m \geq 0, \qquad n + m - l \geq 0.$$

Consequently

$$C_{\mathrm{Borch}} = \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3.$$

*Proof.* The displayed decomposition is

$$n(2f_2 - f_3) + m(2f_{-2} - f_3) + (n + m - l)f_3 = 2nf_2 - lf_3 + 2mf_{-2}.$$

The effective chamber gives $n, m \geq 0$. Since $\phi_{0,1}$ is weak of index one, $f(nm, l) \neq 0$ implies $4nm - l^2 \geq -1$. If $n = 0$ or $m = 0$, this gives $l^2 \leq 1$, and the boundary cases in $\Gamma_{\text{eff}}$ give $l \leq n + m$. If $n, m > 0$ and $l \geq n + m + 1$, then

$$l^2 \geq (n + m + 1)^2 = 4nm + (n - m)^2 + 2(n + m) + 1 > 4nm + 1,$$

contradicting $l^2 \leq 4nm + 1$. Hence $n + m - l \geq 0$. The reverse inclusion follows from the active triples

$$(1, 0, 0), \qquad (0, 0, 1), \qquad (0, -1, 0),$$

whose degrees are $\delta_1 + \delta_3$, $\delta_2 + \delta_3$, and $\delta_3$, and from the active triples

$$(1, 1, 0), \qquad (0, 1, 1),$$

whose degrees are $\delta_1$ and $\delta_2$. □

For $n, l, m \in \mathbb{Z}$ with $n \equiv m \equiv l \equiv 1 \mod 2$, $n, m > 0$ and $4nm - l^2 > 0$, set

$$a = (n - 1)f_2 - \frac{l - 1}{2} f_3 + (m - 1)f_{-2}, \qquad m(a) = -\frac{1}{64} c_\Delta(n, l, m).$$

[30, Theorem 2.1]

$$\rho + a = nf_2 - \frac{l}{2} f_3 + mf_{-2},$$

and

$$\begin{aligned} &\frac{1}{64} c_\Delta(n, l, m) \exp(\pi i(nz_1 + lz_2 + mz_3)) \\ &= \frac{1}{64} c_\Delta(n, l, m) \exp\left(-\pi i\left(nf_2 - \frac{l}{2} f_3 + mf_{-2}, z\right)\right) \\ &= -m(a) \exp(-\pi i(\rho + a, z)), \end{aligned}$$

with

$$a \in (\Lambda^{2,1})^* = \frac{1}{2} \Lambda^{2,1}_{II}.$$

The normalization $c_\Delta(1, 1, 1) = 64$ fixes $m(0) = -1$. Since $c_\Delta(n, l, m) \in 64\mathbb{Z}$, all $m(a)$ are integral. The inequality $4nm - l^2 > 0$ implies

$$\rho + a \in C(\Lambda^{2,1})_+.$$

Let $\lambda = \rho + a$. For nonzero coefficients in the chosen chamber,

$$\lambda \in (\Lambda^{2,1}_{II})^* \cap \mathbb{R}_+ \mathcal{P}_{II}, \qquad (\lambda, \delta_i) \leq 0.$$

If $(\lambda, \delta_i) = 0$, then

$$c_\lambda = -c_\lambda = 0.$$

For nonzero terms, $(\lambda, \delta_i) < 0$. Since $(\rho, \delta_i) = -1$,

$$(a, \delta_i) \leq 0, \qquad a \in \Delta(\Lambda^{2,1}_{II})^*_+.$$

$$a \in \mathbb{R}_{\geq 0} \Delta(\Lambda^{2,1}_{II})^*_+ \cap \overline{C(\Lambda^{2,1})_+} = \mathbb{R}_{\geq 0} \mathcal{P}_{II}.$$

The congruences $m \equiv n \equiv l \equiv 1 \pmod 2$ imply

$$a \in 2(\Lambda^{2,1})^* = \Lambda^{2,1}_{II}.$$

$$a = 0 \qquad \text{or} \qquad a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0} \mathcal{P}_{II}.$$

Lemma 8.6.

$$\frac{1}{64}\Delta_5(Z) = \sum_{w\in W^{(2)}(\Lambda_{II}^{2,1})} \det(w)(\exp(-\pi i(w(\rho),z)) - \sum_{a\in\Lambda_{II}^{2,1}\cap\mathbb{R}_{>0}\mathcal{P}_{II}} m(a)\exp(-\pi i(w(\rho+a),z)))$$

*Proof.* Let $c_\lambda$ be the coefficient of $e^{-\pi i(\lambda,z)}$ in $64^{-1}\Delta_5$. The leading term has

$$\lambda = \rho, \qquad c_\rho = 1.$$

The other exponents in the chosen chamber have

$$\lambda = \rho + a, \qquad a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0}\mathcal{P}_{II}, \qquad c_{\rho+a} = -m(a).$$

Anti-invariance:

$$c_{w\lambda} = \det(w)c_\lambda.$$

If $\lambda$ lies on a simple wall, then

$$s_{\delta_i}\lambda = \lambda, \qquad c_\lambda = -c_\lambda = 0.$$

Summing the signed Weyl translates proves the formula. □

The primitive isotropic rays of $\overline{\mathcal{P}_{II}}$ are generated by

$$2f_2, \qquad 2f_{-2}, \qquad 2f_2 - 2f_3 + 2f_{-2}.$$

The group $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ is transitive on these three vectors. For $a_0 = 2f_2$:

$$1 + \frac{1}{64}\sum_{t\geq 1} c_\Delta(1+2t,1,1)q^t = \prod_{k\geq 1}(1-q^k)^9$$

In denominator notation,

$$1 - \sum_{t\geq 1} m(ta_0)q^t = \prod_{t\geq 1}(1-q^t)^{\tau(ta_0)}, \qquad \tau(ta_0) = 9 \quad (t \geq 1).$$

By $S_3$-symmetry,

$$\tau(ta) = 9 \qquad (t \geq 1)$$

for every primitive isotropic ray

$$\mathbb{Z}_{>0}a \subset \overline{\mathcal{P}_{II}} \cap \Lambda_{II}^{2,1}.$$

On isotropic rays, the coefficients $m(ta)$ are determined by the eta-product expansion; the signed simple-root supermultiplicity is $\tau(ta)$. The imaginary simple roots of $\mathfrak{g}_{\Delta_5}$ are these isotropic and timelike rays in $\overline{\mathcal{P}_{II}} \cap \Lambda_{II}^{2,1}$, with multiplicities $\tau(a)$ on isotropic rays and $m(a)$ on negative-norm rays [30, Theorem 2.1], [31, (5.1), Section 5.1.1].

The real root datum is the wall set

$$\mathcal{P}_{II,prim} = \{\delta_1, \delta_2, \delta_3\}$$

with Gram matrix [30, Theorem 4.1] (and [31, Section 5.1.1, case (t=1,II,0)] as secondary)

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$$

*Definition* 8.7 *(Simple real roots and perturbative walls).*

$$\Delta^{re}_{\mathrm{simp}} = \{\delta_1, \delta_2, \delta_3\}, \qquad \Delta^{re} = W^{(2)}(\Lambda^{2,1}_{II})\Delta^{re}_{\mathrm{simp}},$$

$$\mathcal{W}_{\mathrm{pert}} = \bigcup_{\delta \in \Delta^{re}_{\mathrm{simp}}} \mathbb{H}_\delta.$$

## 9 The denominator algebra

$$A = ((\delta_i, \delta_j))_{1 \le i,j \le 3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$$

$$A \text{ symmetric generalized Cartan}, \qquad \mathfrak{g}(A) \text{ Kac–Moody}.$$

$$\mathcal{A}_{\mathrm{den}} := \mathfrak{g}_{\Delta_5}.$$

This is the target-internal algebraic denominator square root. It is the automorphic correction of $\mathfrak{g}(A)$ constructed by Gritsenko–Nikulin [30, Sections 3–4], with root spaces graded by the Gram-degree map

$$\gamma = (n, l, m) \longmapsto \alpha(\gamma) = 2nf_2 - lf_3 + 2mf_{-2}.$$

The Borcherds correction character in the coefficient chamber is

$$C_{\mathrm{im}}(z) = 1 - \sum_{a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}} m(a) e^{-2\pi i (a,z)}.$$

$$e^{-2\pi i(\rho,z)} C_{\mathrm{im}}(z).$$

Signed root-superdimension convention; Harvey–Moore is the external physical interpretation [32]:

$$\mathrm{smult}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{0}} - \dim \mathfrak{g}_{\alpha,\bar{1}}.$$

Let $\mathcal{R}$ be the root set of $\mathfrak{g}_{\Delta_5}$, and let $\mathcal{R}_+$ be the positive root set in the $\mathcal{P}_{II}$-chamber.

*Definition* 9.1 *(Denominator Lie superalgebra).* For $\gamma \in \Gamma_{\mathrm{ind}}$ set

$$\mathcal{A}_{\mathrm{den},\gamma} := \begin{cases} \Lambda^{2,1}_{II} \otimes \mathbb{C}, & \gamma = 0, \\ \mathfrak{g}_{\Delta_5,\alpha(\gamma)}, & \alpha(\gamma) \in \mathcal{R}, \\ 0, & \alpha(\gamma) \notin \mathcal{R}. \end{cases}$$

The zero degree is the Cartan subalgebra and is not governed by $f(0,0)$. The bracket is the Borcherds–Kac–Moody bracket of $\mathfrak{g}_{\Delta_5}$:

$$[\ ,\ ] : \mathcal{A}_{\mathrm{den},\gamma} \otimes \mathcal{A}_{\mathrm{den},\gamma'} \longrightarrow \mathcal{A}_{\mathrm{den},\gamma+\gamma'}.$$

Theorem 9.2 *(Denominator algebra identity).* The denominator Lie superalgebra has the following properties.

$$\mathcal{A}_{\mathrm{den}} = \mathcal{A}_{\mathrm{den},0} \oplus \bigoplus_{\substack{\gamma \in \Gamma_{\mathrm{ind}} \\ \alpha(\gamma) \in \mathcal{R}}} \mathcal{A}_{\mathrm{den},\gamma} = \mathcal{A}_{\mathrm{den},0} \oplus \mathcal{A}^{\mathrm{bulk}}_{\mathrm{den}} \oplus \mathcal{A}^{\mathrm{cusp}}_{\mathrm{den}}, \qquad [\mathcal{A}_{\mathrm{den},\gamma}, \mathcal{A}_{\mathrm{den},\gamma'}] \subset \mathcal{A}_{\mathrm{den},\gamma+\gamma'},$$

where $\mathcal{A}_{\mathrm{den}}^{\mathrm{bulk}} = \bigoplus_{\alpha(\gamma)\in\mathcal{R},\, m(\gamma)>0} \mathcal{A}_{\mathrm{den},\gamma}$ records the $m > 0$ Borcherds bulk and $\mathcal{A}_{\mathrm{den}}^{\mathrm{cusp}} = \bigoplus_{\alpha(\gamma)\in\mathcal{R},\, m(\gamma)=0} \mathcal{A}_{\mathrm{den},\gamma}$ records the $m = 0$ Weyl-chamber cusp boundary, in parallel with the $\mathcal{V} = \mathcal{V}_{\mathrm{bulk}} \oplus \mathcal{V}_{\mathrm{cusp}}$ splitting at the determinant level above. Its visible positive signed Gram-degree multiplicities are the K3 half-index coefficients:

$$\mathrm{smult}(\alpha(n,l,m)) = f(nm,l) \qquad ((n,l,m) \in \Gamma_{\mathrm{act}}).$$

Its Weyl–Kac–Borcherds denominator is

$$\mathrm{den}(\mathcal{A}_{\mathrm{den}}) = 64^{-1}\Delta_5(2Z).$$

The theorem concerns the Borcherds denominator algebra. A microscopic BPS Hilbert space and its operator product require separate construction.

*Proof.* Gritsenko–Nikulin [31, Theorem 2.1, (2.1)–(2.3)] construct $\mathfrak{g}_{\Delta_5}$ as the generalized Borcherds–Kac–Moody superalgebra whose real simple roots are $\delta_1$, $\delta_2$, $\delta_3$ and whose imaginary simple-root multiplicities are read from the Borcherds correction character. A product exponent gives the signed multiplicity of a visible positive root; the simple imaginary-root datum is Gritsenko–Nikulin's $m(a)$ for negative-norm $a$ and $\tau(a)$ on isotropic rays. These are different pieces of the denominator construction. A root-graded Lie superalgebra has

$$[\mathfrak{g}_{\Delta_5,\alpha}, \mathfrak{g}_{\Delta_5,\beta}] \subset \mathfrak{g}_{\Delta_5,\alpha+\beta}.$$

Here $\alpha \in \Lambda_{II}^{2,1} \otimes \mathbb{C}$, and the addition $\alpha + \beta$ is linear inside the Cartan; this is the linear root grading of a Borcherds–Kac superalgebra. The Gram-degree map

$$\gamma = (n,l,m) \longmapsto \alpha(\gamma) = 2nf_2 - lf_3 + 2mf_{-2} \in \Lambda_{II}^{2,1} \otimes \mathbb{C}$$

is the inverse correspondence of root indexing: it is additive in $(n,l,m)$ and recovers the linear addition in $\Lambda_{II}^{2,1} \otimes \mathbb{C}$. This linear additivity is the relevant statement for the denominator algebra of $\Delta_5$, where the grading lattice is $\Lambda_{II}^{2,1}$ and $\Gamma_{\mathrm{ind}} = \mathbb{Z}^3$ is the Gram parametrization of that grading. In a hypothetical Hall source, the underlying lattice is the physical $\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S$, the Hall bracket is graded by $\Gamma_X^{\mathrm{phys}}$ (additive), and the $\Gamma_{\mathrm{ind}}$-grading of the denominator algebra is the $\Pi$-image, well defined modulo the cocycle $B$ of Definition 6.1. The denominator identity reads

$$\frac{1}{64}\Delta_5(2Z) = e^{-2\pi i(\rho,z)} \prod_{\alpha\in\mathcal{R}_+} (1 - e^{-2\pi i(\alpha,z)})^{\mathrm{smult}(\alpha)}.$$

Comparing this with the Borcherds product gives

$$\mathrm{smult}(\alpha(n,l,m)) = f(nm,l) \qquad ((n,l,m) \in \Gamma_{\mathrm{act}}).$$

□

## 10 Compact Dirac–Igusa realization problem

The denominator algebra is the target. A compact $K3 \times E$ theory would be a source. The distinction is between observables and states. The formal current envelope below is an observable algebra attached to the imported Borcherds–Kac–Moody target; it is not a compact BPS state space and it contains no Hall extension correspondences. Its target-side bar–cobar counit remains internal to that observable algebra. A genuine Dirac–Igusa realization must supply the state-side Hall object, the orientation/Pfaffian line, the normal-ordered Gram descent, and the chiral Koszul comparison before one may identify its primitives with $\mathfrak{g}_{\Delta_5}$.

*Definition* 10.1 *(Formal current envelope).* Let $E$ be the elliptic curve in $X = S \times E$. Put

$$\mathrm{Cur}_E(\mathcal{A}_{\mathrm{den}}) := \mathcal{A}_{\mathrm{den}} \otimes \mathcal{O}_E$$

with the level-zero current bracket

$$[a \otimes f, b \otimes g] = [a, b] \otimes f g.$$

Its chiral enveloping algebra in the sense of Beilinson–Drinfeld [3] is

$$\mathsf{Den}_{\Delta_5,E} := U_E^{\mathrm{ch}}(\mathrm{Cur}_E(\mathcal{A}_{\mathrm{den}})).$$

Proposition 10.2 *(Formal current envelope).* $\mathsf{Den}_{\Delta_5,E}$ is the formal completed ind Beilinson–Drinfeld current envelope on $E$ obtained from the denominator Lie superalgebra. Its constant-current Lie superalgebra is $\mathcal{A}_{\mathrm{den}}$, and the denominator of that constant-current bracket is

$$64^{-1}\Delta_5(2Z).$$

A compact $E_3$/Hall/chiral BPS realization requires additional microscopic input.

*Proof.* In the completed ind setting, the current construction sends a Lie superalgebra to a Lie-* algebra on a smooth curve, and the chiral enveloping construction sends a Lie-* algebra to a chiral algebra [3]; equivalently, in vertex-algebra language, this is the universal current vertex algebra at level zero [36]. Constant sections of $\mathcal{A}_{\mathrm{den}} \otimes \mathcal{O}_E$ recover the original Lie superalgebra. The denominator statement is exactly Theorem 9.2. □

*Definition* 10.3 *(Holomorphic $E_3$ quantum upgrade).* The determinant and denominator results above are protected Euler-characteristic statements. The expected quantum upgrade does not start from the scalar determinant. It would start from a supplied holomorphic $E_3$-factorization algebra on the Calabi–Yau threefold

$$\mathcal{A}_X^{E_3}, \qquad X = K3 \times E.$$

By a holomorphic $E_3$-factorization algebra is meant here: a factorization algebra on $X$ in the sense of Costello–Gwilliam ([15], [16] esp. vol. II §10–11), locally constant in real directions and holomorphic in three complex directions, whose factorization structure maps are governed by holomorphic polydiscs in $\mathbb{C}^3$; equivalently a factorization algebra in the holomorphic $E_3$-operadic sense, together with whatever homotopy/framing datum is required by the chosen operadic model. No identification with an $SO(6)$ or $SO(3)$ framed topological operad is asserted here. This symbol, and the placeholder $\Phi_3^{\mathrm{FA}}(X)$, denote the source object required in the Calabi–Yau-to-chiral construction; they are not a constructed functor evaluated on $X$. The candidate concrete realization, supplied as proposal-only schema, is the local protected observable algebra $\mathcal{A}_X(U)$ of Definition 2.1. A compact holomorphic-Chern–Simons/BV model, if constructed with the required descent, quantum master equation, and anomaly cancellation, would be a candidate source for this datum in the completed charge category. Such a source also requires a chosen $E_3$-formality point, a chain-level homotopy/framing datum for the chosen operadic model, compact-support control, and finite-stage HN quotients on which all operations are defined before passing to the completed inverse system.

The curve-level BPS algebra would be obtained only after a supplied chain-level specialization along the $K3$-direction. Split the triple identification into three named pieces, each carrying a separately labelled comparison map:

$$\mathcal{A}_X^{E_3} \xrightarrow{\mathrm{Sp}^{\mathrm{ch}}_{K3,E}} \mathcal{F}_X^{(0)} \xrightarrow{\kappa_E} \mathcal{F}_X^{(1)} \xrightarrow{\pi_*^{\mathrm{ch}}} (p_E)_*\mathcal{A}_X^{E_3}.$$

Here: $\mathrm{Sp}^{\mathrm{ch}}_{K3,E}$ is the Beilinson–Drinfeld chiral pushforward along the $K3$-fibration $p_E : K3 \times E \to E$ with K3-integration in the Costello–Gwilliam sense: pointwise on $E$, the fibre $K3$-direction of the factorization algebra is integrated to a chiral shadow on $E$; its output $\mathcal{F}_X^{(0)}$ is a chiral algebra on $E$ (equivalently a factorization algebra on $\mathrm{Ran}(E)$). $\kappa_E$ is the chiral-Koszul comparison $\mathcal{F}_X^{(0)} \to \mathcal{F}_X^{(1)}$, where $\mathcal{F}_X^{(1)}$ is the candidate identification of the chiral pushforward with the ordinary derived pushforward $(p_E)_*\mathcal{A}_X^{E_3}$ regarded as a sheaf-of-cochain factorization algebra on $\mathrm{Ran}(E)$; identification of chiral and ordinary pushforward is the open chiral-Koszul comparison. $\pi_*^{\mathrm{ch}}$ is the closing identification of $\mathcal{F}_X^{(1)}$ with $(p_E)_*\mathcal{A}_X^{E_3}$ as factorization data, comparing the Costello–Gwilliam push along a proper holomorphic submersion with the topological pushforward; this comparison reduces to the standard fact in the projection-finite sectors and is open in the wrapped sectors. The convenient shorthand $\mathcal{F}_X$ refers to any one of $\mathcal{F}_X^{(0)}$, $\mathcal{F}_X^{(1)}$, or $(p_E)_*\mathcal{A}_X^{E_3}$ once $\kappa_E$ and $\pi_*^{\mathrm{ch}}$ have been supplied. Without the chiral-Koszul comparison and the proper-pushforward identification, the three names are distinct objects. Restrictions to chambers, pushforwards to $E$, and chiral Koszul dualities are then operations on $\mathcal{A}_X^{E_3}$ and on its $E_1$-chiral shadow $\mathcal{F}_X$, not replacements for the $E_3$-source.

After these data have been constructed, the required Igusa statement is the pair of comparisons

$$H^0\,\mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X) \overset{?}{\cong} \mathfrak{g}_{\Delta_3}, \qquad \Omega^{\mathrm{ch}}_E C_X \overset{?}{\simeq} \mathsf{Den}_{\Delta_3,E}.$$

Here $C_X$ is the source coalgebra later formalized in Definition 10.10; in wrapped degree it is attached to the hybrid object $\mathcal{F}_X^{\mathrm{hyb}}$, not merely to the projection-finite Ran shadow. The first comparison would be the bracket theorem. The second would be the chiral Koszul theorem. Both require microscopic data beyond the scalar square and beyond the formal current envelope.

*Definition* 10.4 *(Compact holomorphic $E_3$ source interface).* Fix a stability sector $(\sigma, S)$ and the finite HN height quotients of Proposition 10.39. A compact holomorphic $E_3$ source interface for $X = K3 \times E$ is not, by itself, a compact quantum field theory. It is the following finite-first chain datum:

$$\mathbb{A}^{E_3}_{X,\sigma,S} = \left\{\mathcal{A}^{E_3}_{X,\sigma,S,\leq R},\ \rho_{R'R},\ m_{\mathcal{U},R},\ \mathrm{Sp}^{\mathrm{ch}}_{K3,E,R}\right\}_{R>0,\ R'\geq R}.$$

Here $\mathcal{A}^{E_3}_{X,\sigma,S,\leq R}$ is a charge-supported cochain-level holomorphic $E_3$-factorization object with charges in $\Gamma^{HN}_{\sigma,S}(R)$; $\rho_{R'R}$ are transition maps for the HN inverse system; $m_{\mathcal{U},R}$ are the structure maps attached to disjoint holomorphic polydiscs $\mathcal{U} \subset X$; and $\mathrm{Sp}^{\mathrm{ch}}_{K3,E,R}$ is the finite-stage K3-to-$E$ specialization map. The interface is required to carry:

(i) descent, QME, and anomaly-cancellation witnesses at every finite height $R$;
(ii) a fixed $E_3$-formality point and a homotopy/framing witness for the chosen holomorphic $E_3$-operadic model;
(iii) compatibility of $m_{\mathcal{U},R}$, specialization, and the transition maps $\rho_{R'R}$;
(iv) compact-support control and continuity for the HN inverse system;
(v) the rule that the K3-to-$E$ specialization is an ordinary $\mathrm{Ran}(E)$-factorization object only on the projection-finite part; positive elliptic degree must be routed through the hybrid wrapped datum of Definition 10.18.

If all finite stages and transition maps are supplied, the completed source is the inverse limit

$$\mathcal{A}^{E_3}_X := \varprojlim_R \mathcal{A}^{E_3}_{X,\sigma,S,\leq R}.$$

This equation is a construction from the displayed finite data, not an existence theorem for those data.

*Open Problem* 10.5 *(Compact holomorphic $E_3$ source).* Construct a nonempty interface $\mathbb{A}^{E_3}_{X,\sigma,S}$ in the sense of Definition 10.4, and hence the completed source object $\mathcal{A}^{E_3}_X$, for $X = K3 \times E$. A candidate compact holomorphic-Chern–Simons/BV construction must prove, before any comparison with $\Delta_5$, the following finite-chain data:

(i) descent for local observables on $X$ and a quantum master equation after compact charge completion;

(ii) anomaly cancellation, a fixed $E_3$-formality point in the sense of Tamarkin/Kontsevich formality for $E_n$-operads, and a chain-level homotopy/framing datum for the chosen holomorphic $E_3$-operadic model; compact-support control;

(iii) a finite-first sectorial completion compatible with Harder–Narasimhan quotients, with all operations defined on finite stages before completion;

(iv) the chain-level $K3$-to-$E$ specialization $\mathrm{Sp}^{\mathrm{ch}}_{K3,E}$;

(v) the projection-finite local sector and the hybrid wrapped repair for positive elliptic degree;

(vi) a strong orientation, protected integration, and primitive recognition certificate.

Existing BV, CoHA, and chiral-algebra formalisms give the language for these requirements. They do not, by themselves, construct this compact source for $K3 \times E$. The external open data are: the $E_3$-formality point (Tamarkin/Kontsevich formality at a chosen associator, of which the Drinfeld–Kontsevich associator is the canonical choice; rationality of formality is open over $\mathbb{Q}$ but known over $\mathbb{R}$), and the chain-level homotopy/framing datum for the chosen holomorphic $E_3$-operadic model; chain-level constructibility for the candidate concrete realization $\mathcal{A}_X(U)$ of Definition 2.1 is open).

*Remark* 10.6 *(Where the quantum comparison would live).* Assume the source algebra $\mathcal{A}^{E_3}_X$, the specialization $\mathrm{Sp}^{\mathrm{ch}}_{K3,E}$, a conilpotent coalgebra $C_X \overset{?}{\simeq} \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_X)$, and a quasi-isomorphism

$$\Theta_{\mathrm{Kos}} : \Omega^{\mathrm{ch}}_E C_X \overset{?}{\simeq} \mathsf{Den}_{\Delta_5,E}$$

have all been supplied. Then $\mathcal{F}^{\mathrm{hyb}}_X$ is the place where the protected primitive bracket and the chiral Koszul comparison must be formulated. Costello–Gwilliam locality explains why a holomorphic quantum theory would produce factorization products before any trace or determinant is taken [14], [15], [16]; Beilinson– Drinfeld chiral operations are the curve-level algebraic form of collision maps [3]. None of this is obtained from the scalar square or from the denominator identity alone.

*Remark* 10.7 *(Pushforward, restriction, Koszul duality).* The second step is specialization, not inversion. A restriction to a chamber or sector is a restriction of $\mathcal{A}^{E_3}_X$ before, or of $\mathcal{F}_X$ after, the $K3$-specialization. Pushforward along $p_E$ is the geometric notation for the same specialization. A compact Koszul theorem would require a supplied conilpotent coalgebra $C_X$, an identification

$$C_X \overset{?}{\simeq} \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_X),$$

and a further quasi-isomorphism

$$\Omega^{\mathrm{ch}}_E C_X \overset{?}{\simeq} \mathsf{Den}_{\Delta_5,E}.$$

These are not procedures for reconstructing the holomorphic $E_3$-source from the denominator alone.

Proposition 10.8 *(Target-internal bar–cobar counit).* Choose an augmented filtered model of the formal target $\mathsf{Den}_{\Delta_5,E}$ whose finite charge truncations lie in the conilpotent chiral setting where Francis–Gaitsgory chiral Koszul duality applies. Then the target has its own bar–cobar counit:

$$\Omega_E^{\mathrm{ch}} \bar{B}_E^{\mathrm{ch}}(\mathsf{Den}_{\Delta_5,E}) \longrightarrow \mathsf{Den}_{\Delta_5,E}.$$

The filtration used here is target-internal: it is the charge filtration on the imported current envelope, together with the chosen augmentation. It is not the source filtration $F^{\mathrm{co}}_{R,\bullet}$, and its bar coalgebra is not $C_{X,R}$. Cf. Kac [36] for the underlying vertex algebra structure, Beilinson–Drinfeld [3] §3.7 for the chiral bar–cobar formalism, and Francis–Gaitsgory (2012) Chiral Koszul Duality, Theorem 6.2.2, for the inversion in the conilpotent setting. The counit reconstructs the already defined target from its target bar coalgebra. It is not the compact Koszul comparison

$$\Omega_E^{\mathrm{ch}} C_X \overset{?}{\simeq} \mathsf{Den}_{\Delta_5,E}$$

unless $C_X$ has first been constructed from the compact $K3 \times E$ factorization object and identified with $\bar{B}_E^{\mathrm{ch}}(\mathcal{F}_X^{\mathrm{hyb}})$. The missing datum is the source coalgebra, its collision coproduct, and the source comparison map, not a homotopy inverse to the target-side counit.

*Proof.* Beilinson–Drinfeld bar–cobar formalism is an adjunction on chiral algebras and conilpotent chiral coalgebras on the curve [3, §3.7]. Applying this formalism to the known target $\mathsf{Den}_{\Delta_5,E}$, after choosing the displayed augmented filtered model, produces a tautological counit map back to that target. The finite charge filtration is used only to place this target calculation in the conilpotent formal category. No moduli stack of compact objects, source collision coproduct, orientation system, protected integration map, or primitive representative enters this construction. Those are precisely the extra data in Definition 10.44. □

Lemma 10.9 *(Source–target separation for the Koszul lane).* Fix a finite height $R$. The target counit

$$\varepsilon_{\Delta,R} : \Omega_E^{\mathrm{ch}} \bar{B}_E^{\mathrm{ch}}(\mathsf{Den}_{\Delta_5,E,\leq R}) \longrightarrow \mathsf{Den}_{\Delta_5,E,\leq R}$$

is a morphism internal to the imported current envelope. A source Koszul certificate is different data: it must first construct

$$\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}, \qquad C_{X,R}, \qquad F^{\mathrm{co}}_{R,\bullet}, \qquad \Delta_R^{\mathrm{ch}},$$

from the finite hybrid correspondence category $\mathsf{Corr}_R^{E,\mathrm{hyb}}$, its quotient functor $Q_{E,R}$, orientations, and full flag atlas; only then may it compare

$$b_{X,R} : C_{X,R} \simeq \bar{B}_E^{\mathrm{ch}}(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}), \qquad \Theta_{\mathrm{Kos},R} : \Omega_E^{\mathrm{ch}} C_{X,R} \simeq \mathsf{Den}_{\Delta_5,E,\leq R}.$$

Thus setting $C_{X,R} = \bar{B}_E^{\mathrm{ch}}(\mathsf{Den}_{\Delta_5,E,\leq R})$, or choosing a homotopy inverse to $\varepsilon_{\Delta,R}$, does not produce a source certificate. It sets the separation residual $\mathfrak{o}_R^{\mathrm{sep}}$ to zero only after an independent source construction of $C_{X,R}$, $F^{\mathrm{co}}_{R,\bullet}$, $\Delta_R^{\mathrm{ch}}$, and $b_{X,R}$ has been supplied and identified with the target-side model.

*Proof.* The domain of $\varepsilon_{\Delta,R}$ contains only the target augmentation, target charge filtration, and target chiral operations. It does not contain the rigidified wrapped prequotients, anchors, mixed local/wrapped correspondences, quotient-after-correspondence functor, orientation transport, protected integration, or primitive representatives. These are the data from which the source coalgebra and collision coproduct are required to be built. Therefore a target bar–cobar inverse can be compared with a source coalgebra after that coalgebra exists; it cannot define the source coalgebra or its primitive restriction. □

*Definition* 10.10 *(Chiral Koszul source certificate).* Fix $(\sigma, S, R)$, the finite HN support $\widehat{\Gamma}_R$, and the Gram image $\Gamma_R^{\Pi}$. Let

$$\mathsf{Den}_{\Delta_5,E,\leq R}$$

denote the charge-truncation of the formal current envelope $\mathsf{Den}_{\Delta_5,E}$ to the finite set of Gram degrees $\Gamma_R^{\Pi}$. A finite-stage chiral Koszul source certificate is a datum

$$\mathfrak{K}^{\mathrm{Kos}}_{X,R} = \left(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}, C_{X,R}, F^{\mathrm{co}}_{R,\bullet}, \Delta^{\mathrm{ch}}_R, b_{X,R}, \Theta_{\mathrm{Kos},R}\right)$$

with the following properties. Every entry is source-side finite data except for the terminal comparison with the imported target $\mathsf{Den}_{\Delta_5,E,\leq R}$. In particular, neither $\bar{B}^{\mathrm{ch}}_E(\mathsf{Den}_{\Delta_5,E,\leq R})$ nor a homotopy inverse to the target counit is allowed to define $C_{X,R}$, $F^{\mathrm{co}}_{R,\bullet}$, or $\Delta^{\mathrm{ch}}_R$. This is the source/target separation of Lemma 10.9.

(i) *Source chiral object.* $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ is the finite hybrid object of Definition 10.18. Its $b_R = 0$ part comes from the $K3$-specialized $E_3$-source; its $b_R > 0$ part is present only through the rigidified wrapped prequotient, the ordered two-sided mixed operations, the full flag atlas, quotient-after-correspondence descent, and protected integration. Equivalently, the hybrid residual $\mathfrak{O}_{\mathrm{hyb},R}$ has been trivialized at this height. A detached global Fock or Hecke factor does not supply this entry unless it acts through the displayed mixed correspondences before the reduced $E$-quotient.

(ii) *Conilpotent chiral coalgebra.* $C_{X,R}$ is a coaugmented conilpotent chiral coalgebra on $E$, with coaugmentation $\eta^C_R : k \to C_{X,R}$ and counit $\varepsilon^C_R : C_{X,R} \to k$, filtered by a finite increasing source charge filtration

$$0 = F^{\mathrm{co}}_{R,-1} \subset F^{\mathrm{co}}_{R,0} \subset \cdots \subset F^{\mathrm{co}}_{R,N_R} = C_{X,R}$$

whose associated graded pieces are finite dimensional in each $\Gamma_R^{\Pi}$-degree. Its reduced iterated coproduct is nilpotent on each finite stage. The chiral coproduct $\Delta^{\mathrm{ch}}_R$ is the collision coproduct attached to the finite Hall/factorization correspondences, including local-local, all ordered local-wrapped, wrapped-wrapped, and eight-word flag stacks. It is a finite pull-push construction on the collision correspondence before $Q_{E,R}$, and after quotient it satisfies the coalgebra identities

$$(\Delta^{\mathrm{ch}}_R \otimes 1)\Delta^{\mathrm{ch}}_R = (1 \otimes \Delta^{\mathrm{ch}}_R)\Delta^{\mathrm{ch}}_R, \qquad (\varepsilon^C_R \otimes 1)\Delta^{\mathrm{ch}}_R = (1 \otimes \varepsilon^C_R)\Delta^{\mathrm{ch}}_R = \mathrm{id}.$$

For $R' \geq R$ the transition map $\rho^C_{R'R} : C_{X,R'} \to C_{X,R}$ preserves the filtration and makes the finite coproduct square commute:

$$\begin{tikzcd} C_{X,R'} \arrow[r, "\Delta^{\mathrm{ch}}_{R'}"] \arrow[d, "\rho^C_{R'R}"'] & C_{X,R'} \otimes C_{X,R'} \arrow[d, "\rho^C_{R'R} \otimes \rho^C_{R'R}"] \\ C_{X,R} \arrow[r, "\Delta^{\mathrm{ch}}_R"'] & C_{X,R} \otimes C_{X,R}. \end{tikzcd}$$

This finite diagram is the source of the completed coproduct on $C_X = \varprojlim_R C_{X,R}$. The coproduct is not transported from $\bar{B}^{\mathrm{ch}}_E(\mathsf{Den}_{\Delta_5,E,\leq R})$ along $\Theta_{\mathrm{Kos},R}$, and no completed tensor statement is made before these finite squares commute.

(iii) *Bar comparison.* There is a charge-preserving quasi-isomorphism of conilpotent chiral coalgebras

$$b_{X,R} : C_{X,R} \xrightarrow{\simeq} \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}).$$

It is compatible with primitive projection, the completed Hopf pairing, the reduced $E$-quotient, and the normal-ordered descent $\overline{\Pi}^{\Theta}_{X,*}$. This is a source-side comparison map, not the tautological bar construction of the target envelope. Its target is the bar construction of the supplied hybrid source object, not the bar construction of $\mathsf{Den}_{\Delta_5,E,\le R}$.

(iv) *Cobar comparison with the target envelope.* There is a charge-preserving quasi-isomorphism of augmented chiral algebras

$$\Theta_{\mathrm{Kos},R} : \Omega^{\mathrm{ch}}_E C_{X,R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E,\le R}.$$

On constant-current primitives, after the Hopf-radical quotient, this map restricts to the primitive recognition morphism

$$\iota^{\mathrm{Prim}}_R : H^0\,\mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}) \longrightarrow \mathfrak{g}_{\Delta_5,\le R}.$$

Here $\mathfrak{g}_{\Delta_5,\le R}$ is the same finite $\Gamma^{\Pi}_R$-truncation on constant currents. The target is the imported Beilinson–Drinfeld current envelope of the Gritsenko–Nikulin denominator algebra. The map does not construct the compact source and is not obtained from the scalar square. Its direction is source-to-target. A chosen homotopy inverse, if one exists after localization or completion, is not part of the certificate and cannot be used to define the source coalgebra or its primitive representatives. The displayed primitive restriction is a condition on already supplied source primitives after the closed Hopf-radical quotient.

(v) *Finite residual.* The exact Koszul obstruction tuple at height $R$ is

$$\mathfrak{O}_{\mathrm{Kos},R} = \Big(\mathfrak{o}^C_R, \mathfrak{o}^{\mathrm{fil}}_R, \mathfrak{o}^{\mathrm{conil}}_R, \mathfrak{o}^{\Delta,\mathrm{ch}}_R, \{\mathfrak{o}^{C,\mathrm{tr}}_{R'R}\}_{R'\ge R}, \mathfrak{o}^b_R, \mathfrak{o}^{\Omega}_R, \mathfrak{o}^{\mathrm{hyb}}_R, \mathfrak{o}^{\Pi}_R, \mathfrak{o}^{\mathrm{sep}}_R, \mathfrak{o}^{\mathrm{Prim}}_R\Big).$$

Here the finite coalgebra defects are

$$\mathfrak{o}^{\mathrm{fil}}_R := \Big(F^{\mathrm{co}}_{R,-1},\ C_{X,R}/F^{\mathrm{co}}_{R,N_R},\ \{d_C(F^{\mathrm{co}}_{R,p}) \to C_{X,R}/F^{\mathrm{co}}_{R,p}\}_p,\ \{\Delta^{\mathrm{ch}}_R(F^{\mathrm{co}}_{R,p}) \to (C_{X,R}\otimes C_{X,R})/\sum_{i+j\le p} F^{\mathrm{co}}_{R,i}\otimes F^{\mathrm{co}}_{R,j}\}_p\Big),$$

$$\overline{\Delta}^{\mathrm{ch}}_R := \Delta^{\mathrm{ch}}_R - \big((\eta^C_R\varepsilon^C_R)\otimes \mathrm{id} + \mathrm{id}\otimes(\eta^C_R\varepsilon^C_R)\big),$$

and

$$\mathfrak{o}^{\mathrm{conil}}_R := (\overline{\Delta}^{\mathrm{ch}}_R)^{N_R+1},$$

$$\mathfrak{o}^{\Delta,\mathrm{ch}}_R := \Big((\Delta^{\mathrm{ch}}_R\otimes 1)\Delta^{\mathrm{ch}}_R - (1\otimes\Delta^{\mathrm{ch}}_R)\Delta^{\mathrm{ch}}_R,\ (\varepsilon^C_R\otimes 1)\Delta^{\mathrm{ch}}_R - \mathrm{id},\ (1\otimes\varepsilon^C_R)\Delta^{\mathrm{ch}}_R - \mathrm{id}\Big),$$

and

$$\mathfrak{o}^{C,\mathrm{tr}}_{R'R} := \Big((\rho^C_{R'R}\otimes\rho^C_{R'R})\Delta^{\mathrm{ch}}_{R'} - \Delta^{\mathrm{ch}}_R\rho^C_{R'R},\ \{\rho^C_{R'R}(F^{\mathrm{co}}_{R',p}) \to C_{X,R}/F^{\mathrm{co}}_{R,p}\}_p\Big).$$

Here $d_C$ is the finite-stage coalgebra differential, and a displayed map to a quotient is zero exactly when the corresponding subobject inclusion holds. The entries record, respectively: existence of the source coalgebra; bounded exhaustive source filtration; conilpotence of the reduced iterated coproduct; finite collision-coproduct construction with coassociativity and counit; transition-strict compatibility of $(C_{X,R}, F^{\mathrm{co}}_{R,\bullet}, \Delta^{\mathrm{ch}}_R)$; the bar quasi-isomorphism $b_{X,R}$; the cobar quasi-isomorphism $\Theta_{\mathrm{Kos},R}$; compatibility with the hybrid local/wrapped operations, meaning $\mathfrak{O}_{\mathrm{hyb},R} = 0$; compatibility with normal-ordered descent and the reduced $E$-quotient; source/target

separation, meaning that the source coalgebra, filtration, coproduct, and bar comparison are supplied before the target comparison and are not reconstructed from the target counit; and the primitive restriction after the closed Hopf-radical quotient. A full Koszul certificate is the compatible vanishing and trivialization of $\mathfrak{O}_{\mathrm{Kos},R}$ for all $R$, followed by passage to the HN inverse limit.

Proposition 10.11 *(Canonical finite source-bar coalgebra).* Fix $R$ and assume the finite hybrid residual $\mathfrak{O}_{\mathrm{hyb},R}$ vanishes, with the transition and properness data of Definition 10.18. Then there is a canonical source-side choice of the finite coalgebra in Definition 10.10:

$$C^{\mathrm{bar}}_{X,R} := \bar{B}^{\mathrm{ch}}_{E}(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}).$$

This is the reduced chiral bar coalgebra of the supplied hybrid source, not the bar coalgebra of $\mathsf{Den}_{\Delta_5,E,\leq R}$. Equivalently, its length $p$ summand is generated by ordered $p$-fold non-vacuum hybrid inputs whose total HN height is $\leq R$, with all outputs of height $> R$ set to zero:

$$C^{\mathrm{bar}}_{X,R} = k\mathbf{1} \oplus \bigoplus_{1\leq p\leq N_R} \bar{B}^{\mathrm{ch},p}_{E}(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}).$$

The finite source filtration is the bar-length filtration

$$F^{\mathrm{co}}_{R,p}C^{\mathrm{bar}}_{X,R} := k\mathbf{1} \oplus \bigoplus_{1\leq j\leq p} \bar{B}^{\mathrm{ch},j}_{E}(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}), \qquad 0 \leq p \leq N_R,$$

where $N_R$ is any finite upper bound for the number of positive-height non-vacuum inputs whose total height is $\leq R$. The coaugmentation is the inclusion of the vacuum summand and the counit is projection to that summand. For a homogeneous bar generator $x = [x_1|\cdots|x_p]$, the collision coproduct is

$$\Delta^{\mathrm{bar}}_{R}(x) = \mathbf{1}\otimes x + x\otimes \mathbf{1} + \sum_{i=1}^{p-1}[x_1|\cdots|x_i]\otimes[x_{i+1}|\cdots|x_p],$$

with the displayed cuts realized by the corresponding finite collision and flag pull-push maps in $\mathsf{Corr}^{E,\mathrm{hyb}}_{R}$. The eight-word flag coherences and the four-input pentagon give coassociativity; the vacuum collision flags give the counit identities; and, for the reduced coproduct obtained by removing the two vacuum terms, the bar-length filtration gives

$$\left(\overline{\Delta}^{\mathrm{bar}}_{R}\right)^{N_R+1} = 0.$$

Thus, after replacing $C_{X,R}$ by $C^{\mathrm{bar}}_{X,R}$, the finite residuals

$$\mathfrak{o}^{C}_{R}, \qquad \mathfrak{o}^{\mathrm{fil}}_{R}, \qquad \mathfrak{o}^{\mathrm{conil}}_{R}, \qquad \mathfrak{o}^{\Delta,\mathrm{ch}}_{R}$$

vanish. The bar comparison $b_{X,R}$ may be chosen to be the identity of $\bar{B}^{\mathrm{ch}}_{E}(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R})$. For $R' \geq R$, the transition residual $\mathfrak{o}^{C,\mathrm{tr}}_{R'R}$ vanishes exactly when the finite hybrid transition functor carries ordered collision flags, quotient compatibilities, and the bar-length filtration at height $R'$ to the corresponding data at height $R$.

*Proof.* The construction applies the Beilinson–Drinfeld chiral bar construction to the already supplied source chiral object $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$. Since the finite HN quotient kills charges of height $> R$, only finitely

many non-vacuum bar lengths occur. The reduced bar construction has a vacuum coaugmentation, a projection counit, and the standard deconcatenation coproduct. In the present correspondence model, each deconcatenation is implemented by cutting the ordered collision flag; admissibility of the pull-push maps and Thom–Sebastiani transport are part of $\mathfrak{D}_{\mathrm{hyb},R} = 0$. The associativity and pentagon axioms for the finite flag stacks are precisely the equality of the two iterated cut maps. Hence the displayed coproduct is coassociative and counital, and the reduced coproduct lowers bar length, so its $(N_R + 1)$-fold iterate is zero. The construction has used only the source hybrid correspondence category, not the target current envelope. □

Corollary 10.12 *(Remaining source-to-target cobar obstruction).* Under the hypotheses of Proposition 10.11, the source coalgebra, finite filtration, collision coproduct, conilpotence, and bar comparison are no longer the Koszul obstruction. The remaining chiral Koszul obstruction is the source-to-target comparison

$$\Theta_{\mathrm{Kos},R} : \Omega_E^{\mathrm{ch}} C_{X,R}^{\mathrm{bar}} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E,\leq R}.$$

The source bar–cobar counit gives only

$$\varepsilon_{\mathrm{bar},R}^{\mathrm{src}} : \Omega_E^{\mathrm{ch}} C_{X,R}^{\mathrm{bar}} \longrightarrow \mathcal{F}_{X,\sigma,S,\leq R}^{\mathrm{hyb}}.$$

Therefore a source-to-target cobar comparison may be supplied, in this source-bar model, by a chiral algebra quasi-isomorphism

$$\vartheta_R : \mathcal{F}_{X,\sigma,S,\leq R}^{\mathrm{hyb}} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E,\leq R}$$

for which the following triangle commutes in the finite homotopy category:

$$\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon_{\mathrm{bar},R}^{\mathrm{src}}.$$

For this model the exact finite defect is

$$\mathfrak{o}_R^{\Omega} := \Big( d_\Delta \vartheta_R - \vartheta_R d_{\mathrm{hyb}},\ \vartheta_R \mu_R^{\mathrm{hyb}} - \mu_{\Delta,R}(\vartheta_R \otimes \vartheta_R),\ \vartheta_R \eta_R^{\mathrm{hyb}} - \eta_{\Delta,R},\ \vartheta_R \circ \tau_{R'R}^{\mathrm{hyb}} - \pi_{R'R}^{\Delta} \circ \vartheta_{R'},\ \vartheta_R|_{\mathrm{Prim}} - \iota_R^{\mathrm{Prim}} \Big).$$

Thus the finite source-bar construction kills the coalgebra part of $\mathfrak{D}_{\mathrm{Kos},R}$ after the hybrid source is supplied, but it does not identify compact primitives with the imported Gritsenko–Nikulin current envelope. That identification remains a source-to-target theorem, not a target-internal counit.

Proposition 10.13 *(Consequence of the chiral Koszul source certificate).* Assume the finite-stage certificates $\mathfrak{K}_{X,R}^{\mathrm{Kos}}$ are supplied for all $R$, with compatible transition maps, in particular with $\mathfrak{o}_{R'R}^{C,\mathrm{tr}} = 0$ for every $R' \geq R$. Then the inverse limit

$$C_X := \varprojlim_R C_{X,R}$$

is a completed pro-conilpotent chiral coalgebra attached to the hybrid compact source, conilpotent on every finite HN quotient, and the maps $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$ induce

$$b_X : C_X \xrightarrow{\simeq} \bar{B}_E^{\mathrm{ch}}(\mathcal{F}_X^{\mathrm{hyb}}), \qquad \Theta_{\mathrm{Kos}} : \Omega_E^{\mathrm{ch}} C_X \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E}.$$

This identifies the supplied source cobar algebra with the formal target current envelope. It does not construct the $E_3$-source, the hybrid wrapped Hall correspondences, the orientation line, the primitive recognition data, or $C_X$ itself from the target-internal counit.

*Proof.* The inverse limit exists by the transition compatibility, bounded exhaustive finite charge filtration, finite coassociative and counital collision coproduct, and continuity hypotheses in the certificate. The coradical filtration is finite on every $\Gamma_R^{\Pi}$-truncation and continuous in the HN completion; the limit is therefore pro-conilpotent, with finite-stage conilpotence preserved. The coproduct on $C_X$ is the inverse limit of the finite $\Delta_R^{\mathrm{ch}}$. The quasi-isomorphisms $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$ commute with the transition maps, hence pass to the completed inverse limit. The final sentence is the definition of the certificate: the source object, mixed correspondences, orientation, primitive presentation, and source/target-separation datum are separate entries of $\mathfrak{K}_X$. □

*Remark* 10.14 *(Locality).* In a compact realization, the required local object would be a finite-stage holomorphic $E_3$-factorization algebra on $X$, and its $K3$-specialization would be a finite-stage chiral factorization algebra on $\mathrm{Ran}(E)$. This assertion is local only for projection-finite sectors over $E$. The automorphic product is a protected global trace after charge completion, wall-chamber choice, and summation over compact $K3$-modes. Duality monodromy, elliptic-degree wrapping, and the Borcherds chamber are therefore global features of the trace. They do not replace the local factorization maps on disjoint opens.

*Definition* 10.15 *(Projection-finite, wrapped, and hybrid Ran strata).* Let $p_E : K3 \times E \to E$ be projection. A compact Hall or chiral sector is projection-finite over $E$ if every object $A$ in it satisfies

$$p_E(\mathrm{Supp}\, A) \subset F$$

for some finite subset $F \subset E$. A one-dimensional sector is wrapped over $E$ if its curve class has positive elliptic degree, i.e. if the pushforward of its one-cycle to $E$ is $b[E]$ with $b > 0$. For such one-dimensional Hall sectors write

$$b(A) = \deg(p_E)_*[\mathrm{Supp}_1 A] \in \mathbb{Z}_{\geq 0}.$$

The same letter $b(\gamma)$ denotes the elliptic-degree component of a charge $\gamma$. Then projection-finite objects have $b(A) = 0$, while wrapped objects have $b(A) > 0$.

The hybrid base used below is a finite-stage coloured correspondence base, not an ordinary Ran space with one scalar attached. At height $R$, after the certificate below is supplied, an object consists of a finite projection-local support $F \subset E$, finitely many wrapped colours $\eta_j \in \Gamma_R^{\mathrm{wr}}$, and rigidified wrapped objects $W_j$ carrying anchor values $\lambda_{\eta_j,R}(W_j) \in E$. Its arrows are generated by finite collisions of $F$, by extension of local colours, by mixed local/wrapped correspondences, and by wrapped/wrapped correspondences. The notation $\mathrm{Ran}^{\mathrm{hyb}}(E)$ denotes this finite-coloured system through the inverse system in $R$. Its underlying degree shadow is the displayed disjoint union

$$\mathrm{Ran}^{\mathrm{hyb}}(E) = \bigsqcup_{b \geq 0} \mathrm{Ran}(E)_{[b]}, \qquad \mathrm{Ran}(E)_{[b]} := \mathrm{Ran}(E) \times \{b\}.$$

The stratum $b = 0$ is the ordinary finite-support Ran stratum. A stratum $b > 0$ records finite local collision points together with a global wrapped elliptic leg of degree $b$; it is not an assertion that the wrapped support is finite over $E$.

Lemma 10.16 *(Projection-to-E support-locality obstruction).* Let $C \subset K3 \times E$ be a one-dimensional subscheme whose curve class has positive elliptic degree:

$$(p_E)_*[C] = b[E], \qquad b > 0.$$

Then $p_E(C) = E$. In particular, $C$ is not localized at finitely many points of $E$, and it is not contained in $p_E^{-1}(U)$ for any proper open $U \subsetneq E$.

*Proof.* If $p_E(C)$ were zero-dimensional, then the pushforward of the one-cycle $[C]$ to the curve $E$ would vanish. This contradicts $(p_E)_*[C] = b[E]$ with $b > 0$. Hence $p_E(C)$ is a one-dimensional closed subset of the irreducible curve $E$, since $p_E$ is proper, and therefore $p_E(C) = E$. □

Corollary 10.17 *(Naive Ran locality misses the elliptic-degree direction).* A support-local factorization algebra on $\mathrm{Ran}(E)$, defined only by separating supports over disjoint opens of $E$, sees the projection-finite sector. It does not by itself realize the full $s$-direction of the Igusa product.

*Proof.* The variable $s$ is the Borcherds variable corresponding to the OP elliptic-degree direction in Proposition 11.1. By Lemma 10.16, positive elliptic degree projects onto all of $E$. Such objects cannot be separated by choosing disjoint opens in $E$. Therefore ordinary support-local factorization on $\mathrm{Ran}(E)$ has no locality mechanism for the wrapped elliptic-degree sector. □

*Definition* 10.18 *(Finite hybrid local/wrapped factorization certificate).* Fix $(\sigma, S, R)$ as in Definition 10.4. A finite-stage hybrid local/wrapped factorization certificate is a datum

$$\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R} = \Big(\mathcal{C}_{\sigma,S,\leq R},\ b_R,\ \mathcal{F}^{\mathrm{loc}}_R,\ \mathcal{G}^{\mathrm{wr}}_R,\ \mathfrak{E}^{\mathrm{loc}}_R,\ \mathfrak{E}^{\mathrm{hyb}}_R,\ \mathfrak{E}^{\mathrm{wr}}_R,\ \mathfrak{Flag}^{\mathrm{hyb}}_R,\ \mathsf{Corr}^{E,\mathrm{hyb}}_R,\ \lambda_R,\ Q_{E,R},\ \theta^Q_R,\ o_R,\ I^{\mathrm{prot}}_R\Big)$$

with the following finite data and axioms. Every stack, sheaf, operation, and coherence named below lives at the fixed height $R$; the inverse limit appears only after all finite stages and transition maps have been supplied.

(i) *Finite Hall stage and elliptic degree.* $\mathcal{C}_{\sigma,S,\leq R}$ is an extension-closed finite HN quotient of the candidate compact Hall category, with finite-type charge substacks $\mathfrak{M}_{\gamma,R}$ for $\gamma \in \Gamma^{HN}_{\sigma,S}(R)$, vanishing-cycle coefficients $\Phi_{\gamma,R}$, and orientation lines $o_{\gamma,R}$. The map

$$b_R : \Gamma^{HN}_{\sigma,S}(R) \longrightarrow \mathbb{Z}_{\geq 0}$$

is the elliptic-degree component. Write $\Gamma^{\mathrm{loc}}_R = \{\alpha \mid b_R(\alpha) = 0\}$ and $\Gamma^{\mathrm{wr}}_R = \{\eta \mid b_R(\eta) > 0\}$. The ideal of height $> R$ is already killed at this stage, and all operations below are understood to be zero when their output has height $> R$. The map $b_R$ is additive on extensions that remain in the quotient:

$$b_R(0) = 0, \qquad b_R(\gamma + \gamma') = b_R(\gamma) + b_R(\gamma')$$

whenever $\gamma, \gamma', \gamma + \gamma' \in \Gamma^{HN}_{\sigma,S}(R)$; if the sum has height $> R$, the corresponding operation is zero in the finite quotient.

(ii) *Projection-finite local sector.* For $U \subset E$ open and $\alpha \in \Gamma^{\mathrm{loc}}_R$,

$$\mathcal{M}^{\mathrm{loc}}_{\alpha,R}(U) = \big\{A \in \mathfrak{M}_{\alpha,R} \,\big|\, p_E(\mathrm{Supp}\, A) \subset U\big\}$$

is supplied as an open substack of $\mathfrak{M}_{\alpha,R}$, and

$$\mathcal{F}^{\mathrm{loc}}_{\alpha,R}(U) := H^{\mathrm{BM}}_*\Big(\mathcal{M}^{\mathrm{loc}}_{\alpha,R}(U), \Phi^{\mathrm{loc}}_{\alpha,R} \otimes o^{\mathrm{loc}}_{\alpha,R}\Big).$$

The direct sum over $\alpha \in \Gamma^{\mathrm{loc}}_R$ is $\mathcal{F}^{\mathrm{loc}}_R(U)$. For disjoint opens this is an ordinary projection-finite Ran factorization object.

(iii) *Wrapped prequotient sector and rigidification.* For $\eta \in \Gamma_R^{\mathrm{wr}}$, a prequotient stack

$$\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}} \xrightarrow{\rho_{\eta,R}} \mathcal{M}_{\eta,R}^{\mathrm{wr}} \subset \mathfrak{M}_{\eta,R}$$

is supplied, together with an $E$-equivariant anchor

$$\lambda_{\eta,R} : \mathcal{M}_{\eta,R}^{\mathrm{wr,rig}} \longrightarrow E, \qquad \lambda_{\eta,R}(t \cdot W) = t + \lambda_{\eta,R}(W).$$

Its objects have support dominating $E$. The pair $(\rho_{\eta,R}, \lambda_{\eta,R})$ is the rigidification: an explicit chosen device, for example an Abel–Jacobi center, determinant trivialization, framed marked point, or a named equivalent, retaining the relative $E$-position needed by mixed extensions before the reduced $E$-quotient. No canonical rigidification is asserted. Put

$$\mathcal{G}_{\eta,R}^{\mathrm{wr}} := H_*^{\mathrm{BM}}\Big(\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}}, \Phi_{\eta,R}^{\mathrm{wr}} \otimes o_{\eta,R}^{\mathrm{wr}}\Big), \qquad \mathcal{G}_R^{\mathrm{wr}} := \bigoplus_{\eta \in \Gamma_R^{\mathrm{wr}}} \mathcal{G}_{\eta,R}^{\mathrm{wr}}.$$

If no such prequotient memory is supplied, the wrapped part of the certificate is absent.

(iv) *Local, mixed, and wrapped extension correspondences.* For disjoint opens $(U_i) \subset E$, local charges $\alpha_i$, and wrapped charges $\eta$, $\eta_1$, $\eta_2$, the finite stage contains correspondences

$$\prod_i \mathcal{M}_{\alpha_i,R}^{\mathrm{loc}}(U_i) \xleftarrow{p} \mathfrak{E}_{\alpha_1,\dots,\alpha_r,R}^{\mathrm{loc}}(U_1,\dots,U_r) \xrightarrow{q} \mathcal{M}_{\sum_i \alpha_i,R}^{\mathrm{loc}}\Big(\bigcup_i U_i\Big),$$

$$\prod_i \mathcal{M}_{\alpha_i,R}^{\mathrm{loc}}(U_i) \times \mathcal{M}_{\eta,R}^{\mathrm{wr,rig}} \xleftarrow{p} \mathfrak{E}_{\alpha_1,\dots,\alpha_r;\eta,R}^{\mathrm{hyb}}(U_1,\dots,U_r) \xrightarrow{q} \mathcal{M}_{\eta+\sum_i \alpha_i,R}^{\mathrm{wr,rig}},$$

and

$$\mathcal{M}_{\eta_1,R}^{\mathrm{wr,rig}} \times \mathcal{M}_{\eta_2,R}^{\mathrm{wr,rig}} \xleftarrow{p} \mathfrak{E}_{\eta_1,\eta_2,R}^{\mathrm{wr}} \xrightarrow{q} \mathcal{M}_{\eta_1+\eta_2,R}^{\mathrm{wr,rig}}.$$

The second display is only the one-sided shorthand. Since Hall convolution is ordered, the datum contains, for every ordered mixed word with local inputs on both sides of one wrapped input, correspondences

$$\prod_{i \in I_-} \mathcal{M}_{\alpha_i,R}^{\mathrm{loc}}(U_i) \times \mathcal{M}_{\eta,R}^{\mathrm{wr,rig}} \times \prod_{j \in I_+} \mathcal{M}_{\beta_j,R}^{\mathrm{loc}}(V_j) \xleftarrow{p} \mathfrak{E}_{\alpha_-;\eta;\beta_+,R}^{\mathrm{hyb}}((U_i),(V_j)) \xrightarrow{q} \mathcal{M}_{\eta+\sum_{i\in I_-}\alpha_i+\sum_{j\in I_+}\beta_j,R}^{\mathrm{wr,rig}}.$$

The displayed mixed correspondence above is the case $I_+ = \varnothing$. The case $I_- = \varnothing$ is the right local action on a wrapped input. Both are part of the certificate. Similarly $\mathfrak{E}_{\eta_1,\eta_2,R}^{\mathrm{wr}}$ is ordered in $(\eta_1, \eta_2)$; no symmetry of wrapped inputs is assumed unless an extra symmetry datum is supplied. The middle stacks classify exact triangles or exact sequences with the displayed associated graded pieces. They are constructed in the unreduced $E$-equivariant theory, with proper-support or compact-support witnesses for $q$, admissibility witnesses for $p^*$ and $q_!$ on the chosen vanishing-cycle/Borel–Moore theory, and Thom–Sebastiani identifications for the coefficient systems. The mixed stacks use the anchor $\lambda_{\eta,R}$ to remember the relative position between the finite local supports $U_i$ and the wrapped leg; replacing this memory by a scalar Fock coefficient is not this datum. The wrapped-wrapped stacks are likewise anchored: before quotienting they carry the map

$$\mathfrak{E}_{\eta_1,\eta_2,R}^{\mathrm{wr}} \longrightarrow E \times E \times E, \qquad \xi \longmapsto \big(\lambda_{\eta_1,R}(W_1), \lambda_{\eta_2,R}(W_2), \lambda_{\eta_1+\eta_2,R}(W)\big),$$

where $W_1, W_2, W$ are the two inputs and the output of $\xi$. The finite correspondence keeps this relation, not only the total degree $b_R(\eta_1 + \eta_2)$.
These object stacks, extension stacks, and flag stacks generate a finite $E$-equivariant correspondence category

$$\mathsf{Corr}_R^{E,\mathrm{hyb}}.$$

Its objects are the finitely many local and rigidified wrapped charge stacks at height $R$. Its arrows are generated by the displayed correspondences, with composition by derived fibre product and with every summand of height $> R$ identified with zero. The flag stacks below are the specified 2-morphisms witnessing equality of the two composites. The finite operation maps are the displayed pull-push maps

$$\mu_R^\bullet = q_! \circ \mathrm{TS}_R^\bullet \circ p^*,$$

namely

$$\mu^{\mathrm{loc}}_{\alpha_1,\dots,\alpha_r,R} : \bigotimes_i \mathcal{F}^{\mathrm{loc}}_{\alpha_i,R}(U_i) \longrightarrow \mathcal{F}^{\mathrm{loc}}_{\sum_i \alpha_i,R}\left(\bigcup_i U_i\right),$$

$$\mu^{\mathrm{hyb}}_{\alpha_1,\dots,\alpha_r;\eta,R} : \bigotimes_i \mathcal{F}^{\mathrm{loc}}_{\alpha_i,R}(U_i) \otimes \mathcal{G}^{\mathrm{wr}}_{\eta,R} \longrightarrow \mathcal{G}^{\mathrm{wr}}_{\eta+\sum_i \alpha_i,R},$$

and

$$\mu^{\mathrm{wr}}_{\eta_1,\eta_2,R} : \mathcal{G}^{\mathrm{wr}}_{\eta_1,R} \otimes \mathcal{G}^{\mathrm{wr}}_{\eta_2,R} \longrightarrow \mathcal{G}^{\mathrm{wr}}_{\eta_1+\eta_2,R}.$$

The two-sided mixed correspondences give, in the same notation,

$$\mu^{\mathrm{hyb}}_{\alpha_-;\eta;\beta_+,R} : \bigotimes_{i\in I_-} \mathcal{F}^{\mathrm{loc}}_{\alpha_i,R}(U_i) \otimes \mathcal{G}^{\mathrm{wr}}_{\eta,R} \otimes \bigotimes_{j\in I_+} \mathcal{F}^{\mathrm{loc}}_{\beta_j,R}(V_j) \longrightarrow \mathcal{G}^{\mathrm{wr}}_{\eta+\sum_i \alpha_i+\sum_j \beta_j,R}.$$

An output with at least one wrapped input is wrapped. The mixed correspondence is also required to record the surviving terms in $\mathrm{RHom}(A, W)$ and $\mathrm{RHom}(W, A)$ for local $A$ and wrapped $W$; replacing it by a scalar global mode loses this data.

(v) *Full flag associativity and coherence.* Write L for a projection-finite local input and W for a rigidified wrapped input. For every word

$$w = (\epsilon_1, \epsilon_2, \epsilon_3) \in \{\mathrm{L}, \mathrm{W}\}^3$$

and every compatible ordered triple of charges and local opens, the datum contains a two-step flag stack

$$\mathfrak{Flag}^w_{\xi_1,\xi_2,\xi_3,R}$$

classifying filtrations with the prescribed associated graded pieces. Here $\mathfrak{E}_{\xi_i,\xi_j,R}$ means the corresponding ordered local-local, mixed local/wrapped, or wrapped-wrapped extension stack from the previous clause. The flag stack carries comparison maps to the two iterated correspondence fibre products

$$\mathfrak{E}_{\xi_1,\xi_2,R} \times_{\mathcal{M}_{\xi_1+\xi_2,R}} \mathfrak{E}_{\xi_1+\xi_2,\xi_3,R}, \qquad \mathfrak{E}_{\xi_2,\xi_3,R} \times_{\mathcal{M}_{\xi_2+\xi_3,R}} \mathfrak{E}_{\xi_1,\xi_2+\xi_3,R}.$$

The eight words

$$\mathrm{LLL},\ \mathrm{LLW},\ \mathrm{LWL},\ \mathrm{WLL},\ \mathrm{LWW},\ \mathrm{WLW},\ \mathrm{WWL},\ \mathrm{WWW}$$

are all part of the datum. In the mixed and wrapped words the flag stack retains the input, intermediate, and output anchors before quotienting by $E$. The two parenthesizations give the same map in the finite quotient:

$$\mu_{\xi_1+\xi_2,\xi_3,R} \circ (\mu_{\xi_1,\xi_2,R} \otimes 1) = \mu_{\xi_1,\xi_2+\xi_3,R} \circ (1 \otimes \mu_{\xi_2,\xi_3,R}).$$

Here $\mu$ means one of the displayed local, mixed, or wrapped maps. The equality is required after vanishing cycles, Thom–Sebastiani transport, orientation signs, and the quotient by height $> R$. Equivalently, the two pull-push maps are induced by the same Borel–Moore class on the corresponding two-step flag stack. This is a finite-stage coherence axiom; no associativity is imported from the completed inverse limit. The defect class of this equality for the word $w$ is denoted

$$\mathfrak{o}^{\mathrm{assoc}}_{w,R}.$$

The certificate requires $\mathfrak{o}^{\mathrm{assoc}}_{w,R} = 0$ for all eight words, with the chosen 2-morphisms satisfying the usual four-input pentagon on the finite flag-of-flags stacks.

(vi) *Reduced $E$-quotient order.* All stacks and correspondences above are $E$-equivariant before quotienting. Apply vanishing-cycle Borel–Moore chains to $\mathsf{Corr}^{E,\mathrm{hyb}}_R$, with the orientation lines included, and denote the resulting finite correspondence module category by $\mathsf{BM}^{E,\mathrm{hyb}}_R$. The symbol $Q_{E,R}$ denotes a supplied functor

$$Q_{E,R} : \mathsf{BM}^{E,\mathrm{hyb}}_R \longrightarrow \mathsf{BM}^{\mathrm{red,hyb}}_R$$

from the finite unreduced equivariant vanishing-cycle/Borel–Moore category generated by the stacks above to the finite reduced category. It is allowed only after the local, wrapped, and mixed extension correspondences have been formed and the chosen rigidification has either descended or has been integrated out with a specified reduced kernel. For every generator $e$ of $\mathsf{Corr}^{E,\mathrm{hyb}}_R$, the quotient package contains the reduced correspondence $\overline{e}$, the quotient maps from $e$ to $\overline{e}$, admissibility witnesses for $\overline{p}^*$ and $\overline{q}_!$, proper-support witnesses for $\overline{q}$, and orientation-descent isomorphisms for the coefficient systems. If the anchor is integrated out rather than descended, the datum also contains the reduced kernel $K_{\lambda,R}$ and its compatibility with composition. The quotient construction is therefore a functor on the finite correspondence category, not a quotient of the object spaces followed by a new Hall product. For each operation $\mu^{\bullet}_R = q_! \,\mathrm{TS}\, p^*$ the datum contains a specified comparison isomorphism

$$\theta^{Q}_{\mu,R} : \; Q_{E,R} q_! \,\mathrm{TS}\, p^* \xrightarrow{\sim} \overline{q}_! \overline{\mathrm{TS}}\, \overline{p}^* (Q^{\boxtimes k}_{E,R}),$$

which is compatible with the flag-stack coherences. The finite-stabilizer gerbe classes needed for the orientation line must vanish on every stack entering these squares. For each two-step flag, the two composites of $\theta^{Q}_{\mu,R}$ agree with the descended flag 2-morphism; for $R' \geq R$, the transition functors commute with $Q_{E,R}$ through a specified natural isomorphism. A quotient-first construction is not this datum. The four quotient residuals are

$$\mathfrak{o}^{Q,\mathrm{form}}_R, \qquad \mathfrak{o}^{Q,\mathrm{adm}}_R, \qquad \mathfrak{o}^{Q,\mathrm{flag}}_R, \qquad \mathfrak{o}^{Q,\mathrm{tr}}_{R'R}.$$

They record, respectively, existence of the functor $Q_{E,R}$ on the finite correspondence category, admissibility/properness after quotient, compatibility with all flag 2-morphisms, and compatibility with HN transition maps. The certificate requires their vanishing and chosen null-homotopies.

(vii) *Protected integration and the Igusa s-direction.* The finite stage carries a protected integration map

$$I_R^{\mathrm{prot}} : Q_{E,R}\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R} \longrightarrow \mathbb{C}[\Gamma_{\mathrm{eff}}]_{\leq R}$$

where the subscript denotes the image of the finite $h_S$-box. The map is compatible with the three reduced products $\overline{\mu}^{\bullet}_R$, coproducts, primitive projection, and transition maps. This means a chain-level character of the finite reduced correspondence algebra, not only the assignment of an $s$-degree. Let $m_{\Gamma,R}$, $\Delta_{\Gamma,R}$, and $P^{\mathrm{Prim}}_{\Gamma,R}$ denote the finite target multiplication, coproduct, and primitive projector on the chosen decategorified monomial model. The integration defects are

$$\mathfrak{o}_R^{I,\mathrm{ch}} := I_R^{\mathrm{prot}} d_R,$$

$$\mathfrak{o}_{w,R}^{I,\mathrm{prod}} := I_R^{\mathrm{prot}}\overline{\mu}_R^w - m_{\Gamma,R}(I_R^{\mathrm{prot}})^{\otimes k}, \qquad w \in \{\mathrm{L},\mathrm{W}\}^k,$$

$$\mathfrak{o}_R^{I,\mathrm{co}} := \Delta_{\Gamma,R} I_R^{\mathrm{prot}} - (I_R^{\mathrm{prot}} \otimes I_R^{\mathrm{prot}})\overline{\Delta}_R, \qquad \mathfrak{o}_R^{I,\mathrm{Prim}} := P^{\mathrm{Prim}}_{\Gamma,R} I_R^{\mathrm{prot}} - I_R^{\mathrm{prot}} P_R^{\mathrm{Prim}}.$$

They are required to vanish, compatibly with $Q_{E,R}$, the orientation trivializations, and transitions in $R$. Equivalently, for every reduced operation $\overline{\mu}_R^w$ and homogeneous inputs $x_i$ of charges $\gamma_i$,

$$I_R^{\mathrm{prot}}\big(\overline{\mu}_R^w(x_1,\dots,x_k)\big) = \prod_i I_R^{\mathrm{prot}}(x_i)$$

whenever the product is defined at height $R$; if the output charge has height $> R$, both sides are zero in the finite quotient. If a homogeneous class $x$ has charge $\gamma$, and $\mathcal{F}^{\mathrm{hyb}}_{\gamma,R}$ denotes the corresponding local or wrapped summand, then every nonzero monomial in $I_R^{\mathrm{prot}}(Q_{E,R}x)$ has $s$-degree $b_R(\gamma)$:

$$I_R^{\mathrm{prot}}(Q_{E,R}x) \in s^{b_R(\gamma)}\mathbb{C}[q^{\pm 1}, r^{\pm 1}]$$

under the OP/Borcherds variable convention of Proposition 11.1. Thus positive powers of the Igusa $s$-variable can only come from the wrapped colour $b_R > 0$. For products, the displayed character law and additivity of $b_R$ give

$$s^{b_R(\gamma_1+\cdots+\gamma_k)} = \prod_i s^{b_R(\gamma_i)}$$

inside the finite HN quotient. Hence the $s$-direction is coupled to the actual local/wrapped operation, not appended as an external Fock determinant. For $R' \geq R$, let $\tau^{\mathrm{hyb}}_{R'R}$ be the finite hybrid transition and let $\pi^{\Gamma}_{R'R}$ be the induced truncation of the monomial target. The integration transition defect is

$$\mathfrak{o}_{R'R}^{I,\mathrm{tr}} := I_R^{\mathrm{prot}} Q_{E,R}\tau^{\mathrm{hyb}}_{R'R} - \pi^{\Gamma}_{R'R} I_{R'}^{\mathrm{prot}} Q_{E,R'}.$$

It is required to vanish. Thus protected integration is compatible with the HN inverse system, not merely with one finite trace. The coefficient of a monomial $q^n r^l s^{b_R(\gamma)}$ is a protected finite-stage trace coefficient. Equality of these coefficients with the Borcherds exponents $f(nm,l)$, and the primitive Borcherds–Kac–Moody bracket test, are additional recognition requirements, not part of the definition.

(viii) *Hybrid residual tuple.* The exact finite obstruction tuple for the hybrid lane is

$$\mathfrak{O}_{\mathrm{hyb},R} = \Big(\mathfrak{o}_R^{\mathrm{stage}},\ \mathfrak{o}_R^{\lambda},\ \mathfrak{o}_R^{\mathrm{corr,LL}},\ \mathfrak{o}_R^{\mathrm{corr,mix}},\ \mathfrak{o}_R^{\mathrm{corr,WW}},\ \{\mathfrak{o}_{w,R}^{\mathrm{assoc}}\}_{w\in\{\mathrm{L},\mathrm{W}\}^3},\ \mathfrak{o}_R^{Q,\mathrm{form}},\ \mathfrak{o}_R^{Q,\mathrm{adm}},\ \mathfrak{o}_R^{Q,\mathrm{flag}},\ \mathfrak{o}_{R'R}^{Q,\mathrm{tr}},\ \mathfrak{o}_R^{I,\mathrm{ch}},\ \{\mathfrak{o}_{w,R}^{I,\mathrm{prod}}\}_w,\ \mathfrak{o}$$

These entries record finite HN existence, anchor rigidification, local-local correspondences, all ordered mixed correspondences, wrapped-wrapped correspondences, full flag coherence, quotient formation/admissibility/flag and transition descent, chain/product/coproduct/primitive integration laws, integration transition compatibility, and HN transition compatibility. A finite hybrid certificate means the compatible vanishing and trivialization of this tuple for every $R' \geq R$. Without $\mathfrak{O}_{\mathrm{hyb},R} = 0$, the notation $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ names an obstruction problem, not a source object.

(ix) *Finite HN completion.* The construction is made only for charges in $\Gamma^{HN}_{\sigma,S}(R)$. For $R' \geq R$, transition maps

$$\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R'} \longrightarrow \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$$

must be induced by functors $\mathsf{Corr}^{E,\mathrm{hyb}}_{R'} \to \mathsf{Corr}^{E,\mathrm{hyb}}_{R}$ compatible with all three kinds of correspondences, with the flag associativity constraints, with the rigidifying anchors, with $Q_{E,R}$, with the comparison isomorphisms $\theta^{Q}_{R}$, and with protected integration. They preserve elliptic degree: if a homogeneous summand of charge $\gamma'$ at height $R'$ maps to a nonzero summand of charge $\gamma$ at height $R$, then

$$b_R(\gamma) = b_{R'}(\gamma').$$

When all finite stages and transitions have been supplied, the completed hybrid object is

$$\mathcal{F}^{\mathrm{hyb}}_{X} := \varprojlim_{R} \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}.$$

It is the charge-completed hybrid source specified by those finite certificates over $\mathrm{Ran}^{\mathrm{hyb}}(E)$. Its restriction to $\mathrm{Ran}(E)_{[0]}$ is ordinary projection-finite Ran factorization. Its $b > 0$ strata are wrapped modules acted on by the local sector through the mixed Hall correspondences. This inverse-limit formula is a construction from the displayed finite certificates; it is not an existence theorem for those certificates.

Proposition 10.19 *(Finite consequences of the hybrid certificate).* Assume the finite-stage data of Definition 10.18 are supplied for all $R$, with compatible transition maps. Then:

(i) the $b = 0$ restriction is an ordinary support-local factorization object on $\mathrm{Ran}(E)$;

(ii) every operation with a wrapped input lands in a wrapped stratum;

(iii) the displayed ordered mixed maps, including the two-sided maps $\mu^{\mathrm{hyb}}_{\alpha_-;\eta;\beta_+,R}$, define the only available Hall actions of projection-finite insertions on wrapped elliptic-degree sectors in $\mathsf{Corr}^{E,\mathrm{hyb}}_{R}$ before the reduced $E$-quotient;

(iv) vanishing of the eight classes $\mathfrak{o}^{\mathrm{assoc}}_{w,R}$ gives an associative finite-stage reduced Hall product after vanishing cycles, orientations, HN quotienting, the quotient functor, and the comparison isomorphisms $\theta^{Q}_{\mu,R}$;

(v) a protected trace or primitive bracket obtained from this object is compatible with reduced $E$-integration only through the specified $Q_{E,R}$-compatibility isomorphisms, their flag coherences, finite-stabilizer conditions, and the vanishing of the integration defects $\mathfrak{o}^{I,\mathrm{ch}}_{R}$, $\mathfrak{o}^{I,\mathrm{prod}}_{w,R}$, $\mathfrak{o}^{I,\mathrm{co}}_{R}$, and $\mathfrak{o}^{I,\mathrm{Prim}}_{R}$, compatibly with $\mathfrak{o}^{I,\mathrm{tr}}_{R'R} = 0$;

(vi) for homogeneous $x \in Q_{E,R}\mathcal{F}^{\mathrm{hyb}}_{\gamma,R}$,

$$I^{\mathrm{prot}}_{R}(x) \in s^{b_R(\gamma)}\mathbb{C}[q^{\pm 1}, r^{\pm 1}],$$

hence the positive $s$-direction cannot be supplied by the projection-finite Ran stratum; on products this law is multiplicative because $b_R$ is additive inside the finite quotient.

A decoupled global Fock or Hecke factor without $\mathsf{Corr}_R^{E,\mathrm{hyb}}$, anchored mixed correspondences, full flag coherence, quotient residuals, and protected-integration residuals may reproduce a determinant-level product, but it does not define the local-wrapped primitive brackets needed for recognition of $\mathfrak{g}_{\Delta_5}$.

*Proof.* The first two assertions are built into the elliptic-degree stratification and Lemma 10.16. The third assertion is the definition of the mixed operation maps: before the reduced quotient they are the only displayed operations remembering relative $E$-position between finite local support and a wrapped leg, including the left and right ordered actions. The fourth is exactly the full flag-stack axiom transported through $\mathsf{Corr}_R^{E,\mathrm{hyb}}$. The fifth is the quotient-order axiom, the coherent comparison isomorphism $\theta^Q_{\mu,R}$, and the vanishing of the quotient residuals. The last statement is the degree law and character property in the protected integration clause, together with the vanishing of the named integration defects and their transition residual. No compact source object, orientation system, quotient functor, or protected integration map is constructed in this proposition. □

Corollary 10.20 *(No quotient-first universal hybrid repair).* Fix $R$. A candidate finite correspondence category for the wrapped sector gives the source object $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ only if it contains the unreduced local, wrapped, and ordered mixed source stacks of Definition 10.18, their anchors, the eight-word flag atlas, the quotient functor $Q_{E,R}$ formed after those correspondences, the comparison maps $\theta^Q_{\mu,R}$, and protected integration with all entries of $\mathfrak{O}_{\mathrm{hyb},R}$, including its transition entries, trivialized. A quotient of object stacks formed before the mixed and wrapped correspondences, or a determinant/Fock category carrying only the trace degree $s^{b_R}$, does not define the hybrid source.

*Proof.* By Lemma 10.16, a positive elliptic-degree sector is not a finite-support Ran sector. The only displayed mechanism retaining its relative $E$-position is the rigidified wrapped prequotient with anchor, together with the unreduced mixed and wrapped correspondences. If one quotients first, these anchors are no longer available as source data for $\mathfrak{E}^{\mathrm{hyb}}_R$, the two-sided maps $\mu^{\mathrm{hyb}}_{\alpha_-;\eta;\beta_+,R}$, or the flag 2-morphisms. Then $Q_{E,R}$, $\theta^Q_{\mu,R}$, and the product, coproduct, primitive, and transition integration residuals are not defined as finite source data. A scalar determinant or Fock trace can record coefficients after decategorification; it cannot supply the correspondence-level primitive brackets required by the certificate. □

Proposition 10.21 *((O2)/hybrid compatibility obstruction).* Fix a finite HN height $R$ and one of the three type-II simple walls. Under the degree map $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$ the wall labels are

| wall | $(n,l,m)$ | elliptic-degree shadow |
|---|---|---|
| $\delta_1 = 2f_2 - f_3$ | $(1,1,0)$ | $m = 0$ |
| $\delta_2 = 2f_{-2} - f_3$ | $(0,1,1)$ | $m = 1$ |
| $\delta_3 = f_3$ | $(0,-1,0)$ | $m = 0.$ |

A finite compact construction which claims both the hybrid source $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ of Definition 10.18 and the type-II Pfaffian wall model of Definition 3.10 must supply a compatibility package at the same height $R$. It consists of:

(i) a wall object $x_{\delta_i,R}$ in the finite local or rigidified wrapped moduli used by $\mathsf{Corr}_R^{E,\mathrm{hyb}}$, mapping to Gram degree $\delta_i$;

(ii) a finite restriction of the unreduced hybrid deformation theory to the wall object, producing a self-Ext complex $K^{E}_{\delta_i,R}$;

(iii) a quotient-compatible comparison diagram

$$\begin{tikzcd}
\mathsf{BM}^{E,\mathrm{hyb}}_{R} \arrow[r, "\ell_{\delta_i,R}"] \arrow[d, "Q_{E,R}"'] & K^{E}_{\delta_i,R} \arrow[r, "\mathrm{split}_{\delta_i,R}"] \arrow[d, "Q^{K}_{\delta_i,R}"] & K^{E,\mathrm{tan}}_{\delta_i,R} \oplus K^{E,\mathrm{nor}}_{\delta_i,R} \arrow[d, "Q^{\mathrm{tan}}_{\delta_i,R}\oplus Q^{\mathrm{nor}}_{\delta_i,R}"] \\
\mathsf{BM}^{\mathrm{red,hyb}}_{R} \arrow[r, "\overline{\ell}_{\delta_i,R}"'] & K^{\mathrm{red}}_{\delta_i,R} \arrow[r, "\overline{\mathrm{split}}_{\delta_i,R}"'] & K^{\mathrm{tan}}_{\delta_i,R} \oplus K^{\mathrm{nor}}_{\delta_i,R},
\end{tikzcd}$$

where the bottom row is the reduced wall chart of Proposition 3.11;

(iv) an invariant normal Pfaffian unit and rank-one normal model

$$\mathrm{Pf}(K^{\mathrm{nor}}_{\delta_i,R}) = \upsilon_{\delta_i,R}u_{\delta_i,R}, \qquad s_{\delta_i}(\upsilon_{\delta_i,R}) = \upsilon_{\delta_i,R},$$

after the quotient orientation trivializations used by $Q_{E,R}$;

(v) protected integration compatibility: if $\mathrm{pf}_{\delta_i,R}$ is the local Pfaffian class of the normal crossing, then every nonzero monomial of

$$I^{\mathrm{prot}}_{R}(Q_{E,R}\,\mathrm{pf}_{\delta_i,R})$$

has $s$-degree $m$, where $(n, l, m)$ is the displayed Gram label, and the equality is compatible with the transition residual $\mathfrak{o}^{I,\mathrm{tr}}_{R'R}$.

Equivalently, the finite residual

$$\mathfrak{O}^{\mathrm{O2/hyb}}_{\delta_i,R} = \left(\mathfrak{o}^{\mathrm{wall}}_{\delta_i,R}, \mathfrak{o}^{\text{Ext-split}}_{\delta_i,R}, \mathfrak{o}^{\mathrm{unit}}_{\delta_i,R}, \mathfrak{o}^{Q,\mathrm{wall}}_{\delta_i,R}, \mathfrak{o}^{I,\mathrm{wall}}_{\delta_i,R}\right)$$

must be trivialized. The $\delta_2$ row has $m = 1$, hence its wall object cannot be supplied by the projection-finite Ran stratum alone. It must pass through the rigidified wrapped prequotient and the ordered mixed correspondences before the reduced $E$-quotient. A local rank-one chart detached from $\mathsf{Corr}^{E,\mathrm{hyb}}_{R}$, or a Fock/Hecke factor carrying only the scalar $s$-degree, does not make this residual vanish.

*Proof.* The three Gram labels are the inverse calculation recorded in the type-II root section:

$$\delta_1 \leftrightarrow (1, 1, 0), \qquad \delta_2 \leftrightarrow (0, 1, 1), \qquad \delta_3 \leftrightarrow (0, -1, 0).$$

The local Pfaffian proposition requires a reduced wall chart, a reduced self-Ext complex, a tangent/normal splitting, an invariant normal unit, and $N^{\mathrm{Pf}}_{\delta_i} = 1$. The hybrid certificate requires unreduced local, wrapped, and ordered mixed correspondences, followed only then by $Q_{E,R}$ and by protected integration. If both claims are claims about the same compact source, the wall complex must be the localization of the finite hybrid deformation theory and its splitting, unit, quotient descent, and protected trace must commute with the hybrid operations. The displayed diagram and residual tuple are precisely this finite compatibility requirement.

For $\delta_2$ the target monomial has $s$-degree one. By Lemma 10.16, positive elliptic degree is not supported at finitely many points of $E$. Thus an ordinary projection-finite Ran object cannot supply the wall object whose local Pfaffian class integrates to the $\delta_2$ label. The wrapped prequotient, its anchor, the mixed correspondences, and the quotient-after-correspondence functor are not optional decoration; they are the data needed even to state the compatibility of the local Pfaffian calculation with the hybrid $s$-direction. □

*Open Problem* 10.22 *(Hybrid wrapped factorization).* Construct the finite-stage certificates $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ of Definition 10.18 and prove that their inverse limit exists in the completed Hall/factorization category. The exact obstruction is the simultaneous construction of:

(i) the compact extension-closed category and stability/HN sector whose local and wrapped moduli are finite type in bounded $h_S$-height;

(ii) a rigidified wrapped prequotient $\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \to \mathcal{M}^{\mathrm{wr}}_{\eta,R}$ retaining relative $E$-position without inserting a positive elliptic-degree object at a finite Ran point;

(iii) local-local, all ordered two-sided local-wrapped, and wrapped-wrapped extension stacks with admissible vanishing-cycle pull-push functors and proper-support witnesses for the $q$-maps;

(iv) Thom–Sebastiani-compatible strong orientations on all three kinds of correspondences and on the full eight-word two-step flag atlas, proving the displayed parenthesization identities and the four-input pentagon in finite HN quotients;

(v) descent through the reduced $E$-quotient after forming correspondences, including the finite-stabilizer gerbe checks and the $Q_{E,R}$-compatibility isomorphisms, with $\mathfrak{o}^{Q,\mathrm{form}}_R = \mathfrak{o}^{Q,\mathrm{adm}}_R = \mathfrak{o}^{Q,\mathrm{flag}}_R = \mathfrak{o}^{Q,\mathrm{tr}}_{R'R} = 0$;

(vi) protected integration whose decategorified trace sends elliptic degree $b$ to $s^b$, recovers the Igusa $s$-direction coefficientwise, vanishes on the chain/product/coproduct/primitive defects $\mathfrak{o}^{I,\bullet}_R$, has $\mathfrak{o}^{I,\mathrm{tr}}_{R'R} = 0$ for all $R' \geq R$, and whose primitive bracket passes the Borcherds–Kac–Moody presentation test.

Equivalently, one must trivialize $\mathfrak{O}_{\mathrm{hyb},R}$ for every finite HN height $R$, including the quotient, integration, and HN transition components for every $R' \geq R$. Until these six items are supplied, $\mathcal{F}^{\mathrm{hyb}}_X$ is a finite categorical datum whose existence remains open, not a compact QFT constructed in this paper and not a detached determinant factor. Ordinary disjoint-open factorization on $\mathrm{Ran}(E)$ is only the projection-finite $b = 0$ part.

*Open Problem* 10.23 *(Normal-ordered descent of the protected bracket).* Let $\Gamma^{\mathrm{phys}}_X = \Lambda_S \oplus \Lambda_S$, $\Pi_X : \Gamma^{\mathrm{phys}}_X \to \Gamma_{\mathrm{gram}}$, $B : \Gamma^{\mathrm{phys}}_X \times \Gamma^{\mathrm{phys}}_X \to \Gamma_{\mathrm{gram}}$ be as in Definition 6.1. Suppose a Hall lift $\widetilde{P}_X$ of the protected primitive algebra is supplied with a $\Gamma^{\mathrm{phys}}_X$-graded bracket. Construct a cocycle-corrected descent $\overline{\Pi}^{\Theta}_{X,*}$ categorifying the normal-ordered homomorphism $\overline{\Pi}_X : \widehat{\Gamma}_X \to \Gamma_{\mathrm{gram}}$. The raw Hall bracket is concentrated in physical charge $c + c'$, whose raw Gram shadow is

$$\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c').$$

The coefficient category, central translations, Hochschild cocycle $B_{\mathrm{ch}}$, and cochain $\Theta_\Pi(c) = \mathsf{T}_{-\Pi_X(c)}$ are those of Definition 10.24. The datum $\Theta_\Pi$ must lift the lattice identity

$$d_{\mathrm{Hoch}}\Theta_\Pi = B$$

to the chain identity

$$d_{\mathrm{Hoch}}\Theta_\Pi = B_{\mathrm{ch}}$$

in the Picard dg groupoid of central translations. Its triple homotopies must be compatible with the Hall extension correspondences, the Joyce–Upmeier orientation cocycle, the Hopf pairing, the reduced $E$-quotient, the finite fibre products over $\overline{\Pi}_X$, the completed HN tensor topology, and the $S^1$-action

on the cyclic complex. Without such a lift, the raw bracket has the wrong Gram degree and the ordinary pushforward along $\Pi_X$ is not a recognition target.

*Definition* 10.24 *(Normal-ordering coefficient dg category).* Fix a finite HN quotient $R$ of the protected Hall sector, before passing to the inverse limit. Let

$$\mathsf{Ch}^{\Gamma_{\mathrm{gram}}}_{k,R,E}$$

be the dg category of $\Gamma_{\mathrm{gram}}$-graded complexes of protected vanishing-cycle sheaves after the cosection-reduced $E$-quotient, with differential preserving the Gram degree and with support projected to the finite test window of the stage. The transition functors for $R' \geq R$ are required to be strict continuous maps for the HN filtration. They define the completed separated dg category

$$\mathsf{Ch}^{\Gamma_{\mathrm{gram}}}_{k,\widehat{\mathrm{HN}},E} := \varprojlim_{R} \mathsf{Ch}^{\Gamma_{\mathrm{gram}}}_{k,R,E}.$$

Every completed tensor product in this section is taken in this topology:

$$V \widehat{\otimes} W := \varprojlim_{R} (V_R \otimes W_R)$$

for compatible finite stages. No tensor product or coproduct is formed directly on an unfiltered infinite direct sum. For $\eta \in \Gamma_{\mathrm{gram}}$ define the central translation dg autoequivalence

$$(\mathsf{T}_\eta V)_\gamma := V_{\gamma+\eta}, \qquad d_{\mathsf{T}_\eta V} = d_V.$$

Then $\mathsf{T}_0 = \mathrm{id}$ and $\mathsf{T}_\eta \mathsf{T}_{\eta'} = \mathsf{T}_{\eta+\eta'}$ strictly. Let $\mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}$ denote the Picard dg groupoid generated by these translations and their chain homotopies.

For physical charges $c, c' \in \Gamma_X^{\mathrm{phys}}$ put

$$B_{\mathrm{ch}}(c, c') := \mathsf{T}_{B(c,c')}.$$

This is the translation-valued Hochschild 2-cocycle representing the normal-ordering defect. A chain-level normal-ordering datum is a normalized 1-cochain

$$\Theta_\Pi(c) = \mathsf{T}_{-\Pi_X(c)} \in C^1_{\mathrm{Hoch}}(\Gamma_X^{\mathrm{phys}}; \mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}})$$

together with natural quasi-isomorphisms

$$\vartheta_{c,c'} : \mathsf{T}_{-\Pi_X(c)} \mathsf{T}_{-\Pi_X(c')} \mathsf{T}_{\Pi_X(c+c')} \xrightarrow{\sim} \mathsf{T}_{B(c,c')}.$$

Equivalently,

$$d_{\mathrm{Hoch}} \Theta_\Pi = B_{\mathrm{ch}}.$$

The translation cochain itself is formal once the finite coefficient dg category exists. The non-formal datum is its realization on the compact Hall correspondence targets, with orientation, quotient, cyclic, and HN transition compatibility. For a finite $\Gamma_X^{\mathrm{phys}}$-supported complex $V = \bigoplus_c V_c$, the uncompleted shorthand is

$$(\overline{\Pi}^{\Theta}_{*,\mathrm{fin}} V)_\gamma = \prod_{\Pi_X(c)=\gamma} V_c,$$

with the product finite. Its product and bracket structure is transported by the $\Theta_\Pi$-quasi-isomorphisms. Thus a raw bracket output in charge $c + c'$, whose Gram shadow is $\Pi_X(c + c')$, is read after $\mathsf{T}_{B(c,c')}$ in degree

$$\Pi_X(c + c') - B(c, c') = \Pi_X(c) + \Pi_X(c').$$

At a finite stage the actual coefficient object is defined on $\widehat{\Gamma}_R$. If

$$V_R = \bigoplus_{(c,T)\in\widehat{\Gamma}_R} (V_R)_{(c,T)}$$

is a finite $\widehat{\Gamma}_R$-graded complex, set

$$(\overline{\Pi}^{\Theta}_{R,*}V_R)_\gamma = \prod_{\substack{(c,T)\in\widehat{\Gamma}_R \\ \overline{\Pi}_X(c,T)=\gamma}} (V_R)_{(c,T)}.$$

This product is finite; when the fibre is empty it is 0. The $\Gamma_X^{\mathrm{phys}}$-graded formula above is only the special case $T = 0$ before the counterterm has been recorded. For a compatible inverse system $V = \varprojlim_R V_R$, the completed descent is

$$(\overline{\Pi}^{\Theta}_{X,*}V)_\gamma := \varprojlim_R (\overline{\Pi}^{\Theta_R}_{R,*}V_R)_\gamma.$$

Thus a product over a fibre of $\overline{\Pi}_X$ means an inverse limit of finite fibre products. The notation is not a licence to multiply over an uncontrolled infinite fibre. Products, brackets, coproducts, pairings, and radical quotients after $\overline{\Pi}^{\Theta}_{X,*}$ are defined only when their finite-stage maps commute with the transition maps and are continuous for the displayed HN completion.

Lemma 10.25 *(Kunneth-pure multiplicative orientation criterion).* Adopt the notation of Definition 6.5 and Definition 4.3. Suppose a projection-finite Kunneth-pure substack of the reduced compact protected derived moduli stack of $K3 \times E$ is supplied, closed under the Hall extension correspondences under consideration. Suppose also that on this substack:

(i) the full self-Ext determinant admits a functorial Kunneth decomposition, including the cross terms for direct sums;
(ii) the elliptic factor carries a chosen Costello–Li flat trivialization compatible with the reduced $E$-quotient;
(iii) the K3 factors carry supplied Joyce–Upmeier multiplicative orientation isomorphisms with strict pentagon; and
(iv) these choices are $S^1$-equivariant on the cyclic complexes.

Then the tensor product of these supplied data gives multiplicative orientation isomorphisms

$$\mu_{c,c'} : o_{c+c'} \xrightarrow{\sim} o_c \otimes o_{c'}$$

on that Kunneth-pure sector, natural for the restricted Hall correspondences and compatible with the reduced $E$-quotient. No assertion is made here for the full compact or wrapped $K3 \times E$ protected sector.

*Proof. Conditional tensor product.* On the supplied Kunneth-pure substack, the full self-Ext determinant has the assumed factorization into K3 factors and the flat elliptic factor, including all direct-sum cross terms. Tensor the supplied factorwise Joyce–Upmeier multiplicative orientation maps with the chosen Costello–Li trivialization on the elliptic factor. This gives the displayed maps $\mu_{c,c'}$ on precisely that restricted sector.

Naturality, the strict pentagon, and reduced- $E$-compatibility follow from the corresponding supplied properties on the factors. The $S^1$-statement is exactly the assumed equivariance of the factorwise cyclic data.

The proof uses no global product decomposition of the full compact moduli stack and no projection formula for wrapped branes. Therefore it proves (O) only on the explicitly supplied Kunneth-pure sector. The full compact/wrapped orientation problem remains the open obligation recorded in Open Problem 5.7. □

*Remark* 10.26 *(Conditional orientation comparison).* The criterion above explains how a multiplicative orientation would factor after one has already supplied the projection-finite Kunneth-pure sector and all factorwise orientation data. It does not construct the reduced compact $K3 \times E$ orientation gerbe, its cyclic equivariance, or its HN-completed higher-rank extension. Those remain part of the Dirac–Igusa realization certificate.

Theorem 10.27 *(PTVV–Joyce normal-ordered descent of the primitive bracket; certificate and obstruction).* Adopt the notation of Definition 6.1 and Definition 6.5: physical lattice $\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S$, Gram shadow $\Pi_X : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}} = \mathbb{Z}^3$, symmetric quadratic cocycle $B(c, c') = (Q \cdot Q',\ Q \cdot P' + Q' \cdot P,\ P \cdot P')$ satisfying $\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c')$. Suppose:

(P) *(PTVV host.)* The protected primitive layer $\widetilde{P}_X = \bigoplus_c \widetilde{P}_{X,c}$ is realized as $\widetilde{P}_X = H^0 \operatorname{Prim}_{\mathrm{prot}}(\mathcal{F}_X)$ for an oriented critical Hall/factorization object $\mathcal{F}_X$ on a $(-1)$-shifted symplectic derived moduli stack $\mathfrak{M}_{X,c}$ of compactly supported B-branes on $K3 \times E$ in the sense of Pantev–Toen–Vaquie–Vezzosi [51] (cf. the local schema of Definition 2.1), graded by physical charge $c \in \Gamma_X^{\mathrm{phys}}$, with reduced $E$-quotient compatible with the projection $p_E : K3 \times E \to E$ of Definition 5.4. The factorization product on $\mathcal{F}_X$ is $A_\infty$ (homotopy $E_3$): the BBDJS [9] Thom–Sebastiani isomorphism on disjoint d-critical charts is a derived natural isomorphism, and downstream consumption acts on the $H^0 \operatorname{Prim}_{\mathrm{prot}}$ truncation, on which the bracket descends strictly. The binary Hall operations are represented, before the reduced quotient, by correspondences

$$\mathfrak{M}_{X,c} \times \mathfrak{M}_{X,c'} \xleftarrow{p_{c,c'}} \mathfrak{E}_{c,c'} \xrightarrow{q_{c,c'}} \mathfrak{M}_{X,c+c'},$$

and the two parenthesizations of a triple product are compared on the two-step flag stack $\mathfrak{Flag}_{a,b,c}$. These correspondences are part of the input; the Borcherds product does not construct them.

(E) *(Reduced $E$-quotient and completion.)* The Hall correspondences, orientation lines, Hopf pairing, and central translation functors $\mathsf{T}_\eta$ are constructed before quotienting by translations of $E$, then descend to the cosection-reduced $E$-quotient and to the sectorial HN inverse system. In particular $\mathsf{T}_\eta$ commutes with the reduced $E$-quotient, with HN completion, and with the local-local, wrapped-wrapped, and mixed local/wrapped extension correspondences of Open Problem 10.22. The completed primitive object is the inverse limit of finite stages:

$$P_X^\Pi = \varprojlim_R P_R^\Pi, \qquad P_X^\Pi \widehat{\otimes} P_X^\Pi = \varprojlim_R (P_R^\Pi \otimes P_R^\Pi).$$

The maps $m_R$, $[\,\cdot\,,\,\cdot\,]_R$, $\Delta_R$, $\langle\,\cdot\,,\,\cdot\,\rangle_R$, and $\overline{\Pi}^{\Theta}_{R,*}$ commute with all transition maps. Thus every product over a fibre of $\overline{\Pi}_X$, and every completed tensor product used by the Hopf structure, is the inverse limit of finite operations. This is a hypothesis; it is not a formal consequence of the lattice homomorphism $\overline{\Pi}_X$.

(O) *($S^1$-equivariant multiplicative orientation cocycle.)* The collection $\{\mu_{c,c'}\}$ is an $S^1$-equivariant multiplicative orientation cocycle on $(-1)$-shifted symplectic stacks satisfying the strict pentagon. Concretely: an orientation datum $\{o_c\}_{c\in\Gamma_X^{\mathrm{phys}}}$ in the sense of Definition 4.3 is supplied with *Joyce–Upmeier multiplicative isomorphisms* [35]

$$\mu_{c,c'} \;:\; o_{c+c'} \;\xrightarrow{\sim}\; o_c \otimes o_{c'}, \qquad c, c' \in \Gamma_X^{\mathrm{phys}},$$

satisfying the strict cocycle pentagon $\mu_{c+c',c''} \circ \mu_{c,c'} = \mu_{c,c'+c''} \circ \mu_{c',c''}$ on the Cartesian extension correspondence, compatible with the reduced $E$-quotient and with the HN/sectorial completion of Lemma 10.39, and *equivariant under the loop-rotation $S^1$-action on the cyclic complex of $\mathfrak{M}_{X,c}$*. This last condition lifts the orientation cocycle from a Hochschild trivialization to a *cyclic* trivialization, and is exactly the kind of orientation technology Joyce–Upmeier [35] supply once the reduced compact sector and its cyclic model have been constructed. The $S^1$-equivariance is not inferred from the scalar determinant; it is an input on the Hall side. Equivalently, the collection $\{o_c\}$ is an $S^1$-equivariant multiplicative section of the determinant gerbe over the moduli of objects in the protected sector.

(Θ) *(Cocycle-corrected degree shifts.)* A coherent family of central degree-shift data is supplied on the primitive bracket targets in the coefficient dg category of Definition 10.24. The chain representative of the Gram defect is the normalized translation-valued Hochschild 2-cocycle

$$B_{\mathrm{ch}}(c, c') := \mathsf{T}_{B(c,c')} \in Z^2_{\mathrm{Hoch}}\big(\Gamma_X^{\mathrm{phys}}; \mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}\big).$$

A normalized 1-cochain

$$\Theta_\Pi \in C^1_{\mathrm{Hoch}}\big(\Gamma_X^{\mathrm{phys}}; \mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}\big), \qquad \Theta_\Pi(c) = \mathsf{T}_{-\Pi_X(c)}$$

is supplied with coherent natural quasi-isomorphisms

$$d_{\mathrm{Hoch}}\Theta_\Pi = B_{\mathrm{ch}}.$$

The formula for $\Theta_\Pi$ is the formal central translation cochain. The hypothesis is stronger: these translations must act on the actual Hall correspondence targets and must be compatible with the orientation, reduced quotient, cyclic model, and finite HN transitions. Equivalently, for every pair $c, c'$ the raw bracket target of Gram degree $\Pi_X(c + c')$ is translated by $\mathsf{T}_{B(c,c')}$ to the additive Gram degree $\Pi_X(c) + \Pi_X(c')$. For triples $a, b, c$, the two composites over $\mathfrak{Flag}_{a,b,c}$,

$$(\vartheta_{a,b} \otimes 1) \circ \vartheta_{a+b,c}, \qquad (1 \otimes \vartheta_{b,c}) \circ \vartheta_{a,b+c},$$

agree after the Hall associator, Thom–Sebastiani transport, and orientation signs. Their quotient in $\pi_1\mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}$ is the triple obstruction

$$\mathrm{Pent}_\Theta(a, b, c) := \big[(\vartheta_{a,b} \otimes 1) \circ \vartheta_{a+b,c}\big] - \big[(1 \otimes \vartheta_{b,c}) \circ \vartheta_{a,b+c}\big],$$

with the Hall associator and orientation transport inserted in the middle. Hypothesis $(\Theta)$ requires $\mathrm{Pent}_\Theta(a,b,c)=0$ for every two-step Hall flag. On degrees this is exactly

$$B(a,b)+B(a+b,c)=B(b,c)+B(a,b+c),$$

i.e. $d_{\mathrm{Hoch}}B_{\mathrm{ch}}=0$, but the displayed equality is required as a chain homotopy, not only as a lattice identity.

(F) *(Frobenius pairing.)* The protected Hall bracket $[\,\cdot\,,\,\cdot\,]$ and the protected Hopf pairing $\langle\,\cdot\,,\,\cdot\,\rangle$ on $\widetilde{P}_X$ (the latter furnished by the squaring isomorphism $o_c^{\otimes 2}\simeq\det\mathrm{RHom}(C,C)$ of Definition 4.3 and the PTVV [51] $(-1)$-shifted symplectic form on $\mathfrak{M}_{X,c}$, trivialized along the reduced $E$-quotient via the Costello–Li flat Calabi–Yau pairing [14]) satisfy the graded-Frobenius identity

$$\langle[x,y],z\rangle+(-1)^{|x|\cdot|y|}\langle y,[x,z]\rangle=0$$

for homogeneous $x,y,z\in\widetilde{P}_X$ of physical charges $c_x,c_y,c_z$ with $c_x+c_y+c_z=0$. After applying $\overline{\Pi}^{\Theta}_{X,*}$ the same identity is imposed for the transported bracket $[\,\cdot\,,\,\cdot\,]_\Theta$.

(Cyc) *(Negative-cyclic lift.)* The Hochschild cochain $\Theta_\Pi$ has an $S^1$-equivariant lift $\Theta^-_\Pi$ in the negative-cyclic complex of the protected Hall sector, and the lifted equation

$$(b+uB_{\mathrm{C}})\Theta^-_\Pi=B^-_{\mathrm{ch}}$$

holds, where $B_{\mathrm{C}}$ is the Connes operator and $B^-_{\mathrm{ch}}$ is a negative-cyclic representative whose forgetful image in Hochschild cochains is $B_{\mathrm{ch}}$. The forgetful map sends the cyclic equation to the Hochschild equation in $(\Theta)$, so the cyclic lift is a refinement, not a replacement, of normal ordering.

Then the raw physical-charge bracket

$$[\widetilde{P}_{X,c},\widetilde{P}_{X,c'}]\subset\widetilde{P}_{X,c+c'}$$

induces a cocycle-corrected Gram-graded bracket

$$[\,\cdot\,,\,\cdot\,]_\Theta:(\overline{\Pi}^{\Theta}_{X,*}\widetilde{P}_X)_{\Pi_X(c)}\otimes(\overline{\Pi}^{\Theta}_{X,*}\widetilde{P}_X)_{\Pi_X(c')}\longrightarrow(\overline{\Pi}^{\Theta}_{X,*}\widetilde{P}_X)_{\Pi_X(c)+\Pi_X(c')}.$$

Thus $B$ is not erased; it is the coherent degree-shift by which the raw target $\Pi_X(c+c')=\Pi_X(c)+\Pi_X(c')+B(c,c')$ is transported to the additive Gram target. The object

$$P^{\Pi}_X:=\overline{\Pi}^{\Theta}_{X,*}\widetilde{P}_X$$

is therefore a $\Gamma_{\mathrm{gram}}$-graded Lie superalgebra at the stated $H^0$ level, and hypothesis (1b) of Theorem 10.45 is satisfied with the $\Theta$-corrected descent.

If the finite-fibre HN topology of (E), $(\Theta)$, its triple coherence, (F), (Cyc), and radical ideal/coideal stability are not supplied, the minimal computable obstruction is the finite-HN inverse system of classes

$$\mathfrak{o}^{\mathrm{top}}_\Theta:=\left\{\mathfrak{o}^{\mathrm{fib}}_{\Theta,R'R},\mathfrak{o}^{m}_{\Theta,R'R},\mathfrak{o}^{\Delta}_{\Theta,R'R},\mathfrak{o}^{\langle,\rangle}_{\Theta,R'R}\right\}_{R'\geq R},$$

where $\rho_{R'R}$ is the source transition, $\rho^{\Pi}_{R'R}$ is the descended primitive transition, and these are the finite Hom-space defects

$$\mathfrak{o}^{\mathrm{fib}}_{\Theta,R'R}:=\rho^{\Pi}_{R'R}\overline{\Pi}^{\Theta_{R'}}_{R',*}-\overline{\Pi}^{\Theta_R}_{R,*}\rho_{R'R},$$

$$\mathfrak{v}^{m}_{\Theta,R'R} := \rho^{\Pi}_{R'R} m^{\Theta}_{R'} - m^{\Theta}_{R}(\rho^{\Pi}_{R'R} \otimes \rho^{\Pi}_{R'R}), \qquad \mathfrak{v}^{\Delta}_{\Theta,R'R} := (\rho^{\Pi}_{R'R} \otimes \rho^{\Pi}_{R'R})\Delta^{\Theta}_{R'} - \Delta^{\Theta}_{R}\rho^{\Pi}_{R'R},$$

$$\mathfrak{v}^{\langle\,,\,\rangle}_{\Theta,R'R} := \langle\,\cdot\,,\,\cdot\,\rangle_R(\rho^{\Pi}_{R'R} \otimes \rho^{\Pi}_{R'R}) - \langle\,\cdot\,,\,\cdot\,\rangle_{R'},$$

$$\mathfrak{v}^{\mathrm{Hoch}}_{\Theta} := [B_{\mathrm{ch}} - d_{\mathrm{Hoch}}\Theta_{\Pi}] \in HH^{2}_{\mathrm{red},E}\big(\widetilde{P}^{\mathrm{br}}_{X}; \mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}\big),$$

$$\mathfrak{v}^{\mathrm{tri}}_{\Theta} := [\mathrm{Pent}_{\Theta}] \in HH^{3}_{\mathrm{red},E}\big(\widetilde{P}^{\mathrm{br}}_{X}; \pi_1\mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}\big),$$

$$\mathfrak{v}^{\mathrm{Jac}}_{\Theta} := [\mathrm{Jac}_{\Theta}] \in C^{3}_{\mathrm{CE,red},E}\big(\overline{\Pi}^{\Theta}_{X,*}\widetilde{P}_X; \overline{\Pi}^{\Theta}_{X,*}\widetilde{P}_X\big),$$

$$\mathfrak{v}_{F} := \big[(x,y,z) \mapsto \langle [x,y]_{\Theta}, z\rangle + (-1)^{|x||y|}\langle y, [x,z]_{\Theta}\rangle\big] \in C^{3}_{\mathrm{cyc,red},E}\big(\overline{\Pi}^{\Theta}_{X,*}\widetilde{P}_X; k\big),$$

$$\mathfrak{v}^{\mathrm{cyc}}_{\Theta} := [B^{-}_{\mathrm{ch}} - (b + uB_{\mathrm{C}})\Theta^{-}_{\Pi}] \in HC^{-2}_{\mathrm{red},E}\big(\widetilde{P}^{\mathrm{br}}_{X}; \mathsf{Pic}^{\mathrm{dg}}_{\Gamma_{\mathrm{gram}}}\big),$$

$$\mathfrak{v}^{\mathrm{rad}}_{\Theta} := \big(\mathfrak{v}^{\mathrm{rad,Lie}}_{\Theta}, \mathfrak{v}^{\mathrm{rad,coid}}_{\Theta}\big),$$

defined on the HN-completed reduced $E$-quotient of the protected bracket category, where

$$P^{\Pi}_{X} = \overline{\Pi}^{\Theta}_{X,*}\widetilde{P}_X, \qquad \mathrm{Rad}^{\Pi}_{X} = \{r \in P^{\Pi}_{X} \mid \langle r, w\rangle = 0 \text{ for all } w \in P^{\Pi}_{X}\}.$$

The two components are

$$\mathfrak{v}^{\mathrm{rad,Lie}}_{\Theta} := \big[(x,r,z) \mapsto \langle [x,r]_{\Theta}, z\rangle\big] \in \mathrm{Hom}_{\mathrm{red},E}\big(P^{\Pi}_{X} \otimes \mathrm{Rad}^{\Pi}_{X} \otimes P^{\Pi}_{X}, k\big),$$

and, for the quotient map $\bar{q}_{\Pi} : P^{\Pi}_{X} \to P^{\Pi}_{X}/\overline{\mathrm{Rad}^{\Pi}_{X}}$,

$$\mathfrak{v}^{\mathrm{rad,coid}}_{\Theta}(r) := (\bar{q}_{\Pi}\widehat{\otimes}\bar{q}_{\Pi})\Delta_{\Theta}(r), \qquad r \in \mathrm{Rad}^{\Pi}_{X}.$$

Here $\Delta_{\Theta}$ is the descended completed Hopf coproduct. The condition $\mathfrak{v}^{\mathrm{rad,coid}}_{\Theta} = 0$ says exactly that

$$\Delta_{\Theta}(\mathrm{Rad}^{\Pi}_{X}) \subset \overline{\mathrm{Rad}^{\Pi}_{X} \widehat{\otimes} P^{\Pi}_{X} + P^{\Pi}_{X} \widehat{\otimes} \mathrm{Rad}^{\Pi}_{X}}.$$

Here $\mathfrak{v}^{\mathrm{top}}_{\Theta}$ is the assertion that $\overline{\Pi}^{\Theta}_{*}$, the transported product, the transported coproduct, and the pairing are strict morphisms of the finite HN inverse system; its vanishing is what makes the completed fibre products and $\widehat{\otimes}$-operations well defined. The class $\mathrm{Pent}_{\Theta}(a,b,c)$ is the difference of the two triple composites over $\mathfrak{Flag}_{a,b,c}$, and $\mathrm{Jac}_{\Theta}$ is the signed Jacobiator of the transported bracket. The Hochschild shadow is computed on homogeneous charges by the explicit formula for $B(c,c')$; the triple shadow is the displayed cocycle identity for $B$, hence zero at the lattice level; the cyclic component is computed by applying $B_{\mathrm{C}}$ to the chosen cyclic orientation representative. The class $\mathfrak{v}^{\mathrm{rad}}_{\Theta}$ vanishes exactly when the Hopf radical is both a Lie ideal for the transported bracket and a coideal for the completed descended coproduct. Closedness of the radical in the HN topology is still a separate topological hypothesis.

*Proof. Step 1 (The coefficient category).* The coefficient system for normal ordering is the Picard dg groupoid of central translations of $\mathsf{Ch}^{\Gamma_{\mathrm{gram}}}_{k,\widehat{\mathrm{HN}},E}$, as in Definition 10.24. That definition constructs $\overline{\Pi}^{\Theta}_{X,*}$ as the inverse limit of the finite fibre products $(\overline{\Pi}^{\Theta}_{R,*}V_R)_{\gamma}$. The finite-transition part of hypothesis (E), encoded by $\mathfrak{v}^{\mathrm{top}}_{\Theta} = 0$, is what permits the transported product, coproduct, and pairing to pass from finite

stages to the completed HN object. With the convention $(\mathsf{T}_\eta V)_\gamma = V_{\gamma+\eta}$, applying $\mathsf{T}_{B(c,c')}$ to a raw target of degree $\Pi_X(c+c')$ displays it in degree

$$\Pi_X(c+c') - B(c,c') = \Pi_X(c) + \Pi_X(c').$$

Thus $B_{\mathrm{ch}}(c,c') = \mathsf{T}_{B(c,c')}$ is the exact translation-valued Hochschild 2-cocycle measuring failure of raw Gram additivity.

*Step 2 (The boundary equation).* For the standard additive Hochschild differential on normalized cochains,

$$(d_{\mathrm{Hoch}}\Theta_\Pi)(c,c') = \Theta_\Pi(c) + \Theta_\Pi(c') - \Theta_\Pi(c+c').$$

The underlying degree of the translation $\Theta_\Pi(c) = \mathsf{T}_{-\Pi_X(c)}$ gives

$$d_{\mathrm{Hoch}}\Theta_\Pi(c,c') = -\Pi_X(c) - \Pi_X(c') + \Pi_X(c+c') = B(c,c').$$

Hypothesis $(\Theta)$ is precisely the statement that this arithmetic boundary is represented by actual central translation quasi-isomorphisms on the protected Hall bracket targets.

*Step 3 (Triples, Jacobi, and Frobenius).* For three charges $a, b, c$, the two ways of translating the target of a two-step bracket differ by

$$B(a,b) + B(a+b,c) - B(b,c) - B(a,b+c) = -d_{\mathrm{Hoch}}B(a,b,c) = 0$$

by Lemma 6.9. The supplied pentagon homotopies in $(\Theta)$ lift this equality from degrees to the Hall chain category: the obstruction is precisely $\mathrm{Pent}_\Theta(a,b,c)$. If it vanishes, the normal-ordered product is associative on $H^0$, and the supercommutator bracket inherits the Jacobi identity from the upstairs Hall bracket. The corresponding signed Jacobiator is

$$\mathrm{Jac}_\Theta(x,y,z) = [x,[y,z]_\Theta]_\Theta - [[x,y]_\Theta,z]_\Theta - (-1)^{|x||y|}[y,[x,z]_\Theta]_\Theta;$$

its degree shadow is zero by the same cocycle identity, and its chain class is the image of the Hall associator and $\mathrm{Pent}_\Theta$. Hypothesis (F) is then the separate vanishing of

$$\mathfrak{o}_F(x,y,z) = \langle [x,y]_\Theta, z\rangle + (-1)^{|x||y|}\langle y, [x,z]_\Theta\rangle.$$

It is not implied by the degree calculation.

*Step 4 (Orientation, reduced quotient, and radical quotient).* Hypothesis (O) transports the orientation lines through the same extension correspondences used by the Hall product; hypothesis (E) asserts that the translation action, orientation transport, and mixed local/wrapped correspondences descend through the reduced $E$-quotient and HN completion. Let $\mathrm{Rad}_X^\Pi$ be the radical of the descended Hopf pairing. The exact obstruction to Lie-ideal stability is

$$\mathfrak{o}_\Theta^{\mathrm{rad,Lie}}(x,r,z) = \langle [x,r]_\Theta, z\rangle, \qquad r \in \mathrm{Rad}_X^\Pi.$$

If (F) holds, this expression equals $-(-1)^{|x||r|}\langle r, [x,z]_\Theta\rangle = 0$, so the radical is a Lie ideal. This does not prove coideal stability. Let $\bar{q}_\Pi : P_X^\Pi \to P_X^\Pi/\overline{\mathrm{Rad}_X^\Pi}$. The exact coideal obstruction is

$$\mathfrak{o}_\Theta^{\mathrm{rad,coid}}(r) = (\bar{q}_\Pi \widehat{\otimes} \bar{q}_\Pi)\Delta_\Theta(r).$$

Its vanishing is equivalent to $\Delta_\Theta(r)$ lying in the completed sum

$$\overline{\mathrm{Rad}_X^\Pi \widehat{\otimes} P_X^\Pi + P_X^\Pi \widehat{\otimes} \mathrm{Rad}_X^\Pi}.$$

Closedness of $\mathrm{Rad}_X^\Pi$ in the Hall topology remains a separate input. Only after ideal, coideal, and closedness conditions does the quotient carry the same $\Gamma_{\mathrm{gram}}$-graded Hopf–Lie structure.

*Step 5 (Cyclic lift and obstruction).* The Hochschild construction controls the displayed $H^0$ bracket. For the full chain-level descent, the degree-shift cochain must be fixed by loop rotation. This is exactly hypothesis (Cyc): $(b + uB_{\mathrm{C}})\Theta_\Pi^- = B_{\mathrm{ch}}^-$. If such a negative-cyclic lift is absent, or if the finite-HN topology, triple/Jacobi/Frobenius/radical coherence fails, the seven classes displayed in the theorem are the obstruction. They are minimal for this model: vanishing of $\mathfrak{o}_\Theta^{\mathrm{top}}$ is equivalent to the finite-fibre descent and completed tensor operations commuting with the HN inverse system; vanishing of $\mathfrak{o}_\Theta^{\mathrm{Hoch}}$ is equivalent to existence of the degree-shift trivialization on bracket targets; vanishing of $\mathfrak{o}_\Theta^{\mathrm{tri}}$ is equivalent to compatibility of that trivialization with the two-step Hall flag correspondences; vanishing of $\mathfrak{o}_\Theta^{\mathrm{Jac}}$ is the Jacobi compatibility of the transported bracket; vanishing of $\mathfrak{o}_F$ is the Frobenius/Hopf invariance of the transported bracket; vanishing of $\mathfrak{o}_\Theta^{\mathrm{cyc}}$ is equivalent to choosing the trivialization compatibly with the $S^1$-equivariant orientation data; and vanishing of $\mathfrak{o}_\Theta^{\mathrm{rad}}$, together with closedness in the HN topology, is the condition for passage to the Hopf radical quotient. □

Proposition 10.28 *(Finite verification diagrams for normal ordering).* Fix a successor arrow $R^+ \to R$ in the cofinal HN subsystem of Definition 10.40. Let $P_R^\Pi = \overline{\Pi}_{R,*}^{\Theta_R} \widetilde{P}_R$ be a finite normal-ordered primitive stage, with transition $\rho_{R^+R}^\Pi : P_{R^+}^\Pi \to P_R^\Pi$. A supplied finite system kills the seven normal-ordering obstructions of Theorem 10.27 at this successor arrow exactly when the following finite maps commute, or are supplied with the displayed null-homotopies, in the indicated finite Hom complexes.

(NO1) Finite fibres.

$$\begin{tikzcd} V_{R^+} \arrow[r, "\overline{\Pi}_{R^+,*}^{\Theta_{R^+}}"] \arrow[d, "\rho_{R^+R}"'] & \overline{\Pi}_{R^+,*}^{\Theta_{R^+}} V_{R^+} \arrow[d, "\rho_{R^+R}^\Pi"] \\ V_R \arrow[r, "\overline{\Pi}_{R,*}^{\Theta_R}"'] & \overline{\Pi}_{R,*}^{\Theta_R} V_R. \end{tikzcd}$$

This is $\mathfrak{o}_{\Theta,R^+R}^{\mathrm{fib}} = 0$; it says that fibre products over $\overline{\Pi}_X$ are finite before completion and transition-compatible before the inverse limit is taken.

(NO2) Product, coproduct, pairing.

$$\begin{tikzcd} P_{R^+}^\Pi \otimes P_{R^+}^\Pi \arrow[r, "m_{R^+}^\Theta"] \arrow[d, "\rho_{R^+R}^\Pi \otimes \rho_{R^+R}^\Pi"'] & P_{R^+}^\Pi \arrow[d, "\rho_{R^+R}^\Pi"] \\ P_R^\Pi \otimes P_R^\Pi \arrow[r, "m_R^\Theta"'] & P_R^\Pi \end{tikzcd} \qquad \begin{tikzcd} P_{R^+}^\Pi \arrow[r, "\Delta_{R^+}^\Theta"] \arrow[d, "\rho_{R^+R}^\Pi"'] & P_{R^+}^\Pi \otimes P_{R^+}^\Pi \arrow[d, "\rho_{R^+R}^\Pi \otimes \rho_{R^+R}^\Pi"] \\ P_R^\Pi \arrow[r, "\Delta_R^\Theta"'] & P_R^\Pi \otimes P_R^\Pi. \end{tikzcd}$$

Together with

$$\langle \rho_{R^+R}^\Pi x, \rho_{R^+R}^\Pi y \rangle_R = \langle x, y \rangle_{R^+},$$

these are $\mathfrak{o}_{\Theta,R^+R}^m = \mathfrak{o}_{\Theta,R^+R}^\Delta = \mathfrak{o}_{\Theta,R^+R}^{\langle,\rangle} = 0$.

(NO3) Hochschild and negative-cyclic equations. For every finite charge pair $c, c' \in \widehat{\Gamma}_{R^+}$, the Hochschild boundary and its cyclic refinement are the finite equations

$$d_{\mathrm{Hoch}}\Theta_{\Pi,R}(c,c') = B_{\mathrm{ch},R}(c,c'), \qquad (b + uB_{\mathrm{C}})\Theta^{-}_{\Pi,R} = B^{-}_{\mathrm{ch},R}.$$

They kill $\mathfrak{o}^{\mathrm{Hoch}}_{\Theta,R}$ and $\mathfrak{o}^{\mathrm{cyc}}_{\Theta,R}$. The second equation is imposed in the negative-cyclic complex; it is not obtained by forgetting to Hochschild cochains.

(NO4) Triple flag. For every two-step finite Hall flag over $\mathfrak{Flag}_{a,b,c,R}$, the two translated products must agree after inserting the Hall associator and orientation transport:

$$(\mathsf{A}_{a,b,c} \circ (\vartheta_{a,b} \otimes 1) \circ \vartheta_{a+b,c}) - ((1 \otimes \vartheta_{b,c}) \circ \vartheta_{a,b+c}) = 0.$$

This is the finite representative of $\mathrm{Pent}_{\Theta}(a,b,c)$. Its degree shadow is the bilinear identity for $B$; the chain equality is additional data.

(NO5) Jacobi. The transported bracket must have zero finite Jacobiator

$$\mathrm{Jac}_{\Theta,R}(x,y,z) = 0$$

for all homogeneous $x, y, z \in P^{\Pi}_{R}$.

(NO6) Frobenius. The finite Frobenius defect

$$\mathfrak{o}_{F,R}(x,y,z) = \langle [x,y]_{\Theta}, z\rangle_R + (-1)^{|x||y|}\langle y, [x,z]_{\Theta}\rangle_R$$

must vanish.

(NO7) Hopf radical. Finally, for $\mathrm{Rad}^{\Pi}_{R} = \{r \in P^{\Pi}_{R} \mid \langle r, w\rangle_R = 0\ \forall w \in P^{\Pi}_{R}\}$ and $\bar{q}_{\Pi,R} : P^{\Pi}_{R} \to P^{\Pi}_{R}/\mathrm{Rad}^{\Pi}_{R}$, one must have both

$$\langle [x,r]_{\Theta}, z\rangle_R = 0 \qquad (x, z \in P^{\Pi}_{R},\ r \in \mathrm{Rad}^{\Pi}_{R})$$

and the finite coideal square

$$\begin{tikzcd}
\mathrm{Rad}^{\Pi}_{R} \arrow[r, "\Delta^{\Theta}_{R}"] \arrow[d, "0"'] & P^{\Pi}_{R} \otimes P^{\Pi}_{R} \arrow[d, "\bar{q}_{\Pi,R}\otimes\bar{q}_{\Pi,R}"] \\
0 \arrow[r] & (P^{\Pi}_{R}/\mathrm{Rad}^{\Pi}_{R}) \otimes (P^{\Pi}_{R}/\mathrm{Rad}^{\Pi}_{R}).
\end{tikzcd}$$

These are the finite Lie-ideal and coideal parts of $\mathfrak{o}^{\mathrm{rad}}_{\Theta,R} = 0$. Closedness of the inverse limit radical is the additional HN topological assertion that the kernels $\mathrm{Rad}^{\Pi}_{R}$ form a transition-compatible closed subsystem.

Consequently, if (NO1)–(NO7) hold for every successor arrow and the non-successor transition maps are their composites, the inverse limit kills

$$\mathfrak{o}^{\mathrm{top}}_{\Theta}, \quad \mathfrak{o}^{\mathrm{Hoch}}_{\Theta}, \quad \mathfrak{o}^{\mathrm{tri}}_{\Theta}, \quad \mathfrak{o}^{\mathrm{Jac}}_{\Theta}, \quad \mathfrak{o}_{F}, \quad \mathfrak{o}^{\mathrm{cyc}}_{\Theta}, \quad \mathfrak{o}^{\mathrm{rad}}_{\Theta}.$$

If one displayed finite diagram fails, its homotopy class is the surviving obstruction. Neither non-degeneracy of the pairing nor the scalar Borcherds determinant removes (NO3), (NO6), or the coideal part of (NO7).

Lemma 10.29 *(Hopf–Frobenius certificate and obstruction).* Adopt hypotheses (P) and (O) of Theorem 10.27, and grant the Hopf-pairing/radical hypothesis of Theorem 10.45: the protected Hopf pairing $\langle\,\cdot\,,\,\cdot\,\rangle$ on the protected primitive layer $\widetilde{P}_X = H^0\,\mathrm{Prim}_{\mathrm{prot}}(\mathcal{F}_X)$ is non-degenerate after the standard radical quotient. Non-degeneracy alone does not imply the graded Frobenius identity. The missing datum is the vanishing of the trilinear obstruction

$$\mathfrak{o}_F(x,y,z) := \langle [x,y],z\rangle + (-1)^{|x|\cdot|y|}\langle y,[x,z]\rangle, \qquad x\in\widetilde{P}_{X,c_x},\ y\in\widetilde{P}_{X,c_y},\ z\in\widetilde{P}_{X,c_z}, \quad c_x+c_y+c_z=0,$$

viewed as a class

$$\mathfrak{o}_F \in C^3_{\mathrm{cyc}}\big(\widetilde{P}_X;\,k\big)$$

after the reduced $E$-quotient and radical quotient. Hypothesis (F) of Theorem 10.27 is equivalent to $\mathfrak{o}_F = 0$. It is supplied, for example, by the CY-3 Serre-functor criterion of Lemma 10.33.

*Proof. Step 1 (Exact defect).* For homogeneous $x, y, z$ with $c_x + c_y + c_z = 0$, the displayed cochain $\mathfrak{o}_F(x,y,z)$ is precisely the left side of the graded-Frobenius identity. Hence hypothesis (F) is equivalent, by definition, to $\mathfrak{o}_F = 0$ after the reduced $E$-quotient and the chosen radical quotient. No determinant or non-degeneracy statement alone forces this vanishing.

*Step 2 (What non-degeneracy detects).* After quotienting by the radical, non-degeneracy identifies a bilinear operation $D(x,y)$ with the functional $z \mapsto \langle D(x,y),z\rangle$. Thus if the Frobenius defect is known to vanish after pairing against every $z$, non-degeneracy forces the corresponding operation-level defect to vanish in the quotient. This is detection, not construction: the cyclic or Serre-functor input must still supply the equality being detected.

*Step 3 (Radical consequence).* Let

$$\mathrm{Rad}^{\Pi}_X = \{r \mid \langle r,w\rangle = 0 \text{ for all } w\}.$$

If $\mathfrak{o}_F = 0$, then for every $r \in \mathrm{Rad}^{\Pi}_X$

$$\langle [x,r],z\rangle = -(-1)^{|x||r|}\langle r,[x,z]\rangle = 0.$$

Therefore $\mathrm{Rad}^{\Pi}_X$ is a Lie ideal. The assertion that it is closed and is also a coideal is the separate Hopf-radical hypothesis in Theorem 10.45. Equivalently, the coproduct must separately satisfy

$$(\bar{q}_\Pi \widehat{\otimes} \bar{q}_\Pi)\Delta_\Theta(r) = 0 \qquad (r \in \mathrm{Rad}^{\Pi}_X),$$

with $\bar{q}_\Pi$ the quotient by the closed radical. Lemma 10.33 is one conditional source for $\mathfrak{o}_F = 0$; it is not a source for coideal stability unless the cyclic/Serre datum is also proved compatible with the Hopf coproduct. □

Proposition 10.30 *(Negative-cyclic route to the normal-ordering cochain).* Assume, in addition to hypothesis (P) of Theorem 10.27, that the protected Hall sector has a cyclic model $A^{\mathrm{prot}}_X$ with mixed complex $(C_*(A^{\mathrm{prot}}_X), b, B_{\mathrm{C}})$, a cyclic class $\varpi_X$ representing the PTVV form, and a Connes-contraction cochain $\iota_{\varpi_X}\Pi_X$ satisfying

$$(b + uB_{\mathrm{C}})(\iota_{\varpi_X}\Pi_X) = B^-_{\mathrm{ch}},$$

where $B^-_{\mathrm{ch}}$ is a negative-cyclic lift whose Hochschild image is $B_{\mathrm{ch}}$, and $B_{\mathrm{ch}}$ maps to the Gram-defect cocycle $B$ under the Loday–Quillen–Tsygan comparison

$$HC^2_{\mathrm{strict}} \longrightarrow HH^2$$

[41], [55], [40, §2]. If the induced two-step flag homotopies have $\mathrm{Pent}_\Theta = 0$, then the cochain $\Theta^-_\Pi := \iota_{\mathscr{B}_X}\Pi_X$ supplies the negative-cyclic part of (Cyc) and the Hochschild part of hypothesis ($\Theta$). It supplies none of the Frobenius identity, the finite-HN topology, or the Hopf-radical ideal/coideal condition.

*Proof.* After forgetting to Hochschild cochains, the displayed equation becomes $d_{\mathrm{Hoch}}\Theta_\Pi = B_{\mathrm{ch}}$, whose degree shadow is $B$, and the cocycle identity $\delta B = 0$ gives the degree shadow of the triple pentagon. The actual Hall flag coherence is still the vanishing of $\mathrm{Pent}_\Theta$. This route is a second construction of the $\Theta$-datum once the cyclic model and the triple homotopies are available. It is not a consequence of (P) alone. It does not assert compatibility of fibre products over $\overline{\Pi}_X$ with the HN transitions, and it contains no information about the completed coproduct on a radical element. □

*Remark* 10.31 *(Comparison of the two proofs).* $[B] = [B_{\mathrm{ch}}]$ under the Loday–Quillen–Tsygan comparison $HC^*_{\mathrm{strict}} \to HH^*$ of [41], [55], [40, §2], with witnessing cochains $\Theta$ (PTVV–Joyce) and $\iota_{\mathscr{B}_X}\Pi_X$ (cyclic homology), whenever the cyclic model and comparison cochain are supplied.

*Remark* 10.32 *(Status of the descent criterion).* Theorem 10.27 is a conditional descent criterion. Its inputs are the PTVV host (P), the reduced quotient and HN completion (E), including finite-fibre compatibility of $\overline{\Pi}^{\Theta}_*$ and continuity of $\widehat{\otimes}$, the supplied $S^1$-equivariant multiplicative orientation cocycle (O), the degree-shift cochain $\Theta$, the Frobenius pairing (F), the negative-cyclic lift (Cyc), and Hopf-radical stability. Lemma 10.25 constructs (O) only on a supplied projection-finite Kunneth-pure sector; it does not construct the full compact or wrapped reduced $K3 \times E$ protected sector. Thus the remaining obligations are geometric and chain-level: construct (P), (E), and (O) beyond the Kunneth-pure sector including the HN/higher-rank extension, prove that the supplied $\Theta$ identifies the raw Gram defect $B$ with canonical degree shifts in the protected primitive bracket, kill $\mathfrak{d}^{\mathrm{top}}_\Theta$, $\mathfrak{d}_F$, and prove closed Hopf-radical coideal stability. The scalar Borcherds determinant and the denominator algebra do not discharge these obligations.

Lemma 10.33 *(Graded Frobenius identity from CY-3 Serre functor).* Suppose that the protected observable algebra $\mathcal{A}_X$ of Definition 2.1, after the reduced $E$-quotient and the Costello–Li flat trivialization [14] along the elliptic factor, is realized as a Calabi–Yau $A_\infty$-category of dimension $d = 3$, i.e. it carries a Serre functor $S \simeq [3]$ and a non-degenerate cyclically invariant pairing

$$\langle\,\cdot\,,\,\cdot\,\rangle_S : \mathrm{Hom}_{\mathcal{A}_X}(X, Y) \otimes \mathrm{Hom}_{\mathcal{A}_X}(Y, X[3]) \longrightarrow k,$$

in the sense of Kontsevich–Soibelman [39] (cyclic $A_\infty$-structure) and PTVV [51]/BBDJS [9] (orientation data realising the shift [3] on the moduli of objects of $\mathcal{A}_X$). Let $\widetilde{P}_X \subset H_*(\mathcal{A}_X)$ be the protected primitive subspace, equipped with the Hall bracket $[\,\cdot\,,\,\cdot\,]$ and the Hopf pairing $\langle\,\cdot\,,\,\cdot\,\rangle$ of Theorem 10.45. Suppose, moreover, that at every finite HN stage the primitive Hall bracket is represented by the anti-symmetrisation of the finite product $m_{2,R}$, the cyclic trace descends through the reduced $E$-quotient, and the trace, pairing, and primitive projection commute with the transition maps. Then the graded Frobenius identity holds:

$$\langle [x, y], z\rangle + (-1)^{|x|\cdot|y|}\langle y, [x, z]\rangle = 0, \qquad x, y, z \in \widetilde{P}_X \text{ homogeneous},$$

i.e. hypothesis (F) of Theorem 10.27 is met.

*Proof.* *Step 1 (Serre pairing and cyclic trace).* A CY-$d$ $A_\infty$-category $\mathcal{A}_X$ is, by [39] (§ 10), an $A_\infty$-category equipped with a quasi-isomorphism $S \xrightarrow{\sim} [d]$ of bimodules and a non-degenerate cyclically symmetric

pairing $\langle\,\cdot\,,\,\cdot\,\rangle_S$ on $\mathrm{Hom}_{\mathcal{A}_X}(-,-)$, of total degree $-d$. Equivalently, there is a chain-level cyclic trace $\mathrm{tr}_S : \mathrm{Hom}_{\mathcal{A}_X}(X,X) \to k[-d]$ satisfying $\mathrm{tr}_S(f\circ g) = (-1)^{|f|\cdot|g|}\,\mathrm{tr}_S(g\circ f)$ for composable homogeneous $f, g$ (loc. cit., Lemma 10.2.4). PTVV [51] produce the underlying $(-d)$-shifted symplectic form on $\mathfrak{M}_{X,c}$; BBDJS [9] extend the orientation/d-critical datum that realises $S \simeq [d]$ at chain level on the perfect-complex moduli. Costello–Li [14] flat trivialization along the elliptic factor reduces $d = 4$ to $d = 3$, so the discussion takes $d = 3$ and the identification $\langle x, y\rangle = \mathrm{tr}_S(x\cdot y)$, where $\cdot$ is the cup product on the protected primitive subspace.

*Step 2 (Frobenius from cyclic invariance).* Specialise to homogeneous primitives $x, y, z \in \widetilde{P}_X$ and expand the Hall bracket as the graded commutator $[x, y] = x\cdot y - (-1)^{|x|\cdot|y|}\,y\cdot x$ (the bracket on primitives in a connected graded Hopf algebra is the antisymmetrised product, as part of the primitive Lie-superalgebra structure assumed in Theorem 10.45). Then

$$\begin{aligned}\langle [x,y], z\rangle &= \mathrm{tr}_S\big((x\cdot y - (-1)^{|x|\cdot|y|}\,y\cdot x)\cdot z\big)\\ &= \mathrm{tr}_S(x\cdot y\cdot z) - (-1)^{|x|\cdot|y|}\,\mathrm{tr}_S(y\cdot x\cdot z).\end{aligned}$$

Cyclic invariance of $\mathrm{tr}_S$ applied to the second term gives $\mathrm{tr}_S(y\cdot x\cdot z) = (-1)^{|y|\cdot(|x|+|z|)}\,\mathrm{tr}_S(x\cdot z\cdot y)$, and applied to the first gives $\mathrm{tr}_S(x\cdot y\cdot z) = (-1)^{|x|\cdot(|y|+|z|)}\,\mathrm{tr}_S(y\cdot z\cdot x)$. Substituting and recollecting,

$$\langle [x,y], z\rangle = -(-1)^{|x|\cdot|y|}\,\mathrm{tr}_S\big(y\cdot(x\cdot z - (-1)^{|x|\cdot|z|}\,z\cdot x)\big) = -(-1)^{|x|\cdot|y|}\langle y, [x,z]\rangle,$$

which is (F).

*Step 3 (Finite-stage chain-level lift).* The cyclic trace $\mathrm{tr}_S$ is a strict invariant on the $A_\infty$-homotopy category by [39], § 10.2; on $d$-critical thickenings it lifts to perfect-complex chain level by [9] (the cyclic structure on the vanishing-cycle dg algebra) and PTVV [51] (the shifted symplectic form descends to the cotangent complex of $\mathfrak{M}_{X,c}$). By the finite-stage hypotheses, this cyclic trace, the product, the pairing, and the primitive projector all descend through the reduced $E$-quotient and commute with HN transitions. The identity of Step 2 therefore holds at each finite $R$ and passes to the completed protected primitive subspace $\widetilde{P}_X$. A cyclic trace on an auxiliary model would not prove the Frobenius identity for the Hall bracket unless these finite-stage compatibilities were supplied. □

*Remark* 10.34. Lemma 10.33 derives (F) from the CY-3 Serre functor of $\mathcal{A}_X$, parallel to the derivation through the squaring isomorphism $o_c^{\otimes 2} \simeq \det \mathrm{RHom}(C, C)$ of Definition 4.3 and the Hopf pairing of Theorem 10.45. Both derivations meet at $\langle\,\cdot\,,\,\cdot\,\rangle = \mathrm{tr}_S(\,\cdot\,\cdot\,\cdot\,)$, the BBDJS [9] cyclic upgrade of the PTVV [51] $(-1)$-shifted symplectic form, and produce the same sign convention $(-1)^{|x|\cdot|y|}$ on the Gram degree $c_x + c_y + c_z = 0$. It proves only $\mathfrak{v}_F = 0$; it does not prove $\mathfrak{v}_\Theta^{\mathrm{top}} = 0$, coideal stability of the Hopf radical, a no-extra-relations theorem, or PBW compatibility. It is therefore one input to the full Dirac–Igusa realization certificate, not a substitute for it.

*Remark* 10.35 *(Unitarity).* This paper uses the protected index, the virtual determinant, and the Borcherds denominator algebra. A holomorphic $E_3$-factorization algebra in cochains is part of the open Dirac–Igusa realization certificate, not a positive Hilbert-space construction already in hand. Negative coefficients $f(nm, l)$ record an odd excess in a protected supertrace or in a $K_0$-representative; actual parity dimensions require an additional Hilbert-space or denominator-algebra parity datum. If the untwisted compactification is unitary, the statement that survives the twist is right-moving cohomology, protected integration, and the signed index. Reflection positivity is not a hypothesis of the Borcherds denominator theorem.

Lemma 10.36 *(Cyclotomic-spectrum construction of the multiplicative orientation cocycle; alternative source for hypothesis (O) of Theorem 10.27).* Let $\mathcal{F}_X$ be the protected observable algebra of Definition 2.1, graded by physical charge $c \in \Gamma_X^{\mathrm{phys}}$, and let $\{o_c\}_{c\in\Gamma_X^{\mathrm{phys}}}$ be the orientation lines of Definition 4.3. Suppose:

(C1) *(Cyclotomic host.)* Each charged sector $\mathcal{F}_{X,c}$ is realised as a connective cyclotomic spectrum in the sense of Nikolaus–Scholze [44]; the $p$-typical cyclotomic structure map $\varphi_p : \mathcal{F}_{X,c} \to (\mathcal{F}_{X,c})^{tC_p}$ is the Frobenius inherited from the chiral product on $\mathrm{Ran}(E)$. Equivalently, the underlying spectrum $HH(\mathcal{F}_{X,c})$ carries the natural $S^1$-action by the Connes operator $B$ of Goodwillie [23], with genuine $S^1$-fixed points $HC^-(\mathcal{F}_{X,c}) := HH(\mathcal{F}_{X,c})^{hS^1}$.

(C2) *(Determinant track.)* The orientation line $o_c$ lifts to a unit $\widehat{o}_c \in \pi_0 HC^-(\mathcal{F}_{X,c})$ representing the graded determinant $\det \mathrm{RHom}(\mathcal{C},\mathcal{C})^{1/2}$ of Definition 4.3 in the $S^1$-equivariant periodic Picard groupoid, in the sense of Joyce–Upmeier [35].

(C3) *(Strict tensor compatibility.)* The external product on $\{\mathcal{F}_{X,c}\}$ is symmetric monoidal as a map of cyclotomic spectra, $\mathcal{F}_{X,c} \boxtimes \mathcal{F}_{X,c'} \to \mathcal{F}_{X,c+c'}$, intertwining $\varphi_p$ on each side.

Then there exists a multiplicative orientation cocycle

$$\mu_{c,c'} \;:\; o_{c+c'} \;\xrightarrow{\sim}\; o_c \otimes o_{c'}, \qquad c, c' \in \Gamma_X^{\mathrm{phys}},$$

with the following properties:

(i) (Strict pentagon.) $\mu_{c+c',c''} \circ \mu_{c,c'} = \mu_{c,c'+c''} \circ \mu_{c',c''}$ as maps of $HC^-$-modules, with no higher coherence remainder.

(ii) ($S^1$-equivariance.) Each $\mu_{c,c'}$ is an $S^1$-equivariant map of cyclotomic spectra; the Connes operator $B$ intertwines source and target.

(iii) (PTVV compatibility.) The Tate construction $(-)^{tC_p}$ carries $\mu_{c,c'}$ to a comparison of Frobenius twists $\mathcal{F}^{(p)}_{X,c} \otimes \mathcal{F}^{(p)}_{X,c'} \to \mathcal{F}^{(p)}_{X,c+c'}$ compatible with the PTVV [51] $(-1)$-shifted symplectic form on $\mathfrak{M}_{X,c}$, in the sense that $\mu_{c,c'}$ restricts on the underlying graded determinants to the symplectic-Lagrangian comparison map of [35].

In particular, hypothesis (O) of Theorem 10.27 is satisfied, and the construction of $\{\mu_{c,c'}\}$ is independent of the Costello–Li flat Calabi–Yau pairing and of any Künneth decomposition of $\mathcal{F}_X$ along $K3 \times E$.

*Proof. Step 1 (Construction of $\mu_{c,c'}$ in $HC^-$).* Apply the $S^1$-fixed-points functor $(-)^{hS^1}$ to the symmetric monoidal external product of (C3): the resulting map

$$HC^-(\mathcal{F}_{X,c}) \otimes HC^-(\mathcal{F}_{X,c'}) \;\longrightarrow\; HC^-(\mathcal{F}_{X,c+c'})$$

is a strict map of $\mathbb{E}_\infty$-algebras in cyclotomic spectra by Nikolaus–Scholze [44, Cor. II.1.7, Thm. II.6.9], since $(-)^{hS^1}$ is lax symmetric monoidal and the cyclotomic Frobenius $\varphi_p$ is itself symmetric monoidal. Take $\widehat{o}_c$ and $\widehat{o}_{c'}$ as in (C2); their product under this map lifts the right-hand side of the desired isomorphism $o_c \otimes o_{c'}$, and divides the lift $\widehat{o}_{c+c'}$ of $o_{c+c'}$ to define $\mu_{c,c'} := \widehat{o}_{c+c'} \cdot (\widehat{o}_c \boxtimes \widehat{o}_{c'})^{-1}$. The Connes operator $B$ acts trivially on units of $\pi_0$, so $\mu_{c,c'}$ is $S^1$-equivariant on the nose.

*Step 2 (Strict pentagon from $\mathbb{E}_\infty$-associativity).* The pentagon $\mu_{c+c',c''} \circ \mu_{c,c'} = \mu_{c,c'+c''} \circ \mu_{c',c''}$ is the $\pi_0$-shadow of the strict associator of the $\mathbb{E}_\infty$-algebra structure on $\bigoplus_c HC^-(\mathcal{F}_{X,c})$ in cyclotomic spectra. Strictness follows because Nikolaus–Scholze [44, Thm. I.1.13] present the cyclotomic $\infty$-category as a strict $\mathbb{E}_\infty$-monoidal limit along the Frobenius diagram; the homotopy-coherent pentagon collapses to an

on-the-nose identity at the level of $\pi_0$-units. No higher coherence datum is required, because $\pi_0 HC^-$ is a discrete commutative monoid and units satisfy strict associativity in any commutative monoid.

*Step 3 ($S^1$-equivariance built in).* By construction $HC^- = HH^{hS^1}$ is the genuine $S^1$-fixed-points spectrum, so every map produced by Step 1 is $S^1$-equivariant. The Connes operator $B$ is the generator of the $S^1$-action on $HH$ [23], and $B$ acts as a derivation on $HC^-$; it kills $\pi_0$-units, hence kills $\mu_{c,c'}$ as a higher homotopy.

*Step 4 (Tate-construction comparison with PTVV).* Apply $(-)^{tC_p}$ to the equation defining $\mu_{c,c'}$. The Frobenius $\varphi_p$ intertwines source and target by (C1) and (C3), so $\mu_{c,c'}^{tC_p}$ is the comparison of Frobenius twists $\mathcal{F}_{X,c}^{(p)} \otimes \mathcal{F}_{X,c'}^{(p)} \to \mathcal{F}_{X,c+c'}^{(p)}$. Restrict to the underlying graded determinants: the symplectic structure on $\mathfrak{M}_{X,c}$ of PTVV [51] is the $(-1)$-shifted symplectic form whose square root is $o_c$ (Definition 4.3); under $(-)^{tC_p}$ this form is sent to the Tate-Frobenius pairing on $\det \mathrm{RHom}(C, C)^{(p)}$, which by Joyce–Upmeier [35, multiplicative orientation construction] agrees with the symplectic-Lagrangian comparison map. Compatibility (iii) follows.

The data $\{\mu_{c,c'}\}$ is a multiplicative orientation cocycle satisfying (i)–(iii), so hypothesis (O) of Theorem 10.27 holds. The construction factors through the cyclotomic-spectrum machinery of Nikolaus–Scholze [44] and the Connes-$B$ calculus of Goodwillie [23]; it does not invoke Costello–Li chiral pairings, Künneth decompositions of $\mathcal{F}_X$ along $K3 \times E$, or any compactness datum beyond (C1)–(C3). □

*Remark* 10.37. Hypotheses (C1), (C2), (C3) are open for $K3 \times E$: (C1) asks $\mathcal{F}_{X,c}$ to carry a cyclotomic structure in the Nikolaus–Scholze sense [44]; (C2) asks orientation lines to lift to $\pi_0 HC^-$ units, refining Definition 4.3; (C3) asks symmetric monoidality of $\{\mathcal{F}_{X,c}\}$ at the cyclotomic level. The cocycle $\mu_{c,c'}$ is built from the Hochschild–cyclic machinery of $\mathcal{F}_X$ alone, supplying an alternative construction of hypothesis (O) disjoint from the Costello–Li flat Calabi–Yau pairing and from any Künneth decomposition.

*Definition* 10.38 *(Sectorial Hall truncation).* Suppose an extension-closed compact Hall category graded by the additive physical lattice $\Gamma_X^{\mathrm{phys}}$ of Definition 6.5, together with the Gram map $\Pi_X$, a stability/HN/support datum $\sigma = (Z_\sigma, \mathcal{P}_\sigma)$, and semistable stacks $\mathfrak{M}_{\sigma,S,c}^{ss}$ for $c \in \Gamma_X^{\mathrm{phys}}$. Fix a strict sector $S \subset \mathbb{C}^\times$ of angular width $< \pi$, with no active semistable ray on its boundary. Choose $\vartheta$ so that

$$h_S(c) := \Re(e^{-i\vartheta} Z_\sigma(c)) > 0$$

on every nonzero semistable class with central charge in $S$. For $R > 0$, set

$$F_{\sigma,S}(R) = \{c \mid \mathfrak{M}_{\sigma,S,c}^{ss} \neq \varnothing,\ 0 < h_S(c) \leq R\},$$

and

$$\Gamma_{\sigma,S}^{HN}(R) = \left\{ c_1 + \cdots + c_k \;\middle|\; c_i \in F_{\sigma,S}(R),\ \sum_i h_S(c_i) \leq R \right\}.$$

If an oriented critical Hall object $\mathcal{H}_{\sigma,S}^{or}$ has been constructed, put

$$\mathcal{H}_{\sigma,S}^{>R} = \prod_{h_S(c)>R} \mathcal{H}_{\sigma,S,c}^{or}, \qquad \mathsf{F}_R \mathcal{H}_{\sigma,S}^{or} = \mathcal{H}_{\sigma,S}^{or} / \mathcal{H}_{\sigma,S}^{>R}.$$

Lemma 10.39 *(Conditional finite Hall quotient).* A nonzero additive heart cannot have finite nonzero charge support in $\Gamma_X^{\mathrm{phys}}$. The pushed-forward active support $\Gamma_{\mathrm{act}} \subset \Gamma_{\mathrm{gram}}$ is not additive. The finite Hall object is therefore not a finite active-support heart.

Assume the support property, the HN property, finite type of the semistable stacks in bounded $h_S$-height, and the oriented critical Hall correspondences. Then $F_{\sigma,S}(R)$ and $\Gamma^{HN}_{\sigma,S}(R)$ are finite, $\mathcal{H}^{>R}_{\sigma,S}$ is a two-sided Hall ideal, and $\mathsf{F}_R\mathcal{H}^{or}_{\sigma,S}$ is a finite Hall quotient. The compact sectorial Hall object is the completed inverse system

$$\widehat{\mathcal{H}}^{or}_{\sigma,S} = \varprojlim_{R} \mathsf{F}_R\mathcal{H}^{or}_{\sigma,S}.$$

*Proof.* If a heart contains an object of nonzero charge $c$, it contains all finite direct sums of that object, hence all charges $kc$. Since $\Gamma^{\mathrm{phys}}_X$ is torsion-free, the charge support is infinite. Also $(0,-1,0) \in \Gamma_{\mathrm{act}}$, while $(0,-2,0) \notin \Gamma_{\mathrm{act}}$, so active support is not additive.

For semistable classes in $S$, the strict-sector condition gives $h_S(c) \geq \kappa|Z_\sigma(c)|$ for some $\kappa > 0$. The support property and discreteness of the numerical lattice make $F_{\sigma,S}(R)$ finite. The function $h_S$ is positive and additive on extensions in the sector; hence HN sums of total height $\leq R$ form a finite set and the subspace of height $> R$ is a two-sided Hall ideal. The inverse limit is the $h_S$-adic Hall completion. □

*Definition* 10.40 *(First-order Dirac–Igusa $D_0$-certificate).* Fix a stability/HN sector $(\sigma, S)$ as in Definition 10.38. Write $\Gamma^{HN}_R := \Gamma^{HN}_{\sigma,S}(R)$. For $c \in \Gamma^{HN}_R$, put

$$\mathcal{T}_R(c) = \left\{ \sum_{1\leq i<j\leq k} B(c_i, c_j) \;\middle|\; c = c_1 + \cdots + c_k,\ c_i \in F_{\sigma,S}(R),\ \sum_i h_S(c_i) \leq R \right\}.$$

The empty decomposition is allowed for $c = 0$, giving $T = 0$. The finite normal-ordered support at height $R$ is

$$\widehat{\Gamma}_R = \{(c,T) \mid c \in \Gamma^{HN}_R,\ T \in \mathcal{T}_R(c)\} \subset \widehat{\Gamma}_X, \qquad \Gamma^{\Pi}_R := \overline{\Pi}_X(\widehat{\Gamma}_R) \subset \Gamma_{\mathrm{gram}}.$$

For nonzero $\gamma = (n,l,m) \in \Gamma_{\mathrm{gram}}$, set the finite target exponent

$$a_\Delta(\gamma) = \begin{cases} f(nm,l), & \gamma \in \Gamma_{\mathrm{eff}}, \\ 0, & \gamma \notin \Gamma_{\mathrm{eff}}. \end{cases}$$

Fix also a proper height

$$h_\Delta : \Gamma_{\mathrm{gram}} \longrightarrow \mathbb{Z}_{\geq 0}, \qquad h_\Delta(n,l,m) = |n| + |l| + |m|$$

on the Gram-index lattice. The $D_0$ bookkeeping also fixes a nondecreasing cofinal target-window schedule

$$N^\Delta_R \in \mathbb{Z}_{\geq 0}, \qquad R' \geq R \Rightarrow N^\Delta_{R'} \geq N^\Delta_R, \qquad \sup_R N^\Delta_R = \infty.$$

This schedule is finite-window bookkeeping, not a relation between HN height and automorphic height. To keep the stage data genuinely finite, fix a cofinal discrete subsystem

$$\mathcal{R} = \{R_0 < R_1 < R_2 < \cdots\} \subset \mathbb{R}_{>0}, \qquad R_\nu \to \infty,$$

and take all inverse limits below over $\mathcal{R}$. Cofinality does not change the completed HN object or the target exhaustion. For $R = R_\nu$ write $R^+ = R_{\nu+1}$. Transition maps for non-successor arrows are composites of the successor maps. Thus the independent transition obstruction at a stage is a finite obstruction, not an

infinite product of defects over all larger heights. We continue to write $R$ for an element of this cofinal subsystem.

Define the finite target and test windows

$$\Gamma_{\Delta,R} := \{\gamma \in \Gamma_{\mathrm{eff}} \setminus \{0\} \mid h_\Delta(\gamma) \le N_R^\Delta\}, \qquad \Gamma_R^{\mathrm{test}} := (\Gamma_R^\Pi \setminus \{0\}) \cup \Gamma_{\Delta,R}.$$

The set $\widehat{\Gamma}_R$, and hence $\Gamma_R^\Pi$, $\Gamma_{\Delta,R}$, and $\Gamma_R^{\mathrm{test}}$, is finite by Lemma 10.39 and properness of $h_\Delta$. Cofinality of $N_R^\Delta$ makes the target windows exhaust $\Gamma_{\mathrm{eff}} \setminus \{0\}$. For $R' \ge R$ the inclusions

$$\widehat{\Gamma}_R \subset \widehat{\Gamma}_{R'}, \qquad \Gamma_R^\Pi \subset \Gamma_{R'}^\Pi, \qquad \Gamma_{\Delta,R} \subset \Gamma_{\Delta,R'}$$

hold. A first-order Dirac–Igusa $D_0$-certificate for $X = K3 \times E$ is the choice of the schedule $N_R^\Delta$ above, together with a compatible inverse system of finite-stage state/observable packages

$$D_{0,R} = \big(\mathcal{H}_R^{or}, \mathcal{F}_R^{D_0}, K_R^\Pi, P_R^\Pi, m_R, \Delta_R, [\,\cdot\,,\,\cdot\,]_R, I_R^{\mathrm{prot}}, \Theta_{\Pi,R}, \Theta_{\Pi,R}^-, \mathfrak{D}_{X,R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \iota_R^{obs}\big), \qquad R \in \mathcal{R},$$

together with successor transition maps whose composites give every non-successor arrow. If the transition residuals below vanish and the composite transition identities hold, the inverse limit is denoted

$$D_0(X;\sigma,S) = \big(\widehat{\mathcal{H}}_{\sigma,S}^{or}, \mathcal{F}_X^{D_0}, K_X^\Pi, P_X^\Pi, \mathfrak{D}_X, \Theta_\Pi, \Theta_\Pi^-, \mathscr{L}_{\mathrm{Pf},X}, \mathrm{pf}_X, \iota_X^{obs}\big).$$

The clauses are the following.

(Do-F) *Finite-stage maps and transitions.* At height $R$, all state, observable, primitive, Pfaffian, and comparison data are defined on the finite support $\widehat{\Gamma}_R$ and its Gram image $\Gamma_R^\Pi$. The stage carries Hall product, coproduct, primitive bracket, and protected integration maps

$$m_R, \qquad \Delta_R, \qquad [\,\cdot\,,\,\cdot\,]_R, \qquad I_R^{\mathrm{prot}},$$

and a charge-preserving observable comparison

$$\iota_R^{obs} : \mathcal{F}_R^{D_0} \longrightarrow \mathsf{Den}_{\Delta_s,E}.$$

For the successor arrow $R^+ \to R$ there are transition maps

$$\rho_{R^+R}^H : \mathcal{H}_{R^+}^{or} \to \mathcal{H}_R^{or}, \qquad \rho_{R^+R}^F : \mathcal{F}_{R^+}^{D_0} \to \mathcal{F}_R^{D_0}, \qquad \rho_{R^+R}^K : K_{R^+}^\Pi \to K_R^\Pi, \qquad \rho_{R^+R}^P : P_{R^+}^\Pi \to P_R^\Pi,$$

and compatible maps $\rho_{R^+R}^{\mathrm{Pf}}$ on Pfaffian lines. These maps are degree-truncation maps: after projection to $\widehat{\Gamma}_R$, $\Gamma_R^\Pi$, and $\Gamma_R^{\mathrm{test}}$, all summands of HN height $> R$ are zero. For a non-successor $R' \ge R$, the map $\rho_{R'R}$ is the composite of the successor maps. Let $\pi_{R^+R}^\Delta$ denote the finite target truncation on $\mathsf{Den}_{\Delta_s,E}$ and on $\mathbb{C}[\Gamma_{\Delta,R^+}] \to \mathbb{C}[\Gamma_{\Delta,R}]$. The transition residual

$$\mathfrak{d}_{R^+R}^{\mathrm{tr}}$$

is the finite tuple of defect maps, in the corresponding finite Hom spaces over $\Gamma_R^{\mathrm{test}}$, asserting compatibility of $\rho_{R^+R}$ with

$$m, \quad \Delta, \quad [\,\cdot\,,\,\cdot\,], \quad I^{\mathrm{prot}}, \quad \Theta_\Pi, \quad \Theta_\Pi^-, \quad \mathfrak{D}, \quad \iota^{obs}, \quad \mathrm{pf}\,.$$

For example it includes the defects

$$\rho^F_{R^+R} m_{R^+} - m_R(\rho^F_{R^+R} \otimes \rho^F_{R^+R}), \qquad \rho^P_{R^+R}[\,\cdot\,,\,\cdot\,]_{R^+} - [\,\rho^P_{R^+R}(\cdot), \rho^P_{R^+R}(\cdot)\,]_R,$$

$$I^{\mathrm{prot}}_R \rho^F_{R^+R} - \pi^\Delta_{R^+R} I^{\mathrm{prot}}_{R^+}, \qquad \iota^{obs}_R \rho^F_{R^+R} - \pi^\Delta_{R^+R} \iota^{obs}_{R^+}, \qquad \rho^K_{R^+R} \mathfrak{D}_{X,R^+} - \mathfrak{D}_{X,R} \rho^K_{R^+R}, \qquad \rho^{\mathrm{Pf}}_{R^+R}(\mathrm{pf}_{X,R^+}) - \mathrm{pf}_{X,R}\,.$$

The certificate requires $\mathfrak{o}^{\mathrm{tr}}_{R^+R} = 0$ for every successor arrow. The zero means a chosen finite equality or null-homotopy in the indicated Hom complex. Non-successor compatibility is then the corresponding composite equality; if non-successor maps are recorded separately, the evident cocycle identity for $R'' \geq R' \geq R$ is also part of the finite transition datum. These equalities are finite-stage data, not formal consequences of the inverse-limit notation.

(Do-S) *State object; conditional datum, open construction.* A certificate supplies an oriented critical Hall/factorization inverse system whose completed state side is

$$\widehat{\mathcal{H}}^{or}_{\sigma,S} = \varprojlim_R \mathsf{F}_R \mathcal{H}^{or}_{\sigma,S}$$

formed first on the finite HN quotients of Definition 10.38. Each finite quotient carries finite-type semistable substacks, extension correspondences, vanishing-cycle coefficients, orientation lines, Hall product and coproduct maps, and protected integration. All products, coproducts, brackets, pairings, and comparison maps are defined on the finite quotients and are continuous for the inverse limit. This clause is a conditional datum; Lemma 10.39 proves only the finite-first formal shape once the Hall object and the HN/support hypotheses are supplied.

(Do-Q) *Reduced quotient and orientation obstruction classes; conditional trivializations.* Let $\mathsf{St}_R$ be the finite stable-stratum index set in the chosen finite-type atlas at height $R$. For every $C \in \mathsf{St}_R$, with stable stratum $\mathfrak{M}^{\mathrm{st}}_{R,C} \subset \mathcal{H}^{or}_R$, the certificate includes the computed obstruction tuple

$$\mathfrak{o}^{or}_{R,C} = \Big(\alpha^{\mathrm{red}}_{R,C},\ \alpha^{E,\mathrm{free}}_{R,C},\ \{(\beta^{E,N}_{R,C}, \lambda^{E,N}_{R,C})\}_{N \in \mathcal{N}_R(C)}\Big),$$

where

$$\alpha^{\mathrm{red}}_{R,C} \in H^2(\mathfrak{M}^{\mathrm{red}}_{R,C}; \mathbb{F}_2), \qquad \alpha^{E,\mathrm{free}}_{R,C} \in H^2(BE; \mathbb{F}_2),$$

and

$$\beta^{E,N}_{R,C} \in H^2(BE[N]; \mathbb{F}_2)$$

is present for each finite stabilizer $E[N]$ appearing at height $R$. After $\beta^{E,N}_{R,C}$ has been null-trivialized, the certificate also records the finite linearization character

$$\lambda^{E,N}_{R,C} \in H^1(BE[N]; \mathbb{F}_2)$$

and requires the zero character. The certificate supplies null-trivializations of all degree-two classes and zero finite degree-one characters, compatible with the transition maps. These are exactly the ordinary reduced gerbe, connected $BE$-linearization, finite-stabilizer gerbe, and finite-stabilizer $H^1$-character obstructions of Lemma 5.8. Their vanishing gives the reduced quotient and orientation line needed for the $D_0$ Pfaffian. It does not compute the Weyl monodromy character; that is the separate content of $(\mathrm{O1})^+$ and (O2) in Theorem 3.12.

(Do-C) *Charge completion; proved lattice input.* The finite quotients are graded by the normal-ordered charge extension

$$\widehat{\Gamma}_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}, \qquad (c,T) \star (c',T') = (c+c', T+T'+B(c,c')),$$

and the observable Gram degree is

$$\overline{\Pi}_X(c,T) = \Pi_X(c) - T.$$

The unrenormalized physical sector $c$ is placed in degree $(c,0)$. The bracket of homogeneous states satisfies

$$[\widetilde{P}_{X,(c,T)}, \widetilde{P}_{X,(c',T')}] \subset \widetilde{P}_{X,(c,T)\star(c',T')}.$$

At height $R$ this statement is read inside $\widehat{\Gamma}_R$, with targets of height $> R$ killed by the quotient. The finite-support assertion follows from Lemma 10.39; the additivity of $\overline{\Pi}_X$ follows from Lemma 6.9; no chain-level Hall strictification is claimed by this lattice statement.

(Do-O) *State/observable interface; imported target, conditional source.* A certificate supplies, at finite height $R$, source observable shadows $\mathcal{F}_R^{D_\circ}$. Their compatible inverse limit $\mathcal{F}_X^{D_\circ}$ is obtained from $\widehat{\mathcal{H}}^{or}_{\sigma,S}$ by the supplied $K3$-to-$E$ specialization and, in positive elliptic degree, by a hybrid wrapped repair. It is a source object. The target observable algebra is the target current envelope $\mathsf{Den}_{\Delta_5,E}$ of Proposition 10.2. The interface is a charge-preserving comparison morphism

$$\iota_X^{obs} : \mathcal{F}_X^{D_\circ} \longrightarrow \mathsf{Den}_{\Delta_5,E}$$

whose constant-current primitive restriction is

$$\iota_X^{\mathrm{Prim}} : H^0 \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X^{D_\circ}) \longrightarrow \mathfrak{g}_{\Delta_5}.$$

The target side is imported from Gritsenko–Nikulin; the source and the comparison morphism are conditional data. The finite observable residual

$$\mathfrak{o}_R^{obs}$$

is the finite tuple of defects measuring whether $\iota_R^{obs}$ is a morphism for the finite product, coproduct, primitive bracket, protected trace, and constant-current primitive comparison after projection to $\Gamma_R^{\mathrm{test}}$. Here $m_{\Delta_5}$, $\Delta_{\Delta_5}$, $[\,\cdot\,,\,\cdot\,]_{\Delta_5}$, and $\mathrm{tr}_{\Delta_5}$ denote the corresponding finite target operations induced from $\mathsf{Den}_{\Delta_5,E}$. Thus $\mathfrak{o}_R^{obs}$ includes

$$\iota_R^{obs} m_R - m_{\Delta_5}(\iota_R^{obs} \otimes \iota_R^{obs}), \qquad \Delta_{\Delta_5}\iota_R^{obs} - (\iota_R^{obs} \otimes \iota_R^{obs})\Delta_R,$$

$$\iota_R^{\mathrm{Prim}}([x,y]_R) - [\iota_R^{\mathrm{Prim}}x, \iota_R^{\mathrm{Prim}}y]_{\Delta_5}, \qquad I_R^{\mathrm{prot}} - \mathrm{tr}_{\Delta_5}\,\iota_R^{obs}.$$

The certificate requires $\mathfrak{o}_R^{obs} = 0$. This is a finite comparison with the imported observable target. It is not the presentation-level primitive isomorphism, injectivity, surjectivity, or no-extra-relations theorem; those belong to the full recognition certificate. No target truncation, target bar–cobar counit, or homotopy inverse to $\iota_R^{obs}$ is allowed to define $\mathcal{F}_R^{D_\circ}$. The source object is supplied first, and the arrow points source-to-target.

(Do-N) *Exact normal-ordered descent; conditional finite-topology, Hochschild, triple, Jacobi, Frobenius, cyclic, and radical trivializations.* At finite height $R$, the datum $\Theta_{\Pi,R}$ is the central translation cochain

$$\Theta_{\Pi,R}(c) = \mathsf{T}_{-\Pi_X(c)}$$

in the coefficient dg category of Definition 10.24, and $\Theta^{-}_{\Pi,R}$ is its negative-cyclic lift. For $(c,T),(c',T') \in \widehat{\Gamma}_R$, it gives coherent degree-shift quasi-isomorphisms

$$\Theta_{(c,T),(c',T')} : \Pi_X(c+c') - T - T' - B(c,c') \longrightarrow \big(\Pi_X(c) - T\big) + \big(\Pi_X(c') - T'\big)$$

on primitive bracket targets. Equivalently, $\Theta_\Pi$ categorifies the equality

$$\overline{\Pi}_X((c,T) \star (c',T')) = \overline{\Pi}_X(c,T) + \overline{\Pi}_X(c',T')$$

and turns the upstairs physical Hall bracket into a $\Gamma_{\mathrm{gram}}$-graded bracket after $\overline{\Pi}^{\Theta}_{X,*}$. This is exactly the conditional input of Theorem 10.27; constructing $\Theta_\Pi$ on the reduced compact/wrapped sector is open. The finite coefficient pieces are

$$(\overline{\Pi}^{\Theta}_{R,*}V_R)_\gamma = \prod_{\substack{(c,T)\in\widehat{\Gamma}_R \\ \overline{\Pi}_X(c,T)=\gamma}} (V_R)_{(c,T)},$$

and this product is finite. For the successor arrow $R^+ \to R$, the transition maps must commute with these finite fibre products, with $m^{\Theta}_R$, with $\Delta^{\Theta}_R$, and with the finite pairing. The corresponding transition defect is

$$\mathfrak{o}^{\mathrm{top}}_{\Theta,R^+R} = \Big(\mathfrak{o}^{\mathrm{fib}}_{\Theta,R^+R}, \mathfrak{o}^{m}_{\Theta,R^+R}, \mathfrak{o}^{\Delta}_{\Theta,R^+R}, \mathfrak{o}^{\langle\,,\,\rangle}_{\Theta,R^+R}\Big),$$

with the components defined in Theorem 10.27. Its vanishing is the finite statement that the completed fibre products and $\widehat{\otimes}$-operations exist as inverse limits. Equivalently, with

$$B_{\mathrm{ch},R}(c,c') := \mathsf{T}_{B(c,c')},$$

the certificate supplies trivializations

$$d_{\mathrm{Hoch}}\Theta_{\Pi,R} = B_{\mathrm{ch},R}, \qquad (b + uB_{\mathrm{C}})\Theta^{-}_{\Pi,R} = B^{-}_{\mathrm{ch},R},$$

and the triple and Jacobi coherences of Theorem 10.27. Thus the finite-stage obstruction classes

$$\mathfrak{o}^{\mathrm{top}}_{\Theta,R^+R}, \qquad \mathfrak{o}^{\mathrm{Hoch}}_{\Theta,R}, \qquad \mathfrak{o}^{\mathrm{tri}}_{\Theta,R}, \qquad \mathfrak{o}^{\mathrm{Jac}}_{\Theta,R}, \qquad \mathfrak{o}_{F,R}, \qquad \mathfrak{o}^{\mathrm{cyc}}_{\Theta,R}, \qquad \mathfrak{o}^{\mathrm{rad}}_{\Theta,R}$$

vanish in the finite-topology, Hochschild, triple-coherence, Chevalley–Eilenberg, cyclic-Frobenius, negative-cyclic, and Hopf-radical ideal/coideal groups displayed there. The radical class is read as the ordered pair $(\mathfrak{o}^{\mathrm{rad,Lie}}_{\Theta,R}, \mathfrak{o}^{\mathrm{rad,coid}}_{\Theta,R})$. Equivalently, every successor arrow must satisfy the finite tests (NO1)–(NO7) of Proposition 10.28.

(Do-D) *First-order object; conditional.* There are perfect finite-stage $\Gamma^{\Pi}_R$-graded supercomplexes $K^{\Pi}_R$, compatible under $\rho^{K}_{R^+R}$, with finite-dimensional parity pieces. At each finite stage there is a $\mathbb{Z}/2$-odd skew-adjoint first-order operator/algebra

$$\mathfrak{D}_{X,R} : K^{\Pi}_R \longrightarrow K^{\Pi}_R$$

whose primitive symbol is the finite normal-ordered Hall bracket on $P_R^{\Pi}$. The successor transition compatibility

$$\rho^K_{R^+R}\mathfrak{D}_{X,R^+} = \mathfrak{D}_{X,R}\rho^K_{R^+R}$$

is one component of $\mathfrak{o}^{\mathrm{tr}}_{R^+R} = 0$. The inverse limits are

$$K_X^{\Pi} = \varprojlim_R K_R^{\Pi}, \qquad \mathfrak{D}_X = \varprojlim_R \mathfrak{D}_{X,R}.$$

Their protected primitive cohomology is

$$P_X^{\Pi} = H^0\,\mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X^{D_0}),$$

and the primitive symbol of $\mathfrak{D}_X$ is the normal-ordered Hall bracket. The term "first-order" means the Pfaffian/primitive object before scalar squaring; it does not assert an analytic Dirac operator on an already constructed Hilbert space.

(Do-Pf) *Pfaffian line, finite support, and normalization; conditional datum, imported automorphic target.* For every $R$ there is a Pfaffian line $\mathscr{L}_{\mathrm{Pf},R}$ attached to $\mathfrak{D}_{X,R}$ and to the finite reduced orientation line of the stage, with a finite squaring isomorphism

$$\mathscr{L}_{\mathrm{Pf},R}^{\otimes 2} \simeq \mathscr{L}_{\det,R}.$$

The successor maps $\rho^{\mathrm{Pf}}_{R^+R}$ commute with these square isomorphisms, as part of $\mathfrak{o}^{\mathrm{tr}}_{R^+R} = 0$, and identify the inverse limit with a Pfaffian line

$$\mathscr{L}_{\mathrm{Pf},X}, \qquad \mathscr{L}_{\mathrm{Pf},X}^{\otimes 2} \simeq \mathscr{L}_{\det,X},$$

attached to $\mathfrak{D}_X$ and the orientation line on the state object, together with a normalized section $\mathrm{pf}_{X,R}$ at height $R$ and $\mathrm{pf}_X = \varprojlim_R \mathrm{pf}_{X,R}$. Let $x^\gamma = q^n r^l s^m$ for $\gamma = (n,l,m)$. Projected to the finite monomial span of $\Gamma_R^{\Pi}$, the Borcherds-cusp expansion is

$$\mathrm{pf}_{X,R}|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\substack{\gamma\in\Gamma_R^{\Pi}\\ \gamma\neq 0}} (1-x^\gamma)^{\mathrm{sdim}\, P^{\Pi}_{R,\gamma}}.$$

For $\gamma \notin \Gamma_R^{\Pi}$ set $P^{\Pi}_{R,\gamma} = 0$, and write

$$p^{\bar{0}}_{R,\gamma} := \dim P^{\Pi}_{R,\gamma,\bar{0}}, \qquad p^{\bar{1}}_{R,\gamma} := \dim P^{\Pi}_{R,\gamma,\bar{1}}.$$

The finite-stage support, signed-parity, and leading-coefficient defects

$$\mathfrak{o}^{\mathrm{supp}}_{R,\gamma} := \begin{cases} P^{\Pi}_{R,\gamma}, & a_\Delta(\gamma) = 0,\\ 0, & a_\Delta(\gamma) \neq 0,\end{cases} \qquad \mathfrak{o}^{\mathrm{par}}_{R,\gamma} := p^{\bar{0}}_{R,\gamma} - p^{\bar{1}}_{R,\gamma} - a_\Delta(\gamma) \in \mathbb{Z},$$

$$\mathfrak{o}^{\mathrm{lead}}_R := \mathrm{lc}_{\mathfrak{c}_\infty}(\mathrm{pf}_{X,R}) - 64 \in \mathbb{C}$$

are required to satisfy

$$\mathfrak{o}^{\mathrm{supp}}_{R,\gamma} = \mathfrak{o}^{\mathrm{par}}_{R,\gamma} = 0 \quad (\gamma \in \Gamma_R^{\mathrm{test}}), \qquad \mathfrak{o}^{\mathrm{lead}}_R = 0.$$

The support residual is an actual super-vector-space residual, not only a $K_0$ class: its vanishing excludes invisible even–odd cancelling primitive pieces in target-zero degrees. The parity residual is a signed superdimension residual; the Borcherds product fixes this integer, not a canonical split of the nonzero multiplicity into even and odd compact states. Since the target windows $\Gamma_{\Delta,R}$ exhaust $\Gamma_{\mathrm{eff}} \setminus \{0\}$, these residuals rule out both extra source factors and missing denominator factors stage by stage. If they vanish compatibly for all $R$, the limit section has the prescribed expansion

$$\mathrm{pf}_X|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma=(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{\mathrm{sdim}\, P^{\Pi}_{X,\gamma}},$$

with

$$\mathrm{sdim}\, P^{\Pi}_{X,(n,l,m)} = f(nm, l).$$

The equality of the resulting product with $\Delta_5$ is imported from the Borcherds–Gritsenko–Nikulin product. The compact parity pieces realizing these signed dimensions are conditional data, not a consequence of the scalar OP square.

Proposition 10.41 *(Finite Pfaffian window obstruction).* Fix $R \in \mathcal{R}$, and put

$$A_R := \{\gamma \in \Gamma^{\mathrm{test}}_R \mid a_\Delta(\gamma) \neq 0\}, \qquad Z_R := \{\gamma \in \Gamma^{\mathrm{test}}_R \mid a_\Delta(\gamma) = 0\}.$$

For any supplied finite source package $D_{0,R}$, vanishing of $\mathfrak{o}^{\mathrm{supp}}_{R,\gamma}$ and $\mathfrak{o}^{\mathrm{par}}_{R,\gamma}$ on $\Gamma^{\mathrm{test}}_R$ is equivalent to the two finite conditions

$$P^{\Pi}_{R,\gamma} = 0 \quad (\gamma \in Z_R), \qquad \gamma \in \Gamma^{\Pi}_R, \quad p^{\bar{0}}_{R,\gamma} - p^{\bar{1}}_{R,\gamma} = a_\Delta(\gamma) \quad (\gamma \in A_R).$$

In particular every active target degree in the cofinal window must already occur in the source image:

$$A_R \subset \Gamma^{\Pi}_R.$$

The strongest object constructed from the automorphic input alone at this stage is the finite $K_0$-Pfaffian shadow

$$[\mathsf{P}^{K_0}_{\Delta,R}] := \sum_{\gamma\in\Gamma^{\mathrm{test}}_R} a_\Delta(\gamma)[\mathbb{C}_\gamma] \in K_0\big(\Gamma^{\mathrm{test}}_R\text{-graded sVect}_{\mathrm{fd}}\big)$$

and the finite formal section

$$\mathrm{pf}^{K_0}_{\Delta,R} := 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma\in\Gamma^{\mathrm{test}}_R} (1 - x^\gamma)^{a_\Delta(\gamma)}.$$

This shadow is a target invariant. It is not a compact Hall object, not a Pfaffian line on the reduced quotient, and not a first-order operator. For $\gamma \in A_R$, the pair of compact parity dimensions is not determined by the shadow: adding a finite vector space $W_\gamma$ in both parities replaces

$$(P^{\Pi}_{R,\gamma,\bar{0}}, P^{\Pi}_{R,\gamma,\bar{1}}) \quad \text{by} \quad (P^{\Pi}_{R,\gamma,\bar{0}} \oplus W_\gamma, P^{\Pi}_{R,\gamma,\bar{1}} \oplus W_\gamma)$$

and leaves the finite product, support test, signed-parity test, and leading coefficient unchanged. Thus compact parity pieces on the cofinal target windows are extra source data. They cannot be constructed from $\Delta_5$, from the denominator product, or from the OP scalar square.

*Proof.* If $\gamma \in Z_R$, the support residual is $\mathfrak{o}^{\mathrm{supp}}_{R,\gamma} = P^{\Pi}_{R,\gamma}$; its vanishing is exactly the first displayed condition. If $\gamma \in A_R$ and $\gamma \notin \Gamma^{\Pi}_R$, the convention $P^{\Pi}_{R,\gamma} = 0$ gives $\mathfrak{o}^{\mathrm{par}}_{R,\gamma} = -a_\Delta(\gamma) \neq 0$. Hence vanishing forces $\gamma \in \Gamma^{\Pi}_R$, and then the parity residual is exactly

$$p^{\bar{0}}_{R,\gamma} - p^{\bar{1}}_{R,\gamma} - a_\Delta(\gamma) = 0.$$

This proves the equivalence and the active-window coverage.

The finite formal section is obtained by applying the superdeterminant formula to the $K_0$-class $[\mathsf{P}^{K_0}_{\Delta,R}]$. This construction takes place in the Grothendieck group of finite graded super vector spaces. It contains no extension correspondences, no orientation line, and no skew-adjoint finite operator. Finally, the replacement by $W_\gamma \oplus \Pi W_\gamma$ has zero class in $K_0$, so it does not change $p^{\bar{0}}_{R,\gamma} - p^{\bar{1}}_{R,\gamma}$ or any product depending only on signed superdimension. The actual parity pieces are therefore invisible to the finite Pfaffian shadow. □

Proposition 10.42 *(Finite Hall–Dirac atlas and parity-lift obstruction).* Fix $R \in \mathcal{R}$. A genuine construction of the finite stage $D_{0,R}$ from Hall/d-critical data is equivalent to supplying a finite Hall–Dirac atlas

$$\mathfrak{A}_R = \big(\mathfrak{M}_{R,\widehat{c}}, \mathfrak{E}_{R,\widehat{c},\widehat{c}'}, \Phi_{R,\widehat{c}}, o_{R,\widehat{c}}, \mu^{or}_R, \mathcal{F}^{D_0}_R, \Theta_{\Pi,R}, \Theta^{-}_{\Pi,R}, \iota^{obs}_R, K^{\Pi}_R, \mathfrak{D}_{X,R}, \mathscr{L}_{\mathrm{Pf},R}, \ell_R\big)$$

with the following data.

(A1) *Finite extension-closed d-critical source.* For each $\widehat{c} = (c, T) \in \widehat{\Gamma}_R$ there is a finite-type d-critical stack $\mathfrak{M}_{R,\widehat{c}}$ with compactly supported BBDJS vanishing-cycle coefficient complex $\Phi_{R,\widehat{c}}$. The finite union $\mathfrak{M}_R = \coprod_{\widehat{c}\in\widehat{\Gamma}_R} \mathfrak{M}_{R,\widehat{c}}$ is closed under those extensions whose charge remains in $\widehat{\Gamma}_R$. Thus for $\widehat{c}, \widehat{c}'$, with $\widehat{c} \star \widehat{c}' \in \widehat{\Gamma}_R$, there is a finite-type extension correspondence

$$\mathfrak{M}_{R,\widehat{c}} \times \mathfrak{M}_{R,\widehat{c}'} \xleftarrow{p_{R,\widehat{c},\widehat{c}'}} \mathfrak{E}_{R,\widehat{c},\widehat{c}'} \xrightarrow{q_{R,\widehat{c},\widehat{c}'}} \mathfrak{M}_{R,\widehat{c}\star\widehat{c}'},$$

and the correspondence is empty after projection to the quotient when the target charge leaves $\widehat{\Gamma}_R$. The two-step extension stacks for triples in $\widehat{\Gamma}_R$ satisfy the usual flag associativity isomorphisms.

(A2) *Orientation square and quotient linearization.* There are orientation lines

$$o^{\otimes 2}_{R,\widehat{c}} \simeq \det \mathrm{RHom}_{\mathrm{red}}(C_{\widehat{c}}, C_{\widehat{c}})$$

on the finite strata, Thom–Sebastiani orientation isomorphisms $\mu^{or}_{R,\widehat{c},\widehat{c}'}$ over the extension correspondences, and zero quotient-linearization characters for every finite stabilizer appearing in the reduced $E$-quotient. Equivalently the classes $\mathfrak{o}^{or}_{R,C}$ of (Do-Q) vanish with chosen finite trivializations.

(A3) *Finite Hall operations, normal ordering, and protected integration.* The maps $p_{R,\widehat{c},\widehat{c}'}$ and $q_{R,\widehat{c},\widehat{c}'}$ are admissible for vanishing-cycle pull-push and proper on the support used by the protected trace. The resulting Joyce–Kontsevich–Soibelman pull-push operations define

$$m_R, \qquad \Delta_R, \qquad [\,\cdot\,,\,\cdot\,]_R, \qquad I^{\mathrm{prot}}_R,$$

and the source observable shadow $\mathcal{F}^{D_0}_R$, normal-ordering cochains $\Theta_{\Pi,R}$, $\Theta^{-}_{\Pi,R}$, and comparison map $\iota^{obs}_R$ are defined source-first. They satisfy the finite flag, coproduct, trace, observable-comparison, Hochschild, and negative-cyclic diagrams required in (Do-F), (Do-O), and (Do-N).

(A4) *Finite first-order complex.* There is a perfect $\Gamma_R^{\Pi}$-graded supercomplex $K_R^{\Pi}$ and an odd skew-adjoint first-order endomorphism

$$\mathfrak{D}_{X,R} : K_R^{\Pi} \to K_R^{\Pi}$$

whose primitive symbol on

$$P_R^{\Pi} = H^0 \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}_{R,*}^{\Theta} \mathcal{F}_R^{D_0})$$

is the Hall bracket of (A3).

(A5) *Pfaffian square.* On the reduced finite quotient $\mathfrak{Q}_R := (\mathfrak{M}_R/E)^{\mathrm{red}}$ there is a determinant line $\mathscr{L}_{\det,R}$ attached to $K_R^{\Pi}$, a square root

$$\eta_R : \mathscr{L}_{\mathrm{Pf},R}^{\otimes 2} \xrightarrow{\sim} \mathscr{L}_{\det,R},$$

and a finite section $\mathrm{pf}_{X,R}$ whose leading coefficient at the Borcherds cusp is 64.

(A6) *Parity lift on the test window.* For every $\gamma \in \Gamma_R^{\mathrm{test}}$ there are actual compact primitive super-vector spaces

$$P_{R,\gamma}^{\Pi} = P_{R,\gamma,\bar{0}}^{\Pi} \oplus P_{R,\gamma,\bar{1}}^{\Pi}$$

with

$$\ell_R(\gamma) = \big(\dim P_{R,\gamma,\bar{0}}^{\Pi}, \dim P_{R,\gamma,\bar{1}}^{\Pi}\big),$$

such that $P_{R,\gamma}^{\Pi} = 0$ for $a_\Delta(\gamma) = 0$, and

$$\dim P_{R,\gamma,\bar{0}}^{\Pi} - \dim P_{R,\gamma,\bar{1}}^{\Pi} = a_\Delta(\gamma)$$

for $a_\Delta(\gamma) \neq 0$. This includes the active-window source coverage $A_R \subset \Gamma_R^{\Pi}$.

(A7) *Successor transition diagrams.* For $R^+ \to R$ there are truncation maps on the stacks, extension stacks, coefficient complexes, first-order complexes, and Pfaffian lines such that

$$\begin{tikzcd}
\mathfrak{E}_{R^+,\widehat{c},\widehat{c}'} \arrow[r, "q_{R^+}"] \arrow[d, "\tau^E_{R^+R}"'] & \mathfrak{M}_{R^+,\widehat{c}\star\widehat{c}'} \arrow[d, "\tau^M_{R^+R}"] \\
\mathfrak{E}_{R,\widehat{c},\widehat{c}'} \arrow[r, "q_R"'] & \mathfrak{M}_{R,\widehat{c}\star\widehat{c}'}
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathfrak{E}_{R^+,\widehat{c},\widehat{c}'} \arrow[r, "p_{R^+}"] \arrow[d, "\tau^E_{R^+R}"'] & \mathfrak{M}_{R^+,\widehat{c}} \times \mathfrak{M}_{R^+,\widehat{c}'} \arrow[d, "\tau^M_{R^+R} \times \tau^M_{R^+R}"] \\
\mathfrak{E}_{R,\widehat{c},\widehat{c}'} \arrow[r, "p_R"'] & \mathfrak{M}_{R,\widehat{c}} \times \mathfrak{M}_{R,\widehat{c}'}
\end{tikzcd}$$

commute after projection to the finite quotient, and the induced diagrams for $\rho^K_{R^+R}\mathfrak{D}_{X,R^+}$, $\rho^{\mathrm{Pf}}_{R^+R}\eta_{R^+}$, and $\rho^{\mathrm{Pf}}_{R^+R}(\mathrm{pf}_{X,R^+})$ commute. The finite tuple of failures of these diagrams is denoted $\mathfrak{o}^{\mathrm{atlas}}_{R^+R}$; it is one source-level realization of the transition residual $\mathfrak{o}^{\mathrm{tr}}_{R^+R}$.

If $\mathfrak{A}_R$ is supplied and $\mathfrak{o}^{\mathrm{atlas}}_{R^+R} = 0$, then the pull-push operations, orientation square, first-order operator, Pfaffian line, and parity lift construct the finite package $D_{0,R}$ of Definition 10.40. Conversely, every constructed $D_{0,R}$ gives such finite atlas data after forgetting to the finite support and reduced quotient. Thus the missing object is not the target determinant; it is the finite atlas $\mathfrak{A}_R$ and its successor-compatible trivializations.

Moreover the two remaining Pfaffian ambiguities are exact. The Pfaffian square-root choices for a fixed determinant line are empty unless $[\mathscr{L}_{\det,R}]$ is divisible by 2 in $\mathrm{Pic}(\mathfrak{Q}_R)$; when nonempty, their isomorphism classes form a torsor under $\mathrm{Pic}(\mathfrak{Q}_R)[2]$, with the $H^1(\mathfrak{Q}_R;\mathbb{F}_2)$ local-system shadow used

in the finite-stabilizer orientation tests. For an active degree $\gamma \in A_R$, the compact parity lifts form the affine monoid

$$\mathscr{P}_R(\gamma) = \{(u,v) \in \mathbb{Z}^2_{\geq 0} \mid u - v = a_\Delta(\gamma)\}, \qquad (u,v) \mapsto (u+t, v+t).$$

The finite $K_0$-Pfaffian shadow fixes only the difference $u - v$. The no-hidden-cancellation condition

$$\min(u,v) = 0$$

would select the unique minimal lift

$$(u,v) = \begin{cases} (a_\Delta(\gamma), 0), & a_\Delta(\gamma) > 0, \\ (0, -a_\Delta(\gamma)), & a_\Delta(\gamma) < 0, \end{cases}$$

but this is an extra source axiom. It is not implied by $\Delta_5$, by the denominator algebra, by the OP scalar square, or by equality in $K_0$.

*Proof.* Given $\mathfrak{A}_R$, clauses (A1)–(A3) are exactly the finite Joyce–Kontsevich–Soibelman Hall pull-push construction with BBDJS vanishing-cycle coefficients and Joyce–Upmeier orientation transport. Finiteness of the index set makes all sums and products finite, and the two-step flag isomorphisms are precisely the associativity and coproduct-coherence diagrams. Clause (A4) gives (D0-D). Clause (A5) gives the finite Pfaffian square and the leading coefficient condition in (D0-Pf). Clause (A6) is exactly the vanishing of the support and parity residuals on $\Gamma_R^{\mathrm{test}}$, with active-window coverage as in Proposition 10.41. Clause (A7) gives the successor compatibility diagrams; their failure is the displayed finite residual $\mathfrak{o}^{\mathrm{atlas}}_{R^+R}$, which maps into $\mathfrak{o}^{\mathrm{tr}}_{R^+R}$. Hence the data assemble to $D_{0,R}$. The converse is tautological: a constructed finite $D_{0,R}$ already contains the finite source stacks, correspondences, coefficient systems, operator, Pfaffian line, and parity pieces named above, after restriction to its finite support.

The assertion about square roots is the elementary Picard-group classification: a line $L$ admits a square root iff its class lies in $2\,\mathrm{Pic}(\mathfrak{Q}_R)$; the set of square roots, when nonempty, is acted on simply transitively by 2-torsion line bundles. Passing to $\mathbb{F}_2$-local systems gives the $H^1(\mathfrak{Q}_R; \mathbb{F}_2)$ shadow. Finally $\mathscr{P}_R(\gamma)$ is the set of nonnegative integer solutions to one linear equation $u - v = a_\Delta(\gamma)$. Adding $(t,t)$ preserves the $K_0$ class and all product exponents. Therefore signed dimension does not determine compact parity pieces unless the extra condition $\min(u,v) = 0$ is supplied. □

Theorem 10.43 *(Finite first-order $D_0$-certificate theorem).* The first-order statement presently supported by this paper is the following finite-stage certificate statement.

(proved) For every $R \in \mathcal{R}$, the normal-ordered support $\widehat{\Gamma}_R$, its image $\Gamma_R^{\Pi}$, the target window $\Gamma_{\Delta,R}$, and the test window $\Gamma_R^{\mathrm{test}}$ of Definition 10.40 are finite; by cofinality of the target-window schedule, the windows $\Gamma_{\Delta,R}$ exhaust $\Gamma_{\mathrm{eff}} \setminus \{0\}$. The lattice charge completion $\widehat{\Gamma}_X$ and the normal-ordered homomorphism $\overline{\Pi}_X$ exist and are additive, by Lemma 10.39 and Lemma 6.9.

(imported) The target observable algebra $\mathsf{Den}_{\Delta_5,E}$, its constant-current denominator algebra $\mathfrak{g}_{\Delta_5}$, and the product

$$64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5$$

are supplied by the Borcherds–Gritsenko–Nikulin theorem and Proposition 10.2.

(finite residual) Once the finite Hall–Dirac atlas $\mathfrak{A}_R$ of Proposition 10.42 is supplied, including the finite semistable stacks, extension correspondences, vanishing-cycle pull-push maps, protected integration maps, source comparison, first-order complex, Pfaffian square, and compact parity lifts at height $R$, the obstruction tuple recorded by the finite-stage $D_0$ certificate is the vanishing, with compatible trivializations, of

$$\mathfrak{O}_{D_0,R} := \Big(\mathfrak{o}^{\mathrm{tr}}_{R^+R},\ \{\mathfrak{o}^{or}_{R,C}\}_{C\in \mathsf{St}_R},\ \mathfrak{o}^{obs}_R,\ \mathfrak{o}^{\mathrm{top}}_{\Theta,R^+R},\ \mathfrak{o}^{\mathrm{Hoch}}_{\Theta,R},\ \mathfrak{o}^{\mathrm{tri}}_{\Theta,R},\ \mathfrak{o}^{\mathrm{Jac}}_{\Theta,R},\ \mathfrak{o}_{F,R},\ \mathfrak{o}^{\mathrm{cyc}}_{\Theta,R},\ \mathfrak{o}^{\mathrm{rad}}_{\Theta,R},\ \{\mathfrak{o}^{\mathrm{supp}}_{R,\gamma}\}_{\gamma\in\Gamma^{\mathrm{test}}_R},\ \{\mathfrak{o}^{\mathrm{par}}_{R,\gamma}\}_{\gamma\in\Gamma^{\mathrm{test}}_R},\ \mathfrak{o}^{\mathrm{l}}$$

These classes are typed in Definition 10.40 and Theorem 10.27. They are not determined by $\Delta_5$, by the denominator algebra, or by the OP scalar square. Vanishing here means chosen null-trivializations in the displayed finite obstruction groups or complexes, compatible with the successor transition $R^+ \to R$ and hence with its composites; a numerical zero in a Grothendieck group is not such a trivialization. The Pfaffian part has the sharp finite consequence recorded in Proposition 10.41: active target degrees in $\Gamma^{\mathrm{test}}_R$ must lie in the source image, but the compact parity dimensions themselves remain extra source data.

(conditional) If a tuple satisfying Definition 10.40 is supplied, then it supplies hypothesis (D0) of Theorem 3.12. Its primitive bracket is compatible with normal-ordered descent on the nose:

$$[P^{\Pi}_{X,\gamma}, P^{\Pi}_{X,\gamma'}]_\Theta \subset P^{\Pi}_{X,\gamma+\gamma'}, \qquad \gamma,\gamma' \in \Gamma_{\mathrm{gram}},$$

and its Pfaffian section is normalized by

$$\mathrm{pf}_X|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma=(n,l,m)\in\Gamma_{\mathrm{eff}}} (1-q^n r^l s^m)^{f(nm,l)},$$

which the imported Borcherds–Gritsenko–Nikulin theorem identifies with $\Delta_5$ in the normalized cusp expansion.

(usage certificate) Every use of (D0) in Theorem 3.12 factors through one of the displayed clauses of Definition 10.40: finite state and transition data through (D0-F)–(D0-S), reduced orientation descent through (D0-Q), source/observable comparison through (D0-O), normal-ordered grading through (D0-C) and (D0-N), the protected first-order complex through (D0-D), and the Pfaffian target windows, support residuals, signed parity exponents, and leading coefficient through (D0-Pf).

(open) The compact state-side construction of $\widehat{\mathcal{H}}^{or}_{\sigma,S}$, the finite Hall–Dirac atlases $\mathfrak{A}_R$, the finite-stage hybrid wrapped object $\mathcal{F}^{D_0}_R$, the obstruction trivializations $\mathfrak{O}_{D_0,R} = 0$ and $\mathfrak{o}^{\mathrm{atlas}}_{R^+R} = 0$, the compatible inverse system, the Pfaffian line on the reduced compact quotient, and the compact parity pieces $P^{\Pi}_{R,\gamma,\bar{0}}$, $P^{\Pi}_{R,\gamma,\bar{1}}$ on $\Gamma^{\mathrm{test}}_R$, including vanishing in target-zero degrees, are not constructed in this paper. The target $K_0$-shadow supplies only the signed differences; it does not trivialize the Pfaffian square-root torsor or select a compact parity lift.

*Proof.* For fixed $R$, the set $F_{\sigma,S}(R)$ is finite by Lemma 10.39. Since $h_S$ is positive on active semistable charges, the length of a decomposition $c = c_1 + \cdots + c_k$ with $\sum_i h_S(c_i) \le R$ is bounded. Hence only finitely many sums $\sum_{i<j} B(c_i, c_j)$ occur, so $\widehat{\Gamma}_R$ and $\Gamma^{\Pi}_R$ are finite. The function $h_\Delta(n,l,m) = |n| + |l| + |m|$ is proper on the lattice $\Gamma_{\mathrm{gram}}$, so $\Gamma_{\Delta,R}$ is finite. Since $\sup_R N^{\Delta}_R = \infty$, the union over all $R$ is $\Gamma_{\mathrm{eff}} \setminus \{0\}$. Thus $\Gamma^{\mathrm{test}}_R$ is finite. If $R' \ge R$, then $F_{\sigma,S}(R) \subset F_{\sigma,S}(R')$ and every decomposition of height $\le R$ is also a decomposition of height $\le R'$; this gives the displayed inclusions of $\widehat{\Gamma}_R$, $\Gamma^{\Pi}_R$, and

$\Gamma_{\Delta,R}$. Lemma 6.9 gives associativity of $\star$ and $\overline{\Pi}_X((c,T)\star(c',T')) = \overline{\Pi}_X(c,T) + \overline{\Pi}_X(c',T')$. The imported item is Proposition 4.5 together with Theorem 9.2 and Proposition 10.2.

The residual tuple $\mathfrak{O}_{D_0,R}$ is a collection of already typed obstructions. The transition component records compatibility in the finite inverse system. The quotient-orientation component is the criterion of Lemma 5.8. The observable component records that the source/target comparison respects the finite product, coproduct, bracket, trace, and primitive comparison. The next seven are the finite-topology, Hochschild, triple, Jacobi, Frobenius, negative-cyclic, and Hopf-radical ideal/coideal obstruction classes of Theorem 10.27; the final three state exactly the finite target coverage, absence of invisible zero-target primitive support, signed Pfaffian multiplicity, and cusp-leading normalization required in the Pfaffian line. Their vanishing is therefore necessary for the finite stage to be a $D_0$-stage, and, once the finite stacks, correspondences, first-order complex, comparison map, and Pfaffian line have been supplied, is precisely the extra trivialization data demanded by Definition 10.40.

Assume the $D_0$-certificate. Clause (D0-F) gives the finite stages and transition maps. Clause (D0-S) gives the HN-completed state object on which the Hall operations are defined. Clause (D0-Q) supplies the reduced quotient and orientation line used in the Pfaffian construction. Clause (D0-C) places operations in the corrected charge extension. Clause (D0-N) supplies the chain-level shift categorifying the additivity of $\overline{\Pi}_X$; hence the descended primitive bracket is $\Gamma_{\mathrm{gram}}$-graded. Clause (D0-O) supplies the source observable shadow and comparison map. Clauses (D0-D) and (D0-Pf) supply the first-order object and its Pfaffian line. The support, signed-superdimension, target-window exhaustion, and leading-coefficient conditions in (D0-Pf), combined with the imported Borcherds product, give the normalized $\Delta_5$ cusp expansion for $\mathrm{pf}_X$.

It remains to check the stated usage certificate in Theorem 3.12. Step 1 uses only the supplied first-order object, the normal-ordered primitive multiplicities, and the leading Pfaffian normalization; these are (D0-D) and (D0-Pf). Step 2 uses the imported Borcherds–Gritsenko–Nikulin product, not a compact construction. Step 3 uses the Pfaffian cusp normalization from (D0-Pf) and the imported Oberdieck–Pixton scalar normalization. Step 5 uses only the supplied chiral protected first-order theory and its decategorification, supplied by (D0-F)–(D0-Pf). Step 4 is independent of $D_0$ except for the existence of the Pfaffian object; its signs come from (O1), $(\mathrm{O1})^+$, and (O2). Thus the certificate implies every use of (D0) and no stronger compact existence claim is made.

The final item is a status statement: none of the source-side objects or trivializations named there is constructed by the lattice calculation, by the denominator algebra, or by the scalar OP square. □

*Definition* 10.44 *(Dirac–Igusa realization certificate).* A compact realization of the Igusa square root is a Dirac–Igusa realization certificate

$$\mathfrak{K}_X = (L_X, H_X, O_X, D_X, R_X).$$

Its first-order truncation is the $D_0$-certificate of Definition 10.40. The full certificate adds the orientation-monodromy, chiral Koszul, and primitive-recognition data needed to identify the compact state-side algebra with the Gritsenko–Nikulin target. It is not a consequence of the Borcherds product, the scalar Oberdieck–Pixton square, the target current envelope, the target bar–cobar counit, or the existence of orientation lines alone. It is the list of data whose construction would turn the virtual determinant $\mathcal{D}_X = \Delta_5$ into a compact first-order protected realization. The five entries have the following meanings.

$(L_X)$ *Local charge and $E_3$ host.* This entry consists of the additive physical charge lattice $\Gamma_X^{\mathrm{phys}}$, the Gram map $\Pi_X : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{gram}}$, and the cocycle $B$ of Definition 6.5 and Lemma 6.8. The Hall category is graded by $\Gamma_X^{\mathrm{phys}}$, not by $\Gamma_{\mathrm{gram}}$ itself. $(L_X)$ also includes the normal-ordered extension

$$\widehat{\Gamma}_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c, T) = \Pi_X(c) - T,$$

and the primitive saturated lifts of Lemma 6.6. Finally, it includes a supplied charge-completed holomorphic $E_3$-factorization algebra, denoted $\mathcal{A}_X^{E_3}$, on $X = K3 \times E$, finite-first after the sectorial HN quotients of Lemma 10.39. A compact holomorphic-Chern–Simons/BV construction, if it satisfies descent, the quantum master equation, and anomaly cancellation in the completed charge category, is a candidate source for this datum. The datum also includes a fixed $E_3$-formality point, a chain-level homotopy/framing datum for the chosen operadic model, compact-support control, and continuity of the $E_3$ operations across the finite HN inverse system.

$(H_X)$ *Hybrid $K3$-to-$E$ factorization shadow.* This entry consists of a supplied finite-first K3-to-$E$ source interface in the sense of Definition 10.4, a chain-level specialization $\mathrm{Sp}^{\mathrm{ch}}_{K3,E}$, and named comparison maps $\kappa_E$ (chiral-Koszul) and $\pi_*^{\mathrm{ch}}$ (proper-pushforward identification), with the triple zigzag

$$\mathcal{A}_X^{E_3} \xrightarrow{\mathrm{Sp}^{\mathrm{ch}}_{K3,E}} \mathcal{F}_X^{(0)} \xrightarrow{\kappa_E} \mathcal{F}_X^{(1)} \xrightarrow{\pi_*^{\mathrm{ch}}} (p_E)_* \mathcal{A}_X^{E_3}$$

of equivalences in the projection-finite sectors. In the wrapped elliptic-degree sectors, the required filling is the finite-HN inverse system whose limit is the hybrid object $\mathcal{F}_X^{\mathrm{hyb}}$ of Definition 10.18. Thus $(H_X)$ includes the hybrid Ran base $\mathrm{Ran}^{\mathrm{hyb}}(E)$, projection-finite local sectors $\mathcal{F}^{\mathrm{loc}}_{\alpha,R}(U)$, rigidified wrapped prequotient sectors $\mathcal{G}^{\mathrm{wr}}_{\eta,R}$ with anchors $\lambda_{\eta,R}$, finite correspondence categories $\mathsf{Corr}_R^{E,\mathrm{hyb}}$, and the three finite operation families

$$\mu_R^{\mathrm{loc}}, \qquad \mu_R^{\mathrm{hyb}}, \qquad \mu_R^{\mathrm{wr}},$$

where $\mu_R^{\mathrm{hyb}}$ includes the ordered two-sided mixed maps $\mu^{\mathrm{hyb}}_{\alpha_-;\eta;\beta_+,R}$. It also includes the full eight-word two-step flag atlas in the finite HN quotients, the quotient operation $Q_{E,R}$ applied only after the extension correspondences are formed, the quotient comparison isomorphisms $\theta^Q_{\mu,R}$, and protected integration maps $I_R^{\mathrm{prot}}$ whose homogeneous $s$-degree is the elliptic degree $b_R$. Equivalently,

$$\mathfrak{D}_{\mathrm{hyb},R} = 0$$

for every finite height $R$, compatibly with transition maps. A global Fock/Hecke sector is acceptable here only if it is represented by the rigidified wrapped prequotients and acts through the mixed maps $\mu_R^{\mathrm{hyb}}$ before the reduced $E$-quotient. These equivalences and wrapped-sector data are part of the datum, not operations constructed from the determinant.

$(O_X)$ *Oriented compact Hall object and Weyl orientation character.* This entry consists of an oriented compact Hall factorization object $\mathcal{H}_X$ graded by $c \in \Gamma_X^{\mathrm{phys}}$, with finite-type sectorial moduli stacks and extension correspondences

$$\mathfrak{M}_c^\sigma \times \mathfrak{M}_{c'}^\sigma \xleftarrow{p_{c,c'}} \mathfrak{E}^\sigma_{c,c'} \xrightarrow{q_{c,c'}} \mathfrak{M}^\sigma_{c+c'},$$

perfect reduced obstruction theories or critical $(-1)$-shifted symplectic structures, vanishing-cycle coefficients, orientation lines with chosen square roots, Thom–Sebastiani orientation transport coherent for triple extensions, and protected integration maps compatible with HN completion and wall crossing. The orientation entry also includes the exact quotient-descent package. For every charge stratum $C$ used by the Hall correspondences one must specify

$$\alpha_C^{\mathrm{red}} = w_2(\det \mathrm{RHom}_{\mathrm{red}}(C,C)) + \mathrm{cs}^* w_2(\det(\operatorname{coker} \mathrm{cs})) \in H^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2),$$

$$\alpha_C^{E,\mathrm{free}} = a_1(C)u_1 + a_2(C)u_2 \in H^2(BE; \mathbb{F}_2), \qquad \beta_C^{E,N} \in H^2(BE[N]; \mathbb{F}_2),$$

and prove their vanishing on the corresponding free and small-orbit strata. At $N = 2$,

$$\beta_C^{E,E[2]} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2 \in H^2(BE[2]; \mathbb{F}_2),$$

with $E[2] = \langle e_1, e_2 \rangle$, $e_3 = e_1 + e_2$, and

$$\mathrm{res}_{\langle e_i \rangle} \beta_C^{E,E[2]} = r_i(C)\tau_i^2, \qquad (b_{20}, b_{11}, b_{02}) = (r_1, r_1 + r_2 + r_3, r_2).$$

Thus the finite two-torsion residual vanishes exactly when $r_1 = r_2 = r_3 = 0$, but the full quotient descent also requires $\alpha_C^{\mathrm{red}} = 0$ and $a_1(C) = a_2(C) = 0$. After this degree-two two-torsion class vanishes, the chosen $E[2]$-linearization has a degree-one character

$$\lambda_C^{E[2]} = \lambda_1 x_1 + \lambda_2 x_2 \in H^1(BE[2]; \mathbb{F}_2),$$

and the quotient orientation requires $\lambda_1 = \lambda_2 = 0$. Odd $N$ finite stabilizers have no positive-degree mod-2 obstruction; even $N \geq 4$ is detected on the 2-primary stabilizer $(\mathbb{Z}/2^a)^2$, $2^a \parallel N$, where for $a \geq 2$

$$H^*(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2, \quad x_i^2 = 0,$$

$$\beta_{C,(2)}^{E,N} = A_1^{(N)} y_1 + A_{12}^{(N)} x_1 x_2 + A_2^{(N)} y_2.$$

The mixed coefficient $A_{12}^{(N)}$ is not detected by cyclic restrictions; after this degree-two class vanishes, the residual $H^1(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2)$-linearization character must also vanish. Thus the even $N \geq 4$ residual remains part of the descent datum; it is not implied by the $E[2]$ vector. The Hall atlas must include local-local, wrapped-wrapped, and mixed local/wrapped extension correspondences before quotienting by the translation action of $E$; quotienting first forgets the relative $E$-position needed by mixed convolution. If a lifted action of $W^{(2)}(\Lambda_{II}^{2,1})$ on the oriented completed Hall object is supplied, its projective determinant-line cocycle

$$[c_o] \in H^2\Big(W^{(2)}(\Lambda_{II}^{2,1}); \mathbb{F}_2\Big)$$

must vanish, the generator lifts must satisfy $s_{\delta_i}^2 = 1$, and the full quotient package, including the zero finite characters $\lambda_C^{E,N} = 0$, must be transported with zero

$$H^1(\mathfrak{M}_{s_{\delta_i} C}^{\mathrm{red}}; \mathbb{F}_2) \oplus \bigoplus_{N \geq 2} H^1(BE[N]; \mathbb{F}_2)$$

torsor defect, as in Lemma 3.18. Only then does the induced orientation character $\epsilon_o$ satisfy

$$\epsilon_o|_{W^{(2)}(\Lambda^{2,1}_{II})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})}.$$

Constructing the domain of this comparison and the character $\epsilon_o$ is part of the datum. The comparison is a geometric orientation-monodromy condition on the Hall side; it requires computing the determinant-line signs on the three type-II simple reflections and checking the reflection relations. It is not obtained by choosing the scalar normalization of $\Delta_5$.

$(D_X)$ *Normal-ordered descent and chiral Koszul comparison.* This entry consists of the chain-level normal-ordering package

$$\mathsf{Ch}^{\Gamma_{\mathrm{gram}}}_{k,\widehat{\mathrm{HN}},E}, \quad \{\mathsf{T}_\eta\}_{\eta\in\Gamma_{\mathrm{gram}}}, \quad B_{\mathrm{ch}}, \quad \Theta_\Pi, \Theta^-_\Pi, \quad \overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X,$$

as in Definition 10.24 and Theorem 10.27. The protected primitives are $\Gamma_{\mathrm{gram}}$-graded only after applying $\overline{\Pi}^{\Theta}_{X,*}$. Their bracket is induced from the upstairs $\Gamma^{\mathrm{phys}}_X$-graded Hall bracket by translating the raw target degree

$$\Pi_X(c+c') = \Pi_X(c) + \Pi_X(c') + B(c,c')$$

through $\mathsf{T}_{B(c,c')}$, hence by shifting the displayed Gram degree by $-B(c,c')$. The datum includes the triple-coherence pentagon over the two-step Hall flag stacks, Jacobi compatibility of the transported bracket, the $S^1$-equivariant negative-cyclic lift

$$(b + uB_{\mathrm{C}})\Theta^-_\Pi = B^-_{\mathrm{ch}},$$

and compatibility with the Joyce–Upmeier orientation cocycle, the graded Frobenius identity, the completed Hopf pairing, and the reduced $E$-quotient. Equivalently, the seven obstruction classes

$$\mathfrak{o}^{\mathrm{top}}_\Theta, \quad \mathfrak{o}^{\mathrm{Hoch}}_\Theta, \quad \mathfrak{o}^{\mathrm{tri}}_\Theta, \quad \mathfrak{o}^{\mathrm{Jac}}_\Theta, \quad \mathfrak{o}_F, \quad \mathfrak{o}^{\mathrm{cyc}}_\Theta, \quad \mathfrak{o}^{\mathrm{rad}}_\Theta$$

of Theorem 10.27 vanish on every finite HN quotient and compatibly in the inverse limit, with $\mathfrak{o}^{\mathrm{top}}_\Theta$ enforcing finite-fibre descent and completed tensor/coproduct continuity, and with $\mathfrak{o}^{\mathrm{rad}}_\Theta$ read as the Lie-ideal and coideal pair. The finite-stage tests are the seven diagrams (NO1)–(NO7) of Proposition 10.28. Before this descent, the protected primitives of $\mathcal{F}_X$ are graded by the upstairs additive charge $c$, not by a Gram triple. The comparison with $\mathfrak{g}_{\Delta_5,\alpha(n,l,m)}$ is made after passing to $(n,l,m) = \Pi_X(c)$. $(D_X)$ also includes the full chiral Koszul source certificate of Definition 10.10. Thus, at every finite HN height $R$, one must supply a conilpotent chiral coalgebra $C_{X,R}$ attached to the hybrid source $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$, a finite source charge filtration $F^{\mathrm{co}}_{R,\bullet}$, the collision coproduct $\Delta^{\mathrm{ch}}_R$, a charge-preserving bar comparison

$$b_{X,R} : C_{X,R} \xrightarrow{\simeq} \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}),$$

and a chiral cobar quasi-isomorphism

$$\Theta_{\mathrm{Kos},R} : \Omega^{\mathrm{ch}}_E C_{X,R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E,\le R},$$

compatibly in $R$. In the inverse limit this gives

$$b_X : C_X \xrightarrow{\simeq} \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_X), \qquad \Theta_{\mathrm{Kos}} : \Omega^{\mathrm{ch}}_E C_X \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E}.$$

The target-internal counit $\Omega_E^{\mathrm{ch}} \bar{B}_E^{\mathrm{ch}}(\mathsf{Den}_{\Delta_5,E}) \to \mathsf{Den}_{\Delta_5,E}$ does not supply $C_X$, $b_X$, or $\Theta_{\mathrm{Kos}}$. These are source-side compact data. Their finite residual is the vanishing of $\mathfrak{D}_{\mathrm{Kos},R}$ in every finite HN quotient, including $\mathfrak{v}_R^{\mathrm{hyb}}$, the assertion that the finite hybrid source has been built with all correspondences, quotient, and integration laws, $\mathfrak{v}_R^{\Delta,\mathrm{ch}}$ and $\mathfrak{v}_{R'R}^{C,\mathrm{tr}}$, the finite collision-coproduct and transition-continuity assertions, and $\mathfrak{v}_R^{\mathrm{sep}}$, the assertion that the source coalgebra, filtration, collision coproduct, bar comparison, and primitive restriction were supplied independently of the target counit.

$(R_X)$ *Primitive recognition.* This entry is a primitive recognition certificate $\Pi_{\mathrm{rec}}$ for

$$P_X^{\Pi} := H^0 \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X).$$

This certificate is not a determinant datum and is not the Gritsenko–Nikulin target itself. It consists of a source-side Gram-descended Lie superalgebra structure on $P_X^{\Pi}$, the parity split of each homogeneous primitive piece, compact representatives for the real and imaginary simple-root fibres of the Gritsenko–Nikulin presentation, verification of the Borcherds–Kac defining relations in the protected Hall bracket, the completed Hopf pairing, generation by the simple primitives and the Cartan, a no-extra-relations theorem, completed PBW compatibility, and identification of the source Hopf radical with the standard Gritsenko–Nikulin/Kac radical after completion.

Theorem 10.45 *(Presentation-level primitive recognition certificate).* Let $\mathscr{B}_{\Delta_5} = (H, I, p, \alpha_i, (\ ,\ ))$ be the imported Gritsenko–Nikulin Borcherds–Cartan superdatum for the automorphic correction. Thus

$$H = \Lambda_{II}^{2,1} \otimes \mathbb{C}, \qquad \{\delta_1, \delta_2, \delta_3\} \subset \Lambda_{II}^{2,1} \subset H,$$

the three $\delta_i$ are even real simple roots, and

$$((\delta_i, \delta_j))_{1\le i,j\le 3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The remaining fibres of $I \to H^*$ are the imaginary simple-root fibres with the Gritsenko–Nikulin parity split $\mu_{\bar{0}}(a), \mu_{\bar{1}}(a)$ of Definition 10.51. The degree map is

$$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2} = n\delta_1 + m\delta_2 + (n + m - l)\delta_3.$$

Let $\widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5})$ be the completed free Lie superalgebra generated by the even Cartan $H$ and by parity-homogeneous generators $e_i, f_i, i \in I$, with $|e_i| = |f_i| = p(i)$ and $\deg e_i = \alpha_i$, $\deg f_i = -\alpha_i$. Let $J_{\mathrm{BK}}$ be the closed Borcherds–Kac ideal generated by the Cartan, Chevalley, real Serre, orthogonality, and super sign relations. After quotienting by the standard radical of the completed invariant pairing one obtains the target denominator algebra

$$G(\mathscr{B}_{\Delta_5}) = \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5})/\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}} = \mathcal{A}_{\mathrm{den}} = \mathfrak{g}_{\Delta_5}.$$

Here $\mathrm{Rad}_{\mathrm{GN}}$ denotes the preimage in the completed presentation of the standard radical. This target-side statement is imported from the Gritsenko–Nikulin/Kac construction; it is not a compact Hall construction.

Suppose a compact Hall/factorization construction supplies a completed $\widehat{\Gamma}_X$-graded primitive Hall Lie superalgebra

$$\widetilde{P}_X = \prod_{\widehat{c}\in\widehat{\Gamma}_X} \widetilde{P}_{X,\widehat{c}}, \qquad [\widetilde{P}_{X,\widehat{c}}, \widetilde{P}_{X,\widehat{c}'}] \subset \widetilde{P}_{X,\widehat{c}\star\widehat{c}'},$$

and a completed Hopf pairing. Its normal-ordered Gram descent is

$$P_X^{\Pi,\mathrm{raw}} = H^0\,\mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X) = \overline{\Pi}^{\Theta}_{X,*}\widetilde{P}_X$$

and

$$P_X^{\Pi,\mathrm{red}} = P_X^{\Pi,\mathrm{raw}}/\overline{\mathrm{Rad}\langle\,\cdot\,,\,\cdot\,\rangle_X}.$$

Assume the following source-side presentation data are supplied. None of these data is inferred from the Borcherds product, from the determinant, or from signed dimensions alone.

(S1) *Charge and normal-ordered descent.* The chain-level descent $\overline{\Pi}^{\Theta}_{X,*}$ categorifies the normal-ordered homomorphism

$$\overline{\Pi}_X : \widehat{\Gamma}_X \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c,T) = \Pi_X(c) - T,$$

of Definition 6.1. For $\widehat{c} = (c,T)$ and $\widehat{c}' = (c',T')$,

$$\overline{\Pi}_X(\widehat{c}\star\widehat{c}') = \overline{\Pi}_X(\widehat{c}) + \overline{\Pi}_X(\widehat{c}'),$$

so the descended bracket is a $\Gamma_{\mathrm{gram}}$-graded bracket:

$$[\widetilde{P}_{X,\widehat{c}}, \widetilde{P}_{X,\widehat{c}'}] \subset \widetilde{P}_{X,\widehat{c}\star\widehat{c}'} \xrightarrow{\overline{\Pi}^{\Theta}_{X,*}} P^{\Pi,\mathrm{raw}}_{X,\overline{\Pi}_X(\widehat{c})+\overline{\Pi}_X(\widehat{c}')}.$$

Equivalently, the raw quadratic shadow $\Pi_X(c+c') = \Pi_X(c) + \Pi_X(c') + B(c,c')$ has been corrected by the $\Theta$-degree shift before comparison with the denominator algebra. Each descended homogeneous parity piece is finite dimensional. The construction of $(\Theta)$, the orientation datum (O), and the chain-level Frobenius compatibility is exactly the compact content of Theorem 10.27, Open Problem 10.23, and Open Problem 5.7.

(S2) *Cartan and root-degree matching.* There is an even Cartan identification

$$\iota_H : P^{\Pi,\mathrm{red}}_{X,0} \xrightarrow{\sim} \Lambda^{2,1}_{II} \otimes \mathbb{C}.$$

The degree map $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$ identifies the descended Gram grading with the Gritsenko–Nikulin root grading after applying $\overline{\Pi}^{\Theta}_{X,*}$. The upstairs Hall bracket remains $\widehat{\Gamma}_X$-graded; only the descended object is $\Gamma_{\mathrm{gram}}$-graded.

(S3) *Simple primitive representatives and parities.* For every simple-root index $i \in I$ of $\mathscr{B}_{\Delta_5}$, the compact source supplies parity-homogeneous primitive representatives

$$e_i^X \in P^{\Pi,\mathrm{red}}_{X,\alpha_i}, \qquad f_i^X \in P^{\Pi,\mathrm{red}}_{X,-\alpha_i}, \qquad h_i^X \in P^{\Pi,\mathrm{red}}_{X,0}$$

with $|e_i^X| = |f_i^X| = p(i)$ and $h_i^X = \iota_H^{-1}(\alpha_i)$. The representatives preserve the real/imaginary distinction, the multiplicity fibres, and the parity map. The real representatives for $\delta_1, \delta_2, \delta_3$ are even. For each imaginary simple root $a$, the supplied representatives have dimensions

$$\dim\langle e_i^X : \alpha_i = a,\ p(i) = 0\rangle = \mu_{\bar{0}}(a), \qquad \dim\langle e_i^X : \alpha_i = a,\ p(i) = 1\rangle = \mu_{\bar{1}}(a)$$

with the same convention for the negative representatives.

(S4) *Borcherds–Kac relations in the Hall bracket.* In the protected Hall/factorization bracket these representatives satisfy the Borcherds–Kac defining relations for $\mathscr{B}_{\Delta_5}$:

$$[h, h'] = 0, \qquad [h, e_i^X] = (h, \alpha_i) e_i^X, \qquad [h, f_i^X] = -(h, \alpha_i) f_i^X,$$
$$[e_i^X, f_j^X] = \delta_{ij} h_i^X,$$

the real Serre relations

$$(\operatorname{ad} e_i^X)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} e_j^X = (\operatorname{ad} f_i^X)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} f_j^X = 0$$

for real $i$ and $i \neq j$, the Borcherds orthogonality relations

$$(\alpha_i, \alpha_j) = 0 \quad \Longrightarrow \quad [e_i^X, e_j^X] = [f_i^X, f_j^X] = 0,$$

and the super sign rule. In the real rank-three subsystem $(\delta_i, \delta_j) = -2$ for $i \neq j$, so the real Serre exponent is 3; the first brackets $[e_i^X, e_j^X]$ are not forced to vanish. For the primitive isotropic simple roots

$$a_{ij} := \delta_i + \delta_j, \qquad i < j,$$

the source representatives $u_{ij,r}^{+,X} \in P_{X,a_{ij}}^{\Pi,\mathrm{red}}$ and $u_{ij,r}^{-,X} \in P_{X,-a_{ij}}^{\Pi,\mathrm{red}}$, $1 \leq r \leq 9$, satisfy the orthogonality and Chevalley zero relations with the two adjacent real roots,

$$[e_i^X, u_{ij,r}^{+,X}] = [e_j^X, u_{ij,r}^{+,X}] = [f_i^X, u_{ij,r}^{+,X}] = [f_j^X, u_{ij,r}^{+,X}] = 0,$$

with the analogous negative-root relations, and for the complementary root $\{i, j, k\} = \{1, 2, 3\}$ the real-string relations

$$(\operatorname{ad} e_k^X)^5 u_{ij,r}^{+,X} = 0, \qquad (\operatorname{ad} f_k^X)^5 u_{ij,r}^{-,X} = 0.$$

These are relation checks in the compact bracket. They are not consequences of the equality $\operatorname{sdim} = f(nm, l)$.

(S5) *Completed Hopf pairing and radical quotient.* The completed Hopf pairing on the positive and negative primitive halves is continuous for the Hall completion, homogeneous for the $\Gamma_{\mathrm{gram}}$-grading, invariant for the bracket and coproduct, and has closed homogeneous radical

$$\mathrm{Rad}_X = \{x \mid \langle x, y \rangle_X = 0 \text{ for all } y\}$$

which is a Lie ideal and a coideal. After quotienting by $\overline{\mathrm{Rad}_X}$, the source radical quotient is the same as the Gritsenko–Nikulin/Kac quotient by $\overline{\mathrm{Rad}_{\mathrm{GN}}}$. Equivalently, on every finite-height saturated set of roots the positive-negative pairing matrices have the same parity-block ranks as the target invariant pairing matrices, their kernels are carried by the transition maps, and the inverse-limit kernel is exactly $\mathrm{Rad}_X$. At every finite stage one has

$$[P_R^\Pi, \mathrm{Rad}_{X,R}] \subset \mathrm{Rad}_{X,R}, \qquad \Delta_R(\mathrm{Rad}_{X,R}) \subset \mathrm{Rad}_{X,R} \otimes P_R^\Pi + P_R^\Pi \otimes \mathrm{Rad}_{X,R},$$

and the completed statement is obtained by inverse limit and closure. The radical rank is computed from these matrices and kernel maps; it is not read from the signed determinant. Equality of ranks alone is not a Hopf quotient theorem without the Lie-ideal, coideal, closedness, and transition-compatibility assertions.

(S6) *No extra relations.* The Hall primitive algebra satisfies a no-extra-relations theorem: if

$$\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \longrightarrow P_X^{\Pi,\mathrm{red}}$$

is the continuous map determined by the displayed representatives, then

$$\ker \pi = \overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}.$$

Equivalently, the induced map $G(\mathscr{B}_{\Delta_5}) \to P_X^{\Pi,\mathrm{red}}$ is injective. This is required finite stage by finite stage: for every saturated finite root window $W$, the restricted map $\pi_W$ has kernel

$$\ker \pi_W = \big(\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}\big)_W$$

after projection to $W$, and these kernel identifications commute with the HN transition maps. A comparison of signed dimensions or a determinant identity is not this theorem.

(S7) *Generation.* $P_X^{\Pi,\mathrm{red}}$ is topologically generated by $P_{X,0}^{\Pi,\mathrm{red}}$ and the simple primitive representatives $e_i^X, f_i^X$.

(S8) *Completed PBW.* A completed PBW theorem holds for $P_X^{\Pi,\mathrm{red}}$: the PBW filtration on its enveloping algebra is transported to the PBW filtration on $U(\mathfrak{g}_{\Delta_5})$, the associated graded finite-height pieces agree with the target pieces, and the finite-height PBW bases form a closed inverse system in the chosen completion. Concretely, for every finite saturated window $W$, the associated graded map

$$\mathrm{gr}_{\mathrm{PBW}}\, U(P_X^{\Pi,\mathrm{red}})_W \longrightarrow \mathrm{gr}_{\mathrm{PBW}}\, U(\mathfrak{g}_{\Delta_5})_W$$

is an isomorphism, the transition maps preserve the PBW filtration strictly, and inverse limits are taken in the same HN topology as Definition 10.24. Target PBW does not transfer to the compact source until these finite-stage isomorphisms are supplied.

(S9) *Full parity dimensions.* The full root-space parity dimensions in $P_X^{\Pi,\mathrm{red}}$ agree with those of $G(\mathscr{B}_{\Delta_5})$, not only their differences. For every nonzero $\gamma$ with $\alpha(\gamma) \in \mathcal{R}$,

$$\dim P_{X,\gamma,\overline{p}}^{\Pi,\mathrm{red}} = d_{\gamma,\overline{p}}^{\mathrm{GN}}, \qquad d_{\gamma,\overline{p}}^{\mathrm{GN}} := \mathrm{mult}_{\overline{p}}(\alpha(\gamma)) \qquad (p = 0, 1).$$

Thus the source must match two integers in every root degree; equality of their difference is not enough. In the first timelike channel $\gamma = (1, 1, 1)$, Proposition 10.48 gives

$$d_{(1,1,1),\bar{0}}^{\mathrm{GN}} = 29, \qquad d_{(1,1,1),\bar{1}}^{\mathrm{GN}} = 93.$$

These are presentation-level target dimensions, not an inference from $29 - 93 = -64$. If $\alpha(\gamma)$ is not a target root and $\gamma \neq 0$, the corresponding source piece is zero after the radical quotient. On active positive Gram degrees this implies

$$\mathrm{sdim}\, P_{X,(n,l,m)}^{\Pi,\mathrm{red}} = f(nm, l).$$

Then

$$P_X^{\Pi,\mathrm{red}} \cong \mathcal{A}_{\mathrm{den}} = \mathfrak{g}_{\Delta_5}$$

as Gram-graded Lie superalgebras, and its Weyl–Kac–Borcherds denominator is $64^{-1}\Delta_5(2Z)$. If in addition the chiral Koszul source certificate of Definition 10.10 is supplied, then

$$\Omega_E^{\mathrm{ch}} C_X \simeq \mathsf{Den}_{\Delta_5,E}.$$

*Proof.* The simple primitive representatives define a continuous homomorphism

$$\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \longrightarrow P_X^{\Pi,\mathrm{red}}.$$

(S1) and (S2) make this a homomorphism of the descended Gram-graded completions. (S3) fixes the real and imaginary simple generators with their parities and multiplicity fibres. (S4) puts $J_{\mathrm{BK}}$ in the kernel. (S5) identifies the compact Hopf radical with the Gritsenko–Nikulin/Kac radical, so $\pi$ descends to

$$G(\mathscr{B}_{\Delta_5}) \longrightarrow P_X^{\Pi,\mathrm{red}}.$$

(S7) gives surjectivity. (S6) gives injectivity after the completed radical quotient. (S8) says that this finite-height identification survives passage to the inverse limit. (S9) rules out hidden zero-superdimension root spaces and cancelling ideals invisible to the determinant. Hence the descended map is an isomorphism of Gram-graded Lie superalgebras. The denominator statement is Theorem 9.2. The chiral statement is Proposition 10.13; it uses the supplied $C_X$, $b_X$, and $\Theta_{\mathrm{Kos}}$, not the target-internal bar–cobar counit. □

Proposition 10.46 *(Low-degree bracket information).* The denominator algebra gives the following target-side low-degree checks. With

$$\alpha(n,l,m) = n\delta_1 + m\delta_2 + (n+m-l)\delta_3,$$

the active positive rows through height three are

| $\mathrm{ht}\,\beta$ | $\beta$ | $\gamma(\beta)$ | $(\beta,\beta)$ | $\mathrm{smult}(\beta)$ |
|---|---|---|---|---|
| 1 | $\delta_1$ | $(1,1,0)$ | 2 | 1 |
| 1 | $\delta_2$ | $(0,1,1)$ | 2 | 1 |
| 1 | $\delta_3$ | $(0,-1,0)$ | 2 | 1 |
| 2 | $\delta_1+\delta_2$ | $(1,2,1)$ | 0 | 10 |
| 2 | $\delta_1+\delta_3$ | $(1,0,0)$ | 0 | 10 |
| 2 | $\delta_2+\delta_3$ | $(0,0,1)$ | 0 | 10 |
| 3 | $2\delta_1+\delta_2$ | $(2,3,1)$ | 2 | 1 |
| 3 | $\delta_1+2\delta_2$ | $(1,3,2)$ | 2 | 1 |
| 3 | $2\delta_1+\delta_3$ | $(2,1,0)$ | 2 | 1 |
| 3 | $\delta_1+2\delta_3$ | $(1,-1,0)$ | 2 | 1 |
| 3 | $2\delta_2+\delta_3$ | $(0,1,2)$ | 2 | 1 |
| 3 | $\delta_2+2\delta_3$ | $(0,-1,1)$ | 2 | 1 |
| 3 | $\delta_1+\delta_2+\delta_3$ | $(1,1,1)$ | $-6$ | $-64$. |

After choosing even Chevalley generators $e_i, f_i, h_i$ for the real simple roots and setting

$$A_{ij} = (\delta_i,\delta_j), \qquad E_{ij} := [e_i, e_j] \quad (i \neq j),$$

so $E_{ji} = -E_{ij}$, one has

$$[h_i,h_j] = 0, \qquad [h_i,e_j] = A_{ij}e_j, \qquad [h_i,f_j] = -A_{ij}f_j, \qquad [e_i,f_j] = \delta_{ij}h_i,$$

and

$$(\mathrm{ad}\,e_i)^3 e_j = 0 \quad (i \neq j).$$

If $\{i, j, k\} = \{1, 2, 3\}$, then

$$[h_i, E_{ij}] = [h_j, E_{ij}] = 0, \qquad [h_k, E_{ij}] = -4E_{ij},$$

$$[f_i, E_{ij}] = 2e_j, \qquad [f_j, E_{ij}] = -2e_i, \qquad [f_k, E_{ij}] = 0.$$

For ordered $i \neq j$, put $E_{iij} := [e_i, E_{ij}]$. Then

$$[e_i, E_{iij}] = 0, \qquad [f_i, E_{iij}] = 2E_{ij}, \qquad [f_j, E_{iij}] = 0.$$

For $i < j$, the isotropic primitive degree $a_{ij} := \delta_i + \delta_j$ has

$$\operatorname{smult}(a_{ij}) = 10 = 1 + 9 :$$

one even real-real commutator direction $E_{ij}$ and nine even Gritsenko–Nikulin isotropic simple directions

$$u_{ij,1}, \ldots, u_{ij,9} \in \mathfrak{g}_{a_{ij}}, \qquad \tau(ta_{ij}) = 9 \quad (t \geq 1).$$

The orthogonality relations give

$$[e_i, u_{ij,r}] = [e_j, u_{ij,r}] = 0, \qquad [f_i, u_{ij,r}] = [f_j, u_{ij,r}] = 0.$$

Thus the target splits the signed multiplicity 10 at $a_{ij}$ into one real-real commutator direction and nine isotropic simple directions. A compact source must realize this splitting by actual protected Hall primitives; the equality sdim = 10 alone does not choose the summands or the orthogonality constants. In degree $\delta_{123} := \delta_1 + \delta_2 + \delta_3$, the real bracketings

$$T_1 = [e_1, E_{23}], \qquad T_2 = [e_2, E_{31}], \qquad T_3 = [e_3, E_{12}]$$

satisfy $T_1 + T_2 + T_3 = 0$. The brackets $[e_k, u_{ij,r}] \in \mathfrak{g}_{\delta_{123}}$ for $\{i, j, k\} = \{1, 2, 3\}$ are not numerical constants until a basis and parity split of $\mathfrak{g}_{\delta_{123}}$ are chosen. Since $(\delta_k, a_{ij}) = -4$, the same Borcherds–Kac presentation also imposes the real-string relation

$$(\operatorname{ad} e_k)^5 u_{ij,r} = 0$$

and the analogous relation for the negative generators. The first bracket $[e_k, u_{ij,r}]$ is only the first visible map in this string. The product exponent alone gives only

$$\operatorname{smult}(\delta_{123}) = f(1, 1) = -64.$$

Let

$$d^{\mathrm{GN}}_{p,123} := \operatorname{mult}_{\overline{p}}(\delta_{123}), \qquad p = 0, 1.$$

Then

$$d^{\mathrm{GN}}_{0,123} - d^{\mathrm{GN}}_{1,123} = -64,$$

but the product exponent alone does not give the two integers $d^{\mathrm{GN}}_{0,123}$, $d^{\mathrm{GN}}_{1,123}$, the bracket coordinates, or the radical. Proposition 10.48 computes the target presentation values 29 and 93 from the Weyl-sum coefficient and the Borcherds–Kac presentation; the compact source must still reproduce them by

protected primitives and the Hopf pairing. The first unresolved compact test channel is therefore the following: produce parity-homogeneous compact primitives

$$v_1^{\bar{0}}, \dots, v_{29}^{\bar{0}}, \qquad v_1^{\bar{1}}, \dots, v_{93}^{\bar{1}},$$

in the descended degree $(1, 1, 1)$, compute the Hall constants

$$[e_k^X, u_{ij,r}^X] = \sum_{a=1}^{29} C_{k;ijr,a}^{\bar{0}} v_a^{\bar{0}}, \qquad \{i, j, k\} = \{1, 2, 3\},$$

the Jacobi relation $T_1^X + T_2^X + T_3^X = 0$, the real-string relations $(\operatorname{ad} e_k^X)^5 u_{ij,r}^X = 0$, the completed Hopf-pairing matrix between degrees $\delta_{123}$ and $-\delta_{123}$, with even block size 29 and odd block size 93, and the closed radical quotient of Theorem 10.45. The bracket above is even, but the odd basis is still load-bearing: it is part of the full parity-dimension, PBW, and radical comparison. This channel is the first place where the compact Hall source must supply more than the GN signed multiplicity. These are denominator-algebra constants. They are not compact Hall constants.

*Proof.* The table is obtained by writing $\beta = c_1\delta_1 + c_2\delta_2 + c_3\delta_3$ and solving

$$\gamma(\beta) = (c_1, c_1 + c_2 - c_3, c_2).$$

Then $(\beta, \beta) = -2(4nm - l^2)$, and $\operatorname{smult}(\beta) = f(nm, l)$ follows from Theorem 9.2. The displayed bracket constants are the normalized Chevalley, Jacobi and Serre constants for the symmetric Cartan matrix $A = ((\delta_i, \delta_j))$. The signs $E_{ji} = -E_{ij}$, $[f_i, E_{ij}] = 2e_j$, and $[f_j, E_{ij}] = -2e_i$ follow from the graded Jacobi identity and the normalization $[e_i, f_i] = h_i$. The relation $T_1 + T_2 + T_3 = 0$ is the ordinary Jacobi identity for the even real generators $e_1, e_2, e_3$. For $a_{ij} = \delta_i + \delta_j$ and $\{i, j, k\} = \{1, 2, 3\}$, one has $(\delta_k, a_{ij}) = -4$, hence the real Borcherds–Kac Serre exponent is $1 - 2(\delta_k, a_{ij})/(\delta_k, \delta_k) = 5$. The equality $\tau(ta_{ij}) = 9$ on primitive isotropic rays is the Gritsenko–Nikulin isotropic simple-root datum recorded above. No compact Hall structure constant is inferred: a compact computation would have to construct the relevant extension correspondences, protected primitive bases, orientation signs, the $\delta_{123}$ channel constants, and the radical quotient. □

Proposition 10.47 *(The determinant does not determine compact Hall constants).* The scalar OP square, the Borcherds product for $\Delta_5$, and the Gritsenko–Nikulin denominator algebra determine the visible signed exponents

$$\operatorname{smult}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{0}} - \dim \mathfrak{g}_{\alpha,\bar{1}}.$$

They do not determine a compact Hall bracket, compact primitive representatives, full source parity dimensions, the positive-negative Hopf pairing, its closed radical quotient, the no-extra-relations theorem, or completed PBW compatibility.

*Proof.* A $\Gamma_{\text{gram}}$-graded super vector space with the same signed dimensions gives the same determinant. Adding a cancelling pair $W_\gamma \oplus \Pi W_\gamma$ in any active degree leaves every signed exponent unchanged, while changing the full parity dimensions. Equipping the same graded vector space with the zero bracket also leaves the determinant unchanged, but it cannot be the denominator algebra because the target has nonzero real-root brackets, for example $[e_i, e_j] \neq 0$ for $i \neq j$, and the first timelike channel requires the 29|93 presentation split of Proposition 10.48. A compact Hall realization must therefore construct extension correspondences, orientation signs, protected primitive bases, Hall structure constants, pairing matrices, radical kernels, and PBW finite-window isomorphisms. The determinant is a necessary decategorified shadow of these data, not a source for them. □

Proposition 10.48 *(First timelike presentation count).* Let

$$\delta_{123} := \delta_1 + \delta_2 + \delta_3 = 2f_2 - f_3 + 2f_{-2}.$$

In the Gritsenko–Nikulin additive correction coefficient one has

$$m(\delta_{123}) = -93.$$

Consequently the target denominator algebra has full parity dimensions

$$\mathrm{mult}_{\bar{0}}(\delta_{123}) = 29, \qquad \mathrm{mult}_{\bar{1}}(\delta_{123}) = 93.$$

The 29 even directions are the two real-real-real Borcherds–Kac presentation words

$$T_1 = [e_1, [e_2, e_3]], \qquad T_2 = [e_2, [e_3, e_1]], \qquad T_3 = [e_3, [e_1, e_2]], \qquad T_1 + T_2 + T_3 = 0,$$

together with the 27 mixed real-isotropic words

$$[e_k, u_{ij,r}], \qquad \{i, j, k\} = \{1, 2, 3\}, \quad 1 \le r \le 9.$$

The 93 odd directions are the negative-norm imaginary simple generators in degree $\delta_{123}$. This is a target-side presentation count. It is not a compact Hall computation and is not deduced from the signed equality $29 - 93 = -64$.

*Proof.* The root $\delta_{123}$ corresponds in the notation of the Weyl-sum coefficient formula to

$$\delta_{123} = (3-1)f_2 - \frac{3-1}{2} f_3 + (3-1)f_{-2}.$$

Hence $m(\delta_{123}) = -64^{-1} c_\Delta(3,3,3)$. After removing the leading monomial from $D_5 = 64^{-1}\Delta_5$,

$$D_5 = q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma \in \Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)},$$

the finite coefficient calculation in $q$- and $s$-degree one gives

$$[qrs] \prod_{\gamma \in \Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = 93.$$

Explicitly, only the $q$-linear, $s$-linear, and $qs$-linear factors enter. With

$$A(r) = 1 - r^{-1}, \qquad B(r) = -(r^{-1} + 10 + r),$$

$$C(r) = -10r^{-2} + 64r^{-1} - 108 + 64r - 10r^2,$$

one has

$$[qrs] \prod_{\gamma \in \Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = [r]\, A(r)\big(B(r)^2 + C(r)\big)$$

and

$$A(r)\big(B(r)^2 + C(r)\big) = 9r^{-3} - 93r^{-2} + 90r^{-1} - 90 + 93r - 9r^2.$$

Equivalently $c_\Delta(3,3,3) = 64 \cdot 93$, so $m(\delta_{123}) = -93$. Since this coefficient is negative, the Gritsenko–Nikulin parity convention makes these 93 simple generators odd.

It remains to count the lower presentation words in the same total degree. The multidegree $(1,1,1)$ part of the free Lie algebra on the three even real generators has dimension 2; the three displayed bracketings satisfy exactly the Jacobi relation $T_1 + T_2 + T_3 = 0$. For each primitive isotropic root $a_{ij} = \delta_i + \delta_j$, the Gritsenko–Nikulin isotropic multiplicity is $\tau(a_{ij}) = 9$, and the complementary real root $\delta_k$ has $(\delta_k, a_{ij}) = -4$. Thus the first mixed brackets $[e_k, u_{ij,r}]$ are allowed presentation words; the orthogonality relations kill only the adjacent brackets with $\delta_i$ and $\delta_j$, and the real-string relation occurs first at $(\operatorname{ad} e_k)^5 u_{ij,r}$. There are $3 \cdot 9 = 27$ such even words. No other simple-root decomposition has total degree $\delta_{123}$. The Borcherds–Kac presentation and PBW therefore give $2 + 27 = 29$ even and 93 odd target directions in this degree. Their signed difference is

$$29 - 93 = -64 = f(1,1),$$

which checks the denominator exponent but does not determine the compact source parity split, bracket constants, pairing matrix, or radical quotient. □

Proposition 10.49 *(First saturated primitive-recognition window).* Let

$$W_{\leq 3} := \left\{\beta = c_1\delta_1 + c_2\delta_2 + c_3\delta_3 \in \mathcal{R}_+ \ : \ c_i \geq 0, \quad 1 \leq c_1 + c_2 + c_3 \leq 3\right\}.$$

This is the first downward-saturated finite target window containing the real simple roots, the primitive isotropic simple-root fibres, and the first timelike degree. In $G(\mathscr{B}_{\Delta_5})$ the full parity dimensions in this window are

| $\beta$ | $\mathrm{mult}_{\bar{0}}(\beta)$ | $\mathrm{mult}_{\bar{1}}(\beta)$ |
|---|---|---|
| $\delta_i$ $(1 \leq i \leq 3)$ | 1 | 0 |
| $\delta_i + \delta_j$ $(i < j)$ | 10 | 0 |
| $2\delta_i + \delta_j$ $(i \neq j)$ | 1 | 0 |
| $\delta_{123}$ | 29 | 93. |

Consequently a compact source passes primitive recognition on $W_{\leq 3}$ only after supplying the following finite certificate: parity-homogeneous representatives with the displayed dimensions; the Chevalley, real-Serre, orthogonality, and super-sign relations restricted to $W_{\leq 3}$; the Jacobi relation

$$T_1^X + T_2^X + T_3^X = 0;$$

the 27 mixed maps $[e_k^X, u_{ij,r}^X]$ into the even $\delta_{123}$-block; positive-negative Hopf-pairing matrices in every degree of $W_{\leq 3}$ with closed radical kernels stable under the finite transition maps; the finite kernel equality

$$\ker \pi_{W_{\leq 3}} = \left(\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}\right)_{W_{\leq 3}};$$

and the associated-graded PBW isomorphism on the truncated enveloping algebras. If this finite certificate is supplied, the restricted map

$$G(\mathscr{B}_{\Delta_5})_{W_{\leq 3}} \longrightarrow P^{\Pi,\mathrm{red}}_{X,W_{\leq 3}}$$

is an isomorphism of finite Gram-graded Lie superalgebras. Without the kernel equality, the Hopf-radical ideal/coideal assertion, and the PBW associated-graded isomorphism, equality of the displayed dimensions is not recognition.

The next coefficient checks already lie outside this window. The exact finite product computation in `compute/verify_square_root.py` gives

$$[q^2r^2s]\frac{D_5}{(qrs)^{1/2}} = [qr^2s^2]\frac{D_5}{(qrs)^{1/2}} = [qs]\frac{D_5}{(qrs)^{1/2}} = -90,$$

and

$$[q^2r^2s^2]\frac{D_5}{(qrs)^{1/2}} = 540.$$

Thus the Gritsenko–Nikulin additive coefficients are

$$m(2\delta_1 + \delta_2 + \delta_3) = m(\delta_1 + 2\delta_2 + \delta_3) = m(\delta_1 + \delta_2 + 2\delta_3) = 90,$$

$$m(2\delta_{123}) = -540.$$

These are target simple-root coefficients. They do not count the lower PBW words in those degrees and do not supply compact Hall representatives, pairing matrices, radicals, or no-extra-relations data.

The first obstruction to extending the displayed finite recognition statement from $W_{\leq 3}$ to height four is already numerical. The full signed target root supermultiplicities in the three height-four timelike degrees are

$$\mathrm{smult}(2\delta_1 + \delta_2 + \delta_3) = \mathrm{smult}(\delta_1 + 2\delta_2 + \delta_3) = \mathrm{smult}(\delta_1 + \delta_2 + 2\delta_3) = 108,$$

whereas the new negative-norm simple-root coefficients in those same degrees are 90. Thus each of these degrees carries a signed height-four residual

$$108 - 90 = 18$$

coming from lower presentation words after the Borcherds–Kac relations and radical quotient are imposed. This number is a signed residual, not a count of compact PBW basis vectors. It says precisely that the height-four window cannot be certified by appending the additive coefficients alone.

The same phenomenon is visible on the doubled isotropic boundary rays: for $i < j$,

$$\mathrm{smult}(2\delta_i + 2\delta_j) = 10, \qquad \tau(2(\delta_i + \delta_j)) = 9,$$

so the signed non-simple residual is 1 on each doubled isotropic ray. These doubled-ray checks test the isotropic multiplicity convention $\tau(t(\delta_i + \delta_j)) = 9$ against the product exponents; they do not identify compact primitives.

The height-four real-string obstruction is also explicit. If $w_s$, $1 \leq s \leq 93$, denotes an odd negative-norm simple generator in degree $\delta_{123}$, then

$$(\delta_i, \delta_{123}) = -2, \qquad 1 - 2(\delta_i, \delta_{123})/(\delta_i, \delta_i) = 3,$$

and the target presentation contains

$$(\mathrm{ad}\, e_i)^3 w_s = 0, \qquad (\mathrm{ad}\, f_i)^3 w_s^- = 0.$$

The first brackets $[e_i, w_s]$ land in the three height-four timelike degrees above and are odd. A height-four compact recognition certificate must therefore supply those odd maps, the even simple generators of multiplicity 90, the lower-word relations producing the signed residual 18, the positive-negative pairing matrices, the radical kernels, and PBW associated-graded compatibility. Dimension equality alone still does not prove recognition.

*Proof.* The table is exactly the target data of Propositions 10.46 and 10.48: the three real simple roots are even one-dimensional, each primitive isotropic degree has one real-real commutator and nine even isotropic simple directions, the six height-three real roots are even one-dimensional, and $\delta_{123}$ has presentation dimensions 29|93. The finite certificate is the restriction of clauses (S3)–(S9) of Theorem 10.45 to the downward-saturated window $W_{\leq 3}$. The same proof as the theorem then gives the restricted isomorphism. The height-four additive coefficients are computed from the monic Borcherds product and converted to Gritsenko–Nikulin $m(a)$ by the coefficient formula $m(a) = -64^{-1}c_\Delta(n,l,m)$ of [30, Theorem 2.1]; the script performs this conversion with $\alpha(n,l,m) = n\delta_1 + m\delta_2 + (n+m-l)\delta_3$. The full signed height-four target multiplicities are read from the same denominator theorem:

$$\mathrm{smult}(\beta) = f(nm,l) \quad \text{for} \quad \gamma(\beta) = (n,l,m).$$

For $\beta = 2\delta_1 + \delta_2 + \delta_3, \delta_1 + 2\delta_2 + \delta_3, \delta_1 + \delta_2 + 2\delta_3$ this gives respectively $(n,l,m) = (2,2,1), (1,2,2), (1,0,1)$, hence the coefficients $f(2,2) = 108, f(2,2) = 108, f(1,0) = 108$. Subtracting the additive coefficients 90 gives the displayed signed residual 18. For the doubled isotropic degrees $2(\delta_i + \delta_j)$, the denominator exponent is $f(4,4) = f(0,0) = 10$ in the three symmetric chambers, while the isotropic simple-root multiplicity is the Gritsenko–Nikulin constant $\tau(2(\delta_i + \delta_j)) = 9$. Finally, the real-string exponents follow directly from the Gram matrix: $(\delta_i, \delta_{123}) = -2$, $(\delta_i, \delta_i) = 2$, and therefore the exponent is 3. All displayed numerical checks are asserted in `compute/verify_square_root.py`. □

*Open Problem* 10.50 *(State-side construction of $D_0$ and the Dirac–Igusa certificate).* Construct first the $D_0$-certificate of Definition 10.40, and then the full certificate $\mathfrak{K}_X$ of Definition 10.44. The exact open obligations are:

(open $D_0$-state) construct, for every $R \in \mathcal{R}$, the finite stage $D_{0,R}$ on $\widehat{\Gamma}_R$, equivalently the finite Hall–Dirac atlas $\mathfrak{A}_R$ of Proposition 10.42, including finite-type semistable stacks, extension correspondences, vanishing cycles, protected integration, and successor transition maps with $\mathfrak{o}^{\mathrm{atlas}}_{R^+R} = \mathfrak{o}^{\mathrm{tr}}_{R^+R} = 0$, whose composites give the non-successor transitions and whose inverse limit is $\widehat{\mathcal{H}}^{or}_{\sigma,S}$;

(open $D_0$-quotient) compute and trivialize the reduced orientation and quotient obstruction tuple $\mathfrak{o}^{or}_{R,C}$ for every stable charge stratum and finite stabilizer appearing at height $R$, including the zero $H^1(BE[N];\mathbb{F}_2)$-linearization characters after finite gerbe null-trivialization, compatibly in $R$;

(open $D_0$-observable) construct the chain-level $K3$-to-$E$ specialization, the projection-finite chiral shadow, and the hybrid wrapped sector $\mathcal{F}^{D_0}_X$ that recovers positive elliptic degree before quotienting by translation in $E$, together with a source-to-target finite comparison map satisfying $\mathfrak{o}^{obs}_R = 0$ on $\Gamma^{\mathrm{test}}_R$;

(open $D_0$-descent) construct the finite-topology, Hochschild, triple-coherence, Jacobi, Frobenius, negative-cyclic, and Hopf-radical ideal/coideal cochains $\Theta_{\Pi,R}$, $\Theta^-_{\Pi,R}$ with

$$\mathfrak{o}^{\mathrm{top}}_{\Theta,R^+R} = \mathfrak{o}^{\mathrm{Hoch}}_{\Theta,R} = \mathfrak{o}^{\mathrm{tri}}_{\Theta,R} = \mathfrak{o}^{\mathrm{Jac}}_{\Theta,R} = \mathfrak{o}_{F,R} = \mathfrak{o}^{\mathrm{cyc}}_{\Theta,R} = \mathfrak{o}^{\mathrm{rad}}_{\Theta,R} = 0,$$

giving exact compatibility with the normal-ordered homomorphism $\overline{\Pi}_X$ on primitive brackets, equivalently the finite diagrams (NO1)–(NO7) of Proposition 10.28, with compatibility against the Hopf radical quotients at every transition;

(open $D_0$-Pfaffian) construct the Pfaffian line and finite first-order operator/algebra $\mathfrak{D}_{X,R}$ on the reduced compact quotient $\mathfrak{Q}_R = (\mathfrak{M}_R/E)^{\mathrm{red}}$, compatible under $R^+ \to R$, including a trivialization of the Pfaffian square-root torsor in $\mathrm{Pic}(\mathfrak{Q}_R)[2]$, with compact parity lifts $\ell_R(\gamma) \in \mathscr{P}_R(\gamma)$ on every $\gamma \in \Gamma^{\mathrm{test}}_R$ satisfying

$$\mathfrak{o}^{\mathrm{supp}}_{R,\gamma} = 0, \qquad \mathfrak{o}^{\mathrm{par}}_{R,\gamma} = 0,$$

including absence of actual primitive support in target-zero degrees, coverage of every nonzero Borcherds target degree in the exhausting windows $\Gamma_{\Delta,R}$, finite Pfaffian square $\mathscr{L}_{\mathrm{Pf},R}^{\otimes 2} \simeq \mathscr{L}_{\det,R}$, and normalized leading coefficient $\mathfrak{o}_R^{\mathrm{lead}} = 0$;

(open full certificate) construct the strong orientation monodromy and the chiral Koszul source certificate

$$\mathfrak{K}_{X,R}^{\mathrm{Kos}} = (\mathcal{F}_{X,\sigma,S,\leq R}^{\mathrm{hyb}}, C_{X,R}, F_{R,\bullet}^{\mathrm{co}}, \Delta_R^{\mathrm{ch}}, b_{X,R}, \Theta_{\mathrm{Kos},R})$$

of Definition 10.10, with

$$\mathfrak{O}_{\mathrm{Kos},R} = 0$$

compatibly in $R$, including $\mathfrak{O}_{\mathrm{hyb},R} = 0$ for the hybrid source. Equivalently, construct the full finite local/wrapped correspondence atlas, the quotient-after-correspondence functor, protected integration, the conilpotent compact source coalgebra, the finite conilpotent filtration, the source collision coproduct with finite coassociativity, counit, and transition-continuity diagrams, the bar comparison to $\bar{B}_E^{\mathrm{ch}}(\mathcal{F}_X^{\mathrm{hyb}})$, and the cobar quasi-isomorphism to $\mathsf{Den}_{\Delta_s,E}$, rather than using the target-internal bar–cobar counit. The residual components $\mathfrak{o}_R^{\mathrm{hyb}}$ and $\mathfrak{o}_R^{\mathrm{sep}}$ record, respectively, the finite hybrid construction and this source/target separation. Then construct the compact Hall bracket constants, the simple primitive representatives, the Borcherds–Kac relation verification, the full compact parity dimensions, the completed Hopf pairing with its closed radical quotient, the no-extra-relations theorem, and completed PBW compatibility of Theorem 10.45. The first finite bracket test is the $\delta_1 + \delta_2 + \delta_3$ channel of Propositions 10.46 and 10.48: the signed dimension $-64$ must be refined to compact $29|93$ bases, parity split, structure constants, Jacobi/Serre checks, pairing matrices, and radical quotient.

The proved lattice normal-ordering theorem and the imported denominator algebra are inputs to this problem. The scalar square is an orientation-forgetful check. None of the three constructs the compact state-side certificate.

*Definition* 10.51 *(Imaginary simple roots).* For $a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0}\mathcal{P}_{II}$ set

$$\mu_{\bar{0}}(a) = \begin{cases} m(a), & (a,a) < 0 \text{ and } m(a) > 0, \\ \tau(a), & (a,a) = 0 \text{ and } \tau(a) > 0, \\ 0, & \text{otherwise}, \end{cases}$$

and

$$\mu_{\bar{1}}(a) = \begin{cases} -m(a), & (a,a) < 0 \text{ and } m(a) < 0, \\ -\tau(a), & (a,a) = 0 \text{ and } \tau(a) < 0, \\ 0, & \text{otherwise}. \end{cases}$$

This sign split is the Gritsenko–Nikulin BKM parity datum. The product statement alone fixes only a Grothendieck representative for the signed exponent. Define multisets by multiplicity:

$$a \in \Delta_{\bar{0}}^{im} \quad \text{with multiplicity } \mu_{\bar{0}}(a),$$

$$a \in \Delta_{\bar{1}}^{im} \quad \text{with multiplicity } \mu_{\bar{1}}(a).$$

Put

$$\Delta^{im} = \Delta_{\bar{0}}^{im} \cup \Delta_{\bar{1}}^{im}.$$

$$\mathcal{O}^{im}_{\mathrm{den}} = \bigoplus_{a\in\Delta^{im}} \left(\mathfrak{g}_{a,\bar{0}} \ominus \mathfrak{g}_{a,\bar{1}}\right).$$

This is notation inside the denominator algebra; it is not a BPS state space.

The real simple roots are

$$\Delta^{re,\mathrm{simp}}_{\bar{0}} = \Delta^{re}_{\mathrm{simp}} = \mathcal{P}_{II,prim} = \{\delta_1, \delta_2, \delta_3\}, \qquad \Delta^{re,\mathrm{simp}}_{\bar{1}} = \varnothing.$$

Put

$$\Delta_{\bar{p}} = \Delta^{re,\mathrm{simp}}_{\bar{p}} \cup \Delta^{im}_{\bar{p}} \qquad (p = 0, 1), \qquad \Delta = \Delta_{\bar{0}} \cup \Delta_{\bar{1}}.$$

The Cartan data satisfy $(\alpha, \alpha) > 0$ for $\alpha \in \Delta^{re}$, $(\alpha, \alpha) \leq 0$ for $\alpha \in \Delta^{im}$, and $(\alpha, \alpha') \leq 0$ for all distinct $\alpha, \alpha' \in \Delta$. If $(\alpha, \alpha) > 0$, then

$$2\frac{(\alpha, \alpha')}{(\alpha, \alpha)} \in \mathbb{Z},$$

since $(\alpha, \alpha) = 2$ and $\Lambda^{2,\mathrm{I}}_{II} = \mathbb{Z}\delta_1 + \mathbb{Z}\delta_2 + \mathbb{Z}\delta_3$.

*Definition* 10.52 *(Generators and relations).* Let $I$ be an index set equipped with a parity map $p : I \to \mathbb{Z}/2\mathbb{Z}$ and a root map $i \mapsto \alpha_i \in \Delta$, with fibres over an imaginary root of cardinality $\mu_{\bar{p}}(a)$ in parity $p$. If $i \neq j$ and $\alpha_i = \alpha_j$, the two indices are distinct simple roots.

$$\mathrm{ch} : I \to \Lambda^{2,\mathrm{I}}_{II}, \qquad i \mapsto \alpha_i.$$

The superalgebra $\mathfrak{g}_{\Delta_5}$ is generated by

$$h_i,\ e_i,\ f_i, \qquad i \in I.$$

The Cartan subalgebra is identified with $\Lambda^{2,\mathrm{I}}_{II} \otimes \mathbb{C}$ by $h_i \mapsto \alpha_i$. Relations:

- $[h_i, e_j] = (\alpha_i, \alpha_j)e_j$ and $[h_i, f_j] = -(\alpha_i, \alpha_j)f_j$
- $[e_i, f_j] = \delta_{ij}h_i$
- if $\alpha_i \in \Delta^{re}_{\mathrm{simp}}$ and $i \neq j$, then

$$(\mathrm{ad}\, e_i)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} e_j = (\mathrm{ad}\, f_i)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} f_j = 0$$

- if $(\alpha_i, \alpha_j) = 0$, then $[e_i, e_j] = [f_i, f_j] = 0$

The grading is by $\Lambda^{2,\mathrm{I}}_{II}$. Put

$$Q_+ = \sum_{\alpha\in\Delta} \mathbb{Z}_{\geq 0}\alpha \subset \Lambda^{2,\mathrm{I}}_{II}.$$

The triangular decomposition is

$$\mathfrak{g}_{\Delta_5} = \left(\bigoplus_{\alpha\in Q_+\setminus\{0\}} \mathfrak{g}_\alpha\right) \oplus (\Lambda^{2,\mathrm{I}}_{II} \otimes \mathbb{C}) \oplus \left(\bigoplus_{\alpha\in Q_+\setminus\{0\}} \mathfrak{g}_{-\alpha}\right).$$

For $i \in I$,

$$\deg e_i = \alpha_i, \qquad \deg f_i = -\alpha_i, \qquad \deg h_i = 0.$$

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}.$$

$$\mathfrak{g}_0 = \Lambda^{2,\mathrm{I}}_{II} \otimes \mathbb{C}.$$

The positive and negative roots are

$$\mathcal{R}_+ = \{\alpha \in Q_+ \setminus \{0\} \mid \mathfrak{g}_\alpha \neq 0\}, \qquad \mathcal{R}_- = -\mathcal{R}_+.$$

For every root $\alpha$, set

$$\mathrm{mult}_{\bar{0}}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{0}}, \qquad \mathrm{mult}_{\bar{1}}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{1}}.$$

The signed root superdimension is

$$\mathrm{smult}(\alpha) = \mathrm{mult}_{\bar{0}}(\alpha) - \mathrm{mult}_{\bar{1}}(\alpha).$$

The Weyl–Kac–Borcherds denominator is

$$\begin{aligned}\Phi_{\Delta_5}(z) &= \sum_{w \in W^{(2)}(\Lambda_{II}^{2,1})} \det(w) \left( e^{-2\pi i (w(\rho),z)} - \sum_{a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0}\mathcal{P}_{II}} m(a) e^{-2\pi i (w(\rho+a),z)} \right) \\ &= e^{-2\pi i(\rho,z)} \prod_{\alpha \in \mathcal{R}_+} \left(1 - e^{-2\pi i(\alpha,z)}\right)^{\mathrm{smult}(\alpha)}.\end{aligned}$$

$$\Omega(\Lambda_{II}^{2,1}) = \Lambda_{II}^{2,1} \otimes \mathbb{R} + iC(\Lambda_{II}^{2,1})_+.$$

Theorem 10.53 *(Denominator identity).*

$$\frac{1}{64}\Delta_5(2Z) = \Phi_{\Delta_5}(z).$$

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = \frac{1}{64}\Delta_5(2Z).$$

*Proof.*

$$\frac{1}{64}\Delta_5(Z) = \sum_{w \in W^{(2)}(\Lambda_{II}^{2,1})} \det(w) \left( e^{-\pi i (w(\rho),z)} - \sum_{a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0}\mathcal{P}_{II}} m(a) e^{-\pi i (w(\rho+a),z)} \right).$$

$$Z \mapsto 2Z: \qquad e^{-\pi i (w(\rho+a),z)} \mapsto e^{-2\pi i (w(\rho+a),z)}.$$

Weyl–Kac–Borcherds:

$$e^{-2\pi i(\rho,z)} \prod_{\alpha \in \mathcal{R}_+} \left(1 - e^{-2\pi i(\alpha,z)}\right)^{\mathrm{smult}(\alpha)}.$$

□

Proposition 10.54 *(Gram-index cone and denominator cone).*

$$\alpha(\Gamma_{\mathrm{act}}) = \{\alpha(n,l,m) \mid (n,l,m) \in \Gamma_{\mathrm{eff}},\ f(nm,l) \neq 0\} \subset \Lambda_{II}^{2,1}.$$

$$\alpha(\Gamma_{\mathrm{act}}) \subset \mathcal{R}_+^{\mathrm{vis}} := \{\alpha \in \mathcal{R}_+ \mid \mathrm{smult}(\alpha) \neq 0\}.$$

$$\mathrm{smult}(\alpha(n,l,m)) = \dim \mathfrak{g}_{\alpha(n,l,m),\bar{0}} - \dim \mathfrak{g}_{\alpha(n,l,m),\bar{1}} = f(nm,l) \qquad ((n,l,m) \in \Gamma_{\mathrm{act}}).$$

*Proof.*

$$\alpha : \Gamma_{\mathrm{ind}} \to \Lambda^{2,1}_{II}, \qquad (n,l,m) \mapsto 2nf_2 - lf_3 + 2mf_{-2}.$$

$$q^n r^l s^m = \exp(-\pi i(\alpha(n,l,m), z))$$

$$Z \mapsto 2Z.$$

$$\frac{1}{64}\Delta_5(2Z) = e^{-2\pi i(\rho,z)} \prod_{\alpha\in\mathcal{R}_+} \left(1 - e^{-2\pi i(\alpha,z)}\right)^{\mathrm{smult}(\alpha)}.$$

$$\frac{1}{64}\mathcal{D}_X(2Z) = e^{-2\pi i(\rho,z)} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} \left(1 - e^{-2\pi i(\alpha(n,l,m),z)}\right)^{f(nm,l)}.$$

The triples in $\Gamma_{\mathrm{eff}} \setminus \Gamma_{\mathrm{act}}$ have exponent zero and do not contribute visible product factors. The denominator product alone does not prove absence of root spaces with zero signed superdimension.

$$\mathcal{D}_X = \Delta_5 \quad \Longrightarrow \quad \mathrm{smult}(\alpha(n,l,m)) = f(nm,l) \quad ((n,l,m) \in \Gamma_{\mathrm{act}}).$$

□

Proposition 10.55 *(What the denominator determines).* The identity

$$\frac{1}{64}\Delta_5(2Z) = e^{-2\pi i(\rho,z)} \prod_{\alpha\in\mathcal{R}_+} \left(1 - e^{-2\pi i(\alpha,z)}\right)^{\mathrm{smult}(\alpha)}$$

determines the signed integers $\mathrm{smult}(\alpha)$, once the Gritsenko–Nikulin root set and positive root monoid have been fixed. It does not determine invisible zero-superdimension root spaces, the two parity dimensions

$$\mathrm{mult}_{\bar{0}}(\alpha), \qquad \mathrm{mult}_{\bar{1}}(\alpha),$$

and it does not determine the structure constants

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \to \mathfrak{g}_{\alpha+\beta}.$$

It also does not determine simple primitive representatives, the no-extra-relations statement, completed PBW compatibility, or the closed radical of the positive-negative Hopf pairing. Thus signed root multiplicity is necessary input for primitive recognition, but it is not a recognition certificate.

*Proof.* Write $X^\alpha = e^{-2\pi i(\alpha,z)}$. In the completed positive root monoid algebra,

$$\log \prod_{\alpha\in\mathcal{R}_+} (1 - X^\alpha)^{\mathrm{smult}(\alpha)} = -\sum_{\alpha\in\mathcal{R}_+} \sum_{k\geq 1} \frac{\mathrm{smult}(\alpha)}{k} X^{k\alpha}.$$

Thus a logarithmic coefficient mixes divisor terms $k\alpha = \lambda$. The exponents $\mathrm{smult}(\alpha)$ are recovered recursively by height, equivalently by Mobius inversion on each primitive ray. The recovered exponent is only

$$\mathrm{smult}(\alpha) = \mathrm{mult}_{\bar{0}}(\alpha) - \mathrm{mult}_{\bar{1}}(\alpha).$$

The difference $a - b$ does not determine the pair $(a, b)$, and no bilinear bracket, PBW filtration, or Hopf-pairing radical appears in the product. □

Proposition 10.56 *(What the determinant forgets).* Let

$$\mathcal{B} = \bigoplus_{\gamma \in \Gamma_{\mathrm{eff}}} \mathcal{B}_\gamma$$

be any charge-graded super vector space with finite-dimensional homogeneous pieces and

$$\operatorname{sdim} \mathcal{B}_{(n,l,m)} = f(nm, l).$$

Then its Gram-graded Euler/Fock determinant is

$$64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma \in \Gamma_{\mathrm{eff}}} (1 - x^\gamma)^{\operatorname{sdim} \mathcal{B}_\gamma} = \Delta_5(Z).$$

If $\mathcal{B}$ carries a bracket compatible with charge addition, a parity split, simple-root representatives, or a Hopf pairing, the determinant forgets those data and remembers the class of $\mathcal{B}$ in

$$K_0(\Gamma_{\mathrm{eff}}\text{-graded finite super vector spaces}).$$

Thus the determinant of $\mathcal{A}_{\mathrm{den}} = \mathfrak{g}_{\Delta_5}$ is the Grothendieck shadow of its denominator bracket. The Borcherds denominator bracket exists inside $\mathcal{A}_{\mathrm{den}}$ by the Gritsenko–Nikulin construction, but the determinant sees only its charge-graded Grothendieck class.

*Proof.* The Fock determinant lemma gives

$$\operatorname{sdet}_{\mathcal{B}}(1 - x) = \prod_{\gamma \in \Gamma_{\mathrm{eff}}} (1 - x^\gamma)^{\operatorname{sdim} \mathcal{B}_\gamma}.$$

The hypothesis $\operatorname{sdim} \mathcal{B}_{(n,l,m)} = f(nm, l)$ identifies this product with the Borcherds product for $64^{-1}\Delta_5$. Multiplying by the vacuum factor gives $\Delta_5$. A bracket

$$[\mathcal{B}_\gamma, \mathcal{B}_{\gamma'}] \to \mathcal{B}_{\gamma+\gamma'}$$

does not occur in the determinant formula. Neither do simple primitive representatives, parity dimensions, relation constants, completed PBW data, or a Hopf-pairing radical. The determinant is therefore invariant under replacing $\mathcal{B}$ by any other graded super vector space with the same Grothendieck class. In particular the zero bracket on such a space gives the same determinant and the wrong Lie algebra, and adjoining a root-graded cancelling pair $M \oplus \Pi M$ changes possible brackets and radicals without changing the signed determinant. □

## 11 The scalar square

Unitarity is used to localize the right-moving K3 supertrace before the index is taken; the index itself is a charge-graded virtual object that no longer requires Hilbert positivity, so the OP/DT scalar series below records charge-graded coefficients of a virtual character, not dimensions of a positive Hilbert space.

$$Z^X_{\mathrm{DT}}(q, t, p) = \sum_{h=0}^{\infty} \sum_{d=0}^{\infty} \sum_{n \in \mathbb{Z}} \mathbf{DT}^X_{n,(\beta_h,d)} q^{d-1} t^{\frac{1}{2}\langle \beta_h, \beta_h \rangle} (-p)^n.$$

| role | OP variable | DT variable | Borcherds variable |
|---|---|---|---|
| $K3$ norm $h-1$ | $q_{\mathrm{OP}}$ | $t_{\mathrm{DT}}$ | $q_{\mathrm{Bch}}$ |
| elliptic degree $d-1$ | $\widetilde{q}_{\mathrm{OP}}$ | $q_{\mathrm{DT}}$ | $s_{\mathrm{Bch}}$ |
| Do/Fourier index $n$ | $p_{\mathrm{OP}} = e^{iu}$ | $p_{\mathrm{DT}}$ with $(-p_{\mathrm{DT}})^n$ | $r_{\mathrm{Bch}}$ |

Proposition 11.1 *(Oberdieck–Pixton normalization).* Let the Oberdieck–Pixton variables be

$$p_{\mathrm{OP}} = e^{iu}, \qquad q_{\mathrm{OP}}, \qquad \widetilde{q}_{\mathrm{OP}}.$$

The variable map is

$$q = q_{\mathrm{OP}}, \qquad r = p_{\mathrm{OP}}, \qquad s = \widetilde{q}_{\mathrm{OP}}.$$

The product index set is

$$h, d \geq 0, \qquad h > 0 \text{ or } d > 0, \qquad \text{or} \qquad h = d = 0,\ k < 0.$$

The Laurent expansion is compared with the Borcherds cusp chamber after the substitution

$$0 < |\widetilde{q}_{\mathrm{OP}}| \ll |q_{\mathrm{OP}}| \ll |p_{\mathrm{OP}}|^{-1} \ll 1, \qquad r = p_{\mathrm{OP}}.$$

Equivalently, the OP Laurent expansion is read in the reciprocal $p_{\mathrm{OP}}$-direction needed by the Borcherds product. The enumerative identity is the Laurent expansion obtained after $p_{\mathrm{OP}} = e^{iu}$ and $y = -e^{iu}$. For the displayed DT variables,

$$q_{\mathrm{DT}} = \widetilde{q}_{\mathrm{OP}} = s, \qquad t_{\mathrm{DT}} = q_{\mathrm{OP}} = q, \qquad p_{\mathrm{DT}} = p_{\mathrm{OP}} = r.$$

The product denoted $\chi_{10}$ by Oberdieck–Pixton is

$$\chi_{10}^{\mathrm{OP}} = D_5^2 = \left(64^{-1}\Delta_5\right)^2 = 4096^{-1}\Delta_5^2.$$

Consequently their primitive reduced partition function is

$$Z_{\mathrm{OP}}^X = -\left(\chi_{10}^{\mathrm{OP}}\right)^{-1} = -4096\,\Delta_5^{-2}.$$

*Proof.* Oberdieck–Pixton use the primitive expansion [48, Theorem 1]

$$Z_{\mathrm{OP}}^X = \sum_{g,h,d} N_{g,h,d} u^{2g-2} q_{\mathrm{OP}}^{h-1} \widetilde{q}_{\mathrm{OP}}^{d-1}$$

and the stable-pairs substitution, in the coordinate convention of Oberdieck–Pandharipande [47, Section 2],

$$y = -p_{\mathrm{OP}}, \qquad p_{\mathrm{OP}} = e^{iu}.$$

There

$$\Omega = \begin{pmatrix} \tau & z \\ z & \widetilde{\tau} \end{pmatrix}, \qquad p_{\mathrm{OP}} = e^{2\pi i z} = e^{iu}, \qquad q_{\mathrm{OP}} = e^{2\pi i \tau}, \qquad \widetilde{q}_{\mathrm{OP}} = e^{2\pi i \widetilde{\tau}}.$$

The displayed DT convention uses

$$q_{\mathrm{DT}}^{d-1} t_{\mathrm{DT}}^{h-1} (-p_{\mathrm{DT}})^n,$$

and Bryan records that the DT convention interchanges the $q$ and $\widetilde{q}$ variables relative to Oberdieck–Pandharipande [10, Definition 2.1 and Conjecture 2.3]. This gives the stated variable map.

Oberdieck–Pixton define the Igusa product by [48, (3)]

$$\chi_{10}^{\mathrm{OP}}(p, q, \widetilde{q}) = pq\widetilde{q} \prod (1 - p^k q^h \widetilde{q}^d)^{c(4hd-k^2)}$$

in the chamber $h > 0$ or $d > 0$, together with $h = d = 0$, $k < 0$. The exponents are the K3 elliptic-genus coefficients. Since

$$Z_{\mathrm{K3}} = 2\phi_{0,1}, \qquad \phi_{0,1} = \sum f(n,l) q^n r^l,$$

those exponents are $2f(nm, l)$. Therefore the Oberdieck–Pixton monic product is exactly the square of

$$D_5 = 64^{-1}\Delta_5 = q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

Their Theorem 1, equivalently the Laurent expansion in [48, (56)], gives $Z^X_{\mathrm{OP}} = -1/\chi^{\mathrm{OP}}_{10}$. □

Theorem 11.2 *(Reduced DT/PT/OP scalar square).* Let $\beta_h \in H_2(S, \mathbb{Z})$ be primitive and non-zero, with

$$\langle \beta_h, \beta_h \rangle = 2h - 2.$$

Under the variable map

$$p = P = e^{iu}, \qquad t = Q, \qquad q = T, \qquad (q_{\mathrm{Bch}}, r_{\mathrm{Bch}}, s_{\mathrm{Bch}}) = (Q, P, T),$$

one has

$$Z^X_{\mathrm{DT}}(T, Q, P) = Z^X_{\mathrm{OP}}(P, Q, T) = -4096\, \Delta_5(Z)^{-2}.$$

The equality is an equality of Laurent expansions in the OP chamber of Proposition 11.1. It is the theorem-level OP/DT/PT scalar branch attached to the Behrend-weighted quotient integration of Proposition 5.2. The physical protected-index identification is Conjecture 5.3.

*Proof.* Oberdieck's reduced DT/PT theorem for $S \times E$, [46, Theorem 3(i)], gives, for every non-zero $\beta \in H_2(S, \mathbb{Z})$ and every $d \geq 0$,

$$\sum_n \mathbf{DT}^X_{n,(\beta,d)} x^n = \sum_n \mathsf{P}^X_{n,(\beta,d)} x^n.$$

Set $x = -P$. The degree-zero factor is

$$M(-P)^{\chi(S\times E)} = M(-P)^0 = 1.$$

For primitive $\beta_h$, the reduced GW/PT comparison of Oberdieck–Pandharipande [47, Proposition 5] gives

$$\sum_n \mathsf{P}^X_{n,(\beta_h,d)} (-P)^n = \sum_g \mathsf{N}^X_{g,h,d} u^{2g-2}, \qquad P = e^{iu}.$$

Multiplying by $Q^{h-1}T^{d-1}$ and summing over $h, d$ gives

$$Z^X_{\mathrm{DT}}(T, Q, P) = Z^X_{\mathrm{OP}}(P, Q, T).$$

Oberdieck–Pixton [48, Theorem 1] and Proposition 11.1 give

$$Z^X_{\mathrm{OP}}(P, Q, T) = -4096\, \Delta_5(Z)^{-2}.$$

□

*Definition* 11.3 *(OP normalization branch).* The OP/DT normalization branch is the proved enumerative series of Theorem 11.2, displayed as a square in the OP convention:

$$Z^X_{\square,\mathrm{OP/DT}} := Z^X_{\mathrm{DT}} = Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}, \qquad C_{\square,\mathrm{OP/DT}} := -4096.$$

The numerical constant $C_{\square,\mathrm{OP/DT}} = -4096$ is a *normalization* datum, not an enumerative invariant beyond the identification of the series itself: it factors as $-4096 = (-1) \times 64^2$, where the $64^2$ is the squared theta-leading coefficient $([q^{1/2}r^{1/2}s^{1/2}]\Delta_5)^2$ and the leading $-1$ is the OP scalar convention $Z^X_{\mathrm{OP}} = -(\chi_{10}^{\mathrm{OP}})^{-1}$. No additional enumerative content beyond the identification of $Z^X_{\mathrm{OP}}$ with $D_5^{-2}$ is carried by the constant. An external scalar-square series, if introduced later, is a Laurent expansion of the form

$$Z^X_{\square,\mathrm{ext}}(Z) = \frac{C_{\square,\mathrm{ext}}}{\Delta_5(Z)^2}, \qquad C_{\square,\mathrm{ext}} \in \mathbb{C}^\times,$$

and carries no OP/DT enumerative status unless its source is specified.

Proposition 11.4 *(Square-root consequence).* For $\mathfrak{s} \in \{\mathrm{OP}, \mathrm{DT}\}$, with variables normalised as in Theorem 11.2,

$$Z^X_{\mathfrak{s}}(Z) = -4096\,\Delta_5(Z)^{-2}.$$

Thus

$$\mathcal{D}_X(Z)^2 = \frac{-4096}{Z^X_{\mathfrak{s}}(Z)}, \qquad \mathcal{D}_X(Z) = \Delta_5(Z).$$

A choice of virtual square root is then

$$\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}} := \pm\mathcal{D}_X^{-1};$$

the sign is the orientation datum sought in Open Problem 5.7 and is not determined by the OP scalar. The squared scalar $Z^X_{\mathfrak{s}} = -4096(\mathcal{D}_X^{-1})^2$ does not fix this square-root sign because squaring kills it. With either sign choice,

$$Z^X_{\mathfrak{s}} = -4096\,(\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}})^2,$$

$$|\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}}| = \frac{1}{64}|q|^{-1/2}|r|^{-1/2}|s|^{-1/2}|\operatorname{sch}\mathcal{F}(\mathcal{V})|.$$

*Proof.* Theorem 11.2 gives the first display for $\mathfrak{s} = \mathrm{OP}$ and $\mathfrak{s} = \mathrm{DT}$. Hence

$$-4096/Z^X_{\mathfrak{s}} = \Delta_5^2.$$

The Borcherds determinant $\mathcal{D}_X = \Delta_5$ is an automorphic square-root section whose square is the scalar determinant $\Delta_5^2$. The opposite-sign section has the same square. Squaring this choice gives

$$Z^X_{\mathfrak{s}} = -4096(\mathcal{D}_X^{-1})^2 = -4096(\pm\mathcal{D}_X^{-1})^2,$$

the square-root sign being invisible to the scalar. Selecting the sign requires an orientation datum, deferred to Open Problem 5.7. □

*Remark* 11.5 *(Constants are not orientation data).* The number 64 fixes the theta-normalized Igusa section and hence the Weyl monomial in the determinant:

$$\Delta_5 = 64q^{1/2}r^{1/2}s^{1/2}\prod(1 - q^n r^l s^m)^{f(nm,l)}.$$

The monic denominator section is $D_5 = 64^{-1}\Delta_5$. The number 4096 is $64^2$, appearing because Oberdieck–Pixton use

$$\chi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2.$$

Read as a normalization conversion, the OP scalar constant factors as

$$-4096 \;=\; (-1)_{\mathrm{OP}} \cdot 64^2,$$

where the leading $-1$ is the OP scalar convention and $64^2$ is the squared theta-leading coefficient $([q^{1/2}r^{1/2}s^{1/2}]\Delta_5)^2$. It is not the square of an orientation sign; every square-root sign squares to $+1$. The minus sign in

$$Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1}$$

is the OP scalar convention. None of these constants is the Hall orientation character. The automorphic character $\nu_{\Delta_5}$ is fixed by the Maass transformation law, and the geometric character $\epsilon_o$, if it exists, must be produced by orientation-line monodromy as in Open Problem 5.7.

## 12 Coefficient dictionary and main synthesis

The coefficient convention is fixed once and for all by Eichler–Zagier normalization:

$$Z_{\mathrm{K3}} = 2\phi_{0,1}, \qquad \phi_{0,1}(\tau, z) = \sum_{n\geq 0,\, l\in\mathbb{Z}} f(n,l)q^n r^l.$$

Equivalently, if $\phi_{12,1}$ is the first Fourier–Jacobi coefficient of Igusa's weight-12 form $F_{12}$, and if

$$\delta_{12}(\tau) = q\prod_{n\geq 1}(1-q^n)^{24},$$

then

$$\phi_{0,1} = \frac{\phi_{12,1}}{\delta_{12}}.$$

The first terms are

$$\phi_{0,1} = (r^{-1} + 10 + r) + q(10r^{-2} - 64r^{-1} + 108 - 64r + 10r^2)$$
$$+q^2(r^{-3} + 108r^{-2} - 513r^{-1} + 808 - 513r + 108r^2 + r^3) + O(q^3).$$

Thus

$$f(0,-1) = 1, \qquad f(0,0) = 10, \qquad f(1,1) = -64, \qquad f(1,0) = 108.$$

The weak Jacobi support condition is

$$f(n,l) = 0 \quad \text{unless} \quad n \geq 0, \quad 4n - l^2 \geq -1,$$

and $f(n,l) = f(n,-l)$. These integers have three distinct readings in the paper:

| object | meaning of $f(nm,l)$ |
|---|---|
| $\mathcal{V}_{(n,l,m)}$ | virtual $K_0$-superdimension |
| $\mathcal{D}_X$ | Borcherds product exponent |
| $\mathfrak{g}_{\Delta_5,\alpha(n,l,m)}$ | signed root supermultiplicity on active support |

The first reading is a protected Euler characteristic. The second and third readings are automorphic and denominator-algebra incarnations of the same integer. This is not an equality of Hilbert spaces, brackets, or Hall structure constants.

The synthesis is the following spine:

virtual determinant $\longrightarrow$ physical charge and normal ordering $\longrightarrow$ denominator algebra $\longrightarrow$ reduced scalar square $\longrightarrow$ compact realization problem $\longrightarrow$ eight-row certificate extension.

Theorem 12.1 *(Coefficient dictionary and synthesis theorem).* Let $X = K3 \times E$. Let

$$\mathbb{U}_{n,l} \in K_0(\mathrm{sVect}_{\mathrm{fd}}), \qquad \mathrm{sdim}\, \mathbb{U}_{n,l} = f(n,l),$$

and define the virtual $K_0$-determinant object in the cusp chamber by

$$\mathcal{V}_{(n,l,m)} = \mathbb{U}_{nm,l}, \qquad (n,l,m) \in \Gamma_{\mathrm{eff}}.$$

First, the virtual determinant is the theta-normalized Igusa square root:

$$\mathcal{D}_X(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

The numerical factor 64 is the theta-constant leading coefficient $[q^{1/2} r^{1/2} s^{1/2}]\Delta_5$; the monic Borcherds product is

$$D_5 = 64^{-1}\Delta_5.$$

Second, the physical charge lattice is additive but its Igusa Gram shadow is quadratic. For

$$\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S, \qquad c = (Q,P),$$

the map

$$\Pi_X(c) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right)$$

satisfies

$$\Pi_X(c+c') = \Pi_X(c) + \Pi_X(c') + B(c,c').$$

The normal-ordered extension

$$\widehat{\Gamma}_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c,T) = \Pi_X(c) - T$$

is additive, and every formal Gram triple has a primitive saturated physical lift when $\Lambda_S$ contains two orthogonal hyperbolic planes. Raw $\Pi_X$-descent cannot realize the full BKM real-root strings; Theorem 6.12 is the obstruction.

Third, under the Lorentzian degree map

$$\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}, \qquad q^n r^l s^m = \exp(-\pi i(\alpha(n,l,m),z)),$$

the same active coefficients are the visible signed root supermultiplicities of the Gritsenko–Nikulin denominator algebra:

$$\mathrm{smult}\, \mathfrak{g}_{\Delta_5,\alpha(n,l,m)} = f(nm,l) \qquad ((n,l,m)\in\Gamma_{\mathrm{act}}),$$

and

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z).$$

This is the Gritsenko–Nikulin denominator theorem. It determines the signed root supermultiplicities in the fixed Borcherds datum; it does not determine compact Hall constants or microscopic brackets. The normal-ordered charge extension is the necessary degree-level bridge from an additive Hall source to these Gram degrees.

Fourth, for the reduced Hilbert-scheme DT/PT/OP scalar branch,

$$Z^X_{\mathrm{DT}}(T,Q,P) = Z^X_{\mathrm{OP}}(P,Q,T) = -4096\,\Delta_5(Z)^{-2}.$$

Here

$$\chi_{10}^{\mathrm{OP}} = D_5^2 = (64^{-1}\Delta_5)^2 = 4096^{-1}\Delta_5^2, \qquad Z^X_{\mathrm{OP}} = -(\chi_{10}^{\mathrm{OP}})^{-1};$$

hence $-4096 = -(64)^2$ is the OP scalar sign together with the theta-to-monic normalization. It is not an orientation character. Thus, with

$$\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}} = \mathcal{D}_X^{-1},$$

one has

$$Z^X_{\mathrm{DT}} = Z^X_{\mathrm{OP}} = -4096\,(\mathcal{Z}^{\mathrm{vir}}_{\mathrm{Borch}})^2.$$

Fifth, the stronger microscopic comparison

$$H^0\,\mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X)/\overline{\mathrm{Rad}\langle\,\cdot\,,\,\cdot\,\rangle_X} \cong \mathfrak{g}_{\Delta_5}$$

is open unless the Dirac–Igusa realization certificate of Definition 10.44 is constructed. If such a certificate is constructed, Theorem 10.45 identifies its primitive denominator with $64^{-1}\Delta_5(2Z)$.

Sixth, any claimed compact realization must also satisfy three consequences.

(i) A naive support-local factorization object on $\mathrm{Ran}(E)$ sees only projection-finite sectors. If a one-dimensional subscheme of $K3 \times E$ has positive elliptic degree, its projection to $E$ is all of $E$. Thus the full $s$-direction of the Igusa product requires finite-stage local/wrapped data: $b_R$, rigidified wrapped prequotients, mixed extension correspondences in both local/wrapped orders, wrapped/wrapped correspondences, the full eight-word flag atlas in $\mathsf{Corr}_R^{E,\mathrm{hyb}}$, quotient-after-correspondence $Q_{E,R}$-compatibility, and protected integration with trace degree $s^{b_R}$. Equivalently, $\mathfrak{O}_{\mathrm{hyb},R} = 0$ for every finite $R$. A separate global Fock/Hecke mode is only determinant-level unless it is coupled to the local sector by mixed Hall actions before the reduced $E$-quotient.

(ii) The automorphic character $\nu_{\Delta_5}$ is theorem-level data of $\Delta_5$, and $\Delta_5^2$ forgets it. A compact Hall orientation must construct a strong orientation character $\epsilon_o$ and prove

$$\epsilon_o|_{W^{(2)}(\Lambda^{2,1}_{II})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})}.$$

The constants 64, 4096, and the OP minus sign do not solve this orientation-monodromy problem. The Hall-side proof must compute the three simple type-II determinant-line signs and check the reflection relations.

(iii) A primitive comparison with $\mathfrak{g}_{\Delta_5}$ is a presentation-level recognition certificate, not a consequence of signed dimensions. One must supply primitive representatives for the real and imaginary simple roots, their parities, the Chevalley, real Serre and orthogonality relations, generation, full root-space parity dimensions, the completed Hopf pairing and the same closed radical quotient, a

no-extra-relations theorem, and completed PBW compatibility in the finite-height inverse system. Under these hypotheses the Borcherds–Kac universal presentation gives the isomorphism with $\mathfrak{g}_{\Delta_5}$; without them the determinant fixes only a Grothendieck class.

Finally, the diagonal-divisor rows extend this same logic. A row is promoted beyond its automorphic product only by a row certificate fixing the modular group or cover, character, divisor, Jacobi seed, product lattice, Weyl chamber, reflection character, scalar theorem if one is claimed, orientation data, one-particle index, and primitive algebra comparison.

*Proof.* The Fock trace calculation gives

$$\mathcal{D}_X = 64q^{1/2}r^{1/2}s^{1/2}\,\mathrm{sdet}_{\mathcal{V}}(1-x).$$

The Borcherds–Gritsenko–Nikulin product formula for the input $\phi_{0,1}$ identifies this product with $\Delta_5$, as recorded in Proposition 4.5. The leading coefficient 64 is fixed in the theta-normalized Fourier expansion of $\Delta_5$, hence $D_5 = 64^{-1}\Delta_5$ is the monic product.

The physical charge statements are Definition 6.5, Lemma 6.6, Lemma 6.8, Lemma 6.9, and Definition 6.11. They are direct lattice calculations in the Mukai pairing. The raw-descent obstruction is Theorem 6.12; it uses the non-zero second real-root strings recorded in Proposition 10.46.

The Lorentzian monomial identity and pairing computation are in Section 6 and the Weyl-chamber section. The root-supermultiplicity and denominator identity are Theorem 9.2; the argument $2Z$ comes from passing from $\exp(-\pi i(\alpha, z))$ to the usual denominator character $\exp(-2\pi i(\alpha, z))$.

The reduced scalar identity is Theorem 11.2, with the normalization conversion of Proposition 11.1. Since Oberdieck–Pixton use $\chi_{10}^{\mathrm{OP}} = D_5^2$ and prove $Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1}$, the theta normalization gives

$$Z_{\mathrm{OP}}^X = -4096\Delta_5^{-2}.$$

Oberdieck's reduced DT/PT comparison and the primitive Oberdieck–Pandharipande comparison give the same branch for $Z_{\mathrm{DT}}^X$.

The compact comparison is not used in these proofs. It is precisely Open Problem 10.50. The $\mathrm{Ran}(E)$ obstruction follows because a positive elliptic-degree curve has one-dimensional image in the irreducible curve $E$, hence image $E$; support separation on disjoint opens of $\mathrm{Ran}(E)$ cannot see such a sector. The orientation statement follows from the automorphic character computation for $\Delta_5$ and from the definition of a compact orientation datum; rescaling $\Delta_5$ by $64^{-1}$ changes no character. The primitive recognition statement is the universal property of the Borcherds–Kac presentation recorded in the denominator-algebra section, together with the closed Hopf-radical and completed PBW hypotheses in Theorem 10.45. The determinant and scalar square do not contain the missing presentation, orientation, radical, PBW, or Hall correspondence data. The row certificate condition is the same separation applied to the Clery–Gritsenko catalogue in the next part. □

# Part IV

# Diagonal-divisor rows and CHL boundary physics

## 13 Eight diagonal-divisor rows: automorphic and physical status

The Clery–Gritsenko catalogue gives eight diagonal-divisor automorphic products. The Gritsenko–Nikulin and Govindarajan theorems promote these products to denominator identities in the rows where the Weyl data are known. A CHL dyon square, a reduced CY/DT scalar host, and a compact Hall/chiral host are further structures, recorded only where their data have been specified.

$$g = 2.$$

$$t, N \in \mathbb{Z}_+ : \quad \Gamma_t(N) = \left\{ \begin{pmatrix} * & *t & * & * \\ & * & * & *t^{-1} \\ N & *Nt & * & * \\ Nt & *Nt & *t & * \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Q}) \mid * \in \mathbb{Z} \right\}.$$

$$\mathbb{H}_2/\Gamma_t \longleftrightarrow \{(1, t)\text{-polarized abelian surfaces}\}.$$

$$\mathcal{H}_{\text{diag}} = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mid a, b \in \mathbb{H}_1 \right\} \subset \mathbb{H}_2.$$

*Definition* 13.1 *(Physical-host certificate).* For a diagonal-divisor row $F_j$, a physical-host certificate is a five-level certificate table

$$\mathfrak{C}_j = \left( \mathscr{A}_j, \mathscr{D}_j, \mathscr{S}_j, \mathscr{Z}_j, \mathscr{H}_j \right).$$

The entries have the following meanings.

(i) $\mathscr{A}_j$ is the automorphic product certificate: the form $F_j$, the group or multiplier cover, the automorphy cocycle, the character or multiplier with exact order, the diagonal divisor with multiplicities, the product chamber and branch, the Jacobi input at every cusp, the product-degree map, the product normalization, and the source theorem.

(ii) $\mathscr{D}_j$ is the denominator certificate: a Weyl–Kac–Borcherds denominator algebra with root lattice, Weyl vector, simple real roots, Weyl chamber, Weyl group, reflection character, imaginary simple-root coefficients, product side, and Weyl alternating side.

(iii) $\mathscr{S}_j$ is a CHL/dyon-square certificate: a sourced dyon-counting scalar form $\Phi_j$, a row map to the same automorphic row and cover, and an identity $F_j^2 = \Phi_j$ in the stated chamber. This is not a denominator theorem and not a CY/DT partition theorem.

(iv) $\mathscr{Z}_j$ is a reduced CY/DT scalar-host certificate: a compact Calabi–Yau threefold, orbifold Calabi–Yau, or CHL background with oriented reduced obstruction theory, fixed variables and chamber, and a proved identity

$$Z_{\text{red}}^{X_j}(Z) = C_j F_j(Z)^{-2}.$$

(v) $\mathscr{H}_j$ is a compact Hall/chiral certificate: an oriented compact Hall, Drinfeld, or factorization object whose protected one-particle index gives the product exponents and whose protected primitive bracket identifies the denominator algebra with the same orientation/reflection character.

The product row, denominator row, dyon square, reduced scalar host, and compact Hall/chiral host are separate structures. None of the five entries implies another unless a cited theorem constructs the implication. In particular, matching a weight, a square root, or a comparison symbol does not fill a certificate. A row-map theorem must name the host, charge lattice, effective chamber, modular group or cover, product lattice, orientation character, one-particle index, scalar identity when claimed, and primitive Hall/chiral comparison.

Theorem 13.2 *(Clery–Gritsenko).* In the class of Siegel modular forms with character or multiplier system for the Hecke-type congruence subgroups $\Gamma_t(N)$ whose divisor is the union of the $\Gamma_t(N)$-translates of $\mathcal{H}_{\mathrm{diag}}$ with multiplicity one, there are exactly eight rows [28, Theorem 1.4]. The nine-candidate list is [28, Proposition 1.2]; the proof of Theorem 1.4 excludes only the ninth candidate $(t, N; k) = (2, 4; \frac{1}{2})$. The congruence Borcherds product with all cusp coefficients is [28, Theorem 3.1], with the final Section 3 product displays for $\nabla_3$, $\nabla_2$, $\nabla_{3/2}$, and $Q_1$.

| $j$ | $F_j$ | $\Gamma$ | $\mathrm{wt}(F_j)$ | character/multiplier |
|---|---|---|---|---|
| 1 | $\Delta_5$ | $\Gamma_1$ | 5 | $v_\eta^{12} \times v_H$ |
| 2 | $\Delta_2$ | $\Gamma_2$ | 2 | $v_\eta^{6} \times v_H$ |
| 3 | $\Delta_1$ | $\Gamma_3$ | 1 | $v_\eta^{4} \times v_H$ |
| 4 | $\Delta_{1/2}$ | $\Gamma_4$ | $\frac{1}{2}$ | $v_\eta^{3} \times v_H$ (degree 8) |
| 5 | $\nabla_3$ | $\Gamma_0^{(2)}(2)$ | 3 | $\chi_2^{(2)} \times v_H$ |
| 6 | $\nabla_2$ | $\Gamma_0^{(2)}(3)$ | 2 | $\chi_2^{(3)} \times v_H$ |
| 7 | $\nabla_{3/2}$ | $\Gamma_0^{(2)}(4)$ | $\frac{3}{2}$ | order-four multiplier $\chi_4^{(4)} \times v_H$ |
| 8 | $Q_1$ | $\Gamma_2(2)$ | 1 | $\chi_4^{(2)} \times v_H$ |

$$\operatorname{supp} \operatorname{div}_{\mathbb{H}_2}(F_j) = \bigcup_{\gamma \in \Gamma_j} \gamma\langle \mathcal{H}_{\mathrm{diag}} \rangle, \qquad \mathrm{mult}_{\mathcal{H}_{\mathrm{diag}}}(F_j) = 1.$$

$$F_1 = \Delta_5, \qquad \frac{1}{64} F_1(2Z) = \mathrm{den}(\mathfrak{g}_{\Delta_5}).$$

| row | automorphic product | denominator theorem | CHL/dyon square | reduced CY/DT scalar host | compact Hall/chiral host |
|---|---|---|---|---|---|
| 1 | $F_1 = \Delta_5$; Clery–Gritsenko product and Borcherds–Gritsenko–Nikulin lift of $\phi_{0,1}$ | Gritsenko–Nikulin denominator algebra $\mathfrak{g}_{\Delta_5}$; $64^{-1}\Delta_5(2Z)$ normalization | no CHL row map used; $\Delta_5^2$ is the Igusa scalar square, not a boundary dyon certificate | proved OP/DT $K3 \times E$ scalar host: $-4096\Delta_5^{-2}$ | open; the formal determinant is not a compact Hall/chiral source |
| 2 | $F_2 = \Delta_2$; Clery–Gritsenko product | Gritsenko–Nikulin denominator row $\mathfrak{g}_2^{\mathrm{GN}}$ | no certified CHL/dyon row map | no certified curve-counting scalar theorem | no certified Hall or factorization host |
| 3 | $F_3 = \Delta_1$; Clery–Gritsenko product | Gritsenko–Nikulin denominator row $\mathfrak{g}_3^{\mathrm{GN}}$ | no certified CHL/dyon row map | no certified curve-counting scalar theorem | no certified Hall or factorization host |
| 4 | $F_4 = \Delta_{1/2}$; Clery–Gritsenko product on the degree-eight multiplier cover | Gritsenko–Nikulin denominator row on the same cover $\mathfrak{g}_4^{\mathrm{GN}}$ | no certified CHL/dyon row map | no certified curve-counting scalar theorem | no certified Hall or factorization host |
| 5 | $F_5 = \nabla_3$; Clery–Gritsenko product | Govindarajan $\mathcal{G}_1(2)$ CHL/WKB denominator theorem | sourced scalar dyon square $\nabla_3^2 = \Phi_6$ | no reduced CY/DT certificate | no compact Hall/chiral certificate |
| 6 | $F_6 = \nabla_2$; Clery–Gritsenko product | Govindarajan $\mathcal{G}_1(3)$ CHL/WKB denominator theorem | sourced scalar dyon square $\nabla_2^2 = \Phi_4$ | no reduced CY/DT certificate | no compact Hall/chiral certificate |
| 7 | $F_7 = \nabla_{3/2}$; Clery–Gritsenko product on the order-four multiplier cover | Govindarajan $\mathcal{G}_1(4)$ CHL/WKB denominator theorem on the $\mu_4$-cover | sourced scalar dyon square $F_7^2 = \Phi_3$ only after fixing the cover and branch | no reduced CY/DT certificate | no compact Hall/chiral certificate |
| 8 | $F_8 = Q_1$; Clery–Gritsenko $(t, N) = (2, 2)$ product | Govindarajan $\mathcal{G}_2(4)$ CHL/WKB denominator theorem | sourced twisted dyon square $Q_1^2 = \Phi_2^{(2,4)}$; $Q_1$ is the special $\mathcal{G}_2(4)$ row | no reduced CY/DT certificate | no compact Hall/chiral certificate |

By Theorem B.7 the full-paramodular rows $F_2$, $F_3$, $F_4$ carry Gritsenko–Nikulin denominator identities. The cited sources do not construct a reduced curve-counting theorem, Hall object, or factorization host for these three rows. The congruence rows $F_5$, $F_6$, $F_7$, $F_8$ have product-degree lattices in Proposition B.9. The additional denominator data (Weyl chamber, real roots, reflection character, Weyl sum) are supplied by the CHL denominator theorem of Govindarajan for $F_5$, $F_6$, $F_7$, $F_8$. For $F_7$ the theorem lives on the $\mu_4$-multiplier cover determined by the Clery–Gritsenko order-four multiplier. For $Q_1$ the row is the special $\mathcal{G}_2(4)$ twisted CHL denominator row. Neither statement supplies a reduced CY/DT scalar host or compact Hall/chiral certificate.

Proposition 13.3 *(Extension separation).* No statement in this section is used in the $N = 1$ theorem for $\Delta_5$. For every row the product formula itself gives a Gram-indexed primitive product. It gives a Gram-graded Euler/Fock determinant only after a one-particle index and a charge-invariant interpretation have been supplied. A nonabelian denominator theorem is stronger: it additionally requires a Weyl chamber, real simple roots, a reflection character, and a Weyl alternating sum.

*Open Problem* 13.4 *(Certificates beyond the known rows).* For any diagonal-divisor or boundary product not already covered by the Gritsenko–Nikulin denominator theorems or by the Govindarajan CHL/WKB denominator theorem, first construct the product certificate

$$\mathscr{P}_j = (\phi_j, \Gamma^+_{j,\mathrm{prod}}, \Gamma_{j,\mathrm{mod}}, \rho_j, \nu_j, L_j, \alpha_j)$$

with

$$\phi^{(j)}(\tau, z) = \sum c_j(n, l) q^n r^l,$$

and then construct, if possible, the additional denominator data: positive product cone, Weyl chamber, simple real roots, reflection character, Weyl vector, Weyl alternating sum, imaginary simple-root coefficients, and a generalized Borcherds–Kac–Moody superalgebra $\mathfrak{g}_j$ such that

$$F_j(Z) = C_j e^{2\pi i(\rho_j, Z)} \prod_{\gamma \in \Gamma^+_{j,\mathrm{prod}}} (1 - e^{2\pi i(\gamma, Z)})^{c_j(\gamma)},$$

$$\mathrm{den}(\mathfrak{g}_j) = C'_j F_j(a_j Z), \qquad \mathrm{sdim}(\mathfrak{g}_{j,\gamma}) = c_j(\gamma).$$

If a CHL/K3 sector supplies a dyon-counting scalar square, it fills only the $\mathscr{S}_j$ level of Definition 13.1. If a reduced CY/DT scalar identity is claimed, one must separately construct $X_j$, the reduced obstruction theory, orientation, Fourier chamber, variables, and the theorem

$$Z^{X_j}_{\mathrm{red}}(Z) = C_j F_j(Z)^{-2}.$$

If a compact Hall/chiral host is claimed, one must separately construct the protected one-particle index and primitive bracket comparison.

Proposition 13.5 *(CHL squares and denominator signs).* The identities

$$\nabla_3^2 = \Phi_6, \qquad \nabla_2^2 = \Phi_4$$

which occur in the CHL dyon literature do not by themselves give denominator identities for the square-root rows $F_5 = \nabla_3$ and $F_6 = \nabla_2$. They give scalar Siegel modular forms on the CHL side. To descend from a scalar square to a denominator theorem for a root form, one must also construct the reflection

character, the parity, and the Weyl anti-invariant additive side. Govindarajan's twisted CHL theorem supplies exactly this missing structure for

$$F_5 = \nabla_3, \qquad F_6 = \nabla_2, \qquad F_7 = \nabla_{3/2}, \qquad F_8 = Q_1,$$

as recorded in Theorem B.13.

*Proof.* Clery–Gritsenko record the dd-forms $\nabla_3$ and $\nabla_2$ and note that their squares were constructed in the CHL papers as $\Phi_6$ and $\Phi_4$. The square forgets the sign character which a denominator identity needs on simple real reflections. Thus the scalar CHL product is evidence for the automorphic product, not by itself a proof of the BKM datum $(P_j, \mathcal{M}_j, W_j, \epsilon_j, \rho_j, N_j(a))$ for the root form. The additional datum is supplied by [24], building on [25] and [13]. □

A separate comparison tuple has weights

$$(5, 2, 1, 1, \tfrac{1}{2}, 1, \tfrac{1}{4}, 0).$$

It is not the Clery–Gritsenko table. A comparison with the Clery–Gritsenko forms requires the relevant entries of Definition 13.1. Thus

$$\text{Clery–Gritsenko classification} \neq \text{CHL realisation theorem.}$$

$$F_1 : \text{ theorem}, \qquad F_2, F_3, F_4 : \text{ GN denominator theorem},$$

$$F_5, F_6, F_7, F_8 : \text{ Govindarajan CHL/WKB denominator theorem},$$

with $F_7$ understood on the $\mu_4$-multiplier cover determined by its order-four multiplier. Dijkgraaf–Verlinde–Verlinde and Borcherds give the $N = 1$ physical and automorphic background [7, 20]; they do not construct physical-host certificates for the other seven rows.

## A Spin $L$-factor normalisation for $\Delta_5^2$

Set the theta-constant normalisation

$$\Delta_{10}^{\theta} := \Delta_5^2.$$

The Oberdieck–Pixton monic normalisation is

$$\chi_{10}^{\mathrm{OP}} := 2^{-12}\Delta_{10}^{\theta} = 4096^{-1}\Delta_5^2.$$

The spinor $L$-function is attached to the Hecke eigenline

$$\mathbb{C}\Delta_{10}^{\theta} = \mathbb{C}\chi_{10}^{\mathrm{OP}};$$

it is independent of the non-zero scalar used to name its generator. Put

$$\Delta(q) = q\prod_{n\geq 1}(1-q^n)^{24}, \qquad E_6(q) = 1 - 504\sum_{n\geq 1}\sigma_5(n)q^n,$$

$$g := \Delta E_6 = \sum_{n\geq 1} a_n(g)q^n \in S_{18}(\mathbf{SL}_2(\mathbb{Z})).$$

Theorem A.1 *(Andrianov factorisation).* For the Saito–Kurokawa representation generated by $\Delta_{10}^{\theta}$,

$$L_{\mathrm{spin}}(s, \Delta_{10}^{\theta}) = \zeta(s-8)\zeta(s-9)L(s, g).$$

Here $L_{\mathrm{spin}}$ is Andrianov's uncompleted classical spinor $L$-function. For a genus-two form of weight 10, the completed function is

$$L^*_{\mathrm{spin}}(s, F) = (2\pi)^{-2s}\Gamma(s)\Gamma(s-8)L_{\mathrm{spin}}(s, F).$$

*Proof.* The Igusa cusp form $\Delta_{10}^{\theta}$ spans the weight-10 Maass Spezialschar line attached by the Saito–Kurokawa correspondence to the unique normalized form $g = \Delta E_6 \in S_{18}(\mathbf{SL}_2(\mathbb{Z}))$ [59]. This is Andrianov's spinor $L$-factor formula for the degree-2 Saito–Kurokawa lift attached to the weight-18 elliptic cusp form $g$ [1]. Since $\dim S_{18}(\mathbf{SL}_2(\mathbb{Z})) = 1$, the source form is $\Delta E_6$. □

Proposition A.2 *(Central-zero cancellation).* With

$$\Lambda(g, s) := (2\pi)^{-s}\Gamma(s)L(s, g),$$

one has

$$\Lambda(g, s) = -\Lambda(g, 18-s), \qquad L(9, g) = 0.$$

Hence the possible pole of $\zeta(s-8)L(s, g)$ at $s = 9$ is cancelled. At $s = 10$ the residue is

$$\operatorname*{Res}_{s=10} L_{\mathrm{spin}}(s, \Delta_{10}^{\theta}) = \zeta(2)L(10, g) = \frac{\pi^2}{6}L(10, g),$$

with $L(10, g) \neq 0$, so $L_{\mathrm{spin}}(s, \Delta_{10}^{\theta})$ has a simple pole at $s = 10$ with non-zero residue.

*Proof.* For a level-one cusp form of weight 18, the root number is $(-1)^{18/2} = -1$. Thus the completed $L$-function is odd about $s = 9$, and the central value vanishes. The factor $\zeta(s-8)$ has its only possible pole at $s = 9$, so the product $\zeta(s-8)L(s, g)$ is regular there. At $s = 10$ the factor $\zeta(s-9)$ has residue 1, while $\zeta(s-8) = \zeta(2)$. Write

$$L_p(s, g) = (1 - \alpha_p p^{-s})^{-1}(1 - \beta_p p^{-s})^{-1}.$$

Deligne's Ramanujan–Petersson theorem gives

$$|\alpha_p| = |\beta_p| = p^{17/2}.$$

At $s = 10$,

$$|\alpha_p p^{-10}| = |\beta_p p^{-10}| = p^{-3/2},$$

so the Euler product converges absolutely and has no zero. □

Proposition A.3 *(Manin–Deligne period).* There are Manin periods $\Omega^{\pm}(g)$ such that

$$\varepsilon(m) = \begin{cases} +, & m \equiv 0 \pmod 2, \\ -, & m \equiv 1 \pmod 2, \end{cases} \qquad \frac{L(m, g)}{(2\pi i)^m \Omega^{\varepsilon(m)}(g)} \in \mathbb{Q}, \qquad 1 \leq m \leq 17.$$

In particular

$$L(10, g) \in \mathbb{Q}^{\times}(2\pi i)^{10}\Omega^{+}(g).$$

*Proof.* This is the Eichler–Shimura–Manin period theorem for the rational newform $g$ [43], [38], in Deligne's period normalisation [18]. The nonvanishing at $m = 10$ is supplied by the Euler product in Proposition A.2. No identity of the form

$$L(10, g) = q\, a_{10}(g)\Omega^-(g), \qquad q \in \mathbb{Q}^\times,$$

is used. □

Proposition A.4 *(Coefficient check).*

$$a_{10}(g) = 541648800 = 2^5 \cdot 3^2 \cdot 5^2 \cdot 7 \cdot 11 \cdot 977.$$

*Proof.*

$$a_{10}(g) = \tau(10) - 504 \sum_{k=1}^{9} \tau(k)\sigma_5(10-k).$$

$$(\tau(1), \ldots, \tau(10)) = (1, -24, 252, -1472, 4830, -6048, -16744, 84480, -113643, -115920),$$

$$(\sigma_5(1), \ldots, \sigma_5(9)) = (1, 33, 244, 1057, 3126, 8052, 16808, 33825, 59293).$$

Substitution gives $a_{10}(g) = 541648800$. Equivalently, since $g$ is a Hecke eigenform,

$$a_{10}(g) = a_2(g)a_5(g) = (-528)(-1025850) = 541648800.$$

□

*Remark* A.5 *(Scalar-square separations).*

| claim | failure | correct statement |
|---|---|---|
| $\chi_{10}^{\mathrm{OP}} = \Delta_5^2$ | normalisation conflict | $\chi_{10}^{\mathrm{OP}} = 2^{-12}\Delta_5^2$ |
| $L_{\mathrm{spin}} = \zeta(s-8)\zeta(s-9)L(s, g)$ | scalar normalisation | Hecke eigenline |
| unique pole from $\zeta(s-9)$ | $\zeta(s-8)$ also has $s = 9$ | $L(9, g) = 0$ |
| $L(10, g) \sim a_{10}(g)\Omega^-(g)$ | wrong period parity | $L(10, g) \sim \Omega^+(g)$ |
| arithmetic shadow $\Rightarrow$ BKM proof | wrong implication | appendix only |

## B Eight diagonal-divisor boundary rows

Let $F_j$ denote the diagonal-divisor forms in the present $F$-order. The Clery–Gritsenko source order is

$$\Delta_5, \Delta_2, \nabla_3, \Delta_1, \nabla_2, \Delta_{1/2}, \nabla_{3/2}, Q_1.$$

The present order is [28]

| $j$ | $F_j$ | $(t, N)$ | $\Gamma_j$ | wt |
|---|---|---|---|---|
| 1 | $\Delta_5$ | $(1, 1)$ | $\Gamma_1$ | 5 |
| 2 | $\Delta_2$ | $(2, 1)$ | $\Gamma_2$ | 2 |
| 3 | $\Delta_1$ | $(3, 1)$ | $\Gamma_3$ | 1 |
| 4 | $\Delta_{1/2}$ | $(4, 1)$ | $\Gamma_4$ | $\frac{1}{2}$ |
| 5 | $\nabla_3$ | $(1, 2)$ | $\Gamma_0^{(2)}(2)$ | 3 |
| 6 | $\nabla_2$ | $(1, 3)$ | $\Gamma_0^{(2)}(3)$ | 2 |
| 7 | $\nabla_{3/2}$ | $(1, 4)$ | $\Gamma_0^{(2)}(4)$ | $\frac{3}{2}$ |
| 8 | $Q_1$ | $(2, 2)$ | $\Gamma_2(2)$ | 1 |

The plus-groups $\Gamma_t^+$ used below for $t = 2, 3, 4$ are the Fricke/Gritsenko–Nikulin denominator normalizations; they are not the source-group labels in the Clery–Gritsenko table.

$$\operatorname{supp}\operatorname{div}_{\mathbb{H}_2}(F_j) = \bigcup_{\gamma\in\Gamma_j} \gamma\langle\mathcal{H}_{\mathrm{diag}}\rangle, \qquad \operatorname{mult}_{\mathcal{H}_{\mathrm{diag}}}(F_j) = 1.$$

Here $F_7 = \nabla_{3/2}$ is the Clery–Gritsenko row at $(t, N) = (1, 4)$. It is not a physical comparison slot until a separate row map has been constructed.

The characters and multiplier systems are

| $j$ | $\nu_j$ |
|---|---|
| 1 | $v_\eta^{12} \times v_H$ |
| 2 | $v_\eta^{6} \times v_H$ |
| 3 | $v_\eta^{4} \times v_H$ |
| 4 | $v_\eta^{3} \times v_H$ (degree 8) |
| 5 | $\chi_2^{(2)} \times v_H$ |
| 6 | $\chi_2^{(3)} \times v_H$ |
| 7 | $\chi_4^{(4)} \times v_H$ (order 4) |
| 8 | $\chi_4^{(2)} \times v_H$ |

*Definition* B.1 *(Candidate denominator proof datum).* For a row $j$, a candidate denominator proof datum consists of

$$\mathscr{P}_j = (\phi_j, \Gamma^+_{j,\mathrm{prod}}, \Gamma_{j,\mathrm{mod}}, \rho_j, \nu_j, L_j, \alpha_j),$$

where

$$\phi_j(\tau, z) = \sum c_j(n, l) q^n r^l,$$

$$F_j(Z) = C_j e^{2\pi i(\rho_j, Z)} \prod_{\gamma\in\Gamma^+_{j,\mathrm{prod}}} (1 - e^{2\pi i(\gamma, Z)})^{c_j(\gamma)},$$

$$\nu_j : \Gamma_{j,\mathrm{mod}} \to \mathbb{C}^\times, \qquad \operatorname{supp}\operatorname{div}(F_j) = \Gamma_{j,\mathrm{mod}} \cdot \mathcal{H}_{\mathrm{diag}}, \qquad \operatorname{mult}_{\mathcal{H}_{\mathrm{diag}}}(F_j) = 1,$$

and a degree map

$$\alpha_j : \Gamma^+_{j,\mathrm{prod}} \to L_j.$$

Only after a Weyl–Kac–Borcherds denominator theorem has been supplied does this datum include a Lie superalgebra $\mathfrak{g}_j$ with

$$\operatorname{sdim}(\mathfrak{g}_{j,\alpha_j(\gamma)}) = c_j(\gamma)$$

on the corresponding product side. For a product with several cusps this single-source notation is only a template. The congruence-row product datum is the cusp-labelled datum of Proposition B.9.

*Definition* B.2 *(Gritsenko–Nikulin degree lattice).* For $t \in \mathbb{Z}_{>0}$ set

$$M_t := U(4t) \oplus \langle 2\rangle = \mathbb{Z}f_2 \oplus \mathbb{Z}\widehat{f_3} \oplus \mathbb{Z}f_{-2},$$

with Gram matrix

$$\begin{pmatrix} 0 & 0 & -4t \\ 0 & 2 & 0 \\ -4t & 0 & 0 \end{pmatrix}.$$

For $(n, l, m) \in \mathbb{Z}^3$ put

$$\alpha_t(n, l, m) := nf_2 + l\widehat{f_3} + mf_{-2} \in M_t, \qquad e_t^{(n,l,m)} := q^n r^l s^{tm}.$$

Then

$$(\alpha_t(n, l, m), \alpha_t(n, l, m)) = -2(4tnm - l^2).$$

Let $\Gamma_t^{\mathrm{prod}}$ be the Gritsenko–Nikulin product cone: $(n, l, m) > 0$ if $m > 0$, or if $m = 0, n > 0$, or if $m = n = 0, l < 0$.

Lemma B.3 *(Two exponential normalizations).* Let

$$D_5(Z) := 64^{-1}\Delta_5(Z).$$

Gritsenko–Nikulin's Siegel normalization gives

$$D_5(Z) = q^{1/2}r^{1/2}s^{1/2} \prod_{(n,l,m)>0} (1 - q^n r^l s^m)^{f(nm,l)}$$

with leading monomial $q^{1/2}r^{1/2}s^{1/2}$. The denominator normalization used for the main BKM algebra is

$$q^n r^l s^m = \exp(-\pi i(\alpha(n, l, m), z)).$$

Thus replacing $Z$ by $2Z$ converts the product character to the usual $\exp(-2\pi i(\alpha, z))$ denominator character, and

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 64^{-1}\Delta_5(2Z).$$

For rows $F_2, F_3, F_4$, Appendix B uses the Gritsenko–Nikulin exponential

$$e_t^{(n,l,m)} = q^n r^l s^{tm}$$

directly. Hence

$$C'_j = 1, \qquad a_j = 1 \qquad (j = 2, 3, 4)$$

is a statement in the $e_t$ normalization. It is not the doubled $N = 1$ normalization of the main $\Delta_5$ denominator.

*Proof.* The first product is the singular-theta lift of Borcherds [6] together with the explicit paramodular product identity of Gritsenko–Nikulin [31, Theorem 2.1, (2.21)]; the Lorentzian Kac–Moody correction and the identification of $D_5(2Z)$ with the denominator function of the corrected Lorentzian BKM superalgebra is the content of [30], in particular the corrected denominator identity for $\mathfrak{g}_{\Delta_5}$. The rows $F_2, F_3, F_4$ are the separate $e_t^{(n,l,m)} = q^n r^l s^{tm}$ denominator identities of Gritsenko–Nikulin [31, Theorem 2.1, (2.3), (2.11), Lemma 2.5, (2.18), (2.19), Theorem 2.6, (2.20), (2.21), Section 5.1.1]. □

Proposition B.4 *(Full-paramodular rows).* Let

$$\phi_{0,2} = \sum f_2(n, l)q^n r^l, \qquad \phi_{0,3} = \sum f_3(n, l)q^n r^l, \qquad \phi_{0,4} = \sum f_4(n, l)q^n r^l,$$

where [31]

$$\phi_{0,2} = \frac{1}{2}\eta^{-4} \sum_{m,n\in\mathbb{Z}} (3m - n)\left(\frac{-4}{m}\right)\left(\frac{12}{n}\right)q^{(3m^2+n^2)/24}r^{(m+n)/2},$$

$$\phi_{0,3} = \left(\frac{\vartheta(\tau, 2z)}{\vartheta(\tau, z)}\right)^2, \qquad \phi_{0,4} = \frac{\vartheta(\tau, 3z)}{\vartheta(\tau, z)}.$$

Then the rows $F_2$, $F_3$, $F_4$ have the following product data:

| $j$ | $F_j$ | $\phi_j$ | $L_j$ | $\rho_j$ |
|---|---|---|---|---|
| 2 | $\Delta_2$ | $\phi_{0,2}$ | $M_2$ | $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ |
| 3 | $\Delta_1$ | $\phi_{0,3}$ | $M_3$ | $(\frac{1}{6}, \frac{1}{2}, \frac{1}{6})$ |
| 4 | $\Delta_{1/2}$ | $\phi_{0,4}$ | $M_4$ | $(\frac{1}{8}, \frac{1}{2}, \frac{1}{8})$ |

and

$$\alpha_j^{\mathrm{GN}}(n, l, m) := \alpha_t(n, l, m) \quad \text{for } (j, t) = (2, 2), (3, 3), (4, 4).$$

The product identities are

$$\Delta_2(Z) = q^{1/4} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_2^{\mathrm{prod}}} (1 - q^n r^l s^{2m})^{f_2(nm,l)},$$

$$\Delta_1(Z) = q^{1/6} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_3^{\mathrm{prod}}} (1 - q^n r^l s^{3m})^{f_3(nm,l)},$$

$$\Delta_{1/2}(Z) = q^{1/8} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_4^{\mathrm{prod}}} (1 - q^n r^l s^{4m})^{f_4(nm,l)}.$$

The characters are respectively induced by $v_\eta^6 \times v_H$, $v_\eta^4 \times v_H$, and $v_\eta^3 \times v_H$. Equivalently, on the extended paramodular groups,

$$\nu_2 = \chi_4 : \Gamma_2^+ \to \mu_4, \qquad \nu_3 = \chi_6 : \Gamma_3^+ \to \mu_6, \qquad \nu_4 = v_8 : \Gamma_4^+ \to \mu_8.$$

Thus for $(j, t) = (2, 2), (3, 3), (4, 4)$,

$$\Gamma_{j,\mathrm{prod}}^+ = \Gamma_t^{\mathrm{prod}}, \qquad \Gamma_{j,\mathrm{mod}} = \Gamma_t^+.$$

*Proof.* The product formulae are Gritsenko–Nikulin [31], Theorem 2.1 and formulae (2.21), (2.20), (2.11). Their Section 5 puts all formulae of Lie type on the hyperbolic lattice $M_t$ and writes the exponent as $e^{(n,l,m)} = q^n r^l s^{tm}$. The Weyl vector is

$$\rho_{t,II,\bar{0}} = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t}\right),$$

which gives the three displayed prefactors for $t = 2, 3, 4$. □

Proposition B.5 *(Gritsenko–Nikulin chambers for $F_2$, $F_3$, $F_4$).* For $t = 2, 3, 4$ the Weyl group is

$$W_{t,II,\bar{0}} = W^{(2)}(M_t),$$

and all simple real roots in this chamber are even. The Weyl vector is

$$\rho_t = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t}\right).$$

The simple real roots are

$$P_2 = \{(0,-1,0),\ (1,1,0),\ (1,3,1),\ (0,1,1)\},$$

$$P_3 = \{(0,-1,0),\ (1,1,0),\ (2,5,1),\\ (2,7,2),\ (1,5,2),\ (0,1,1)\},$$

and

$$P_4 = \{\delta \in M_4 \mid (\delta,\delta) = 2,\ (\delta,\rho_4) = -1\}.$$

Thus the automorphic denominator chamber for $F_j$ is the chamber in $M_t \otimes \mathbb{R}$ cut out by $(x,\delta) \le 0$ for $\delta \in P_t$, where $(j,t) = (2,2), (3,3), (4,4)$.

*Proof.* This is Gritsenko–Nikulin [31], Section 5.1.1, cases $(t, II, \bar{0})$. The displayed sets are exactly $P_{t,II,\bar{0}}(\mathcal{M}_{t,II,\bar{0}})$ for $t = 2,3,4$. □

Lemma B.6 *(Product cone and chamber compatibility).* Let $t = 2,3,4$. If

$$f_t(nm,l) \neq 0, \qquad (n,l,m) \in \Gamma_t^{\mathrm{prod}},$$

then

$$(\alpha_t(n,l,m),\rho_t) \le 0.$$

For $t = 2,3$ the inequality is strict. For $t = 4$, equality occurs only on the parabolic ray

$$\alpha_t(n,l,m) \in \mathbb{Q}_{>0}\rho_4.$$

*Proof.* The Jacobi support satisfies

$$4tnm - l^2 \ge -1.$$

Since

$$(\alpha_t(n,l,m),\rho_t) = -2n + l - 2m,$$

the chamber inequality is $l \le 2n + 2m$. For $t = 2,3$, Gritsenko– Nikulin Lemma 2.5 gives the required strict chamber condition on the active support. For $t = 4$, the proof of Gritsenko–Nikulin (1.33) gives the same condition, with equality only in the parabolic case. Equivalently, equality and $4tnm - l^2 \ge -1$ force

$$n = m, \qquad l = 4n,$$

so

$$\alpha_4(n,4n,n) = 8n\rho_4.$$

□

Theorem B.7 *(Gritsenko–Nikulin denominator rows).* Let $t = 2,3,4$ and put

| $t$ | $F(t)$ | $\phi_t$ |
|---|---|---|
| 2 | $\Delta_2$ | $\phi_{0,2}$ |
| 3 | $\Delta_1$ | $\phi_{0,3}$ |
| 4 | $\Delta_{1/2}$ | $\phi_{0,4}$ |

Let

$$\mathfrak{g}_t^{\mathrm{GN}} \quad (t = 2, 3, 4)$$

be the generalized Borcherds–Kac–Moody superalgebra whose Lorentzian root lattice is $M_t$, whose Weyl group is $W_t = W^{(2)}(M_t)$, whose Weyl vector is

$$\rho_t = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t}\right),$$

and whose simple real roots are $P_t = P_{t,II,\bar{0}}$ from Proposition B.5. Let $\mathcal{M}_t$ denote the corresponding chamber. Then

$$P_{t,\bar{1}} = \varnothing, \qquad P_{t,\bar{0}} = P_t,$$

so every simple real root is even and $\epsilon_t(s_\delta) = -1$ for every $\delta \in P_t$; equivalently $\epsilon_t(w) = \det(w)$ on $W_t$.

Let

$$\phi_t(\tau, z) = \sum f_t(n, l) q^n r^l.$$

The integers $N_t(a)$ are the additive-side coefficients in Gritsenko–Nikulin (5.1), equivalently the signed supermultiplicities of imaginary simple roots. They are not the product exponents $f_t(nm, l)$. In the Gritsenko–Nikulin exponential normalization $e_t^{(n,l,m)} = q^n r^l s^{tm}$, the Weyl–Kac–Borcherds denominator identity is

$$\begin{aligned}\operatorname{den}(\mathfrak{g}_t^{\mathrm{GN}}) &= \sum_{w \in W_t} \epsilon_t(w) \left( e_t^{w(\rho_t)} + \sum_{a \in \mathbb{R}_{++}\mathcal{M}_t \cap M_t} N_t(a) e_t^{w(\rho_t + a)} \right) \\ &= e_t^{\rho_t} \prod_{(n,l,m) \in \Gamma_t^{\mathrm{prod}}} \left(1 - e_t^{(n,l,m)}\right)^{f_t(nm,l)} = F(t)(Z).\end{aligned}$$

Equivalently,

$$\operatorname{den}(\mathfrak{g}_2^{\mathrm{GN}}) = \Delta_2(Z), \qquad \operatorname{den}(\mathfrak{g}_3^{\mathrm{GN}}) = \Delta_1(Z), \qquad \operatorname{den}(\mathfrak{g}_4^{\mathrm{GN}}) = \Delta_{1/2}(Z).$$

Thus, for rows $j = 2, 3, 4$,

$$C'_j = 1, \qquad a_j = 1, \qquad L_j = M_j, \qquad \alpha_j = \alpha_j^{\mathrm{GN}}$$

in this normalization. The imaginary root spaces are determined here only as signed multiplicity spaces:

$$\operatorname{sdim}(\mathfrak{g}_{t,\alpha_t(n,l,m)}) = f_t(nm, l)$$

on the product side, with imaginary simple-root superdimensions $N_t(a)$ on the Weyl-sum side.

*Proof.* Gritsenko–Nikulin [31, Section 5.1] state that almost all identities proved earlier in their paper give denominator formulae for Lorentzian Kac–Moody algebras. Their formula (5.1) is exactly the displayed Weyl–Kac–Borcherds identity on $M_t = U(4t) \oplus \langle 2 \rangle$, with $e^{(n,l,m)} = q^n r^l s^{tm}$.

For the cases $(t, II, \bar{0})$, $t = 2, 3, 4$, they give

$$W_{t,II,\bar{0}} = W^{(2)}(M_t), \qquad \rho_{t,II,\bar{0}} = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t}\right),$$

and

$$P_{t,II,\bar{0}}(\mathcal{M}_{t,II,\bar{0}}) = P_{t,II,\bar{0}}(\mathcal{M}_{t,II,\bar{0}})_{\bar{0}}.$$

Hence the odd simple-real-root set is empty and the Weyl character is the ordinary reflection sign on the simple real reflections. The roots in $P_t$ have square 2, and the generalized Cartan matrix is $((\delta, \delta'))_{\delta,\delta' \in P_t}$; for $t = 4$ this is the infinite symmetric parabolic matrix recorded by Gritsenko–Nikulin. For real roots $4tn - l^2 = -1$, the relevant Jacobi coefficient is 1: this is Gritsenko–Nikulin Lemma 2.5 for $t = 2, 3$ and the proof of Gritsenko–Nikulin (1.33) for $t = 4$.

The product factors are the full-paramodular products [31, (2.21), (2.20), (2.11)], already displayed in Proposition B.4. Their prefactors are exactly $e_t^{\rho_t}$. Therefore the denominator functions are $\Delta_2$, $\Delta_1$, and $\Delta_{1/2}$ with no further scalar or argument rescaling in the $e_t$ normalization. □

Proposition B.8 *(Congruence products).* For $j = 5, 6, 7, 8$ Clery–Gritsenko construct the following Borcherds products [28]. The weak Jacobi inputs are

| $j$ | $\phi_j$ | $\rho_j^{\mathrm{pref}}$ |
|---|---|---|
| 5 | $4(\xi_{1,0}^{(2)})^2 \in J_{0,1}^w(\Gamma_0(2))$ | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ |
| 6 | $3\xi_{3,1}^{(6)}\xi_{3,5}^{(6)} \in J_{0,1}^w(\Gamma_0(3))$ | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ |
| 7 | $2\xi_{2,1}^{(4)}\xi_{2,3}^{(4)} \in J_{0,1}^w(\Gamma_0(4))$ | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ |
| 8 | $2\xi_{1,0}^{(2)}(\tau, 2z) \in J_{0,2}^w(\Gamma_0(2))$ | $(\frac{1}{4}, \frac{1}{2}, \frac{1}{2})$ |

where $\rho_j^{\mathrm{pref}}$ is read from the literal product prefactor $q^{\rho_\tau} r^{\rho_z} s^{\rho_\omega}$. At this stage it is a prefactor exponent, not a proved Weyl vector. In particular, for $F_8$ the $M_2$ product coordinate corresponding to $s^{1/2}$ is $(1/4, 1/2, 1/4)$ because $e^{\alpha_2(a,b,c)} = q^a r^b s^{2c}$. More precisely, if

$$S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right),$$

and

$$\phi_5 = \sum c_5(n,l) q^n r^l, \qquad \phi_5|_{0,1} S = \sum c_{5,S}(n,l) q^n r^l,$$

then

$$\begin{aligned} F_5(Z) &= \nabla_3(Z) \\ &= q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_1^{\mathrm{prod}}} (1 - q^n r^l s^m)^{c_5(nm,l)} (1 - q^{2n} r^{2l} s^{2m})^{c_{5,S}(nm,l)}. \end{aligned}$$

If

$$\phi_6 = \sum c_6(n,l) q^n r^l, \qquad \phi_6|_{0,1} S = \sum c_{6,S}(n,l) q^n r^l,$$

then

$$\begin{aligned} F_6(Z) &= \nabla_2(Z) \\ &= q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_1^{\mathrm{prod}}} (1 - q^n r^l s^m)^{c_6(nm,l)} (1 - q^{3n} r^{3l} s^{3m})^{c_{6,S}(nm,l)}. \end{aligned}$$

If

$$\phi_7 = \sum c_7(n,l) q^n r^l, \qquad \phi_7|_{0,1} S = \sum c_{7,S}(n,l) q^n r^l, \qquad \phi_7|_{0,1} M = \sum c_{7,M}(n,l) q^n r^l,$$

where $M = \left(\begin{smallmatrix} 1 & -1 \\ 2 & -1 \end{smallmatrix}\right)$, then

$$\begin{aligned} F_7(Z) &= \nabla_{3/2}(Z) \\ &= q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_1^{\mathrm{prod}}} (1 - q^n r^l s^m)^{c_7(nm,l)} \\ &\qquad \cdot (1 - q^{2n} r^{2l} s^{2m})^{c_{7,M}(nm,l)/2} (1 - q^{4n} r^{4l} s^{4m})^{c_{7,S}(nm,l)}. \end{aligned}$$

The exponent $c_{7,M}/2$ is the literal middle-cusp exponent in the $\mu_4$-cover normalization; it is not a scalar-character exponent on the original congruence group. If

$$\phi_8 = \sum c_8(n,l)q^n r^l, \qquad \phi_8|_{0,2}S = \sum c_{8,S}(n,l)q^n r^l,$$

then

$$\begin{aligned} F_8(Z) &= Q_1(Z) \\ &= q^{1/4}r^{1/2}s^{1/2} \prod_{(n,l,m)\in\Gamma_2^{\mathrm{prod}}} (1-q^n r^l s^{2m})^{c_8(nm,l)}(1-q^{2n}r^{2l}s^{4m})^{c_{8,S}(nm,l)}. \end{aligned}$$

These four displays pin the automorphic products. By themselves they do not construct a BKM chamber, parity function, or Weyl-sum identity.

*Proof.* The formulae are Clery–Gritsenko [28, Theorem 3.1] and the four product displays for $\nabla_3$, $\nabla_2$, $\nabla_{3/2}$, and $Q_1$ at the end of Section 3. The second factors in $F_5$, $F_6$, $F_8$ and the second and third factors in $F_7$ come from the non-infinity cusps of $\Gamma_0(N)$ in their congruence Borcherds product theorem. □

Proposition B.9 *(Product degree lattices for $F_5$, $F_6$, $F_7$, $F_8$).* For the product rows $j = 5, 6, 7, 8$ set

$$(t_5, N_5) = (1,2), \qquad (t_6, N_6) = (1,3), \qquad (t_7, N_7) = (1,4), \qquad (t_8, N_8) = (2,2).$$

The Clery–Gritsenko modular host is the orthogonal lattice

$$O_j = U \oplus U(N_j) \oplus \langle -2t_j \rangle$$

of signature $(2,3)$. The single product-degree lattice for the monomial denominator is

$$L_j^{\mathrm{prod}} := M_{t_j}.$$

Define the cusp-labelled degree maps by

$$\alpha_5^{\mathrm{prod}}(n,l,m;\infty) = \alpha_1(n,l,m), \qquad \alpha_5^{\mathrm{prod}}(n,l,m;S) = \alpha_1(2n,2l,2m),$$

$$\alpha_6^{\mathrm{prod}}(n,l,m;\infty) = \alpha_1(n,l,m), \qquad \alpha_6^{\mathrm{prod}}(n,l,m;S) = \alpha_1(3n,3l,3m),$$

$$\alpha_7^{\mathrm{prod}}(n,l,m;\infty) = \alpha_1(n,l,m), \qquad \alpha_7^{\mathrm{prod}}(n,l,m;M) = \alpha_1(2n,2l,2m),$$

$$\alpha_7^{\mathrm{prod}}(n,l,m;S) = \alpha_1(4n,4l,4m),$$

and

$$\alpha_8^{\mathrm{prod}}(n,l,m;\infty) = \alpha_2(n,l,m), \qquad \alpha_8^{\mathrm{prod}}(n,l,m;S) = \alpha_2(2n,2l,2m).$$

The corresponding product-prefactor degrees are

$$\rho_5^{\mathrm{prod}} = \rho_6^{\mathrm{prod}} = \rho_7^{\mathrm{prod}} = \alpha_1(\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}), \qquad \rho_8^{\mathrm{prod}} = \alpha_2(\tfrac{1}{4},\tfrac{1}{2},\tfrac{1}{4}).$$

For $\lambda \in L_j^{\mathrm{prod}}$ let $\mu_j(\lambda)$ be the sum of all cusp coefficients whose labelled degree is $\lambda$:

$$\mu_j(\lambda) = \sum_{\alpha_j^{\mathrm{prod}}(n,l,m;\kappa)=\lambda} \widetilde{c}_{j,\kappa}(nm,l),$$

where

$$\widetilde{c}_{j,\infty} = c_j, \qquad \widetilde{c}_{5,S} = c_{5,S}, \qquad \widetilde{c}_{6,S} = c_{6,S},$$
$$\widetilde{c}_{7,M} = \frac{1}{2}c_{7,M}, \qquad \widetilde{c}_{7,S} = c_{7,S}, \qquad \widetilde{c}_{8,S} = c_{8,S}.$$

For $j = 7$ this is a product datum on the $\mu_4$-multiplier cover; the integrality of the displayed half-exponents is part of the Clery–Gritsenko product theorem in that cover. Until one proves that integrality directly from the Jacobi seed, the row is used only with this sourced cover datum. The product is therefore used as an integral automorphic product only in the sourced multiplier-cover normalization. Let

$$\mathcal{K}_5 = \mathcal{K}_6 = \mathcal{K}_8 = \{\infty, S\}, \qquad \mathcal{K}_7 = \{\infty, M, S\},$$

and

$$\mathcal{D}_j := \Gamma_{t_j}^{\mathrm{prod}} \times \mathcal{K}_j.$$

The cusp-labelled product datum is

$$\mathscr{E}_j^{\mathrm{prod}} := (\mathcal{D}_j, \alpha_j^{\mathrm{prod}}, \mu_j).$$

Then

$$F_j(Z) = e^{\rho_j^{\mathrm{prod}}} \prod_{\lambda \in \alpha_j^{\mathrm{prod}}(\mathcal{D}_j)} \left(1 - e^{\lambda}\right)^{\mu_j(\lambda)} \qquad (j = 5, 6, 7, 8),$$

where $e^{\alpha_1(a,b,c)} = q^a r^b s^c$ and $e^{\alpha_2(a,b,c)} = q^a r^b s^{2c}$.

*Proof.* The displayed maps are the unique monomial-degree maps forced by the three products in Proposition B.8. For $F_5$ the non-infinity cusp contributes the monomials $q^{2n} r^{2l} s^{2m}$, hence degree $\alpha_1(2n, 2l, 2m)$. For $F_6$ the corresponding monomial is $q^{3n} r^{3l} s^{3m}$, hence degree $\alpha_1(3n, 3l, 3m)$. For $F_7$ the two non-infinity factors have monomials $q^{2n} r^{2l} s^{2m}$ and $q^{4n} r^{4l} s^{4m}$; the first has half exponent $c_{7,M}/2$ exactly as in the Clery–Gritsenko product. For $F_8$ the product has index $t = 2$: $e^{\alpha_2(n,l,m)} = q^n r^l s^{2m}$ and the non-infinity cusp contributes $e^{\alpha_2(2n,2l,2m)} = q^{2n} r^{2l} s^{4m}$. If two labelled cusp contributions land on the same lattice degree, exponents add; this is precisely the definition of $\mu_j$.

Clery–Gritsenko [28, Theorem 3.1] realize these products as orthogonal Borcherds products for $U \oplus U(N_j) \oplus \langle -2t_j \rangle$. That theorem supplies the automorphic product and the divisor. It does not supply a Weyl chamber, simple real roots, a parity function, or a Weyl-sum identity on $\mathcal{M}_{t_j}$. Thus $L_j^{\mathrm{prod}}$ and $\alpha_j^{\mathrm{prod}}$ are product data, not a completed BKM denominator datum. The completed denominator datum for $F_5, F_6, F_7, F_8$ comes from the CHL theorem in Theorem B.13. □

*Definition* B.10 *(Formal product algebra of a row).* For a row $j$ with product-degree function $\mu_j$, let

$$\mathcal{V}_j^{\mathrm{prod}} = \bigoplus_{\lambda \in L_j^{\mathrm{prod}},\ \lambda > 0} \mathcal{V}_{j,\lambda}$$

be any graded super vector space with finite-dimensional homogeneous pieces and

$$\operatorname{sdim} \mathcal{V}_{j,\lambda} = \mu_j(\lambda).$$

Choose a Grothendieck representative: the even part has dimension $\mu_j(\lambda)$ when $\mu_j(\lambda) > 0$, the odd part of dimension $-\mu_j(\lambda)$ when $\mu_j(\lambda) < 0$, and the zero space when $\mu_j(\lambda) = 0$. Give it the abelian Lie superalgebra bracket:

$$[\mathcal{V}_{j,\lambda}, \mathcal{V}_{j,\lambda'}] = 0.$$

Its formal current envelope on $E$ is

$$\mathsf{Prod}_{j,E} := U_E^{\mathrm{ch}}\Big(\mathcal{V}_j^{\mathrm{prod}} \otimes \mathcal{O}_E\Big).$$

Proposition B.11 *(Eight formal current rows).* Every Clery–Gritsenko diagonal-divisor row has a formal determinant object. For $j = 1$ this is $\mathsf{Den}_{\Delta_5,E}$ from Proposition 10.2. For $j = 2, 3, 4$ it is the current envelope of the Gritsenko–Nikulin BKM algebra $\mathfrak{g}_j^{\mathrm{GN}}$. For $j = 5, 6, 7, 8$ it is the formal product algebra $\mathsf{Prod}_{j,E}$. For $j = 5, 6, 7, 8$, Theorem B.13 also supplies a nonabelian CHL/WKB denominator algebra; the abelian product algebra in this theorem is the part constructed directly from the Clery–Gritsenko product exponents. For $j = 5, 6, 7, 8$ this is only the tautological virtual product object realizing the Clery–Gritsenko exponents. Its Beilinson–Drinfeld envelope is abelian; it carries no Weyl-sum side, no BKM bracket, and no physical protected BPS interpretation.

In all eight cases the formal determinant is the corresponding monic automorphic product:

$$e^{\rho_j}\,\mathrm{sdet}_{\mathcal{V}_j}(1 - e) = D_j(Z),$$

where $D_1 = 64^{-1}\Delta_5$ and $D_j = F_j$ for the displayed Clery–Gritsenko product normalizations $j > 1$, with the normalizations stated in Propositions B.4, B.8, and B.9.

*Proof.* For $j = 1$, Theorem 9.2 gives the Borcherds denominator algebra and Proposition 10.2 gives the formal current envelope. For $j = 2, 3, 4$, Theorem B.7 gives the BKM brackets and their denominator identities.

For $j = 5, 6, 7, 8$, Proposition B.9 constructs the degree lattice, the positive product support, the prefactor, and the signed exponent $\mu_j$. The super vector space $\mathcal{V}_j^{\mathrm{prod}}$ realizes these signed dimensions. The Fock determinant lemma gives

$$e^{\rho_j^{\mathrm{prod}}} \prod_\lambda (1 - e^\lambda)^{\mathrm{sdim}\,\mathcal{V}_{j,\lambda}} = e^{\rho_j^{\mathrm{prod}}} \prod_\lambda (1 - e^\lambda)^{\mu_j(\lambda)} = F_j(Z).$$

The formal current enveloping construction gives a chiral algebra on $E$ from this abelian Lie superalgebra. It is not a compact BPS factorization algebra. □

Proposition B.12 *(Nonabelian promotion test for $F_5$, $F_6$, $F_7$, $F_8$).* Suppose that one of the congruence product rows

$$j \in \{5, 6, 7, 8\}$$

is the denominator of a generalized Borcherds–Kac–Moody superalgebra on the product lattice $L_j^{\mathrm{prod}}$. Then the following additional data must exist:

$$\Big(P_j, \mathcal{M}_j, W_j, \epsilon_j, \rho_j, N_j(a)\Big).$$

They must satisfy

$$W_j = \langle s_\delta \mid \delta \in P_j\rangle, \qquad \mathcal{M}_j = \{x \mid (x, \delta) \le 0 \text{ for all } \delta \in P_j\},$$

$$(\rho_j, \delta) = -\frac{(\delta, \delta)}{2} \qquad (\delta \in P_j),$$

and the Weyl–Kac–Borcherds identity

$$\begin{aligned} F_j(Z) &= \sum_{w \in W_j} \epsilon_j(w) \left( e^{w(\rho_j)} + \sum_{a \in L_j^{\mathrm{prod}} \cap \mathbb{R}_{>0} \mathcal{M}_j} N_j(a) e^{w(\rho_j + a)} \right) \\ &= e^{\rho_j^{\mathrm{prod}}} \prod_{\lambda \in \alpha_j^{\mathrm{prod}}(\mathcal{D}_j)} \left(1 - e^{\lambda}\right)^{\mu_j(\lambda)}. \end{aligned}$$

In particular the product already fixes the signed root multiplicity function $\mu_j$ and the product-prefactor degree

$$\rho_5^{\mathrm{prod}} = \rho_6^{\mathrm{prod}} = \rho_7^{\mathrm{prod}} = \alpha_1(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), \qquad \rho_8^{\mathrm{prod}} = \alpha_2(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}),$$

but it does not fix $P_j$, $\mathcal{M}_j$, $W_j$, $\epsilon_j$, or the imaginary simple-root coefficients $N_j(a)$.

*Proof.* The equalities in Proposition B.9 construct only the monomial degree and the total exponent $\mu_j(\lambda)$. A denominator theorem has a second side: the Weyl alternating sum. That side requires a reflection chamber, a set of simple real roots, a reflection character, and the imaginary simple-root contribution.

The Gritsenko–Nikulin chambers for $M_1$ and $M_2$ do not follow by notation. The forms $\nabla_3$, $\nabla_2$, $\nabla_{3/2}$, and $Q_1$ are congruence products with non-infinity cusp factors, whereas $\Delta_5$ and $\Delta_2$ are full-paramodular denominator forms. Clery–Gritsenko construct the automorphic product, character, and diagonal divisor on

$$U \oplus U(N_j) \oplus \langle -2t_j \rangle,$$

but their product theorem does not identify the simple real roots or prove a Weyl anti-invariant additive expansion on $M_{t_j}$. Therefore the displayed six-tuple is an additional theorem to be proved, not a consequence of the product formula. □

Theorem B.13 *(Govindarajan CHL/WKB denominator rows).* In the CHL/twisted-dyon normalization of Cheng–Dabholkar and Govindarajan, the rows

$$F_5 = \nabla_3, \qquad F_6 = \nabla_2, \qquad F_7 = \nabla_{3/2}, \qquad F_8 = Q_1$$

are Weyl–Kac–Borcherds denominator forms. The row $F_7$ is taken on the $\mu_4$-multiplier cover determined by the Clery–Gritsenko order-four multiplier. Thus $F_7$ is a denominator on the $\mu_4$-cover, with no physical row map, metaplectic boundary theorem, or CY/DT/chiral host. In the product-degree notation of this appendix the real-root data are

$$L_5 = L_6 = L_7 = M_1, \qquad P_5 = P_6 = P_7 = P^{(1)}, \qquad \rho_5 = \rho_6 = \rho_7 = \alpha_1(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}),$$

where

$$P^{(1)} = \{(0, -1, 0),\ (1, 1, 0),\ (0, 1, 1)\},$$

and

$$L_8 = M_2, \qquad P_8 = P^{(2)}, \qquad \rho_8 = \alpha_2(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}),$$

where

$$P^{(2)} = \{(0, -1, 0),\ (1, 1, 0),\ (1, 3, 1),\ (0, 1, 1)\}.$$

The Weyl chambers and Weyl groups are

$$\mathcal{M}_j = \{x \in L_j \otimes \mathbb{R} \mid (x, \delta) \leq 0 \text{ for every } \delta \in P_j\}, \qquad W_j = \langle s_\delta \mid \delta \in P_j \rangle.$$

Every simple real root is even and the reflection character is

$$\epsilon_j(w) = \det(w).$$

For $j = 5, 6, 7, 8$ the denominator product side is expressed through the cusp-labelled product datum $\mathscr{E}_j^{\text{prod}}$, with root multiplicity function $\mu_j$. In Govindarajan's CHL normalization the corresponding square-root form is anti-invariant under the Weyl reflections, with character $\epsilon_j(w) = \det(w)$. This anti-invariance is part of the Govindarajan WKB denominator theorem, not a consequence of the Clery–Gritsenko product alone; this manuscript does not construct an independent row-by-row embedding of the CHL Weyl group into each Clery–Gritsenko modular group or multiplier cover. The Cartan matrices are, in the CHL/WKB BKM normalization of [13, 24],

$$A^{(1)} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix},$$

for $F_5$, $F_6$, $F_7$, and

$$A^{(2)} = \begin{pmatrix} 2 & -2 & -6 & -2 \\ -2 & 2 & -2 & -6 \\ -6 & -2 & 2 & -2 \\ -2 & -6 & -2 & 2 \end{pmatrix},$$

for $F_8$.

Equivalently, for $x = (u, v, w)$, the two chambers are

$$\mathcal{M}_1 = \{x \mid v \geq 0,\ v \leq 2w,\ v \leq 2u\},$$

and

$$\mathcal{M}_2 = \{x \mid v \geq 0,\ v \leq 4w,\ 3v \leq 4(u + w),\ v \leq 4u\}.$$

There are integral imaginary simple-root coefficients $N_j(a)$, depending on the CHL twist [13, 24], such that

$$\begin{aligned} F_j(Z) &= \sum_{w \in W_j} \det(w) \left( e^{w(\rho_j)} + \sum_{a \in L_j \cap \mathbb{R}_{>0} \mathcal{M}_j} N_j(a) e^{w(\rho_j + a)} \right) \\ &= e^{\rho_j} \prod_{\lambda \in \alpha_j^{\text{prod}}(\mathcal{D}_j)} \left(1 - e^{\lambda}\right)^{\mu_j(\lambda)} \qquad (j = 5, 6, 7, 8). \end{aligned}$$

Thus the six-tuples

$$(P_j, \mathcal{M}_j, W_j, \epsilon_j, \rho_j, N_j(a))$$

are constructed for $j = 5, 6, 7, 8$ by the CHL denominator theorem. They are not consequences of the Clery–Gritsenko product formula alone. In the monic Clery–Gritsenko/Govindarajan product variables of Proposition B.8, the identities have

$$C'_j = 1, \qquad a_j = 1 \qquad (j = 5, 6, 7, 8).$$

This is not the doubled $N = 1$ normalization of Lemma B.3.

*Remark* B.14 *($F_7$ on the $\mu_4$-multiplier cover).* The row $F_7 = \nabla_{3/2}$ is the Clery–Gritsenko diagonal-divisor form of type $(t, N) = (1, 4)$, weight $3/2$, on $\Gamma_0^{(2)}(4)$ with an order-four multiplier. After pullback to the associated $\mu_4$-multiplier cover, Govindarajan's CHL/WKB theorem identifies it with the Weyl–Kac–Borcherds denominator of $\mathcal{G}_1(4)$. It is the $\mu_4$-cover denominator identity, with no physical comparison-row identification, no metaplectic boundary host, and no CY/DT or chiral/Hall theorem.

The square identity $F_7^2 = \Phi_3$ is only a scalar square identity. It does not carry the reflection sign, parity, Weyl chamber, or Weyl-alternating side. The coefficient convention for the additive side is fixed in Lemma B.16.

Lemma B.15 *(Coefficient normalisation for $\nabla_3$ and $\nabla_2$).* Let

$$\psi_5(\tau, z) = \vartheta(\tau, z)\eta(\tau)\eta(2\tau)^4 = \sum g_5(u, v) q^{u/2} r^{v/2},$$

and

$$\psi_6(\tau, z) = \vartheta(\tau, z)\eta(3\tau)^3 = \sum g_6(u, v) q^{u/2} r^{v/2}.$$

For $a = \alpha_1(A, B, C) \in M_1 \cap \mathbb{R}_{\geq 0}\mathcal{M}_1$, put

$$n = 2A + 1, \qquad l = 2B + 1, \qquad m = 2C + 1.$$

In the plus-sign convention of Theorem B.13, the imaginary simple-root coefficients for the rows $F_5$ and $F_6$ are

$$N_5(a) = \sum_{d|(n,l,m)} d^2\, g_5\left(\frac{nm}{d^2}, \frac{l}{d}\right),$$

and

$$N_6(a) = \sum_{d|(n,l,m)} \left(\frac{-3}{d}\right) d\, g_6\left(\frac{nm}{d^2}, \frac{l}{d}\right).$$

Govindarajan–Gopala Krishna write the same coefficients with the opposite sign in the imaginary-root correction term:

$$m_j(a) = -N_j(a) \qquad (j = 5, 6).$$

*Proof.* Govindarajan–Gopala Krishna construct the additive lift whose Weyl alternating side gives the denominator identity for the corresponding CHL square roots [25, (4.2), (4.4), (4.5), (4.8)]. Their Maass lift gives the displayed divisor sums from the Jacobi seeds $\psi_5$ and $\psi_6$, and their Weyl–Kac–Borcherds sum writes these coefficients as the imaginary simple-root correction. The sign difference is only the convention:

$$e^\rho + \sum_a N_j(a) e^{\rho + a},$$

whereas [25] writes the correction with a minus sign. □

Lemma B.16 *(Coefficient normalisation for $\nabla_{3/2}$).* Let

$$\nabla_{3/2}(Z) = \Delta_{3/2}^{(4)}(Z) = \frac{1}{8}\theta_5(2Z)\theta_7(2Z)\theta_9(2Z) = \sum_{n,l,m} A_7(n, l, m) q^{n/2} r^{l/2} s^{m/2},$$

with $A_7(1, 1, 1) = 1$. The first term is

$$\left(\sqrt{r} - \frac{1}{\sqrt{r}}\right)\sqrt{q}\,\sqrt{s} + \cdots,$$

so $A_7(1,-1,1) = -1$. For $a = \alpha_1(A,B,C) \in M_1 \cap \mathbb{R}_{\geq 0}\mathcal{M}_1$, put

$$n = 2A+1, \qquad l = 2B+1, \qquad m = 2C+1.$$

Then the imaginary simple-root coefficient in the plus-sign convention of Theorem B.13 is

$$N_7(\alpha_1(A,B,C)) = A_7(2A+1, 2B+1, 2C+1) = [q^{A+1/2} r^{B+1/2} s^{C+1/2}] \nabla_{3/2}(Z).$$

In Govindarajan–Gopala Krishna's WKB sign convention,

$$m_7(a) = -N_7(a).$$

No Maass divisor-sum formula for $N_7$ is asserted here.

*Proof.* Clery–Gritsenko construct $\nabla_{3/2}$ as the order-four multiplier row. Govindarajan–Gopala Krishna identify the square-root theta product and the square relation

$$F_7^2 = \Phi_3$$

in the composite $N = 4$ CHL case [26]. Factoring the Weyl vector gives

$$e^{\rho_7 + a} = q^{A+1/2} r^{B+1/2} s^{C+1/2},$$

which is exactly the displayed coefficient extraction. The sourced divisor-sum formula belongs to the square $F_7^2$, with its composite $N = 4$ Jacobi seed. A direct prime-type half-index formula for $F_7$ introduces fractional divisor powers at non-primitive chamber points, whereas the theta product has integral coefficients. This is a theta-product normalization identification. It is not a constructed physical host and not a physical row-map equivalence. □

Lemma B.17 *($Q_1$ in $M_2$ notation).* In $M_2$,

$$e^{\alpha_2(n,l,m)} = q^n r^l s^{2m}, \qquad (\alpha_2(n,l,m), \alpha_2(n',l',m')) = 2ll' - 8(nm' + n'm).$$

The four simple real roots of the $Q_1$ row are

$$\delta_1 = (0,-1,0), \quad \delta_2 = (1,1,0), \quad \delta_3 = (1,3,1), \quad \delta_4 = (0,1,1),$$

with Gram matrix $A^{(2)}$. The Weyl vector is

$$\rho_8 = \alpha_2(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}), \qquad e^{\rho_8} = q^{1/4} r^{1/2} s^{1/2},$$

and

$$(\rho_8, \delta_i) = -1 \qquad (1 \leq i \leq 4).$$

Let $A_8(n,l,m)$ be defined by

$$Q_1(Z) = \sum_{n,l,m} A_8(n,l,m) q^{n+1/4} r^{l+1/2} s^{2m+1/2}.$$

Govindarajan's theta-product formula, in his characteristic notation, is

$$Q_1(Z) = \frac{1}{4} \vartheta \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}(Z') \vartheta \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}(Z'), \qquad Z' = \begin{pmatrix} \tau & 2z \\ 2z & 4\sigma \end{pmatrix}.$$

Equivalently,

$$A_8(n,l,m) = \frac{1}{4} \sum_{\mathcal{S}(n,l,m)} (-1)^{j+k+h},$$

where $\mathcal{S}(n,l,m)$ is the set of $(i,j,k,h) \in \mathbb{Z}^4$ satisfying

$$\frac{1}{2}\big((i+\tfrac{1}{2})^2 + (k+\tfrac{1}{2})^2\big) = n + \tfrac{1}{4},$$

$$2(i+\tfrac{1}{2})j + 2(k+\tfrac{1}{2})(h+\tfrac{1}{2}) = l + \tfrac{1}{2},$$

and

$$2j^2 + 2(h+\tfrac{1}{2})^2 = 2m + \tfrac{1}{2}.$$

Then the imaginary simple-root coefficient in the plus-sign convention is

$$N_8(\alpha_2(n,l,m)) = A_8(n,l,m)$$

for the chamber-supported imaginary shifts. Govindarajan's expansion has coefficient $-1$ at the monomials

$$r^{-1}, \qquad qr, \qquad rs^2, \qquad qr^3s^2.$$

After factoring $e^{\rho_8}$ these are precisely the four simple real root monomials for

$$\delta_1, \quad \delta_2, \quad \delta_4, \quad \delta_3,$$

so the signed real-simple-root supermultiplicity is $+1$ in each case. Their even parity and multiplicity-one real-root data are the Cheng–Dabholkar/Govindarajan CHL denominator data.

*Proof.* The first display is the definition of $\alpha_2$ and the Gram matrix on $M_2$. The simple roots and Weyl vector are Cheng–Dabholkar's $N = 2$ chamber data translated through

$$\alpha_2(n,l,m) \longleftrightarrow \begin{pmatrix} 2n & l \\ l & 4m \end{pmatrix}.$$

Govindarajan identifies $Q_1$ with the row $\mathcal{G}_2(4)$ and gives both the product formula and the first Fourier terms. In that CHL normalization the square-root form is anti-invariant under the Weyl reflections. This is the sourced denominator anti-invariance, not an independently constructed restriction of the Clery–Gritsenko character through a physical row map. □

*Proof of Theorem B.13.* Cheng–Dabholkar identify, for CHL level $N = 1, 2, 3$, the Weyl group, simple real roots, Weyl chamber, and Weyl denominator identity of the dyon algebra [13, (2.10)–(2.16), (3.16), (3.22), (3.23), (3.26), (3.42), (5.1), (5.2), (5.8), (5.10), (5.12)–(5.15), Appendix A]. For $N = 1$ the simple real roots are $P^{(1)}$ and the Cartan matrix is $A^{(1)}$; for $N = 2$ the simple real roots are $P^{(2)}$ and the Cartan matrix is $A^{(2)}$. Govindarajan then treats twisted dyons and proves that, for fixed $N$, the algebras $\mathcal{G}_N(M)$ have the same real simple roots, Weyl vector, and Cartan matrix, while the imaginary simple roots depend on $M$ [24, Section 4.2, Section 4.3, Table 3].

Govindarajan's Table 3 gives

$$\nabla_3 \longleftrightarrow \mathcal{G}_1(2), \qquad \nabla_2 \longleftrightarrow \mathcal{G}_1(3), \qquad \nabla_{3/2} \longleftrightarrow \mathcal{G}_1(4), \qquad Q_1 \longleftrightarrow \mathcal{G}_2(4).$$

Section 4.2 of [24] checks the required ingredients: sum and product representations, integrality, correct multiplicity of the simple real roots, and the sign change of the square root under a Weyl reflection. Section 4.3 states the Weyl–Kac–Borcherds denominator formula for $\mathcal{G}_N(M)$. For $F_7 = \nabla_{3/2}$ the theorem is on the $\mu_4$-multiplier cover, with no physical boundary partition identity and no CY/DT/chiral host. For $Q_1$, Govindarajan supplies the WKB denominator data [24, Appendix C, Appendix D]. The dyon-counting additive-seed issue for the $(2,4)$ and $(4,2)$ examples plays no role. The later hyperbolic and umbral extensions of [27] are also outside this proof: for $N = 5, 6$ they replace $W$ by $W_+$, leave the sum side incompletely characterized, and treat some rows as product-only or candidate BKM data [27, Sections 3.3 and 4].

The real-root data translate to this notation as follows. On $M_t$,

$$((n,l,m),(n',l',m')) = 2ll' - 4t(nm' + n'm).$$

With $t = 1$, the set $P^{(1)}$ has Gram matrix $A^{(1)}$ and $\rho = \alpha_1(1/2, 1/2, 1/2)$ satisfies $(\rho, \delta) = -1$ for every $\delta \in P^{(1)}$. With $t = 2$, the set $P^{(2)}$ has Gram matrix $A^{(2)}$ and $\rho = \alpha_2(1/4, 1/2, 1/4)$ satisfies $(\rho, \delta) = -1$ for every $\delta \in P^{(2)}$. These are exactly the Weyl-vector equations $(\rho, \delta) = -(\delta, \delta)/2$ because all displayed real roots have square 2. □

*Remark* B.18 *(Nonabelian row data).* The row data are:

| $j$ | automorphic data $(F_j, \Gamma_j, \nu_j, \rho_j^{\mathrm{pref}})$ | lattice and degree datum |
|---|---|---|
| 1 | constructed here | constructed here |
| 2, 3, 4 | GN denominator theorem | $L_j = M_j,\ \alpha_j = \alpha_j^{\mathrm{GN}}$ |
| 5, 6, 7 | CG product; Govindarajan denominator | $P^{(1)}, M_1, W_1, \epsilon = \det$ (CHL) |
| 8 | CG product; Govindarajan denominator | $P^{(2)}, M_2, W_2, \epsilon = \det$ (CHL) |

Thus all eight rows have formal product/current-envelope objects. Rows 1–4 have the Gritsenko–Nikulin denominator algebras. Rows 5–8 have the Govindarajan CHL/WKB denominator algebras in the CHL normalization. The row $F_7$ lives on the $\mu_4$-multiplier cover and fills no physical-host certificate beyond the sourced denominator level.

*Open Problem* B.19 *(Physical-host certificates and row maps).* Construct, for every claimed physical realization of a Clery–Gritsenko row, a datum

$$(\iota_j, \mathscr{R}_j, \mathscr{A}_j, \mathscr{D}_j, \mathscr{S}_j, \mathscr{Z}_j, \mathscr{H}_j)$$

such that

$$\iota_j : \Phi_j^{\mathrm{phys}} \longleftrightarrow F_i^{\mathrm{CG}}$$

fixes the exact row, parent form when a root is used, group or cover, automorphy cocycle, multiplier and its order, divisor multiplicities, Jacobi seed at every cusp, product-degree lattice, chamber, branch, normalization, and source theorem. The entries $(\mathscr{A}_j, \mathscr{D}_j, \mathscr{S}_j, \mathscr{Z}_j, \mathscr{H}_j)$ are the five levels of Definition 13.1. For rows $F_5, \ldots, F_8$, the Govindarajan CHL/WKB theorem fills $\mathscr{D}_j$, and the sourced scalar square fills $\mathscr{S}_j$ where stated. It does not fill $\mathscr{Z}_j$ or $\mathscr{H}_j$. A boundary partition identity and a compact Hall/chiral source remain open until these certificate entries are constructed.

Proposition B.20 *(Comparison row systems).* The Clery–Gritsenko rows in the present $F_i$-order have weights

$$(5, 2, 1, \tfrac{1}{2}, 3, 2, \tfrac{3}{2}, 1).$$

In triple order

$$(t, N) = (1,1), (2,1), (1,2), (3,1), (1,3), (4,1), (1,4), (2,2),$$

the weights are

$$(5, 2, 3, 1, 2, \tfrac{1}{2}, \tfrac{3}{2}, 1),$$

and the permutation from triple order to present order is

$$(1, 2, 5, 3, 6, 4, 7, 8).$$

Consequently

$$\{j : \mathrm{wt}(F_j) \in \mathbb{Z}\} = \{1, 2, 3, 5, 6, 8\}, \qquad \{j : \mathrm{wt}(F_j) \notin \mathbb{Z}\} = \{4, 7\}.$$

The comparison notation

$$\Phi_1^{\mathrm{cmp}}, \dots, \Phi_8^{\mathrm{cmp}}, \qquad \mathrm{wt}(\Phi_j^{\mathrm{cmp}}) = (5, 2, 1, 1, \tfrac{1}{2}, 1, \tfrac{1}{4}, 0),$$

does not define the Clery–Gritsenko row system and defines no physical atlas in this paper. A row map

$$\Phi_j^{\mathrm{cmp}} \longleftrightarrow F_i$$

requires the relevant entries of Definition 13.1. Consequently the direct identifications

$$\Phi_5^{\mathrm{cmp}} = F_5, \qquad \Phi_7^{\mathrm{cmp}} = F_7, \qquad \Phi_8^{\mathrm{cmp}} = F_8$$

are false already by weight. The eight-row Clery–Gritsenko catalogue contains no weight-1/4 row and no weight-0 row. The CHL square identities, such as $\nabla_3^2 = \Phi_6$ and $\nabla_2^2 = \Phi_4$, are scalar square identities; they are not row maps and do not by themselves carry the denominator sign, parity, or Weyl alternating side.

*Proof.* The Clery–Gritsenko theorem gives the exact weights, groups, characters, and divisors of the eight diagonal-divisor rows [28]. The displayed comparison tuple is not that theorem table. The Borcea–Voisin route [5], [57] and higher-dimensional curve-counting routes [50], [49], [11] require separate construction of $X_j$, the reduced theory, the protected index, and the product identity. None is constructed here. Thus the comparison symbols remain external labels until a certificate identifies them with Clery–Gritsenko rows. □

*Definition* B.21 *(Canonical fractional branch).* Let

$$F(Z) = a\, q^{\alpha} r^{\beta} s^{\gamma} (1 + O(q, r, s)), \qquad a \in \mathbb{R}_{>0},$$

be a product expansion in a fixed positive Weyl chamber. Its canonical $k$-th root in that chamber is

$$F(Z)^{1/k} = a^{1/k} q^{\alpha/k} r^{\beta/k} s^{\gamma/k} \exp\Big(\frac{1}{k} \log(1 + O(q, r, s))\Big),$$

where $a^{1/k}$ is the positive real root and the logarithm is the formal logarithm in the completed product algebra. This defines a formal branch. It does not define a modular form until the relevant metaplectic or multiplier-cover character has been constructed.

Proposition B.22 *(Fractional-power comparison test).* The formal fractional-power comparison symbols

$$\Phi_5^{\mathrm{cmp}}, \qquad \Phi_7^{\mathrm{cmp}}, \qquad \Phi_8^{\mathrm{cmp}}$$

do not label Clery–Gritsenko rows. The only source-compatible statement retained here is the following conditional test.

The displayed comparison weight of $\Phi_5^{\mathrm{cmp}}$ is $1/2$. If it is presented as a square root, the parent form must have weight 1, and the root must live on the pullback of the parent group to the genus-two metaplectic cover with a genuine half-integral multiplier. A weight-5 parent would produce a weight-$5/2$ square root and does not certify this displayed comparison slot.

If the comparison symbol $\Phi_7^{\mathrm{cmp}}$ is supplied with its displayed comparison weight $1/4$, then it must be a genuine quarter-integral-weight form on a multiplier cover. If it is presented as a fourth root of a parent form with exact order-four character, an order-eight cover is not forced by character theory. If a character root $\psi$ with $\psi^4 = \chi$ exists after pullback and $\chi$ has exact order 4, then the character part has exact order 16, unless an explicit automorphy-cocycle relation reduces it. No such relation is constructed here. The $\mu_4$-cover of the Clery–Gritsenko row $F_7 = \nabla_{3/2}$, whose weight is $3/2$, is not a modular-root certificate for a comparison object $\Phi_7^{\mathrm{cmp}}$ of weight $1/4$. That comparison object needs its own parent, cover, cocycle, and character-root computation.

If a comparison form $\Phi_8^{\mathrm{cmp}}$ has weight 0, the canonical branch of Definition B.21 is only a formal logarithmic branch. By itself it gives no finite-order modular multiplier and no chiral square-root theorem.

*Proof.* The first two assertions are necessary metaplectic-cover conditions: a fractional power of an automorphic product is modular only after the automorphy factor, character, and divisor multiplicities lift to a genuine character on a cover; Weil and Shimura give the classical background for such metaplectic phenomena [58], [52]. The stated character-order obstruction is elementary: if $\chi$ has exact order 4 and $\psi^4 = \chi$, then $\psi$ cannot have order 8, since $\psi^4$ would have order at most 2; absent a further cocycle relation, order 16 is the first possible character order. In the weight-zero case the formal branch supplies no metaplectic weight contribution; modularity would have to come entirely from divisor divisibility and a character or automorphy cocycle root. A formal root can be chosen in a chamber, but this is not a modular or chiral construction. □

*Open Problem* B.23 *(Fractional comparison-row physical-host certificate).* For a fractional comparison row, construct a modular-root certificate $\mathscr{R}_j$ and the five-level physical-host certificate of Definition 13.1. The root certificate must fix the parent form, root exponent, exact group or cover, multiplier character and its order, automorphy cocycle, divisor multiplicities, branch, product normalization, and source theorem. The reduced scalar level, if claimed, must produce a compact Calabi–Yau, orbifold Calabi–Yau, or CHL background with oriented reduced obstruction theory and prove

$$Z_j^{\mathrm{red}} = C_j(\Phi_j^{\mathrm{cmp}})^{-2}$$

in the fixed chamber and cover. The compact Hall/chiral level must construct the protected one-particle index and primitive bracket comparison for the same row.

Borcea–Voisin and higher-dimensional curve-counting constructions [5], [57], [50], [49], [11] are possible sources for such a certificate. This paper does not assert that they supply it. Until $\mathscr{R}_j$ and the relevant certificate entries are constructed, the symbols $\Phi_5^{\mathrm{cmp}}, \Phi_7^{\mathrm{cmp}}, \Phi_8^{\mathrm{cmp}}$ are formal comparison symbols only: no Clery–Gritsenko row, finite-order multiplier, denominator identity, CY/DT partition theorem, or chiral power follows.

The following comparison table is restricted to the $N = 1$ Igusa boundary. It contains no row map

$$\Phi_j^{\mathrm{cmp}} \longleftrightarrow F_i$$

and no physical row-map denominator theorem for any congruence row. The separate Govindarajan CHL/WKB denominator theorem remains Theorem B.13.

For $X = K3 \times E$ the protected one-particle index is theorem-level, the virtual $K_0$-determinant is $\Delta_5$ in the theta normalization, and the same active half-coefficients are the visible signed root supermultiplicities of the Gritsenko–Nikulin denominator algebra $\mathfrak{g}_{\Delta_5}$; the reduced scalar branch is $-4096\,\Delta_5^{-2}$ by Oberdieck–Pixton. The compact observable algebra realizing these on $K3 \times E$, with orientation character $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$, is the open recognition target. The seven other Clery–Gritsenko diagonal-divisor rows extend the same separation: each row carries its automorphic product, and the levels $\mathscr{A}_j, \mathscr{D}_j, \mathscr{S}_j, \mathscr{Z}_j, \mathscr{H}_j$ of physical-host certificate are filled only where the cited literature supplies them.

## C $N = 1$ Boundary-Compatibility Conditions

Let a chambered Hall-descent datum supply the compact comparison data of Definition 10.44, so that the criterion of Theorem 10.45 applies. Its $N = 1$ decategorified boundary would then be constrained by the Igusa datum

$$\begin{aligned}\Gamma_{\mathrm{eff}} = \{(n,l,m) \in \mathbb{Z}^3 \mid m > 0,\ n \geq 0,\ l \in \mathbb{Z}\}\\ \cup \{(n,l,0) \in \mathbb{Z}^3 \mid n > 0,\ l \in \mathbb{Z}\}\\ \cup \{(0,l,0) \in \mathbb{Z}^3 \mid l < 0\} \subset \mathbb{Z}^3.\end{aligned}$$

| comparison Hall descent | Igusa boundary |
|---|---|
| Gram index lattice | $\Gamma_{\mathrm{ind}} = \mathbb{Z}^3$ |
| sector completion | $\Gamma_{\mathrm{eff}}$ |
| active denominator support | $\Gamma_{\mathrm{act}} = \{\gamma \in \Gamma_{\mathrm{eff}} \mid f(nm,l) \neq 0\}$ |
| Lorentzian degree | $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$ |
| orientation character | required geometric monodromy comparison $\epsilon_o\vert_{W^{(2)}} = \nu_{\Delta_5}\vert_{W^{(2)}}$ |
| automorphic boundary | $\mathrm{Borch}_\theta(\phi_{0,1}) = \Delta_5$ |
| BKM denominator | $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ |
| denominator algebra | $\mathcal{A}_{\mathrm{den}} = \mathfrak{g}_{\Delta_5}$ |
| formal current envelope | $\mathsf{Den}_{\Delta_5,E} = U_E^{\mathrm{ch}}(\mathfrak{g}_{\Delta_5} \otimes \mathcal{O}_E)$ |
| scalar square | $Z_{\mathrm{OP}}^X = Z_{\mathrm{DT}}^X = -4096\Delta_5^{-2}$ (proved). |

The degree map satisfies

$$(\alpha(\gamma), \alpha(\eta)) = -2\langle \gamma, \eta \rangle_{\mathrm{ind}}, \qquad (\alpha(n,l,m), \alpha(n,l,m)) = -2(4nm - l^2).$$

The real walls are

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

The Weyl vector is

$$\rho = f_2 - \frac{1}{2}f_3 + f_{-2}, \qquad (\rho, \delta_i) = -1.$$

These equations give the denominator polarisation of the same Gram-index lattice whose cusp polarisation gives the Fock determinant. The cone is generated by the active support. Triples in $\Gamma_{\mathrm{eff}}$ with zero Jacobi coefficient fix ordering and contribute no visible product factors. The product alone does not exclude invisible zero-superdimension root spaces.

The comparison is the $N = 1$ normalization

$$\mathrm{Borch}_\theta(\phi_{0,1}) = \Delta_5, \qquad \mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z), \qquad \kappa_{\mathrm{BKM}}(\mathfrak{g}_{\Delta_5}) = \mathrm{wt}(\Delta_5) = 5.$$

It contains no denominator theorem beyond the $N = 1$ theorem and no row map

$$\Phi_j^{\mathrm{cmp}} \longleftrightarrow F_i.$$

The protected scalar has the form

$$Z_\square^X = C_\square \Delta_5^{-2}.$$

The Oberdieck–Pixton and reduced Hilbert-scheme DT branches have constant $-4096$ in the quotient-by-$E$ normalization of Theorem 11.2. The chiral/denominator half is $\Delta_5$.

A compact Hall–Drinfeld double would realize a protected bracket from the compactification only when the comparison package of Definition 10.44 and Theorem 10.45 is supplied: finite-first oriented Hall descent, protected integration, charge additive Hall primitives, Borcherds relations inside the Hall super-commutator, and the completed Hopf pairing with orientation character required to restrict to $\nu_{\Delta_5}$. Homotopy descent must produce the positive half, its completion data, and the Hall-side domain on which the comparison character is defined. This orientation character is not produced by rescaling $\Delta_5$; it must come from monodromy of the chosen square root of the Hall determinant line. The Igusa boundary identifies the denominator algebra $\mathfrak{g}_{\Delta_5}$ and its protected decategorified multiplicities before those comparison data are proved.

For the Clery–Gritsenko rows $F_5$, $F_6$, $F_7$, $F_8$, this table contains no nonabelian chamber, no simple real roots, no reflection character, and no Weyl-sum side. Rows $F_5$, $F_6$, $F_7$, $F_8$ use the separate Govindarajan CHL/WKB denominator theorem when a nonabelian bracket is needed. For $F_7$ this is a theorem on the $\mu_4$-multiplier cover determined by the Clery–Gritsenko order-four multiplier. None of these denominator rows supplies a comparison row map or a CY/DT/chiral boundary host.

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